

CHIRAL KOSZUL DUALITY NONCOMMUTATIVE FACTORISATION D -MODULES, ORDERED QUANTUM FIELDS, AND THE GEOMETRY OF DEFECTS

E_1 -algebras in factorisable D -modules on ordered chiral configurations

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Abstract

The primitive object of this paper is an augmented E_1 -algebra in the symmetric monoidal category of factorisable right D -modules on a smooth complex curve X . We call it a *noncommutative ordered chiral algebra*. It carries two independent local geometries. The factorisation D -module contains the complete singular dependence of fields on X , including every principal part and its covariance under coordinate change. The internal E_1 multiplication records associative, generally noncommutative fusion in a transverse oriented line. In the regular-holonomic analytic realization, the same local structure is a product-generated factorisation algebra on $X^{\text{an}} \times \mathbb{R}$ that is holomorphic in X and locally constant along $\mathbb{R}$.

The Ran category carries two compatible monoidal products: pointwise tensor and disjoint-union chiral convolution. Their Beck–Chevalley interchange makes the relevant subcategory duoidal. A noncommutative ordered chiral algebra is equivalently a bimonoid whose pointwise multiplication is a morphism of factorisation coalgebras. Locality is therefore a bialgebra law between two different products. Its ordered bar inherits a coassociative deconcatenation coproduct and a commutative chiral factorisation coproduct, and the two coproducts commute by the same interchange law.

The natural indexing object is the operadic Grothendieck construction for $E_1 \otimes \text{Ch}_X$. It records an ordered sequence of entire finite chiral configurations, not a linear order on the points of one configuration. Straightening identifies its factorisation objects with internal E_1 -algebras. A Boardman–Vogt resolution is indexed by planar levelled forests with two edge colours, one for transverse composition and one for chiral collision. In every oriented cellular realization, the two edge-contraction differentials anticommute because their determinant lines are independent.

The ordered chiral bar construction is the ordinary simplicial bar performed in the coefficient category:

$$\text{Bar}_X^{\text{ord}}(A) = \left[[r] \mapsto \mathbf{1} \otimes \bar{A}^{\otimes r} \otimes \mathbf{1} \right] \simeq \int_{([0,1], \partial[0,1])} A.$$

Its face maps fuse adjacent transverse factors. The full chiral operator product remains in the coefficient D -module; no logarithmic residue is asked to recover a double, fourth, or arbitrary pole. The associahedral bar is a cofibrant model of this derived interval tensor product, not a termwise replacement of the simplicial bar. On connected positive-weight objects, and on their complete Mittag–Leffler limits, the ordinary associative bar–cobar theorem applies internally and reconstructs A . Separately, the cardinality filtration is pro-nilpotent for the chiral convolution product on $D(\text{Ran } X)$, where the enhanced Francis–Gaitsgory bar–cobar equivalence holds. The two theorems have different monoidal products and are related only by a specified monoidal comparison.

The boundary Koszul dual

$$A^! := \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$$

is the endomorphism algebra of the augmentation boundary condition, and the double dual is the derived augmentation completion for positive finite-type objects. If $C = \mathrm{Bar}_X^{\mathrm{ord}}(A)$, the universal twisting morphism gives an equivalence between augmentation-complete A -modules and conilpotent C -comodules. Its two-sided form sends the diagonal bimodule to the diagonal bicomodule C_Δ , carries derived tensor product to derived cotensor product, and identifies Hochschild theory with coHochschild theory. Thus the interval, the annulus, and the derived centre form one Koszul–Morita square. Quadratic Koszul duality, PBW deformation, curved nonhomogeneous duality, and Verdier duality are comparison constructions to this universal bar resolution; none is identified with another by notation.

Order does not contain exchange. Each chamber $t_1 < \dots < t_n$ of $\mathrm{Conf}_n(\mathbb{R})$ is contractible, so a bare E_1 algebra has no canonical braid operator. Exchange appears after an E_2 enhancement, a flat connection on a two-dimensional configuration space, or equivalent meromorphic tensor data. The internal Hochschild cochains

$$Z_{\mathrm{ord}, \mathrm{ch}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A)$$

form an E_2 algebra in factorisable D -modules and are the universal recipient of central actions. A rigid meromorphic line category with a faithful strong monoidal fibre functor reconstructs a coquasitriangular bialgebra, and a Hopf algebra when rigidity and the relevant continuous duals supply an antipode. A faithful quasi-monoidal fibre functor reconstructs instead a coquasi-bialgebra, or a quasi-Hopf algebra under the corresponding rigidity hypotheses. Yangians, quantum loop algebras, and elliptic quantum groups enter only through additional RTT, monodromy, or representation-theoretic recognition theorems. Knizhnik–Zamolodchikov transport and four-dimensional Chern–Simons theory provide exact examples of this passage.

A square-zero BRST derivation extends factorwise to the ordered bar and anticommutes with its bar differential. The resulting spectral sequence compares the bar of the derived BRST reduction with BRST cohomology of the bar. Quantum Drinfeld–Sokolov reduction is treated in this form; passage to an ordinary degree-zero W -algebra is made only under a cohomological concentration theorem.

There are two genera. A Calabi–Yau trace on the transverse E_1 algebra closes intervals into circles and produces Hochschild chains, annuli, ribbon graphs, and open topological surfaces. A modular sewing structure on the chiral coefficient glues pointed algebraic curves and produces conformal blocks over Deligne–Mumford moduli. We encode their simultaneous formal algebra by a two-edge-coloured Feynman transform. Its master equation is

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_X + \frac{1}{2}[\Theta, \Theta]_\perp + \hbar_X \Delta_X \Theta + \hbar_\perp \Delta_\perp \Theta = 0,$$

where the two loop contractions have different geometric sources and supercommute. Ribbon genus and chiral genus may be related by a physical duality, but they are not equal by definition. Curved Koszul data, determinant-line curvature, BRST anomaly, framing anomaly, and conformal anomaly likewise remain in their native deformation complexes until an explicit comparison identifies them.

The mathematical and physical descriptions meet through factorisation observables. A perturbatively renormalised holomorphic–topological BV theory on $X \times \mathbb{R}$ whose quantum master equation holds determines an ordered chiral algebra. Compactified configuration spaces resolve ultraviolet collisions; factorisation and the BV master equation identify the boundary restrictions of renormalised graph weights; a second topological direction supplies line exchange and the Yang–Baxter equation. The derived centre is the universal algebraic target of the bulk action, while equality with a physical bulk is a separate quasi-isomorphism theorem.

The paper develops this language through diagonals, pole filtrations, partition lattices, Fulton–

MacPherson screens, Ran spaces, ordered intervals, Hochschild and Chevalley constructions, surface braid groups, ribbon surfaces, stable curves, and determinant lines. The constructions are examined on tensor, quiver, matrix, Weyl, Clifford, and enveloping chiral algebras; Heisenberg and affine currents; $\beta\gamma$ and bc systems; Virasoro and principal W -algebras; KZ and KZB connections; Yangians; conformal blocks and Verlinde traces; Chern–Simons boundary algebras; Calabi–Yau pushforwards; Hall products; and Borchers automorphic products, always through the comparison theorem native to the example. Every numerical quantity appears only after a specified trace, index, monodromy, or comparison morphism has been constructed.

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Overture: five geometries of one quantum operation

The diagonal knows how fields may collide. The interval knows in which order operations may be composed. The plane knows how defects may pass one another. The circle knows which operations admit a trace. A nodal curve knows how states may be sewn. These are not metaphors. They are five geometric carriers, and each carrier has its own algebraic operation.

On a smooth complex curve X , the singular product is supported at a diagonal:

$$\mu : j_* j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta! \mathfrak{g}. \quad (1)$$

On an oriented line, ordered fusion is an associative operation

$$m : A \otimes A \longrightarrow A. \quad (2)$$

On a transverse plane, exchange is parallel transport in a configuration space and therefore a representation of a braid group. On a circle, closure is factorisation homology

$$\int_{S^1} A \simeq \mathrm{HH}_*(A). \quad (3)$$

On the boundary of the moduli space of curves, sewing is contraction along the edges of a stable graph and is governed by a modular master equation.

The subject becomes obscure whenever two of these carriers are identified before a map between them has been constructed. It becomes transparent when all five are retained. The basic object is then

$$A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}). \quad (4)$$

The factorisable D -module carries the whole chiral singularity. The E_1 structure carries transverse order. Exchange, trace, and sewing arise only after the corresponding geometric structures have been supplied.

Theorem 0.1 (Five-carrier theorem). *Let A be as in (4). Then the following constructions are functorial and have distinct sources.*

The chiral coefficient is the system of D -modules on powers of X with its factorisation maps and diagonal singularities. The transverse bar is the derived tensor product

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \mathbf{1} \otimes_A^L \mathbf{1}.$$

When the analytic realization is locally constant in the transverse direction and satisfies excision, this object is the factorisation homology of an interval with augmentation boundary labels. An E_2 enhancement of the transverse multiplication, or a flat meromorphic connection on two-dimensional configurations, supplies braid transport. A cyclic trace supplies the circle object and ribbon contractions. A clutching-compatible modular action supplies stable-curve sewing. No one of these last three structures is determined by the preceding one.

Proof. The first assertion is the definition of the coefficient category. The second is the simplicial bar identity; its interval realization is factorisation excision under the stated analytic hypotheses. The third follows because the little two-disks operad contains exchange paths whereas the little one-interval operad does not. The fourth follows because closing an interval requires evaluation against a trace.

The fifth follows because gluing two marked points requires a pairing and a compatible action of the clutching correspondences. The independence statements are witnessed by the standard examples: an arbitrary associative algebra has interval fusion without a braid; an E_2 algebra need not carry a trace; a cyclic algebra need not form conformal blocks over families of curves. $\square$

The theorem gives the following order of dependence:

$$\begin{aligned}
 &\text{chiral coefficient} \xrightarrow{E_1\text{-fusion}} \text{ordered chiral algebra}, \\
 &\text{ordered chiral algebra} \xrightarrow{E_2\text{-lift or transport}} \text{exchange category}, \\
 &\text{ordered chiral algebra} \xrightarrow{\text{cyclic trace}} \text{ribbon amplitudes}, \\
 &\text{chiral coefficient} \xrightarrow{\text{modular sewing}} \text{stable-curve amplitudes}.
 \end{aligned} \tag{5}$$

The last two extensions may coexist and commute; their genera remain independent.

The common law

All five geometries obey one elementary law. Codimension-one boundary strata carry compositions. Codimension-two corners carry the relations among those compositions. The oriented boundary of an oriented boundary is zero. Once coefficient maps have been assigned to the faces, this topological identity becomes an algebraic master equation.

For the interval, the two ends of a one-dimensional associahedral boundary give

$$(ab)c - a(bc) = 0.$$

For three chiral points, the partial diagonals give the chiral Jacobi identity after the coefficient operation is inserted. For a compactified configuration space of Feynman vertices, the same boundary calculus gives the classical or quantum master equation after graph weights and counterterms are inserted. For a stable curve, separating and nonseparating nodes give the bracket and loop terms of the modular equation. Topology supplies the relation. The field theory supplies the coefficient.

Theorem 0.2 (Stokes-to-master theorem). *Let $\overline{C}$ be an oriented stratified space whose codimension-one faces are labelled by elementary contractions, and let E be a coefficient complex. Suppose every oriented face F carries a degree-one contraction map ρ_F on the total coefficient complex E , compatible with restriction to corners and satisfying $d_E \rho_F + \rho_F d_E = 0$. Set*

$$D = d_E + \sum_F \rho_F.$$

Then the component of D^2 supported on a codimension-two stratum is the signed sum of the coefficient composites associated with its incident codimension-one faces. In particular, $D^2 = 0$ exactly when those corner sums vanish.

Proof. Expand D^2 . The square of d_E vanishes. The mixed terms are zero precisely when every face map is a chain map. The remaining terms are ordered pairs of incident faces. Their signs are the boundary orientations induced from $\overline{C}$. Grouping the ordered pairs by their common codimension-two stratum gives the stated corner sum. This is the cellular form of $\partial^2 = 0$ with coefficients. $\square$

The theorem is small enough to be missed and strong enough to organize the whole subject. It explains why the same algebraic shape appears in the bar differential, the Jacobi identity, the Yang–Baxter equation, the Batalin–Vilkovisky equation, and the modular sewing equation. It also explains why topology alone is insufficient: without the maps ρ_F , there is no coefficient equation to satisfy.

Six exact viewpoints

A singular quantum field is a distribution. Its support and wave-front behavior are part of the observable; they are not decorations on a formal product. The D -module in (1) is the algebraic expression of this fact.

A quantum amplitude composes by summing over intermediate states. The bar construction is the derived form of that composition. A bar word is an ordered sequence of intermediate boundary operations, and the bar differential fuses adjacent operations.

A conformal anomaly measures the failure of a quantum theory to descend from metrics to conformal structures without an invertible correction. Its geometric home is a determinant line, an Atiyah algebra, or an invertible field theory. It is not the square of the internal bar differential.

A topological field theory converts isotopy into algebra. Moving line defects in a plane gives braid transport; closing them gives ribbon traces. The braid and the trace are created by the topology of the transverse space.

A theory with boundaries and defects is naturally categorical. Boundary conditions are objects, boundary-changing fields are morphisms, interfaces are bimodules, and junctions are higher morphisms. The algebra (4) is the endomorphism algebra of one such object, not the whole category.

A perturbative quantum field theory is controlled by a Batalin–Vilkovisky complex and by extensions of graph weights across collision strata. When the quantum master equation holds, the renormalized observables form a factorisation algebra. When the theory is holomorphic along X and topological along a line, that factorisation algebra is the analytic realization of (4).

These viewpoints do not compete. They are the distributional, homological, anomalous, topological, categorical, and perturbative realizations of the same geometric syntax.

The invariant sentence

The complete statement of the subject is therefore concise:

A noncommutative chiral field theory on a curve with one ordered topological direction is an E_1 -algebra in factorisable D -modules.

Every other object in this paper is obtained from (6) by a named functor, a named geometric extension, or a named trace. The geometry determines what may be composed. The coefficient determines what is composed. The boundary identity determines the equation.

Part I

The noncommutative chiral object

1 The phenomenon and its invariant language

1.1 Two limits of the same experiment

Place local operators on a complex curve X and line defects on an oriented real line. There are two independent short-distance limits. In the first, two points of X approach one another while their transverse positions remain fixed. The product develops a finite principal part along the diagonal. In the second, two transverse intervals become adjacent while their positions on X remain separated. The product is associative and ordered. A theory in which both limits exist therefore carries two operations of different geometric origin:

$$\text{chiral collision on } X, \quad \text{ordered fusion on } \mathbb{R}. \quad (7)$$

The first is encoded by a factorisation D -module. The second is encoded by an E_1 -multiplication internal to the category of such modules.

This distinction is forced by topology. A punctured complex disc has a nontrivial fundamental group, so analytic continuation of insertions can produce monodromy. An ordered chamber

$$\text{Conf}_n^<(\mathbb{R}) = \{t_1 < \dots < t_n\}$$

is contractible, so order alone produces no exchange operator. The chiral singularity and the associative order are therefore independent data. A braid appears only after the transverse line is thickened to a plane.

Definition 1.1 (Ordered chiral algebra). Let X be a smooth complex algebraic curve and let $\text{FactD}(X)$ be the symmetric monoidal stable category of factorisable right D -modules defined in Section 2. An *ordered chiral algebra* on X is an augmented algebra

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)).$$

Its multiplication is denoted $m : A \otimes A \rightarrow A$, and its augmentation is denoted $\varepsilon : A \rightarrow \mathbf{1}$.

The word *chiral* belongs to the coefficient category. The word *ordered* belongs to the E_1 -operation. Neither adjective is redundant.

1.2 The diagonal and the interval

The diagonal is the carrier of the operator product expansion. Let

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2, \quad \Delta : X \hookrightarrow X^2.$$

A chiral singular product is a morphism from an extension off the diagonal to a distribution supported on the diagonal. In the Lie form of Beilinson and Drinfeld it is

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}. \quad (8)$$

A coordinate expansion of (8) produces all OPE modes. The morphism itself is coordinate-free.

The interval is the carrier of the ordered product. If $I_1 < I_2$ are disjoint subintervals of an oriented interval I , factorisation gives

$$A(I_1) \otimes A(I_2) \longrightarrow A(I).$$

Shrinking I_1 and I_2 without crossing them produces the binary multiplication. Three ordered subintervals produce the two composites $(ab)c$ and $a(bc)$; the contractibility of the space of such shrinkings gives their coherent homotopy. After fixing the endpoints and quotienting by affine reparametrisation, the standard real compactification of ordered points is an associahedron.

The two geometries meet on $X \times \mathbb{R}$. A small rectangle $U \times I$ carries a complex of observables. Chiral factorisation varies U in X ; ordered factorisation varies I in $\mathbb{R}$. Their interchange is the geometric origin of an E_1 -algebra in $\text{FactD}(X)$.

1.3 The unique language

A useful language must satisfy four requirements.

First, it must retain the entire singular product. A logarithmic residue detects a simple normal coefficient. A fourth-order Virasoro pole and a double affine pole require higher normal jets. A factorisation D -module contains all of them at once.

Second, it must retain noncommutative fusion before any symmetrisation. Tensor order, line order, and braid exchange are three different structures. The E_1 -multiplication retains the first two; an E_2 enhancement supplies the third.

Third, it must separate algebraic and analytic completion. The bar construction is defined in the stable category. Trace-class sewing and convergence of conformal blocks are additional analytic theorems.

Fourth, it must type every anomaly. Curvature of a determinant line, failure of a BRST differential, failure of a BV quantum master equation, and curvature in a curved Koszul dual are classes in different complexes. They agree only after a comparison morphism has been constructed.

These requirements lead to the following square:

$$\begin{array}{ccc} \text{Alg}_{E_1}(\text{FactD}(X)) & \xrightarrow{\text{Bar}_X^{\text{ord}}} & \text{Coalg}_{\text{conil}}(\text{FactD}(X)) \\ \text{an} \downarrow & & \downarrow \text{an} \\ \text{Fact}_{\text{hol,lc}}^{\text{rect}}(X^{\text{an}} \times \mathbb{R}) & \xrightarrow{\int_{([0,1],\partial)}} & \text{Fact}_{\text{hol}}(X^{\text{an}}). \end{array} \tag{9}$$

The upper row is the internal associative bar construction. For positive complete objects, cobar reconstructs it. The lower row is interval factorisation of a rectangular holomorphic–topological theory. On regular holonomic objects, Riemann–Hilbert identifies the vertical arrows. The square expresses one phenomenon in algebraic and analytic language.

1.4 What is genuinely noncommutative

A conventional factorisation algebra on a manifold has symmetric multiplication for disjoint opens because disjoint union is unordered. Noncommutativity enters when a codimension-one direction orders the opens. A boundary condition, an interface, or a family of parallel line defects supplies such

a direction. The resulting multiplication is not the chiral OPE. It is composition in the category of defects.

Let $\mathcal{C}$ be a factorisation category on X and let b be an object representing a boundary condition. The internal endomorphism object

$$A_b = \mathrm{RHom}_{\mathcal{C}}(b, b) \quad (10)$$

is an E_1 -algebra in the factorisation coefficient category. Composition of endomorphisms gives the E_1 product. Collision of their insertion points gives the chiral singularity. Formula (10) is the native source of noncommutative chiral algebras.

A simple algebraic model is obtained from a factorisation object V and an associative algebra B . If V is commutative as a factorisation object, then

$$A = B \otimes V$$

is an ordered chiral algebra with multiplication $m_B \otimes m_V$. Matrix-valued free fields and Chan–Paton factors have this form locally. The bar construction sees the noncommutative resolution of B while every term remains a chiral coefficient system on X .

1.5 The bar is transverse

The reduced bar construction of an augmented algebra is generated by ordered words

$$[a_1 | \cdots | a_r],$$

and its differential multiplies adjacent letters. The word order is the order of intervals on $\mathbb{R}$. Chiral singularities are not extracted by this differential. They are already present in each coefficient a_i and in the factorisation maps of $A^{\otimes r}$.

This observation removes the original ambiguity between collision residues and the associative bar differential. The de Rham differential of a configuration space, the residue differential along a collision divisor, and the simplicial bar differential are three distinct operators. They can coexist in a total complex, but their source and target degrees must be written separately.

Proposition 1.2 (Low-arity bar identity). *Let A be a dg E_1 -algebra in a stable symmetric monoidal category. On homogeneous elements of the augmentation ideal,*

$$b_2[a|b|c] = (-1)^\alpha[(ab)|c] + (-1)^\beta[a|(bc)],$$

with the standard suspension signs, and

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

Hence the multiplication part of the bar differential squares to zero exactly by associativity.

Proof. There are two length-two contractions of a word of length three. The suspension orientation gives opposite signs. The two coefficient operations agree by associativity. $\square$

The complete chiral identities are tested elsewhere: by the chiral operad on the formal disc, by factorisation on the Ran space, or by the partition collision complex. Associativity of the transverse product and the Borchers identity of the chiral coefficient are compatible but not identical.

1.6 From order to exchange

The distinction between order and exchange is the first physical test of the language. Parallel line defects in a topological line can be multiplied, but cannot pass through one another. A second topological direction allows an exchange path. The homotopy class of that path is a braid. Thus

$$E_1 \text{ fusion} \xrightarrow{\text{one more topological direction}} E_2 \text{ exchange.}$$

Dunn additivity identifies the second structure with an E_1 -algebra object in E_1 -algebras. The universal E_2 object acting on an E_1 algebra is its Hochschild centre.

This yields a precise physical dictionary:

ordered intervals	fusion of parallel defects,
chiral diagonals	singular OPE on the spectral curve,
second transverse direction	braid exchange,
Hochschild centre	universal bulk or exchange algebra,
Tannaka coend	reconstructed quantum symmetry.

No arrow in this table is an equality of vector spaces. Each arrow is a functor determined by geometry.

1.7 The two notions of modularity

A trace on an E_1 algebra closes an interval into a circle. Iterated gluing produces ribbon surfaces. This is the modularity of open topological field theory and Hochschild homology.

A chiral algebra on a varying algebraic curve has a different modularity. Plumbing a node changes the complex curve. The relevant spaces are $\overline{\mathcal{M}}_{g,n}$, and the amplitudes are conformal blocks. The first genus is transverse and ribbon; the second is complex and Deligne–Mumford.

When an ordered chiral algebra carries both sewing structures and their mixed interchange maps, the two contractions supercommute with the product orientation. The resulting master equation has two loop operators. The separation of the two loop parameters is essential in gauge theory: one counts topological colour loops, the other counts handles of the chiral curve.

1.8 Mathematics and physics

The mathematical object is an E_1 algebra in factorisable D -modules. The physical object is a holomorphic–topological factorisation algebra with a boundary or line defect. The comparison is not a metaphor. It is a morphism of factorisation algebras obtained by taking quantum observables and restricting them to the defect.

Let $\text{Obs}_{\mathcal{T}}$ be the quantum observables of a renormalised BV theory $\mathcal{T}$ on $X \times \mathbb{R}_{\geq 0}$. A boundary condition gives a factorisation algebra Obs_{∂} on $X \times \mathbb{R}$. When holomorphic dependence admits a factorisable D -module realization and the boundary theory is locally constant in the ordered direction, Theorem 10.5 gives

$$\text{Obs}_{\partial} \in \text{Alg}_{E_1}(\text{FactD}(X)). \quad (11)$$

The renormalized quantum master equation makes the observable differential square to zero and makes the factorisation maps compatible with ultraviolet contractions. A nonzero BV anomaly obstructs this quantum factorisation object.

The perturbative protected sectors of Chern–Simons theory, four-dimensional Chern–Simons theory, suitable holomorphic twists of supersymmetric gauge theories, and open string field theory furnish instances of this pattern after their boundary or defect observable comparisons are constructed. Their different quantum symmetries arise from different exchange geometries and spectral curves, not from different definitions of the bar construction.

1.9 The theorem spine

The paper proves the following sequence of statements.

- (i) Ordered block-Ran factorisation objects are equivalent to E_1 -algebras in $\text{FactD}(X)$.
- (ii) Their ordered bar is interval factorisation homology and has an associahedral cofibrant model.
- (iii) On connected positive-weight objects and their complete Mittag–Leffler limits, the internal associative bar and cobar reconstruct one another. Separately, Ran cardinality is pro-nilpotent for chiral convolution.
- (iv) The augmentation boundary has endomorphism algebra $A^!$, and the double dual is the augmentation completion.
- (v) An E_1 algebra has no canonical braid; its Hochschild centre is the universal E_2 algebra acting on it.
- (vi) Rigid meromorphic line categories reconstruct quantum symmetries by Tannaka coends.
- (vii) BRST derivations extend to the bar and yield a convergent comparison spectral sequence under boundedness and completeness.
- (viii) Cyclic transverse sewing and chiral curve sewing form a bi-modular master equation with two independent loop operators.
- (ix) Under the algebrisation, locality, gauge-fixing, and anomaly-cancellation hypotheses stated in Part V, factorisation observables realize the preceding constructions in perturbative holomorphic-topological field theory.

The remainder of the paper develops the geometry and the exact examples behind these statements. The purpose is not to place many subjects beside one another. It is to show that they are functorial realizations of a single ordered chiral object.

2 Factorisable D -modules and the complete chiral coefficient

2.1 The finite-set presentation

Let X be a smooth complex algebraic curve. Write Fin^{surj} for the category of nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines a diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \quad (x_j)_{j \in J} \longmapsto (x_{\alpha(i)})_{i \in I}.$$

The Ran prestack is the colimit

$$\mathrm{Ran}(X) = \operatorname{colim}_{I \in (\mathrm{Fin}^{\mathrm{surj}})^{\mathrm{op}}} X^I.$$

A right D -module on $\mathrm{Ran}(X)$ is represented by a compatible family $\mathcal{F}_I \in D\text{-mod}(X^I)$ with transition morphisms along the diagonals. All functors below are derived.

For a decomposition $I = I_1 \sqcup \cdots \sqcup I_r$, let

$$U(I_1, \dots, I_r) \subset X^I$$

be the open locus on which points belonging to different blocks are disjoint; points in one block may collide. Denote the inclusion by $j_{I_\bullet}$.

Definition 2.1 (Factorisable D -module). A factorisable right D -module $\mathcal{F}$ on X is a compatible Ran family $\{\mathcal{F}_I\}$ together with equivalences

$$j_{I_\bullet}^! \mathcal{F}_I \simeq j_{I_\bullet}^! (\mathcal{F}_{I_1} \boxtimes \cdots \boxtimes \mathcal{F}_{I_r}) \quad (12)$$

for every decomposition of I , compatible with refinement, permutation, and diagonal transition maps.

The equivalence (12) expresses independence on separated supports. The extension of $\mathcal{F}_I$ across partial diagonals carries the singular terms of the operator product. The definition uses neither a coordinate nor a propagator.

Let $\mathrm{FactD}(X)$ denote a stable presentable enhancement of this category. Its *coefficient tensor product* is the componentwise $!$ -tensor product

$$(\mathcal{F} \otimes^{\mathrm{pt}} \mathcal{G})_I := \mathcal{F}_I \otimes^! \mathcal{G}_I. \quad (13)$$

The unit is the vacuum factorisation module $\mathbf{1}$.

Proposition 2.2. *Assume that the chosen Ran enhancement is closed under componentwise $!$ -tensor product and componentwise colimits. Then (13) makes $\mathrm{FactD}(X)$ a presentable stable symmetric monoidal category, and tensor product preserves colimits separately.*

Proof. The assertion holds on each X^I . The diagonal compatibilities and the restrictions in (12) are preserved by $!$ -tensor product. The stated closure hypotheses therefore descend the componentwise structure to the factorisation subcategory. $\square$

This is the symmetric monoidal category in which the transverse E_1 multiplication is formed. It must be distinguished from the chiral tensor product on $D(\mathrm{Ran} X)$, introduced below.

2.2 The complete singular coefficient

Consider the binary component $\mathcal{F}_{\{1,2\}}$ on X^2 . Off the diagonal, factorisation identifies it with $\mathcal{F}_{\{1\}} \boxtimes \mathcal{F}_{\{2\}}$. The extension across the diagonal has local cohomology supported on $\Delta(X)$. In a formal coordinate it is expressed by principal parts

$$\sum_{m \geq 0} \frac{c_m(w)}{(z - w)^{m+1}}.$$

The coefficients depend triangularly on the coordinate; the filtered distribution does not.

For $D = \Delta(X) \subset X^2$, the pole filtration is

$$F_m \mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}(mD),$$

and

$$\mathrm{gr}_m^F \mathcal{O}_{X^2}(*D) \simeq N_{D/X^2}^{\otimes m}. \quad (14)$$

Thus the associated graded of a pole is a tensorial normal symbol; the filtered object retains the coordinate-dependent mixing of modes.

Principle 2.3 (Complete coefficient principle). *The chiral coefficient of an ordered chiral algebra is the factorisable D -module itself. Residues, normal jets, formal modes, and distributional kernels are realizations of this coefficient, not substitutes for it.*

For an ordered chiral algebra, the transverse multiplication

$$m : A \otimes^{\mathrm{pt}} A \longrightarrow A$$

is a morphism of factorisable D -modules. It therefore respects the whole diagonal filtration and every disjoint-factorisation map. In coordinates it acts on the complete singular expansion, not merely on the simple-pole coefficient.

2.3 Two Lie structures that must not be confused

Restriction along the operad map $\mathrm{Lie} \rightarrow E_1$ gives the graded commutator

$$[a, b]_{\perp} = m(a, b) - (-1)^{|a||b|} m(b, a). \quad (15)$$

Proposition 2.4 (Internal commutator). *Restriction of operads defines*

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}) \longrightarrow \mathrm{Alg}_{\mathrm{Lie}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}).$$

The bracket (15) satisfies the graded Jacobi identity.

Proof. It is the commutator bracket in an associative algebra object of a symmetric monoidal category. The usual cancellation proof of Jacobi is operadic and therefore applies internally. $\square$

The bracket $[-, -]_{\perp}$ is a Lie operation on the *transverse multiplication*. A Beilinson–Drinfeld chiral Lie bracket is instead a morphism

$$j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

in the chiral pseudo-tensor category. The factorisation coefficient of A may carry such a chiral operation, but it is not created by taking (15). A comparison between the two Lie structures is an additional morphism of operadic deformation problems. This distinction is essential in boundary field theory: the chiral bracket measures collision on X , while $[-, -]_{\perp}$ measures the discontinuity between two transverse orders.

2.4 Augmentations and boundary conditions

An augmentation $\varepsilon : A \rightarrow \mathbf{1}$ is a morphism of E_1 algebras in $\text{FactD}(X)$. It makes $\mathbf{1}$ a left and right A -module, and the augmentation ideal is

$$\bar{A} = \text{Fib}(A \xrightarrow{\varepsilon} \mathbf{1}).$$

For a right module R and a left module L , the two-sided bar $\text{Bar}(R, A, L)$ computes interval factorisation homology with endpoint labels R and L . The reduced bar is the case $R = L = \mathbf{1}$.

The module formulation is physically primary. A boundary condition is an object, not an augmentation map; its local endomorphisms form an E_1 algebra. The augmentation is the special boundary state represented by the unit.

2.5 Two monoidal products on the Ran space

There are two different tensor products in the subject.

The coefficient tensor $\otimes^{\text{pt}}$ of (13) is componentwise. Its support satisfies

$$\text{supp}(\mathcal{F} \otimes^{\text{pt}} \mathcal{G}) \subset \text{supp}(\mathcal{F}) \cap \text{supp}(\mathcal{G}).$$

It does not add the cardinalities of finite subsets. Consequently the cardinality filtration is not a pro-nilpotence filtration for this tensor product.

The *chiral tensor product* on $D(\text{Ran } X)$ is defined by the disjoint-union correspondence

$$\text{Ran}(X) \times \text{Ran}(X) \supset (\text{Ran} \times \text{Ran})_{\text{disj}} \longrightarrow \text{Ran}(X).$$

For reduced objects supported on subsets of cardinality at least r and s , respectively, chiral convolution is supported in cardinality at least $r + s$:

$$D(\text{Ran } X)^{\geq r} \otimes^{\text{ch}} D(\text{Ran } X)^{\geq s} \subset D(\text{Ran } X)^{\geq r+s}. \quad (16)$$

Theorem 2.5 (Ran chiral pro-nilpotence). *For the chiral tensor product $\otimes^{\text{ch}}$, the quotient by objects supported in cardinality $> N$ is nilpotent on reduced objects, and the cardinality completion is pro-nilpotent. This statement concerns $D(\text{Ran } X)$ with chiral convolution; it does not concern the pointwise tensor product of $\text{FactD}(X)$.*

Proof. Equation (16) implies that an $(N + 1)$ -fold chiral product of reduced objects is supported in cardinality at least $N + 1$ and hence vanishes in the N th quotient. Taking the inverse limit gives the pro-nilpotent category. This is the support argument used in chiral Koszul duality on the Ran space [38]. $\square$

The two products answer different questions. The pointwise tensor permits an E_1 algebra of defects with a fixed chiral coefficient. The chiral tensor assembles disjoint finite subsets and is the ambient product for the Francis–Gaitsgory Lie–commutative-coalgebra equivalence. Any comparison between their bar constructions must name the monoidal functor that relates them.

2.6 Regular holonomic realization

Assume that every $\mathcal{F}_I$ is regular holonomic and that the chosen subcategory is closed under the tensor operations used above. After the standard de Rham shifts, Riemann–Hilbert identifies $\mathcal{F}_I$ with a constructible complex on $X^{I,\text{an}}$. Factorisation equivalences become constructible factorisation equivalences on the disjoint loci.

Proposition 2.6 (Analytic realization). *Under these regular-holonomic closure hypotheses, Riemann–Hilbert gives a symmetric monoidal realization from $(\text{FactD}(X), \otimes^{\text{pt}})$ to constructible factorisation sheaves on X^{an} . It carries internal E_1 algebras, modules, bar constructions, and Hochschild cochains to their constructible realizations whenever it commutes with the required geometric realizations and internal Homs.*

The analytic realization reveals monodromy already present in a local system on $\text{Conf}_n(X)$. It does not manufacture a braid from the transverse E_1 order.

2.7 Factorisation categories and Morita origin

Let $\mathcal{C}$ be a D -linear factorisation category on X , and let b be a compatible object. Its internal endomorphisms are

$$A_b(U) = \text{RHom}_{\mathcal{C}(U)}(b|_U, b|_U).$$

Composition is ordered, while factorisation in U supplies the chiral coefficient.

Theorem 2.7 (Morita source). *Let $\mathcal{C}$ be a presentable D -linear factorisation category and let b be a dualisable factorisation object. Then $A_b = \text{RHom}_{\mathcal{C}}(b, b)$ is an E_1 algebra in $\text{FactD}(X)$. If b is compact and generates each local category, and these local Morita equivalences commute with factorisation descent, then*

$$\text{RHom}_{\mathcal{C}}(b, -) : \mathcal{C} \longrightarrow \text{Mod}_{A_b}(\text{FactD}(X))$$

is a factorisation Morita equivalence.

Proof. Composition gives the E_1 multiplication. Factorisation of b on disjoint opens gives the coefficient factorisation maps. The compact-generator Morita theorem applies locally; the final hypothesis identifies the local equivalences on overlaps and descends them. $\square$

This is the source encountered in gauge theory: a boundary condition or line defect is b ; local endomorphisms form A_b ; transverse composition is noncommutative; singular dependence on the holomorphic insertion point is carried by the factorisable D -module.

2.8 Historical position

Dirac’s distributional formalism made the singular support of products of quantum fields unavoidable. Wilson organised short-distance singularities by the operator product expansion. Borchers and Kac encoded their algebraic identities. Beilinson and Drinfeld replaced coordinate series by D -module morphisms on diagonals. Factorisation algebras expressed the same local-to-global law for quantum observables. The present object adds an oriented topological direction and retains its associative order before any exchange or symmetrisation is imposed.

3 The duoidal law: locality as a bialgebra identity

The primitive object admits a second, equivalent description which makes its geometry visible at once. The Ran category carries a pointwise tensor product and a convolution product along disjoint union. One product combines coefficients supported at the same finite subset; the other combines observables supported at disjoint finite subsets. Their compatibility is duoidal. An ordered chiral algebra is a monoid for the first product and a factorisation coalgebra for the second. The assertion that multiplication is a morphism of factorisation objects is therefore a bialgebra identity, but the multiplication and comultiplication belong to different monoidal structures.

3.1 The two products and their interchange

Let

$$\mathcal{D}_X = D(\mathrm{Ran} X).$$

Write $\otimes^{\mathrm{pt}}$ for the pointwise $!$ -tensor product. Write $\otimes^{\mathrm{ch}}$ for chiral convolution along the correspondence

$$\mathrm{Ran} X \times \mathrm{Ran} X \xleftarrow{j} (\mathrm{Ran} X \times \mathrm{Ran} X)_{\mathrm{disj}} \xrightarrow{u} \mathrm{Ran} X, \quad F \otimes^{\mathrm{ch}} G := u_! j^!(F \boxtimes G).$$

The diagonal of the Ran space is compatible with disjoint union. The projection formula and Beck–Chevalley base change therefore give a natural interchange morphism

$$(F_1 \otimes^{\mathrm{ch}} F_2) \otimes^{\mathrm{pt}} (G_1 \otimes^{\mathrm{ch}} G_2) \longrightarrow (F_1 \otimes^{\mathrm{pt}} G_1) \otimes^{\mathrm{ch}} (F_2 \otimes^{\mathrm{pt}} G_2), \quad (17)$$

with the middle factors permuted by the symmetry of $\otimes^{\mathrm{pt}}$. In a six-functor realization for which the relevant base-change and projection-formula maps are equivalences, (17) is an equivalence on the disjoint locus. Without those hypotheses it is the corresponding lax interchange map.

Theorem 3.1 (Duoidal Ran theorem). *Fix a sheaf theory on $\mathrm{Ran} X$ for which pointwise $!$ -tensor, chiral convolution, the projection formula, and disjoint-union base change are defined. Then*

$$(\mathcal{D}_X, \otimes^{\mathrm{pt}}, \otimes^{\mathrm{ch}})$$

is a lax duoidal stable category. On the full subcategory on which the base-change morphisms in (17) are equivalences, it is a strong duoidal category.

Proof. Associativity and symmetry of $\otimes^{\mathrm{pt}}$ are inherited componentwise from $D(X^I)$. Associativity of $\otimes^{\mathrm{ch}}$ is the associativity of union of pairwise disjoint finite subsets. Apply the two chiral coproduct correspondences, tensor their outputs pointwise, and then use the diagonal base-change square to regroup the first and third, and the second and fourth, factors. The resulting natural transformation is (17). The coherence diagrams reduce to the associativity, symmetry, and base-change diagrams of the six-functor formalism. This is precisely the definition of a lax duoidal category; invertibility of the interchange maps gives the strong form. $\square$

The theorem is an instance of the general theory of duoidal ∞ -categories [79]. Its particular content here is geometric: the two products are the diagonal product and disjoint-union convolution on the Ran space.

3.2 Factorisation coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\text{ch}}$. Its coproduct restricts, for disjoint finite subsets S and T , to a map

$$\Delta_{S,T} : F_{S \cup T} \longrightarrow F_S \boxtimes F_T.$$

It is a factorisation coalgebra when this map is an equivalence on every disjointness locus. Francis and Gaitsgory identify factorisation algebras with the corresponding full subcategory of chiral commutative coalgebras [38].

Let

$$\iota : \text{FactD}(X) \hookrightarrow \text{CoCAlg}^{\text{ch}}(\mathcal{D}_X)$$

denote this realization. The pointwise tensor of two factorisation coalgebras is again a factorisation coalgebra: its coproduct is obtained by applying (17) to their two coproducts.

Theorem 3.2 (Ordered chirality as a duoidal bimonoid). *Assume the strong duoidal hypotheses of Theorem 3.1 on the chosen factorisation subcategory. For an object $A \in \text{FactD}(X)$, the following data are equivalent:*

- (i) *an E_1 multiplication on A internal to $\text{FactD}(X)$;*
- (ii) *an E_1 monoid structure for $\otimes^{\text{pt}}$ on ιA such that its multiplication and unit are morphisms of chiral commutative coalgebras;*
- (iii) *a bimonoid in the duoidal category $(\mathcal{D}_X, \otimes^{\text{pt}}, \otimes^{\text{ch}})$ whose chiral coproduct is factorising on disjoint loci.*

The compatibility is the duoidal bialgebra identity

$$\Delta_{\text{ch}} \circ m = (m \otimes^{\text{ch}} m) \circ \chi \circ (\Delta_{\text{ch}} \otimes^{\text{pt}} \Delta_{\text{ch}}), \quad (18)$$

where χ is the interchange morphism.

Proof. An E_1 algebra in the full subcategory of factorisation coalgebras is, by definition, an E_1 multiplication whose structure maps remain in that subcategory. Saying that multiplication is a morphism of coalgebras is exactly (18); saying that the unit is a coalgebra map is the corresponding unit identity. These are the axioms of a bimonoid in a duoidal category. The factorisation condition is the requirement that the chiral coproduct become an equivalence on disjoint loci. $\square$

Equation (18) is the exact algebraic form of locality for the noncommutative theory. Separate two collections of insertions in the chiral curve and fuse them in the ordered direction. One obtains the same operation by fusing inside each separated region and then applying factorisation. The equation is not cocommutativity of m , and it is not an R -matrix relation.

3.3 The bar retains factorisation

The simplicial ordered bar is levelwise a chiral commutative coalgebra:

$$B_r(A) = \mathbf{1} \otimes^{\text{pt}} \bar{A}^{\otimes^{\text{pt}} r} \otimes^{\text{pt}} \mathbf{1}.$$

Every face is built from m or the augmentation. By Theorem 3.2, these are coalgebra maps. Every degeneracy is built from the unit and is likewise a coalgebra map.

Theorem 3.3 (Double-coalgebra bar theorem). *Suppose geometric realization in $\mathcal{D}_X$ preserves chiral commutative coalgebras and commutes with the pointwise tensor products appearing in the simplicial bar. Then*

$$C = \text{Bar}_X^{\text{ord}}(A)$$

has two compatible coalgebra structures:

- (i) *the coassociative deconcatenation coproduct of the E_1 bar;*
- (ii) *the commutative chiral factorisation coproduct inherited levelwise from A .*

Deconcatenation is a morphism of chiral factorisation coalgebras. Equivalently, C is a coassociative coalgebra object in chiral commutative coalgebras.

Proof. The simplicial bar is a simplicial object in chiral commutative coalgebras because every simplicial operator is a coalgebra morphism. Geometric realization gives the second coproduct. Deconcatenation is defined by cutting an ordered word. The chiral coproduct acts independently on each letter, so cutting and factorising commute by the interchange law. The resulting square is (18) applied to the iterated pointwise tensor powers of A . $\square$

This theorem explains why the ordered bar is simultaneously an interval object and a Ran-space coefficient. The coassociative coproduct remembers where the interval is cut. The chiral coproduct remembers where the curve is separated. Their compatibility is not extra structure added after the bar; it is inherited from locality of the original ordered chiral algebra.

3.4 A Fubini theorem

Let $p : X^{\text{an}} \times \mathbb{R} \rightarrow X^{\text{an}}$ and $q : X^{\text{an}} \times \mathbb{R} \rightarrow \mathbb{R}$ be the projections. Factorisation pushforward along p integrates the locally constant real direction; pushforward along q integrates the chiral direction when the corresponding chiral homology is defined. Product excision gives a Fubini comparison.

Theorem 3.4 (Ordered chiral Fubini). *Let A be an ordered chiral algebra whose analytic realization is rectangularly generated. Assume the relevant factorisation homologies exist and that the pushforwards preserve the geometric realizations of the bar. Then there are natural equivalences*

$$\int_X \text{Bar}_X^{\text{ord}}(A) \simeq \text{Bar}^{\text{ord}}\left(\int_X A\right), \quad (19)$$

$$\int_X \text{HH}_*(A) \simeq \text{HH}_*\left(\int_X A\right). \quad (20)$$

More generally, factorisation homology over a product rectangle may be computed in either order.

Proof. The bar is a geometric realization of the simplicial object $[r] \mapsto \bar{A}^{\otimes^{\text{pt}} r}$. Chiral factorisation homology is a colimit over chiral disks. Under the stated preservation hypotheses, the two colimits commute, giving (19). Hochschild chains are factorisation homology over the transverse circle, or equivalently the cyclic realization of the two-sided bar. Commuting that realization with chiral factorisation homology gives (20). On the analytic product, both assertions are the Fubini theorem for factorisation homology and follow from excision on product-disk covers. $\square$

The first equivalence says that one may integrate the chiral curve before or after resolving ordered fusion. The second says the same after the interval has been closed into a circle. In conformal field theory this is the equality between first forming the open-string trace and then taking chiral coinvariants, or first taking chiral coinvariants and then closing the ordered state space. Analytic convergence of either trace remains a separate theorem.

3.5 Primitives and quantum symmetry

The chiral factorisation coproduct in Theorem 3.3 is a separation map. It is not the coproduct of a quantum group. A quantum-group coproduct acts on tensor products of line representations and requires a tensor-compatible exchange theory. The two coproducts may be compared in a gauge-theoretic example, but neither determines the other.

The primitive part of the chiral coalgebra records connected support. The primitive part of the coassociative bar coalgebra records indecomposable interval states. Their intersection consists of states that are connected in the curve and indecomposable along the interval. This bidegree is the natural home of a binary collision kernel or a first line-defect interaction. An R -matrix arises only after an exchange functor transports these primitives around a two-dimensional collision.

3.6 Physical meaning

The two products of the duoidal category are two experimental operations. Pointwise tensor places two coefficient systems on the same chiral support and then fuses them in transverse order. Chiral convolution places them on separated supports and then forgets the separation by union. Equation (18) says that a local experiment performed in two separated laboratories is independent of whether fusion is described before or after the laboratories are separated.

This is the geometric content of locality in the ordered theory. It is stronger than associativity, because it constrains the dependence on support. It is weaker than braiding, because it contains no exchange path. It is the point at which Beilinson's factorisation language and the noncommutative language of defects become one sentence.

4 The block-ordered chiral configuration category

4.1 What is ordered

Let Ch_X denote a fixed operadic enhancement of the Beilinson–Drinfeld chiral pseudo-tensor structure whose algebras in the chosen derived system of right D -modules are the factorisable D -modules of Theorem 2.1. The noncommutative chiral operad is

$$\text{OCh}_X := E_1 \otimes \text{Ch}_X, \quad (21)$$

where $\otimes$ is the Boardman–Vogt tensor product of ∞ -operads. An algebra over (21) has a chiral factorisation structure and an associative transverse multiplication, with the interchange relation built into the operad.

The word *ordered* refers to the E_1 inputs. It does not impose a linear order on the points of one

chiral configuration. A typical operation is represented by an ordered sequence

$$(I_1, \dots, I_r), \quad (22)$$

where the finite set I_a labels the chiral insertions carried by the a th transverse interval. Points belonging to different blocks may have the same coordinate on X because their transverse coordinates are distinct. Collisions inside the chiral direction are carried by the D -module extension across diagonals; composition in the transverse direction is carried by E_1 .

4.2 The operadic Grothendieck construction

The block-ordered chiral configuration category $\mathbf{BRan}_X$ is the operadic Grothendieck construction of the associative operad acting on the chiral configuration category. Its objects are pairs

$$([r]; I_1, \dots, I_r),$$

with $[r]$ a finite ordinal and each I_a a finite chiral label set. Its generating morphisms are of three kinds:

- (i) chiral refinements, collisions, and permutations inside one I_a ;
- (ii) adjacent composition of two E_1 slots;
- (iii) insertion or removal of the unit slot.

The relations are the chiral relations, the simplicial identities of the associative nerve, and the interchange squares between the two kinds of operation.

The object space of a fixed block profile (22) is

$$X^{I_1} \times \dots \times X^{I_r}. \quad (23)$$

This product is a carrier for the operations, not by itself the monoidal product of $\text{FactD}(X)$. The coefficient morphism attached to an adjacent block composition is obtained from the E_1 multiplication after the chiral inputs have been placed in the common output profile. In a six-functor model, the required pull–tensor–push correspondence is part of the chosen operadic enhancement. There is no canonical theorem obtained by putting arbitrary D -modules on $\text{Ran}(X)^r$ and declaring the union map to be multiplication.

Definition 4.1 (Block-ordered factorisation D -module). A block-ordered factorisation D -module is an algebra over $\text{OCh}_X = E_1 \otimes \text{Ch}_X$ in the chosen derived D -module system. Equivalently, it is a coCartesian operadic section over $\mathbf{BRan}_X$ whose restriction to each block is chiral factorisable, whose restriction to the ordinal direction is Segal, and whose mixed squares satisfy interchange.

Write $\text{OrdFactD}(X)$ for this category. The definition includes the coefficient correspondences needed to compare the different spaces (23); it is therefore stronger than the datum of a simplicial family of sheaves on their underlying products.

Theorem 4.2 (Block-ordered recognition). *There are natural equivalences*

$$\text{Alg}_{E_1}(\text{FactD}(X)) \simeq \text{Alg}_{E_1 \otimes \text{Ch}_X}(\mathcal{V}_X) \simeq \text{OrdFactD}(X), \quad (24)$$

where $\mathcal{V}_X$ is the chosen finite-set D -module system presenting $\text{FactD}(X)$.

Proof. By definition, $\text{FactD}(X) \simeq \text{Alg}_{\text{Ch}_X}(\mathcal{V}_X)$. The exponential law for ∞ -operads gives

$$\text{Alg}_{E_1}(\text{Alg}_{\text{Ch}_X}(\mathcal{V}_X)) \simeq \text{Alg}_{E_1 \otimes \text{Ch}_X}(\mathcal{V}_X).$$

Unstraightening an algebra over the tensor-product operad gives a coCartesian section over its operadic Grothendieck construction. The restrictions to the two operadic factors give the chiral and Segal conditions; the tensor-product relation gives interchange. Straightening reverses the construction. $\square$

The theorem is the invariant meaning of factorisation over an ordered chiral configuration space. The block picture is a geometric presentation of the tensor-product operad, not a replacement for it.

4.3 The strict and homotopy-coherent forms

In a strict dg model, an ordered chiral algebra is a factorisable D -module A with maps

$$m : A \otimes^{\text{pt}} A \longrightarrow A, \quad \eta : \mathbf{1} \longrightarrow A,$$

satisfying strict associativity and unitality. In the invariant ∞ -category it is an E_1 algebra. A cofibrant A_∞ model replaces the single multiplication by operations indexed by planar trees. The chiral coefficient is unchanged.

Proposition 4.3 (Interchange). *Let γ be a chiral operation and let m be the transverse multiplication. Whenever the two composites are defined by one product-disk embedding, their images satisfy*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \tag{25}$$

with the symmetry isomorphism that places equal-coloured inputs together.

Proof. Equation (25) is the defining interchange relation of $E_1 \otimes \text{Ch}_X$. In the analytic product-disk model, both sides are the factorisation map associated with the same embedding, evaluated by composing first in opposite coordinate directions. $\square$

The proposition is stronger than the coexistence of two operations and weaker than braiding. It says that holomorphic collision and transverse composition are compatible. It does not provide a path that exchanges two transverse factors.

4.4 Analytic chamber model

For $r \geq 1$, let

$$C_r^< = \{(t_1, \dots, t_r) \in \mathbb{R}^r \mid t_1 < \dots < t_r\}.$$

This chamber is contractible. A profile (22) has analytic carrier

$$X^{I_1, \text{an}} \times \dots \times X^{I_r, \text{an}} \times C_r^<. \tag{26}$$

Local constancy in $C_r^<$ erases the distances while retaining the order. The X -coordinates remain holomorphic and may approach any partial diagonal allowed by the coefficient D -module.

Compactifying the configurations of subintervals in one interval modulo translation and positive dilation gives associahedra. Their strata are indexed by planar rooted trees. This compactification resolves the homotopy coherence of the E_1 multiplication.

Proposition 4.4 (Associahedral resolution). *For a locally constant factorisation algebra on an oriented line, cellular chains on compactified ordered interval configurations act by the A_∞ bar operations. Contracting every associahedral cell to its corolla gives the ordinary simplicial bar construction through a quasi-isomorphism of operadic resolutions.*

Proof. Factorisation assigns an operation to every ordered embedding of disjoint intervals. The corresponding embedding spaces are contractible, and their compactified boundary strata are planar nested collisions. Stokes' formula is the sum of edge contractions, hence the Stasheff identity. The cellular chains form a cofibrant resolution of the associative operad, while the simplicial bar is the standard resolution of an algebra over that operad. The comparison is the augmentation of the cofibrant resolution. $\square$

The simplex and the associahedron therefore play different roles. The simplex is the indexing object of the simplicial bar. The associahedron is a geometric cofibrant resolution of associative coherence.

4.5 Mixed collision geometry

A chain model for OCh_X may be built from two-coloured forests. Chiral edges describe nested partial diagonals in X ; transverse edges describe nested intervals in the ordered line. A face contracts one edge of one colour. On a total complex, the two boundary operators anticommute by the product orientation.

Lemma 4.5 (Mixed square-zero identity). *Let d_X be the differential supplied by a chain model of the chiral coefficient and let $b_\perp$ be the E_1 bar differential. If both are morphisms for the interchange structure, then*

$$d_X b_\perp + b_\perp d_X = 0$$

after the standard totalization sign. Hence

$$(d_X + b_\perp)^2 = d_X^2 + b_\perp^2.$$

Proof. The two operators are boundaries in independent factors of the mixed operadic resolution. Exchanging their conormal orientation lines contributes the totalization sign. Compatibility with interchange identifies the two coefficient composites. $\square$

Associativity proves $b_\perp^2 = 0$. The chiral identities prove $d_X^2 = 0$ in the chosen chiral model. Neither identity repairs the other.

4.6 Modules, interfaces, and marked blocks

A right module is an algebra over the two-coloured operad with ordered algebra inputs followed by one module input. A left module uses the opposite order, and a bimodule has both boundary colours. For a right module R and a left module L , the simplicial two-sided bar is

$$B_r(R, A, L) = R \otimes A^{\otimes r} \otimes L. \quad (27)$$

Its realization is interval factorisation homology with endpoint labels R and L .

Proposition 4.6 (Marked-block recognition). *The category of right, left, and bi-modules over A is equivalent to the corresponding category of marked coCartesian sections over the coloured block-ordered configuration category. Relative tensor product of bimodules is gluing of marked intervals.*

Proof. Apply the straightening argument of Theorem 4.2 to the standard coloured associative operads for modules and bimodules. Excision for the glued interval computes the relative tensor product. $\square$

This is the native geometry of boundary-changing fields and defect junctions.

4.7 Opposite order

Reflection of the transverse line reverses the ordinal factor while leaving X unchanged. It induces

$$\text{rev} : \text{OCh}_X \longrightarrow \text{OCh}_X$$

and sends multiplication to its opposite.

Proposition 4.7 (Geometric opposite). *Under Theorem 4.2, transverse reflection sends A to A^{op} . Its chiral coefficient is unchanged.*

This is the canonical inversion available at the E_1 level. Inverse braiding requires an E_2 structure and is not implied by reflection alone.

4.8 Three distinct forgetful operations

There are three operations that must not be called symmetrisation indiscriminately.

First, forgetting the E_1 multiplication leaves the underlying factorisable D -module. This functor loses all transverse products.

Second, derived commutativisation is the left adjoint

$$\text{Alg}_{E_1}(\text{FactD}(X)) \longrightarrow \text{Alg}_{E_\infty}(\text{FactD}(X))$$

when it exists in the selected presentable category. It imposes coherent commutativity on the transverse product. It is not termwise Reynolds averaging of tensor words.

Third, descent from labelled to unlabelled configurations concerns monodromy of a coefficient local system on $\text{Conf}_n(X)$. It asks for extension from the pure braid group to the full braid group; symmetric exchange further asks that the action factor through Σ_n . This is independent of the existence of a transverse E_1 multiplication.

The separation of these three operations is the first safeguard of noncommutative chiral Koszul duality.

5 The mixed operad of chiral collision and ordered fusion

5.1 The invariant operad

Let Ch_X be an ∞ -operad whose algebras in the chosen coefficient category are factorisable D -modules on X . The ordered chiral operad is defined by the Boardman–Vogt tensor product

$$\mathrm{OrdCh}_X := E_1 \otimes \mathrm{Ch}_X. \quad (28)$$

The tensor product is characterized by interchange: an operation in the E_1 direction commutes with an operation in the chiral direction after the canonical permutation and base-change maps.

Theorem 5.1 (Operadic additivity). *Let $\mathcal{V}$ be a presentable symmetric monoidal ∞ -category for which the indicated algebra categories exist. Then*

$$\mathrm{Alg}_{\mathrm{OrdCh}_X}(\mathcal{V}) \simeq \mathrm{Alg}_{E_1}(\mathrm{Alg}_{\mathrm{Ch}_X}(\mathcal{V})). \quad (29)$$

For the D -module realization of Ch_X , the right-hand side is $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. Equation (29) is the universal property of the tensor product of ∞ -operads. It identifies an algebra carrying two compatible operadic actions with an algebra over their tensor product. $\square$

For locally constant operations, Dunn additivity identifies $E_1 \otimes E_n$ with E_{n+1} [28, 54]. The chiral operation on X is not locally constant across collision diagonals, so (28) retains a holomorphic singular direction and an ordered topological direction.

5.2 Three geometric models

Three spaces occur for three different reasons.

The simplex indexes the monadic simplicial bar. The associahedron resolves the coherent multiplication of an E_1 algebra. The Fulton–MacPherson compactification resolves relative chiral configurations near collision diagonals. None can replace the other.

Fix an ordered partition $I = I_1 \sqcup \cdots \sqcup I_r$. On that stratum, a local product chart has the form

$$K_r \times \prod_{a=1}^r \mathrm{FM}_{I_a}(X), \quad (30)$$

up to the usual base and normal-bundle factors. When a transverse face merges two blocks, the partition changes and the corresponding Fulton–MacPherson factors are replaced by the factor for their union. Thus the global resolution is assembled from the charts (30) over the category of ordered partitions; it is not a single product $\mathrm{FM}_I(X) \times K_r$.

5.3 Two-coloured levelled forests

Choose a cofibrant dg-operadic presentation

$$\mathrm{Q}_X \longrightarrow \mathrm{Ch}_X$$

whose chiral cells are represented locally by Fulton–MacPherson incidence cells with their screen-chain coefficients. Apply a functorial Boardman–Vogt construction to the tensor-product operad:

$$W(E_1 \otimes \mathbf{Q}_X) \longrightarrow E_1 \otimes \mathbf{Ch}_X. \quad (31)$$

A cellular basis element of this resolution is represented by a rooted levelled forest with two kinds of internal edge:

- a transverse edge, recording composition of consecutive E_1 blocks;
- a chiral edge, recording a nested collision inside a chiral coefficient.

A vertex carries the screen-chain coefficient prescribed by $\mathbf{Q}_X$. The determinant line is

$$\mathrm{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\mathrm{ch}}(F)). \quad (32)$$

Definition 5.2 (Mixed forest complex). For a finite label set I , let

$$W\mathrm{OrdCh}_X(I) = \bigoplus_F C_*(\mathrm{Screen}(F)) \otimes \mathrm{or}(F), \quad (33)$$

where F ranges over the two-coloured levelled forests in the chosen cellular presentation. The differential is the sum of the internal screen differential and the signed contractions of one transverse or one chiral edge.

This definition is a chain model of the operadic resolution. It does not assert the existence of a single smooth compactification whose strata are literally all the summands of (33). Such a compactification, when available, gives a geometric realization of the same incidence data.

5.4 The boundary identity

Write

$$d = d_{\mathrm{scr}} + d_\perp + d_{\mathrm{ch}}.$$

Theorem 5.3 (Mixed boundary theorem). *With the orientation convention (32),*

$$d_\perp^2 = 0, \quad d_{\mathrm{ch}}^2 = 0, \quad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0,$$

and the screen differential supercommutes with both edge-contraction differentials. Hence $d^2 = 0$.

Proof. A codimension-two forest is obtained either by contracting two transverse edges, two chiral edges, or one edge of each colour. In the first two cases, the two orders of contraction induce opposite boundary orientations in the corresponding one-colour Boardman–Vogt complex. In the mixed case, the final forest is the same in either order, while exchanging the two one-dimensional determinant factors in (32) changes the sign. The screen differential is compatible with every operadic face map and contributes the standard total-complex signs. $\square$

When endomorphism operations are attached to vertices, the transverse part of this identity becomes the Stasheff relations, the chiral part becomes the identities of the chosen chiral coefficient model, and the mixed part says that transverse multiplication is a morphism of chiral objects.

5.5 Cofibrant comparison

Theorem 5.4 (Mixed Boardman–Vogt resolution). *Assume that chain operads are taken in an admissible monoidal model category, that $\mathbf{Q}_X$ is cofibrant, and that the tensor-product operad and functorial W -construction preserve weak equivalences between the relevant cofibrant objects. Then the augmentation*

$$W(E_1 \otimes \mathbf{Q}_X) \longrightarrow E_1 \otimes \mathbf{Ch}_X \quad (34)$$

is a cofibrant resolution. Its algebras are homotopy-coherent ordered chiral algebras. When rectification holds in the coefficient model category, their localized category is equivalent to $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. The W -construction replaces each internal edge by a length parameter and has contractible length fibres. Under the stated model-categorical hypotheses it is cofibrant and its edge-contraction augmentation is an aritywise weak equivalence. The tensor-product universal property identifies its strict algebras with compatible E_1 and chiral structures. Rectification is the standard change-of-operad equivalence along a weak equivalence between admissible cofibrant operads. $\square$

The hypotheses in this theorem are part of the mathematical statement. In particular, one does not infer cofibrancy of a tensor product merely from cofibrancy of two unrelated resolutions.

5.6 Incidence and screen geometry

A collision forest has an incidence type and a screen space. The incidence category records which clusters collide and in what order. Its fibre over a fixed root partition has a terminal corolla and hence a contractible nerve. The screen space records relative positions inside each cluster and generally has nontrivial cohomology.

For three points on an affine complex line modulo translation and dilation, the open screen is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore a proper pushforward of the geometric stratum retains more information than the pushforward of its incidence nerve. Arnold classes, configuration-space periods, and propagator forms belong to the screen coefficient in (33); they are not produced by the incidence differential.

5.7 Homotopy transfer and Feynman trees

Let a deformation retract of ordered chiral complexes be given. Homological perturbation transfers the structure by summing over the two-coloured forests: vertices carry original operations, internal edges carry the contracting homotopy, and screen chains carry the chiral coefficient. The resulting operations satisfy the complete forest boundary identity because Theorem 5.3 is the differential of the cofibrant resolution.

This is the precise coincidence between Feynman trees and homotopy transfer. The equality is combinatorial once the propagator and vertex maps have been supplied; it is not an identification of every configuration-space form with an OPE coefficient.

5.8 Renormalised Feynman weights

In perturbative field theory, an edge of a graph carries a propagator and a vertex carries an interaction. Renormalisation extends the resulting distribution to a compactified configuration space. Boundary restriction factors through contracted subgraphs.

Proposition 5.5 (Feynman weight morphism). *Let a holomorphic-topological BV theory on $X \times \mathbb{R}$ have renormalised graph weights extending to a geometric realization of the mixed forest model. Assume that restriction to every boundary stratum factorizes by graph contraction and that the renormalised quantum master equation holds. Then the graph weights define a morphism from the mixed cofibrant operad to the endomorphism operad of the boundary observables.*

Proof. Factorisation on a boundary stratum identifies the weight with the composite of the weights of the contracted graph and the quotient graph. Stokes' formula, including the BV boundary and counterterm contributions, is exactly compatibility with the differential of the mixed forest complex. These two identities are the operadic composition and chain-map axioms. $\square$

5.9 Three reversals

Reflection of the transverse line sends an E_1 algebra to its opposite. Reversal of an exchange path, when an E_2 enhancement exists, inverts the braid. Verdier duality reverses $!$ - and $*$ -extensions of the chiral coefficient. These are three different operations. The two-colour grading keeps their domains visible and prevents one from being inferred from another.

6 The ordered chiral bar construction

6.1 The simplicial object

Let A be an augmented ordered chiral algebra and let $\bar{A} = \text{Fib}(A \rightarrow \mathbf{1})$. The reduced simplicial bar object in $\text{FactD}(X)$ is

$$B_r^{\text{ord}}(A) = \bar{A}^{\otimes r}, \quad r \geq 0. \quad (35)$$

The inner face d_i multiplies the i th and $(i+1)$ st factors. The outer faces apply the augmentation. Degeneracies insert the unit in the unreduced model. The ordered chiral bar is the geometric realization

$$\text{Bar}_X^{\text{ord}}(A) := |B_{\bullet}^{\text{ord}}(A)|. \quad (36)$$

Every term of (35) is a factorisable D -module. The differential is a morphism in $\text{FactD}(X)$.

In cochain notation, after desuspension,

$$\text{Bar}_X^{\text{ord}}(A) \simeq \left(\bigoplus_{r \geq 0} (s^{-1} \bar{A})^{\otimes r}, d_A + b_m \right), \quad (37)$$

where b_m is the coderivation induced by the E_1 multiplication. Completion replaces the direct sum by a product.

6.2 The chiral coefficient is not contracted

Formula (37) contains no residue map. If a and b have a fourth-order OPE pole, that pole belongs to the factorisation D -module $A \otimes A$ and is carried through the multiplication morphism m . The bar face contracts the ordered factors; it does not extract a particular Laurent coefficient.

A residue complex can be paired with the bar as a separate geometric realization. Such a pairing

has the form

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow \Gamma(\mathrm{FM}_I(X), \mathcal{K}_I \otimes A_I),$$

where $\mathcal{K}_I$ is a chosen complex of forms, currents, or local cohomology classes. The map must specify how each principal part is paired with a normal jet. Its square-zero identity uses both the bar differential and the differential of $\mathcal{K}_I$. The intrinsic bar exists before this choice.

No-go theorem 6.1 (Residue substitution). *There is no natural identification of the internal bar differential with Poincaré residue along the chiral collision divisor. Poincaré residue lowers logarithmic form degree and detects a first normal coefficient. The bar differential lowers transverse word length and uses the entire multiplication morphism.*

The no-go theorem is a type statement. It follows from the source and target of the two maps.

6.3 Coalgebra structure

Deconcatenation gives

$$\Delta[a_1 | \cdots | a_r] = \sum_{p=0}^r [a_1 | \cdots | a_p] \otimes [a_{p+1} | \cdots | a_r]. \quad (38)$$

It is coassociative and conilpotent. The bar differential is a coderivation.

Proposition 6.2 (Internal bar coalgebra). *The ordered chiral bar is a conilpotent coassociative coalgebra object in $\mathrm{FactD}(X)$. The coaugmentation is the empty word, and the coradical filtration is the transverse word-length filtration.*

Proof. Deconcatenation is defined componentwise in every symmetric monoidal category. Its coassociativity is the equality of the two ways of cutting a word twice. The multiplication coderivation is local with respect to cuts, so it satisfies the coderivation identity. Iterating the reduced coproduct of a word of length r more than r times gives zero. $\square$

The coalgebra structure records interval splitting. It is unrelated to a chiral coproduct of fields. A bialgebra structure requires an additional multiplication compatible with (38).

6.4 Interval factorisation homology

Let $([0, 1], \partial)$ denote an oriented interval whose endpoints are labelled by the augmentation module $\mathbf{1}$. Excision for factorisation homology gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes_A^L \mathbf{1}. \quad (39)$$

The standard two-sided simplicial resolution of the derived tensor product is precisely (35).

Theorem 6.3 (Bar as interval factorisation homology). *There is a canonical equivalence in $\mathrm{FactD}(X)$*

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \int_{([0,1],\partial)} A. \quad (40)$$

For a right module R and a left module L ,

$$\mathrm{Bar}(R, A, L) \simeq \int_{([0,1],\{0,1\})} (A; R, L).$$

Proof. Choose a Weiss cover of the interval by an ordered chain of small intervals together with collars at the endpoints. Factorisation excision identifies the colimit of the Čech diagram with the derived relative tensor product. Refining the cover gives the simplicial two-sided bar. The argument takes place in the coefficient category $\text{FactD}(X)$ and is therefore compatible with chiral factorisation. $\square$

The theorem is the physical meaning of the bar construction. It is the amplitude of an interval whose interior carries the boundary operator algebra and whose endpoints carry the vacuum boundary state.

6.5 The associahedral model

The simplicial bar is the derived tensor product associated with a strict E_1 multiplication. A homotopy-coherent multiplication is an algebra over a cofibrant resolution $W\text{Ass} \rightarrow \text{Ass}$. Its bar coderivation contains the transferred operations m_k ,

$$b = \sum_{k \geq 1} b_{m_k}, \quad (41)$$

and $b^2 = 0$ is the family of Stasheff identities.

Let K_r denote the associahedron in the convention in which its cells are planar trees with r leaves. The cellular chains $C_*(K_r)$ form one presentation of $W\text{Ass}(r)$. They resolve the operation, not the simplicial degree.

Theorem 6.4 (Associahedral comparison). *Assume an admissible monoidal dg model of $\text{FactD}(X)$, let A be cofibrant as a $W\text{Ass}$ -algebra, and let A^{str} be a strictification along $W\text{Ass} \rightarrow \text{Ass}$. The operadic bar construction of A has a cellular filtration with terms*

$$\bigoplus_{r \geq 0} C_*(K_r) \otimes (s^{-1}\bar{A})^{\otimes r}, \quad (42)$$

whose differential contains the cellular boundary and the operations m_k . Change of operad and the simplicial two-sided bar give a natural zigzag of weak equivalences

$$\text{Bar}_{W\text{Ass}}(A) \simeq \text{Bar}_{\text{Ass}}(A^{\text{str}}) \simeq \text{Bar}_X^{\text{ord}}(A^{\text{str}}).$$

Proof. Cellular chains of associahedra present the Boardman–Vogt resolution of the associative operad. The change-of-operad adjunction is a Quillen equivalence under the stated admissibility and cofibrancy hypotheses. Bar constructions are homotopy invariant on cofibrant algebras and compute the same derived relative tensor product $\mathbf{1} \otimes_A^L \mathbf{1}$. This yields the zigzag. $\square$

The theorem does not identify a simplex with an associahedron. It identifies two homotopy-invariant models of the same interval amplitude.

6.6 Low arity

For a strict dg algebra, the normalized reduced bar begins

$$\bar{A} \xleftarrow{b} (s^{-1}\bar{A})^{\otimes 2} \xleftarrow{b} (s^{-1}\bar{A})^{\otimes 3} \xleftarrow{b} \dots$$

The binary differential is

$$b[a|b] = (-1)^{|a|}[ab].$$

The length-three component of b^2 is the signed associator. In higher length, every term is indexed by two adjacent multiplications; the two orders of contracting the corresponding planar edges occur with opposite bar signs.

For an A_∞ algebra, m_3 fills the homotopy between the two binary composites. At the first two-dimensional associahedral cell, m_4 and composites of m_2, m_3 fill the pentagon. The geometry is exactly the cellular boundary of associahedra.

The chiral coefficient adds an independent filtration. A word $[a|b]$ can carry an arbitrary OPE singularity in the X variables. The bar differential applies the transverse product ab as a morphism of those singular coefficient systems.

6.7 Functoriality

An augmented morphism $f : A \rightarrow B$ of ordered chiral algebras induces

$$\text{Bar}(f) : \text{Bar}_X^{\text{ord}}(A) \longrightarrow \text{Bar}_X^{\text{ord}}(B)$$

by applying f to every bar factor. A homotopy-coherent morphism induces the corresponding morphism of cellular bars.

A square-zero derivation Q of A induces a coderivation Q_{Bar} acting on each factor. Since Q is a derivation of multiplication,

$$[b_m, Q_{\text{Bar}}] = 0.$$

This identity is the basis of BRST descent.

A monoidal functor $F : \text{FactD}(X) \rightarrow \mathcal{V}$ preserving geometric realizations satisfies

$$F(\text{Bar}_X^{\text{ord}}(A)) \simeq \text{Bar}_{\mathcal{V}}(F(A)). \quad (43)$$

Thus restriction to a formal disc, analytic realization, taking a fibre, and factorisation homology commute with the ordered bar when their continuity hypotheses hold.

6.8 Fubini in the two directions

Let $\mathcal{A}$ be the analytic factorisation algebra on $X^{\text{an}} \times \mathbb{R}$ corresponding to A . For a compact one-manifold M with boundary labels, Fubini for factorisation homology gives

$$\int_{U \times M} \mathcal{A} \simeq \int_U \left(\int_M \mathcal{A} \right). \quad (44)$$

Taking $M = [0, 1]$ with vacuum endpoints yields the analytic realization of (40). Taking $M = S^1$ yields internal Hochschild chains whenever the circle factorisation homology exists. A cyclic trace or trace-class analytic completion is additional data needed only to turn those chains into scalar amplitudes.

Formula (44) is the first exact relation between chiral homology and topological factorisation. It does not exchange the two directions; it integrates one while retaining the other as a coefficient system.

6.9 Completion

The bar length filtration has finite quotients

$$\mathrm{Bar}_{\leq N}(A) = \bigoplus_{r=0}^N (s^{-1}\bar{A})^{\otimes r}.$$

The completed bar is

$$\widehat{\mathrm{Bar}}_X^{\mathrm{ord}}(A) = \prod_{r \geq 0} (s^{-1}\bar{A})^{\otimes r}. \quad (45)$$

The coderivation is continuous when every component has finite output in a fixed total filtration degree. This holds automatically for strict E_1 multiplication and for filtered A_∞ operations that raise a complete positive filtration.

If conformal weight is unbounded, one can first restrict to a finite weight window and then complete. The two inverse limits commute under the Mittag–Leffler condition described in [Theorem 7.3](#).

Proposition 6.5 (Cohomology of completion). *Let C_N be the tower of finite bar-length and finite-weight truncations with surjective transition maps on cohomology. Then*

$$H^*(\varprojlim_N C_N) \simeq \varprojlim_N H^*(C_N).$$

Proof. The Milnor exact sequence has a $\varprojlim^1$ term. Surjectivity gives the Mittag–Leffler condition and annihilates that term. $\square$

Completion is therefore a theorem about an inverse system, not a formal decoration on an infinite sum.

6.10 The bar of a boundary endomorphism algebra

Let $A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$ be the endomorphism algebra of a compact boundary condition. The two-sided bar

$$\mathrm{Bar}(b, A_b, b)$$

resolves the identity kernel of the full subcategory generated by b . Under the Morita equivalence of [Theorem 2.7](#), it is the factorisation analogue of the diagonal bimodule resolution.

Applying internal Hom to this resolution computes Hochschild cochains. Applying a trace computes Hochschild chains. Applying a second boundary label computes defect-changing operators. Thus the ordered bar is the common resolution behind the centre, the trace, and the module category.

6.11 The perturbative reading

In a perturbative boundary field theory, a bar word is a sequence of boundary insertions ordered along a collar. The bar differential fuses adjacent insertions. A Feynman propagator on X controls the chiral singularity of that fusion. Counterterms extend the coefficient distribution to the collision diagonal. The two operations are separate but compatible.

For a free Gaussian theory, Wick contraction gives an explicit morphism from the bar to the configuration-space de Rham complex. For an interacting renormalisable theory, Costello’s renormal-

isation constructs the corresponding factorisation map order by order in $\hbar$. The quantum master equation is precisely the condition that the resulting differential squares to zero.

This is the exact sense in which a bar word is an interval amplitude. The bar differential is composition of boundary operations; the propagator supplies the coefficient of that composition.

7 Ordered chiral bar–cobar reconstruction

7.1 The internal adjunction

Put

$$\mathcal{C}_X := (\text{FactD}(X), \otimes^{\text{pt}}, \mathbf{1}).$$

Assume throughout this section that $\mathcal{C}_X$ admits the geometric realizations and totalizations used below, and that tensor product preserves geometric realizations separately. These hypotheses hold in the standard presentable derived enhancements after restricting to the complete subcategories on which the indicated limits exist.

For a coaugmented conilpotent coassociative coalgebra C with coaugmentation coideal $\bar{C}$, define

$$\text{Cobar}_X^{\text{ord}}(C) = \left(\bigoplus_{r \geq 0} (s\bar{C})^{\otimes r}, d_C + d_\Delta \right), \quad (46)$$

where d_Δ is the derivation induced by the reduced coproduct. A complete cobar uses the product over tensor lengths and the continuous extension of the same derivation.

A degree-one morphism $\tau : C \rightarrow A$ is a twisting morphism when

$$d\tau + m(\tau \otimes \tau)\bar{\Delta} = 0. \quad (47)$$

The universal twisting morphisms give natural equivalences of mapping spaces

$$\text{Map}_{\text{Alg}_{E_1}}(\text{Cobar}_X^{\text{ord}} C, A) \simeq \text{Tw}(C, A) \simeq \text{Map}_{\text{Coalg}_{\text{coAss}}} (C, \text{Bar}_X^{\text{ord}} A).$$

Hence

$$\text{Cobar}_X^{\text{ord}} : \text{Coalg}_{\text{coAss}}^{\text{conil}}(\mathcal{C}_X) \rightleftarrows \text{Alg}_{E_1}^{\text{aug}}(\mathcal{C}_X) : \text{Bar}_X^{\text{ord}} \quad (48)$$

is an adjunction. Existence of the adjunction does not assert that its unit and counit are equivalences.

7.2 The connected positive-weight theorem

Definition 7.1 (Admissible positive object). An augmented algebra A in $\mathcal{C}_X$ is admissibly positive if it has a complete weight decomposition

$$A \simeq \mathbf{1} \oplus \prod_{w \geq 1} A_w, \quad A_p \otimes A_q \longrightarrow A_{p+q},$$

with each finite weight truncation dualisable and bounded below after every component evaluation $\text{ev}_I : \text{FactD}(X) \rightarrow D\text{-mod}(X^I)$. The evaluations are required to be conservative and to commute with the bar and cobar constructions. A coalgebra is admissibly positive when its coaugmentation coideal has the analogous positive, complete, conilpotent weight grading.

The positivity condition makes every fixed weight of a bar or cobar contain only finitely many tensor lengths. It is the internal analogue of a connected weight-graded augmented dg algebra.

Theorem 7.2 (Positive ordered bar–cobar reconstruction). *For an admissibly positive augmented ordered chiral algebra A , the counit*

$$\mathrm{Cobar}_X^{\mathrm{ord}} \mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow A \quad (49)$$

is an equivalence. For an admissibly positive conilpotent coalgebra C , the unit

$$C \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}} \mathrm{Cobar}_X^{\mathrm{ord}}(C) \quad (50)$$

is an equivalence.

Proof. Filter the two composites by tensor length and then take a fixed total weight. Positivity makes the resulting filtration finite. After applying ev_I , the associated graded is the ordinary connected associative bar–cobar resolution in complexes; its augmentation has the standard extra-degeneracy contraction. Thus each evaluated unit and counit is a quasi-isomorphism in every finite weight. Conservativity of the evaluations gives the equivalence in $\mathcal{C}_X$, and completeness reconstructs the full object from its finite weight truncations. $\square$

Corollary 7.3 (Completed reconstruction). *Let $A \simeq \varprojlim_N A_{\leq N}$ be an inverse limit of admissibly positive finite truncations, and suppose the transition maps are Mittag–Leffler on cohomology and tensor product commutes with the inverse limit. Then the continuous counit*

$$\widehat{\mathrm{Cobar}}_X^{\mathrm{ord}} \widehat{\mathrm{Bar}}_X^{\mathrm{ord}}(A) \longrightarrow A$$

is an equivalence. The analogous assertion holds for a complete conilpotent coalgebra.

Proof. The theorem applies at every finite stage. The Milnor exact sequence identifies cohomology of the inverse limit with the inverse limit of cohomology because the $\varprojlim^1$ term vanishes under the Mittag–Leffler hypothesis. $\square$

This is the associative bar–cobar theorem internal to the coefficient category. Its convergence is controlled by weight or augmentation completion, not by the cardinality filtration of the pointwise tensor product.

7.3 The separate Ran theorem

The Ran space carries another monoidal structure, the chiral tensor product $\otimes^{\mathrm{ch}}$. By Theorem 2.5, its reduced cardinality quotients are nilpotent. Francis and Gaitsgory prove the enhanced bar–cobar equivalence there for operads satisfying their reduced or pro-nilpotent hypotheses. The reduced associative operad gives the outer E_1 case, while their chiral Lie construction gives the Lie–commutative–coalgebra equivalence [38].

Theorem 7.4 (Enhanced Ran bar–cobar theorem). *In the pro-nilpotent monoidal category $(D(\mathrm{Ran} X), \otimes^{\mathrm{ch}})$, enhanced bar and cobar are inverse for the reduced associative operad, and more generally for the operads covered by the Francis–Gaitsgory enhanced theorem. The corresponding Lie case identifies chiral Lie algebras with the appropriate conilpotent commutative coalgebras. A*

restriction to a factorisation subcategory is valid when that subcategory is closed under the enhanced constructions and the unit and counit remain factorisation equivalences.

Theorem 7.4 and Theorem 7.2 are not the same theorem. The first uses chiral convolution on the Ran space. The second uses the pointwise coefficient tensor and resolves the transverse E_1 multiplication. A comparison between them is a monoidal comparison functor, not a change of notation.

7.4 The augmentation boundary

The unit is a left A -module through the augmentation. Define

$$A^! := \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}). \quad (51)$$

Composition gives $A^!$ an E_1 multiplication. When the bar terms are dualisable, or when continuous internal Hom is used, the bar resolution gives

$$A^! \simeq \mathrm{RHom}_{\mathcal{C}_X}(\mathrm{Bar}_X^{\mathrm{ord}}(A), \mathbf{1}). \quad (52)$$

Theorem 7.5 (Boundary Koszul duality). *Let A be an admissibly positive augmented ordered chiral algebra and let $A^!$ be defined by (51). The adjunction*

$$- \otimes_{A^!}^L \mathbf{1} \dashv \mathrm{RHom}_A(\mathbf{1}, -)$$

restricts to an equivalence between the thick subcategory of A -modules generated by $\mathbf{1}$ and the thick subcategory of right $A^!$ -modules generated by $A^!$. If $\mathbf{1}$ is a compact generator of the selected A -module category, the equivalence extends to its presentable completion.

Proof. The unit of the adjunction is an equivalence on the free rank-one $A^!$ -module, and the counit is an equivalence on $\mathbf{1}$. Exactness and preservation of retracts extend these equivalences to the two thick closures. Compact generation extends the result by Ind-completion. $\square$

Thus $A^!$ is the algebra of local operators on the augmentation boundary condition. It is not the quotient obtained by forgetting transverse order.

7.5 The double dual and completion

There is a canonical map

$$A \longrightarrow A^{!!} := \mathrm{RHom}_{A^!}(\mathbf{1}, \mathbf{1}). \quad (53)$$

Theorem 7.6 (Double dual as augmentation completion). *Let A be admissibly positive, with dualisable finite weight pieces. Then (53) identifies $A^{!!}$ with the derived completion of A for the augmentation ideal. In particular, if A is complete for its positive weight filtration, then $A \simeq A^{!!}$.*

Proof. At every finite weight, the bar resolution is finite and dualisable. Dualising it once computes $A^!$; dualising the dual bar computes the finite augmentation quotient of A . The maps are compatible with the weight tower. Taking the inverse limit gives the derived augmentation completion, and completeness gives the final equivalence. $\square$

The completion is part of the statement. A direct sum and its weight completion need not be isomorphic even when their finite truncations agree.

7.6 Quadratic comparison

Suppose that A has a connected quadratic presentation

$$A = T_{\mathcal{C}_X}(V)/(R)$$

with dualisable positive weight pieces. The quadratic coalgebra

$$A^i = C(s^{-1}V, s^{-2}R) \subset T^c(s^{-1}V)$$

has a canonical twisting morphism and a comparison

$$q_A : A^i \longrightarrow \text{Bar}_X^{\text{ord}}(A). \quad (54)$$

Theorem 7.7 (Quadratic recognition). *Assume that component evaluation is conservative and preserves the quadratic twisting constructions. The following conditions are equivalent:*

- (i) q_A is an equivalence;
- (ii) $\text{Cobar}(A^i) \rightarrow A$ is an equivalence;
- (iii) the left quadratic twisted tensor product is acyclic;
- (iv) the right quadratic twisted tensor product is acyclic;
- (v) the evaluated bar homology is concentrated on the quadratic diagonal in every finite weight.

Proof. After component evaluation this is the fundamental theorem for Koszul twisting morphisms. Conservativity lifts the equivalences to $\mathcal{C}_X$. Finite positive weights justify the passage between the quadratic coalgebra and the full bar. $\square$

Quadratic Koszulness concerns the comparison (54). It is stronger than the universal reconstruction of Theorem 7.2.

7.7 PBW and nonhomogeneous relations

Let $F_\bullet A$ be a complete exhaustive multiplicative filtration and suppose $\text{gr}_F A \cong A_0$. Filter the bar by total PBW degree.

Theorem 7.8 (Filtered PBW comparison). *Assume that the filtered comparison is complete and separated, its spectral sequence converges strongly in every finite chiral-cardinality and conformal-weight window, and A_0 is quadratic Koszul. Then the nonhomogeneous Koszul coalgebra determined by the quadratic, linear, and constant parts of the relations maps equivalently to $\text{Bar}_X^{\text{ord}}(A)$.*

Proof. The associated graded of the comparison is the quadratic equivalence for A_0 . Strong convergence lifts it through the filtration. Linear terms become part of the coalgebra differential, and constant terms become its curvature functional. $\square$

A homogeneous quadratic dual does not by itself compute the bar of an inhomogeneous algebra.

7.8 Curved coalgebras

A curved coalgebra has a coderivation d_C and a curvature functional $h : C \rightarrow \mathbf{1}[2]$ satisfying

$$d_C^2 = (h \otimes \text{id} - \text{id} \otimes h)\bar{\Delta}, \quad hd_C = 0.$$

Its curved cobar has a square-zero total differential because the curvature functional supplies the constant part of the cobar derivation. This curvature is an algebraic feature of an inhomogeneous presentation. It is not a two-form on a moduli space.

When ordinary acyclic complexes are not the correct null system, curved modules belong to the coderived or contraderived category [60]. The choice is determined by the module theory and completion, not by the numerical value of a central charge.

7.9 Verdier duality

Let $\mathbb{D}_{\text{Ran}}$ denote Verdier duality. If the finite bar stages are dualisable and duality commutes with the completion, then

$$A_{\text{Verdier}}^\vee := \mathbb{D}_{\text{Ran}} \text{Bar}_X^{\text{ord}}(A) \quad (55)$$

acquires an algebra structure by dualising deconcatenation.

The Verdier algebra (55) is not automatically the boundary endomorphism algebra (51). Their comparison requires a continuous perfect pairing

$$\text{Bar}_X^{\text{ord}}(A) \otimes A^! \longrightarrow \mathbf{1}.$$

Proposition 7.9 (Three dual constructions). *For an augmented ordered chiral algebra, the following are distinct until an explicit equivalence is proved:*

- (i) *the covariant reconstruction* $\text{Cobar Bar}(A) \rightarrow A$;
- (ii) *the boundary algebra* $A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1})$;
- (iii) *the Verdier dual algebra* $\mathbb{D}_{\text{Ran}} \text{Bar}(A)$, *when defined.*

This separation prevents opposite order, inverse braiding, level reflection, and Verdier duality from being treated as one involution.

7.10 The complete deformation complex

The deformation complex of the transverse E_1 multiplication is the shifted internal Hochschild complex

$$\text{Def}_{E_1}(A) = \text{HH}_{\text{FactD}(X)}^\bullet(A, A)[1]. \quad (56)$$

The chiral coefficient has its own deformation complex. On a formal disc it is the chiral operadic complex governing vertex or nonlocal-vertex operations. The deformation theory of the full ordered chiral object is controlled by a cofibrant resolution of the tensor-product operad $E_1 \otimes \text{Fact}_X$.

Proposition 7.10 (Bigraded deformation complex). *Choose an admissible cofibrant resolution of $E_1 \otimes \text{Fact}_X$. The convolution deformation complex of an ordered chiral algebra has a complete filtration by the number of transverse and chiral vertices. Its associated graded is spanned by two-coloured*

trees; the horizontal axis restricts to the Hochschild deformation complex of the E_1 multiplication, the vertical axis restricts to the deformation complex of the chiral coefficient, and the mixed components impose the interchange law.

Proof. The cellular Boardman–Vogt resolution is filtered by the two edge colours. Taking convolution with the endomorphism operad sends its cellular boundary to the deformation differential. Forests containing only transverse edges give Hochschild cochains; forests containing only chiral edges give coefficient deformations; forests of both colours give the mixed compatibility terms. $\square$

A Morita equivalence identifies the Hochschild axis. It identifies the full ordered chiral deformation problem only when it also identifies the chiral coefficient and the mixed interchange maps.

8 The Koszul–Morita square and the universal boundary state

The bar coalgebra is not only a resolution of the augmentation. It is the coalgebraic form of the universal boundary state. Once this point is made explicit, the interval, the diagonal bimodule, the annulus, Hochschild theory, and the derived centre become different shadows of one two-sided construction.

Throughout this section let

$$\mathcal{C}_X = (\text{FactD}(X), \otimes^{\text{pt}}, \mathbf{1}),$$

let A be an admissibly positive augmented E_1 algebra in $\mathcal{C}_X$, and put

$$C = \text{Bar}_X^{\text{ord}}(A).$$

Let $\tau : C \rightarrow A$ be the universal twisting morphism. All tensor and cotensor products below are derived in the complete subcategories selected by the positive weight filtration.

8.1 One-sided Koszul transforms

For a left A -module M , define the twisted left C -comodule

$$\mathbb{K}_A(M) = (C \otimes M, d_C + d_M + d_\tau), \quad (57)$$

where d_τ applies the reduced coproduct to C , evaluates one factor by τ , and lets the result act on M . For a conilpotent left C -comodule N , define

$$\mathbb{L}_A(N) = (A \otimes N, d_A + d_N + d_\tau). \quad (58)$$

The Maurer–Cartan equation for τ is exactly the statement that both displayed differentials square to zero.

Theorem 8.1 (Module–comodule Koszul equivalence). *The functors $\mathbb{K}_A$ and $\mathbb{L}_A$ are inverse equivalences between the derived category of augmentation-complete left A -modules and the coderived category of conilpotent left C -comodules. The same statement holds for right modules and right comodules.*

Proof. Filter both composites by the coradical degree of C . In each fixed positive weight, the filtration is finite. The associated graded twisting differential vanishes, and the unit and counit reduce to the

ordinary bar–cobar contractions. The extra degeneracy of the simplicial bar gives the contraction at each finite weight. Conservativity of component evaluation and completeness lift the equivalence to $\mathcal{C}_X$. When the coalgebra differential is curved or infinite products are essential, the same filtered contraction has coacyclic cone and therefore gives the asserted coderived equivalence. $\square$

The theorem is the module form of ordered chiral Koszul duality. It is stronger than an equality of Ext groups and weaker than a Morita equivalence between all A -modules and all modules over $A^!$. It identifies precisely the sector generated from the augmentation boundary.

8.2 The two-sided transform

For an A -bimodule M , set

$$\mathbb{K}_A^{\text{bi}}(M) = C \otimes_{\tau} M \otimes_{\tau} C. \quad (59)$$

The two outer factors carry the left and right C -coactions. The differential has four terms: the differentials of C and M , the left twisting action, and the right twisting action. The two twisting terms anticommute because the left and right A -actions commute.

Conversely, for a conilpotent C -bicomodule N , set

$$\mathbb{L}_A^{\text{bi}}(N) = A \otimes_{\tau} N \otimes_{\tau} A. \quad (60)$$

Theorem 8.2 (Bimodule–bicomodule equivalence). *The functors $\mathbb{K}_A^{\text{bi}}$ and $\mathbb{L}_A^{\text{bi}}$ are inverse equivalences between augmentation-complete A -bimodules and conilpotent C -bicomodules in the corresponding derived or coderived categories.*

Proof. Apply Theorem 8.1 first to the left action and then to the right action. The two transforms commute because their twisting coderivations act on different outer coalgebra factors and the bimodule identity makes their mixed commutator zero. Equivalently, filter (59) by the sum of the two coradical degrees and use the two-sided extra-degeneracy contraction on the associated graded. $\square$

8.3 The diagonal bicomodule

The diagonal A -bimodule is A itself. The diagonal C -bicomodule is the coalgebra C equipped with the two coactions

$$C \xrightarrow{\Delta} C \otimes C, \quad C \xrightarrow{\Delta} C \otimes C,$$

one read as a left coaction and the other as a right coaction. Denote it by C_{Δ} .

Theorem 8.3 (Diagonal correspondence). *There is a canonical equivalence of C -bicomodules*

$$\mathbb{K}_A^{\text{bi}}(A) \simeq C_{\Delta}. \quad (61)$$

Proof. The two-sided transform of A is

$$C \otimes_{\tau} A \otimes_{\tau} C.$$

The bar–cobar counit contracts the middle A successively against the left and right twisting cochains. Filter by the total coradical degree of the two outer factors. On the associated graded, the twisting terms vanish and the contraction identifies the complex with one copy of C . Under this identification

the two surviving coactions are the two legs of Δ_C . Finite positive weights and completeness lift the associated-graded equivalence. $\square$

The theorem gives a precise mathematical form to the phrase “universal boundary state.” The diagonal bimodule is the identity kernel for tensoring over A ; its Koszul transform is the identity kernel for cotensoring over C .

Corollary 8.4 (Composition unit). *For every conilpotent C -bicomodule N ,*

$$C_\Delta \square_C^L N \simeq N, \quad N \square_C^L C_\Delta \simeq N.$$

8.4 Tensor becomes cotensor

Let M be an A – B bimodule and N a B – D bimodule, with all algebras admissibly positive and all modules complete. Relative tensor product composes defects. The coalgebraic operation dual to it is derived cotensor product.

Theorem 8.5 (Tensor–cotensor gluing). *Let C_A, C_B, C_D be the ordered bar coalgebras. Then there is a canonical equivalence*

$$\mathbb{K}^{\text{bi}}(M \otimes_B^L N) \simeq \mathbb{K}^{\text{bi}}(M) \square_{C_B}^L \mathbb{K}^{\text{bi}}(N). \quad (62)$$

Proof. Choose two-sided bar resolutions for the relative tensor product and cobar resolutions for the cotensor product. After inserting the universal twisting morphisms, both total complexes are indexed by the same strings of B -operations between M and N . The Eilenberg–Zilber comparison identifies the two bisimplicial totalizations. In fixed positive weight the comparison is finite and is a quasi-isomorphism by the standard two-sided Koszul contraction. Completion gives (62). $\square$

This theorem is the algebraic form of cutting and regluing an interval. Tensoring sums over intermediate open states. Cotensoring imposes equality of the two coalgebraic boundary states.

8.5 Hochschild and coHochschild theories

Let

$$C^e = C \otimes C^{\text{cop}}.$$

For a C -bicomodule N , define

$$\text{coHH}_*(C, N) := C_\Delta \square_{C^e}^L N,$$

and

$$\text{coHH}^*(C, N) := \text{RHom}_{C^e}(C_\Delta, N)$$

in the selected coderived category.

Theorem 8.6 (Chiral Hochschild–coHochschild duality). *For every complete A -bimodule M there are natural equivalences*

$$\text{HH}_*(A, M) \simeq \text{coHH}_*(C, \mathbb{K}_A^{\text{bi}}(M)), \quad (63)$$

$$\text{HH}^*(A, M) \simeq \text{coHH}^*(C, \mathbb{K}_A^{\text{bi}}(M)). \quad (64)$$

In particular,

$$\begin{aligned} \mathrm{HH}_*(A) &\simeq C_\Delta \square_{C^e}^L C_\Delta, \\ \mathrm{HH}^*(A) &\simeq \mathrm{RHom}_{C^e}(C_\Delta, C_\Delta). \end{aligned}$$

Proof. Hochschild homology is the derived tensor product of the diagonal bimodule with M over A^e . Transport this expression through Theorem 8.2. By Theorem 8.3, the diagonal bimodule maps to C_Δ , and by Theorem 8.5, derived tensor becomes derived cotensor. This proves (63). Derived Hom is preserved by an equivalence of derived categories, which proves (64). $\square$

The annulus is therefore the self-cotensor of the universal boundary state. The derived centre is its coalgebraic self-extension. The two descriptions are not analogous; they are related by an equivalence of categories.

8.6 The Koszul–Morita square

The preceding theorems assemble into a commutative square of functors:

$$\begin{array}{ccc} D(A^e\text{-mod})_{\mathrm{comp}} & \xrightarrow[\sim]{\mathbb{K}_A^{\mathrm{bi}}} & D^{\mathrm{co}}(C^e\text{-comod})_{\mathrm{conil}} \\ A \otimes_{A^e}^L (-) \downarrow & & \downarrow C_\Delta \square_{C^e}^L (-) \\ D(\mathcal{C}_X) & \xlongequal{\quad\quad\quad} & D(\mathcal{C}_X). \end{array} \tag{65}$$

The corresponding square for derived Hom computes the centre. This is the Koszul–Morita square. Its horizontal maps change algebraic presentation. Its vertical maps close an interval into a circle. The square says that duality commutes with closure.

Corollary 8.7 (Centre as the closed endomorphism of the boundary state). *The ordered chiral centre has the equivalent descriptions*

$$Z_{\mathrm{ord}, \mathrm{ch}}(A) \simeq \mathrm{RHom}_{A^e}(A, A) \simeq \mathrm{RHom}_{C^e}(C_\Delta, C_\Delta).$$

Its E_2 operations are transported by the bimodule–bicomodule equivalence to the brace operations of the diagonal bicomodule.

8.7 Calabi–Yau transport

Suppose the perfect A -module category carries a d -Calabi–Yau trace. Equivalently, the diagonal bimodule has a nondegenerate cyclic pairing of degree d compatible with composition. Transporting the pairing through Theorem 8.2 gives a cyclic copairing on C_Δ .

Theorem 8.8 (Transport of the open trace). *Assume that the bimodule–bicomodule equivalence lifts to cyclic objects and commutes with geometric realization of the cyclic bar constructions. Then a perfect d -Calabi–Yau structure on the augmentation-generated A -module category induces a nondegenerate degree- d cyclic structure on the diagonal bicomodule C_Δ . Under the resulting S^1 -equivariant Hochschild–coHochschild equivalence, the Connes operator and the pair-of-pants product correspond; when the framed circle action is defined, so do the induced Batalin–Vilkovisky operators.*

Proof. A Calabi–Yau trace is a cyclic natural transformation on the Morita category of perfect modules. Theorem 8.5 transports composition. By the assumed lift to cyclic objects, the cyclic bar construction, its rotation, and its gluing maps are carried to their coHochschild counterparts. The Connes and pair-of-pants operations therefore agree under the equivalence. The same argument applies to the BV operator when the framed circle action has been supplied. $\square$

This is the exact bridge from ordered Koszul duality to ribbon geometry. It does not create the modular geometry of varying complex curves. It creates the open topological surfaces generated by the transverse trace.

8.8 Physical interpretation

Let a factorization-homology realization identify a boundary condition b with the endomorphism algebra $A = \mathrm{RHom}(b, b)$. Under interval excision, the bar coalgebra C is the state object of an interval ending on the augmentation boundary. An A – B bimodule is an interface. Relative tensor product composes interfaces. The Koszul transform turns the same interface into a bicomodule of boundary states, and cotensor product performs the same gluing in coalgebraic variables.

Closing an interface into an annulus gives Hochschild homology. Inserting a local bulk operation gives Hochschild cohomology. Theorem 8.6 says that both amplitudes can be computed before or after ordered chiral Koszul duality. The operation is the same; only the variables have changed.

The square (65) is therefore the precise mathematical form of open–closed complementarity on the augmentation-generated sector. It is categorical, not numerical. A scalar equality follows only after a trace is applied to this square.

9 Exchange begins beyond the ordered line

9.1 Order is not braiding

For every n , the chamber

$$\mathrm{Conf}_n^<(\mathbb{R}) = \{t_1 < \cdots < t_n\}$$

is convex. It remembers the order of factors but has trivial fundamental groupoid.

Theorem 9.1 (No braid from order). *The topology of ordered configurations on an oriented line supplies no nontrivial braid operator functorially from an E_1 algebra. A braid action requires an E_2 enhancement, a specified flat connection on a two-dimensional configuration space, or equivalent exchange data.*

Proof. Parallel transport inside a contractible chamber is canonically trivial. A braid generator is represented by a path interchanging points in a plane or another two-dimensional configuration space; no such path exists on the line. $\square$

Noncommutativity and braiding are therefore different structures. An associative algebra need not possess an R -matrix, and a braid representation need not arise from multiplication.

9.2 The E_2 enhancement

Replacing the transverse line by an oriented plane gives a forgetful functor

$$\mathrm{Alg}_{E_2}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)).$$

An E_2 lift supplies exchange paths. Three-disk isotopy gives the braid relation. Passing to module categories produces the pentagon and hexagon coherences. Analytically, the lift is a factorization algebra holomorphic on X and locally constant in two real transverse directions.

9.3 Internal Hochschild cochains

For an E_1 algebra A in $\mathrm{FactD}(X)$, define

$$Z_{\mathrm{ord},\mathrm{ch}}(A) := \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A). \quad (66)$$

The diagonal bar resolution gives the cup product, braces, and Gerstenhaber operations internally in the coefficient category.

Theorem 9.2 (Universal E_2 centre). *Assume that the diagonal bimodule admits a homotopically correct bar resolution in $\mathrm{FactD}(X)$. Then $Z_{\mathrm{ord},\mathrm{ch}}(A)$ is an E_2 algebra. For every E_2 algebra B acting on A through the Swiss-cheese operad, there is a contractibly unique E_2 morphism*

$$B \longrightarrow Z_{\mathrm{ord},\mathrm{ch}}(A)$$

that induces the given action.

Proof. The higher Deligne theorem equips endomorphisms of the unit object A in the monoidal category of A – A bimodules with an E_2 structure. The Swiss-cheese universal property identifies a central action with a morphism to these endomorphisms. The mapping space of such factorizations is contractible by the universal property of the endomorphism object. $\square$

The theorem does not identify a physical bulk with the centre. It identifies the centre as the universal recipient of every central bulk action.

9.4 Bimodules and categorical centres

The category $\mathrm{Bimod}_A(\mathrm{FactD}(X))$ is monoidal under relative tensor product, with unit the diagonal bimodule A .

Proposition 9.3 (Hochschild cochains as unit endomorphisms). *There is a natural equivalence*

$$\mathrm{End}_{\mathrm{Bimod}_A}(A) \simeq Z_{\mathrm{ord},\mathrm{ch}}(A),$$

and the E_2 structure on the right is the endomorphism E_2 structure of the unit in the monoidal bimodule category. The Drinfeld centre $Z(\mathrm{Bimod}_A)$ is braided and acts on this unit endomorphism algebra. An equivalence

$$Z(\mathrm{Bimod}_A) \simeq \mathrm{Mod}_{Z_{\mathrm{ord},\mathrm{ch}}(A)}$$

requires a separate monoidal-affineness theorem and is not asserted in general.

Left A -modules alone do not form a monoidal category without additional structure. The correct categorical centre is therefore attached to the bimodule category or to another explicitly monoidal line-defect category.

9.5 Meromorphic exchange

Suppose line defects are parametrized by points of a spectral curve. A meromorphic exchange is a family

$$R_{L,M}(z, w) : L_z \otimes M_w \longrightarrow M_w \otimes L_z \quad (67)$$

defined away from the collision divisor. Isotopy of three lines gives

$$R_{12}(z_1, z_2)R_{13}(z_1, z_3)R_{23}(z_2, z_3) = R_{23}(z_2, z_3)R_{13}(z_1, z_3)R_{12}(z_1, z_2). \quad (68)$$

The poles are chiral singularities on the spectral curve; the exchange path is topological in the transverse plane.

9.6 Pure-braid discrepancy

Let M_n be a complex with a B_n action and let $P_n = \ker(B_n \rightarrow \Sigma_n)$. Its derived pure-braid invariants and coinvariants are

$$M_n^{hP_n} = \mathrm{RHom}_{\mathbb{C}[P_n]}(\mathbb{C}, M_n), \quad (M_n)_{hP_n} = \mathbb{C} \otimes_{\mathbb{C}[P_n]}^L M_n,$$

with residual Σ_n actions.

Definition 9.4 (Pure-braid discrepancy). Writing $q : B_n \rightarrow \Sigma_n$ and q^* for inflation, define

$$\mathfrak{C}_n^{\mathrm{inv}}(M) := \mathrm{Cofib}(q^* M_n^{hP_n} \longrightarrow M_n), \quad (69)$$

$$\mathfrak{C}_n^{\mathrm{coinv}}(M) := \mathrm{Fib}(M_n \longrightarrow q^*(M_n)_{hP_n}). \quad (70)$$

The two objects are different derived measures of what is lost when pure-braid transport is removed. They are defined only after a braid action has been supplied.

Proposition 9.5 (Relative group model). *The homology of (70) is computed by the reduced group-bar complex of P_n with coefficients in M_n . The cohomology of (69) is computed by reduced group cochains. The residual Σ_n action is induced by conjugation in B_n .*

A partition complex may filter this group complex only after a map from collision strata to braid cells has been constructed. It is not the group-bar complex by definition.

9.7 Meromorphic Tannaka reconstruction

Let K be the function field of the spectral curve. Let $\mathcal{L}$ be a small rigid K -linear tensor category with finite-dimensional Hom spaces. A fibre functor may carry either strong monoidal or only quasi-monoidal comparison maps.

Assume first that

$$F : \mathcal{L} \longrightarrow \mathrm{Vect}_K$$

is faithful, exact, and strong monoidal, and that the coend

$$\mathcal{H} = \int^{L \in \mathcal{L}} F(L)^\vee \otimes F(L) \quad (71)$$

exists. A meromorphic braiding on $\mathcal{L}$ is understood over the complement of the collision divisor.

Theorem 9.6 (Meromorphic Tannaka reconstruction). *The strong monoidal pair $(\mathcal{L}, F)$ reconstructs a bialgebra $\mathcal{H}$ over K ; rigidity supplies an antipode whenever the coend lies in the rigid finite or continuous-dual class. A braiding reconstructs a coquasitriangular structure. If the fibre functor is only quasi-monoidal, the same coend carries a coquasi-bialgebra structure whose coassociator is the transported tensor constraint. When the relevant dual exists, these statements dualize to quasitriangular Hopf or quasi-Hopf algebras, respectively.*

Proof. The coend represents natural endomorphisms of the fibre functor. The strong tensor constraints define a coassociative coproduct and counit; rigidity defines the antipode. The braiding gives the universal R -form, and the hexagon identities give coquasitriangularity. For a quasi-monoidal fibre functor, the defect of strict coassociativity is exactly the transported associator, which satisfies the pentagon and hence defines a coquasi-bialgebra. $\square$

A presentation as a Yangian, quantum loop algebra, elliptic quantum group, or Hall double is an additional recognition theorem.

9.8 Morita invariance

Derived Morita equivalent E_1 algebras have equivalent bimodule categories and equivalent Hochschild cochains as E_2 algebras.

Theorem 9.7 (Centre transport through a Morita context). *Let A and B be ordered chiral algebras joined by an invertible A – B bimodule in the Morita $(\infty, 2)$ -category of complete factorisation algebras. Then*

$$Z_{\text{ord, ch}}(A) \simeq Z_{\text{ord, ch}}(B)$$

as E_2 algebras, and the induced monoidal equivalence of bimodule categories identifies their Drinfeld centres.

For $B = A^!$, the conclusion applies only when the boundary Koszul equivalence of Theorem 7.5 extends to such an invertible bimodule Morita context. Equivalence of the two augmentation-generated thick subcategories alone does not imply the conclusion.

The theorem is the precise algebraic content of two boundary presentations having the same universal central bulk. It is not a numerical complementarity law, and it is not automatic for a Koszul-dual pair such as a polynomial algebra and its exterior dual.

9.9 Topological enhancement

The centre is E_2 in the two transverse directions and remains holomorphic on the complex curve. Suppose both real translations underlying the holomorphic coordinate become null-homotopic, coherently with all E_2 operations.

Proposition 9.8 (Dimension of the topological centre). *Under this coherent topologization, the analytic centre is locally constant on the four-real-dimensional manifold $X^{\text{an}} \times \mathbb{R}^2$. It therefore determines an E_4 algebra.*

Proof. The null-homotopies make the factorization algebra locally constant in the two real directions of X . It was already locally constant in two transverse directions. The recognition theorem for locally constant factorization algebras gives an E_4 algebra; equivalently, Dunn additivity combines the two E_2 structures. $\square$

The result is E_4 , not E_3 , when the whole complex curve is topologized. Topologizing only one additional real direction gives an E_3 intermediate structure.

9.10 The bulk-to-centre map

Let a bulk factorization algebra Obs_{bulk} act on the boundary algebra A . Bringing a bulk observable to the boundary gives a central action and hence a map

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord, ch}}(A). \quad (72)$$

Criterion 9.9 (Algebraically complete boundary). *The boundary condition reconstructs the chosen local bulk observables exactly when (72) is an equivalence in the relevant factorization category. Without this equivalence, the centre is only the universal recipient of the bulk action.*

A decoupled bulk factor maps trivially and shows why a boundary algebra cannot reconstruct arbitrary global physics without a generation or nondegeneracy theorem.

10 Holomorphic–topological recognition

10.1 Product-generated local observables

Let $M = X^{\text{an}} \times \mathbb{R}$. A product disk is an open set $U \times I$, with $U \subset X^{\text{an}}$ a complex disc and $I \subset \mathbb{R}$ an oriented interval. Let $\text{Disk}_{X \times \mathbb{R}}^{\text{rect}}$ be the symmetric monoidal category generated by finite disjoint unions of product disks and their embeddings. An arbitrary holomorphic–topological factorization algebra need not be determined by this subcategory; we therefore isolate the precise local class used in the recognition theorem.

Definition 10.1 (Rectangular holomorphic–topological factorization algebra). A rectangular holomorphic–topological factorization algebra is a factorization algebra $\mathcal{F}$ on $X^{\text{an}} \times \mathbb{R}$ such that:

- (i) $\mathcal{F}$ is the factorization left Kan extension of its restriction to $\text{Disk}_{X \times \mathbb{R}}^{\text{rect}}$;
- (ii) for fixed I , the dependence on U is the analytic realization of a factorisable right D -module;
- (iii) for fixed U , the dependence on I is locally constant and orientation preserving.

Write $\text{Fact}_{\text{hol, lc}}^{\text{rect}}(X \times \mathbb{R})$ for this category.

The first condition is the generation statement needed to pass from product disks to all opens. The second is the chiral regularity. The third gives the little-interval action. None uses a metric.

10.2 The exponential law

The local operation category of product disks is the tensor product of the chiral disk category with the little-interval operad:

$$\mathrm{Disk}_X^{\mathrm{ch}} \otimes E_1.$$

The tensor-product universal property gives an exponential law for algebras.

Theorem 10.2 (Holomorphic–topological recognition). *There is a natural equivalence*

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \simeq \mathrm{Fact}_{\mathrm{hol,lc}}^{\mathrm{rect}}(X \times \mathbb{R}). \quad (73)$$

On the analytic side, the statement is restricted to coefficient systems in the image of the chosen Riemann–Hilbert realization. On the algebraic side, it is an equivalence of local factorization structures generated by product disks.

Proof. An object on the left is an algebra over E_1 in algebras over the chiral disk category. By the tensor-product universal property,

$$\mathrm{Alg}_{E_1}(\mathrm{Alg}_{\mathrm{Disk}_X^{\mathrm{ch}}}(\mathcal{V})) \simeq \mathrm{Alg}_{E_1 \otimes \mathrm{Disk}_X^{\mathrm{ch}}}(\mathcal{V}).$$

The latter is exactly a factorization algebra on the rectangular disk category that is chiral in U and locally constant in I . Factorization left Kan extension gives the right-hand side of (73). Restriction back to product disks is inverse because rectangular generation is part of its definition. Riemann–Hilbert identifies the regular-holonomic algebraic coefficient with its analytic constructible realization. $\square$

The theorem identifies local algebraic structures. It neither classifies all field theories on $X \times \mathbb{R}$ nor asserts that every ordered chiral algebra comes from a Lagrangian.

10.3 The two short-distance limits

Let A correspond to $\mathcal{F}$. For disjoint intervals $I_1 < I_2 \subset I$ and discs $U_1, U_2 \subset U$, factorization gives

$$\mathcal{F}(U_1 \times I_1) \otimes \mathcal{F}(U_2 \times I_2) \longrightarrow \mathcal{F}(U \times I). \quad (74)$$

Approaching the diagonal in $U_1 \times U_2$ probes the singular chiral coefficient. Bringing I_1 and I_2 together without crossing probes the E_1 multiplication. The two operations satisfy interchange because they are the two factors of one product-disk operation.

Proposition 10.3 (Interchange of short-distance limits). *Let m be the transverse multiplication and let γ be a chiral factorization map. On every product-disjoint locus where both composites are defined,*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \quad (75)$$

with the canonical permutation of coefficient factors.

Proof. Both sides are the image of one embedding of four product disks into a larger product disk. They are equal by functoriality of the factorization algebra. $\square$

This interchange is stronger than the coexistence of two products and weaker than braiding. Braiding requires motion in a second transverse direction.

10.4 Boundary and line-defect origins

A boundary condition b in a holomorphic-topological theory has a factorisable endomorphism object

$$A_b = \mathrm{RHom}(b, b).$$

Composition in the normal or topological direction is E_1 multiplication; collision of insertion points on X is the chiral coefficient. Boundary-changing operators are bimodules.

Parallel line defects in a theory on $X \times \mathbb{R}^2$ give the same structure after their transverse positions are restricted to an oriented line. Allowing the lines to move in the full plane may promote fusion to an E_2 structure and supply exchange. The E_1 fusion exists before that promotion.

Principle 10.4 (Defect interpretation). *The E_1 product is composition of boundary operators or line defects. The factorisable D -module records holomorphic collision. An exchange operator is additional codimension-two data.*

10.5 Classical and quantum observables

Let $(\mathcal{E}, Q, \{-, -\})$ be the local fields of a classical BV theory. A local interaction I satisfies

$$QI + \frac{1}{2}\{I, I\} = 0. \quad (76)$$

The classical observables have differential $Q + \{I, -\}$. A perturbative quantization is represented by $I_\hbar = I + \hbar I_1 + \dots$ satisfying the renormalized quantum master equation

$$QI_\hbar + \frac{1}{2}\{I_\hbar, I_\hbar\} + \hbar \Delta I_\hbar = 0. \quad (77)$$

The equation makes the renormalized observable differential square to zero and makes graphwise factorization compatible with contraction of ultraviolet subgraphs.

Theorem 10.5 (BV realization of an ordered chiral algebra). *Let a perturbatively renormalized BV theory on $X \times \mathbb{R}$ satisfy the quantum master equation. Assume that its quantum observables are rectangularly generated, locally constant in the real direction, and admit a factorisable D -module realization in the holomorphic direction. Then they determine*

$$\mathrm{Obs}_{X \times \mathbb{R}}^q \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X)).$$

A weak equivalence of renormalized theories that induces an objectwise quasi-isomorphism of these factorization observables induces an equivalence of ordered chiral algebras.

Proof. The quantum master equation supplies the square-zero differential and compatibility of renormalized factorization maps. The three additional hypotheses place the observable factorization algebra in the right-hand side of Theorem 10.2. Applying that equivalence gives the ordered chiral algebra. Objectwise quasi-isomorphisms become equivalences after localization. $\square$

The converse requires more data: fields, gauge symmetry, a BV pairing, a real form, counterterm choices, state spaces, and integration cycles are not reconstructible from local observables alone.

10.6 Interval evolution

With augmentation boundary conditions at both ends of an interval, excision gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes_A^L \mathbf{1} \simeq \text{Bar}_X^{\text{ord}}(A). \quad (78)$$

The ordered bar is therefore the complex of interval amplitudes between the two chosen boundaries. Cutting the interval gives deconcatenation.

Corollary 10.6 (Orientation reversal). *Reflection of the interval identifies the underlying bar objects of A and A^{op} by reversing every word and inserting the Koszul sign of the reversal permutation.*

The corollary produces opposite multiplication. It does not invert a braid, because a braid is absent in one real dimension.

Part II

The chiral coefficient and its geometric resolutions

11 The diagonal as the primitive chiral coefficient

11.1 From singular products to geometry

A quantum field is not multiplied at a point. Two fields are first placed at distinct points, where their product is defined, and then the product is extended toward coincidence. The singular part of this extension is the operator product expansion. The regular part is its renormalized continuation. Both are carried by the diagonal.

Let X be a smooth complex curve and let

$$j_2 : X^2 \setminus \Delta(X) \hookrightarrow X^2, \quad \Delta : X \hookrightarrow X^2$$

be the open complement and the diagonal embedding. A chiral operation on a right D_X -module $\mathfrak{g}$ is a morphism

$$\mu : j_{2,*} j_2^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}. \quad (79)$$

The source allows arbitrary finite poles along the diagonal. The target records that the singular product is supported where the two points meet. Antisymmetry and Jacobi make $\mathfrak{g}$ a chiral Lie algebra in the sense of Beilinson and Drinfeld [5].

Equation (79) is the common language of several classical constructions. In a formal coordinate it is the singular part of a vertex operator. In distribution theory it is a distribution supported on the diagonal. In perturbative quantum field theory it is a local counterterm or contact operation. In factorization theory it is the first nontrivial structure map. In representation theory its horizontal sections produce differential equations for conformal blocks. The categories differ, but the support condition does not.

Principle 11.1 (Diagonal principle). *The invariant datum of a chiral product is the morphism supported on the diagonal. Laurent coefficients, residues, modes, propagators, and contact terms are presentations of that morphism after auxiliary choices.*

The principle has a sharp consequence. No single differential form can encode every operator product coefficient. The logarithmic form $d\log(z - w)$ detects a first normal coefficient. A double pole, a fourth-order pole, and a regular normally ordered term belong to different pieces of the normal filtration. The complete distribution is primary; its coefficients are derived from it.

11.2 The historical line

Dirac's delta distribution made support a mathematical operation. Wilson's operator product expansion made short distance a hierarchy of local fields. Borchers compressed the consistency of that hierarchy into the vertex identity. Beilinson and Drinfeld replaced formal coordinates by D -modules and diagonals. Fulton and MacPherson resolved simultaneous collisions. Segal, Tsuchiya–Ueno–Yamada, and the theory of conformal blocks organized sewing. Getzler and Kapranov encoded stable sewing by modular operads. Drinfeld and Kohno identified braid monodromy with quantum groups. Costello and Gwilliam expressed perturbative quantum field theory as a factorization algebra of observables. The chiral coefficient branch of this paper is the exact overlap of these constructions; the transverse E_1 multiplication is the independent ordered structure developed in Part I.

The overlap becomes visible only when the order of the objects is respected:

$$\text{diagonal operation} \xrightarrow{\text{Chev}^{\text{ch}}} \text{factorization coalgebra on } \text{Ran}(X) \xrightarrow{\text{Prim}} \text{chiral Lie algebra} \quad (80)$$

The same local operation has three independent enlargements:

$$\begin{array}{ccccc}
 & \text{labelled configurations} & & & \\
 & \downarrow \text{monodromy} & & & \\
 \text{formal disc} & \xrightarrow{\text{coordinate expression}} & \text{chiral operation} & \xrightarrow{+ \text{cyclic sewing data}} & \text{stable curves} \\
 & \nwarrow \text{localization} & \downarrow \text{Ran-}\mathfrak{g} \text{ globalization} & & \downarrow \text{master equation} \\
 & & \text{factorization coalgebra} & & \text{modular graph operations}
 \end{array} \quad (81)$$

No arrow in these diagrams is an equality of unlike objects. Each arrow is a functor, a comparison map, or a monodromy construction.

11.3 The central theorem

The first theorem identifies the geometry that lies between the diagonal operation and the Ran coalgebra.

Theorem 11.2 (Geometric collision theorem). *Let X be a smooth algebraic curve over a field of characteristic zero and let $\mathfrak{g}$ be a chiral Lie algebra on X . For every nonempty finite set I there are three canonically related complexes.*

(i) *The partition collision complex*

$$C_I^{\text{coll}}(\mathfrak{g}) = \bigoplus_{\pi \in \text{Part}(I)} \Delta_{\pi,!} \left(\bigotimes_{B \in \pi}^{\text{unord}} \mathfrak{g}[1] \right),$$

with differential given by merging two blocks and applying the suspended chiral bracket.

(ii) *The corolla-normalized forest complex $W_I^{\text{FM}}(\mathfrak{g})$, obtained by resolving each block of a partition by the normalized chains of the rooted-tree category and coupling this vertical resolution to the collision differential.*

(iii) *The labelled component $\text{coChev}_I^{\text{ch}}(\mathfrak{g})$ of the chiral Chevalley coalgebra.*

The square of the differential on $C_I^{\text{coll}}(\mathfrak{g})$ is the suspended chiral Jacobiator. There are canonical quasi-isomorphisms

$$W_I^{\text{FM}}(\mathfrak{g}) \xrightarrow{\sim} C_I^{\text{coll}}(\mathfrak{g}) \xrightarrow{\cong} \text{coChev}_I^{\text{ch}}(\mathfrak{g}). \quad (82)$$

The first map applies the augmentation from each rooted-tree nerve to its terminal corolla. The second identifies the partition decomposition with the cofree commutative coalgebra in the chiral monoidal category.

The theorem is proved in Sections 13, 14 and 17. Its force is conceptual. The partition lattice carries the algebraic differential. The Fulton–MacPherson boundary supplies a geometric resolution of the incidence pattern. The Ran space globalizes the result into a factorization coalgebra. Screen configuration spaces and differential forms are genuine additional coefficients; they are not silently collapsed to a point.

11.4 The formal disc and the full operator product

Choose a formal coordinate z on a disc. A weakly translation-equivariant chiral algebra becomes a vertex algebra, and the morphism (79) becomes the full operator product

$$Y(a, z)Y(b, w) \sim \sum_{n \geq 0} \frac{Y(a_{(n)}b, w)}{(z - w)^{n+1}}. \quad (83)$$

The mode $a_{(n)}b$ is not invariant under a general change of coordinate. The filtered singular distribution is invariant. The quotient of pole order $n + 1$ by pole order n is a tensor power of the normal line to the diagonal. This is the geometric reason that modes transform triangularly.

The algebraic chiral operad $\mathcal{P}^{\text{ch}}$ makes the statement exact. If V is a vector superspace with translation operator, an odd element $X \in W_1^{\text{ch}}(\Pi V)$ determines a nonunital vertex algebra exactly when

$$[X, X] = 0. \quad (84)$$

The associated coderivation obeys

$$d_X^2 = \frac{1}{2}d_{[X, X]}. \quad (85)$$

Thus the full vertex identity is the coordinate expression of the chiral Jacobi identity. Arnold’s relation contributes to the geometry of configuration-space forms, but it does not replace the Borchers identity and does not manufacture higher poles from a logarithmic residue.

11.5 Five exact separations

A single language does not mean a single object. The following separations are structural.

First, the chiral Chevalley coalgebra is not a quadratic dual algebra. It exists without finite-dimensionality. A dual algebra appears only after a duality functor has been proved continuous on the relevant objects.

Second, braid monodromy is not a quantum group presentation. Monodromy gives a braided category or a local-system representation. Tannaka reconstruction with a strong or quasi-monoidal fibre functor produces a bialgebra/Hopf or coquasi/quasi-Hopf object; a further recognition theorem produces a named quantum group.

Third, projective curvature on moduli is not the square of a fiberwise differential. It is the curvature of a connection or the class of an Atiyah algebra. Tensoring by an inverse anomaly line cancels it when that line exists.

Fourth, the derived center is not automatically the physical bulk. It is the universal recipient of compatible algebraic actions. A bulk theory determines a map to the center; equality is a separate theorem.

Fifth, numerical invariants are traces of structures. Central charge, level, rank, Hodge classes, theta coefficients, and entropy coefficients are not interchangeable because they inhabit different functors.

These separations remove false universal statements while preserving the full mathematical and physical content.

11.6 Mathematical conventions

All derived categories are stable and $\mathbb{C}$ -linear. Right D -modules are used on algebraic curves. For a finite set I , X^I is the labelled Cartesian power, $\text{Part}(I)$ is its partition lattice, and $\Delta_\pi : X^\pi \hookrightarrow X^I$ is the partial diagonal associated with π . Tensor products indexed by unordered finite sets are taken in the symmetric monoidal derived category; the suspension [1] carries the Koszul signs. The symbol $\text{FM}_I(X)$ denotes the algebraic Fulton–MacPherson compactification. Real oriented compactifications are used only in the analytic and field-theoretic sections, where Stokes’ theorem is explicit.

For a vertex algebra, modes follow the convention

$$Y(a, z)b = \sum_{n \in \mathbb{Z}} (a_{(n)}b)z^{-n-1}.$$

For a simple Lie algebra $\mathfrak{g}$, the invariant form and the dual Coxeter number are normalized so that the KZ parameter is $\hbar = (k + h^\vee)^{-1}$; other conventions change only the displayed exponent defining q .

12 Chiral operations on a curve

12.1 The chiral tensor product

Let X be a smooth algebraic curve. The ordinary tensor product of right D_X -modules describes fields that occupy the same point. A chiral product must first separate the points. For two right D_X -modules M and N , write

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2.$$

The binary chiral tensor product is represented by $j_*j^*(M \boxtimes N)$, followed by a pushforward to the diagonal when an operation is evaluated. The distinction between j_* and $\Delta_!$ is the distinction between a meromorphic product away from collision and a distribution supported at collision.

For a finite set I , the open configuration locus is

$$U_I = X^I \setminus \bigcup_{i \neq j} \Delta_{ij}, \quad j_I : U_I \hookrightarrow X^I.$$

An I -ary chiral operation with values in a right D_X -module M is a morphism

$$\mu_I : j_{I,*}j_I^*(\boxtimes_{i \in I} M_i) \longrightarrow \Delta_{I,!}M, \quad (86)$$

where $\Delta_I : X \hookrightarrow X^I$ is the small diagonal. Partial compositions are defined by allowing a subset of points to collide first, applying one operation, and then colliding the resulting point with the remaining points. The associativity of these partial compositions is a base-change theorem for the corresponding diagrams of open embeddings and diagonals.

Definition 12.1 (Chiral Lie algebra). A chiral Lie algebra on X is a right D_X -module $\mathfrak{g}$ with an antisymmetric operation

$$\mu : j_*j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_!\mathfrak{g}$$

whose three partial composites satisfy the chiral Jacobi identity.

The word “Lie” refers to the symmetry and Jacobi law of the operation. It does not mean that the coefficients of the operation form an ordinary finite-dimensional Lie algebra. The source contains arbitrary poles and derivatives. On a formal disc, the chiral Jacobi identity becomes the vertex identity.

12.2 The Jacobi identity as a relation among partial diagonals

Let $I = \{1, 2, 3\}$. There are three pair diagonals and one small diagonal. The composite that first collides 1 with 2 and then collides the result with 3 is a morphism supported on Δ_{123} . The two other parenthesizations have the same target. After the symmetry signs are inserted, their sum is the Jacobiator

$$J_\mu = \mu \circ_{12} \mu + \mu \circ_{23} \mu + \mu \circ_{31} \mu. \quad (87)$$

The chiral Jacobi identity is $J_\mu = 0$.

The support geometry already predicts the form of the relation. Two disjoint collisions commute. Three points colliding into one admit three intermediate pair collisions. Every square of a collision differential therefore splits into disjoint-pair terms and triple-collision terms. The former cancel by the Koszul sign; the latter are the Jacobiator. This observation is made precise in Section 13.

12.3 Units and the vacuum

A unital chiral algebra contains a morphism from the dualizing module ω_X that plays the role of the vacuum. The unit axiom says that a field inserted next to the vacuum extends regularly across the diagonal and returns the field. On the formal disc this becomes

$$Y(\mathbf{1}, z) = \text{id}, \quad Y(a, z)\mathbf{1} \in a + zV[[z]].$$

The nonunital operation and its Jacobi identity contain the singular and regular binary product. The vacuum is additional nullary data. This separation is useful because chiral Koszul duality is naturally formulated for nonunital or augmented objects, while vertex operator algebras are ordinarily presented with a vacuum.

12.4 Chiral algebras and factorization algebras

A factorization algebra on X assigns an object to each finite configuration and identifies the object on disjoint unions with the tensor product of the objects on the pieces. A chiral algebra supplies the infinitesimal collision law from which this factorization structure is generated. Beilinson and Drinfeld construct the chiral enveloping algebra and the corresponding factorization algebra; Francis and Gaitsgory place the relation in a derived categorical setting [5, 38].

The relation may be read in two directions. Starting from a chiral Lie algebra, the chiral Chevalley construction forms a commutative factorization coalgebra. Starting from a conilpotent factorization coalgebra, primitives recover the chiral Lie algebra. The equivalence is not a linear duality and does not require the underlying D -module to be finite-dimensional. It is a Koszul duality internal to the chiral monoidal category.

Historical note 12.2. Beilinson and Drinfeld's use of right D -modules incorporates the transformation law of fields into the definition. A coordinate is used to calculate, not to define. This is the decisive step that separates a chiral algebra from a list of formal Laurent series.

12.5 Distributions and contact terms

The map (79) has an analytic analogue. A distribution on X^2 supported on the diagonal is a finite sum of normal derivatives of the delta distribution with coefficients along the diagonal. In a coordinate z ,

$$T(z, w) = \sum_{m=0}^N A_m(w) \partial_z^m \delta(z - w). \quad (88)$$

Cauchy's formula identifies the same coefficients with pole terms in a meromorphic kernel. The formal-distribution identity

$$\partial_w^m \delta(z - w) = \iota_{z,w} \frac{(-1)^m m!}{(z - w)^{m+1}} - \iota_{w,z} \frac{(-1)^m m!}{(z - w)^{m+1}}$$

relates the two expansions. Thus a chiral operation may be read either as a principal part or as a contact distribution. The equality is between two presentations of one diagonal-supported object.

In perturbative field theory, local counterterms are also distributions supported on diagonals. Renormalization extends distributions from configuration space to the compactification. The extension is not unique, but the ambiguity is local and supported on collision strata. Chiral algebra encodes the allowed local extensions and their consistency relations.

12.6 Why ordinary residue is insufficient

Let $D \subset Y$ be a smooth divisor. Poincaré residue is a map

$$\text{Res}_D : \Omega_Y^p(\log D) \longrightarrow \Omega_D^{p-1}.$$

It detects the coefficient of dt/t in a logarithmic form. It does not detect the coefficient of t^{-m} for $m > 1$ in a meromorphic function without further differential or jet data. The following statement is therefore immediate.

No-go theorem 12.3 (No universal logarithmic extractor). *There is no coordinate-free linear map from logarithmic forms along a smooth divisor to all finite principal parts that agrees, in a coordinate t , with extraction of the coefficient of t^{-m} for every $m \geq 1$. Logarithmic forms contain one normal generator dt/t , whereas the successive principal-part quotients are the mutually distinct line bundles $N_{D/Y}^{\otimes m}$.*

Proof. The exact sequence $0 \rightarrow \mathcal{O}_Y((m-1)D) \rightarrow \mathcal{O}_Y(mD) \rightarrow N_{D/Y}^{\otimes m} \rightarrow 0$ identifies the order- m symbol with a section of $N_{D/Y}^{\otimes m}$. Under $t' = ut + O(t^2)$ its coordinate coefficient changes by the m th power of the transition of the normal line. By contrast, Poincaré residue is the quotient map $\Omega_Y^p(\log D) \rightarrow \Omega_D^{p-1}$ arising from the single logarithmic generator dt/t . The logarithmic complex contains no canonical copies of all $N_{D/Y}^{\otimes m}$, so it cannot perform all coefficient extractions. $\square$

The correct replacement is the full pole filtration, constructed in [Section 15](#).

12.7 The affine-line presentation

On $X = \mathbb{A}^1$ with coordinate z , translation-invariant sections of $j_*\mathcal{O}_{X^2 \setminus \Delta}$ are Laurent polynomials in $z_1 - z_2$. A chiral operation can therefore be written as a finite sum

$$\{a_\lambda b\} = \sum_{n \geq 0} \frac{\lambda^n}{n!} a_{(n)} b, \quad (89)$$

combined with the regular product. The relation

$$\{\partial a_\lambda b\} = -\lambda \{a_\lambda b\}, \quad \{a_\lambda \partial b\} = (\partial + \lambda) \{a_\lambda b\}$$

is the coordinate form of D -module covariance. The Jacobi identity becomes

$$\{a_\lambda \{b_\mu c\}\} - (-1)^{p(a)p(b)} \{b_\mu \{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}. \quad (90)$$

The binomial coefficients of the Borchers identity arise by expanding the translated variable $\lambda + \mu$. They are not produced by the Arnold relation. The Arnold relation controls differential forms on configuration space; equation (90) controls the coefficient operation.

12.8 Factorization as locality

Suppose $U_1, \dots, U_n \subset X$ are pairwise disjoint discs contained in a larger disc V . A factorization algebra supplies a product

$$\mathcal{F}(U_1) \otimes \dots \otimes \mathcal{F}(U_n) \longrightarrow \mathcal{F}(V).$$

When the discs move without meeting, these products form a local system. When discs collide, the chiral operation describes the singular extension. This is the precise relation between OPE and locality: the OPE is the boundary value of factorization.

The factorization axiom is stronger than associativity at a point. It compares all decompositions by disjoint subdiscs and is compatible with isotopy. On a complex curve the holomorphic dependence of

the products adds the D -module structure. The result is a geometric object that can be transported, integrated, and sewn.

12.9 The physical reading

In Euclidean quantum field theory, a local observable supported in a small disc is an element of a cochain complex. Disjoint observables multiply. The product depends smoothly or holomorphically on their positions. A quantization satisfying locality therefore determines a factorization algebra [14, 15]. If one direction is holomorphic and the other directions are topological, restriction to a holomorphic boundary produces a chiral algebra. The local operation (79) is then the renormalized collision of boundary observables.

The mathematical and physical statements agree once a comparison map has been supplied between the algebraic D -module and the analytic factorization algebra. Without that map they are parallel realizations. The distinction is essential because a physical theory contains metric, reality, positivity, and global state-space data that a chiral algebra does not remember.

13 The partition collision complex

13.1 Partitions and partial diagonals

Let I be a finite nonempty set. A partition π of I is a collection of disjoint nonempty blocks whose union is I . The refinement order is written $\pi \leq \rho$ when every block of π is contained in a block of ρ . The minimal partition has singleton blocks; the maximal partition has one block.

To π associate the partial diagonal

$$\Delta_\pi : X^\pi \hookrightarrow X^I,$$

where the coordinates indexed by a block are required to coincide. If $\pi \triangleleft \rho$ is a cover relation, then ρ is obtained from π by merging exactly two blocks. Every saturated chain in $\text{Part}(I)$ is an order of pairwise collisions.

Let $s\mathfrak{g} = \mathfrak{g}[1]$. For a partition π , set

$$C_\pi(\mathfrak{g}) = \Delta_{\pi,!} \left(\bigotimes_{B \in \pi}^{\text{unord}} s\mathfrak{g} \right). \quad (91)$$

The unordered tensor product is the coinvariant form of the tensor product in the symmetric monoidal derived category. The suspension is not cosmetic: it turns the antisymmetric chiral bracket into a symmetric degree-one contraction, exactly as in the Chevalley complex of an ordinary Lie algebra.

Definition 13.1 (Partition collision complex). The I -labelled collision object is

$$C_I^{\text{coll}}(\mathfrak{g}) = \bigoplus_{\pi \in \text{Part}(I)} C_\pi(\mathfrak{g}). \quad (92)$$

For a cover $\pi \triangleleft \rho$ that merges blocks B_1 and B_2 , the component

$$d_{\pi,\rho} : C_\pi(\mathfrak{g}) \longrightarrow C_\rho(\mathfrak{g})$$

applies the suspended chiral bracket to the factors indexed by B_1, B_2 and acts identically on the remaining factors. The total differential is the internal differential plus the signed sum over covers.

The sign is determined by moving the two suspended factors next to one another, applying the degree-one bracket, and returning the merged factor to the position determined by the orientation line of the cover. Because the tensor product is unordered, the definition is independent of a chosen linear order on the blocks.

13.2 The square of the collision differential

Every length-two interval $[\pi, \sigma]$ in the partition lattice has one of two forms.

If σ is obtained by merging two disjoint pairs of blocks, there are exactly two intermediate partitions. The coefficient operations act on disjoint tensor factors and commute before suspension. Interchanging the two degree-one contractions changes the sign, so the two paths cancel.

If σ is obtained by merging three blocks into one, there are exactly three intermediate partitions. The three composites are the three terms of the suspended Jacobiator.

Theorem 13.2 (Square of the collision differential). *Let μ be an antisymmetric chiral binary operation on a right D_X -module $\mathfrak{g}$. The collision operator defined from μ satisfies*

$$d_{\text{coll}}^2 = \frac{1}{2}d_{[\mu, \mu]}. \quad (93)$$

Consequently $C_I^{\text{coll}}(\mathfrak{g})$ is a complex for every finite I if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal differential squares to zero and its mixed commutator with d_{coll} vanishes precisely when μ is a chain map. The remaining square is a sum over length-two intervals in $\text{Part}(I)$. Disjoint mergers cancel in pairs by the Koszul sign. Triple mergers sum to the three suspended partial composites of μ . This sum is $\frac{1}{2}[\mu, \mu]$ in the convolution Lie algebra of chiral operations. No other length-two interval occurs. $\square$

Corollary 13.3 (Selection by collisions). *An antisymmetric collection of diagonal-supported binary operations defines a chiral Lie algebra exactly when the corresponding partition collision operators form differentials in all arities.*

This is the correct geometric selection statement. It uses the complete chiral operation. It does not identify the Jacobi identity with a relation among differential forms and does not require a central term to repair an ordinary Lie bracket.

13.3 Arity two and three

For $I = \{1, 2\}$ there are two partitions. The complex is

$$\Delta_{1|2,!}(sg \boxtimes sg) \xrightarrow{s\mu} \Delta_{12,!}(sg).$$

The map is the collision operation.

For $I = \{1, 2, 3\}$, the Hasse diagram is

$$\begin{array}{ccccc}
 & & 1|2|3 & & \\
 & \swarrow & \downarrow & \searrow & \\
 12|3 & & 13|2 & & 23|1 \\
 & \searrow & \downarrow & \swarrow & \\
 & & 123 & &
 \end{array} \tag{94}$$

The square from the top to the bottom is the sum of three composites. In a coordinate presentation these are the three expansions of the vertex Jacobi identity. In the D -module presentation they are three maps with the same source and target, supported on the small diagonal.

13.4 The cofree commutative coalgebra

Let $\mathcal{C}$ be a symmetric monoidal stable category. The cofree conilpotent commutative coalgebra on an object V has underlying object

$$\mathrm{coSym}(V) = \bigoplus_{n \geq 1} (V^{\otimes n})_{\Sigma_n}.$$

In the chiral monoidal category on the Ran space, the labelled component of $\mathrm{coSym}(\mathfrak{g})$ decomposes by the ways in which labels are identified. Those ways are the partitions of I . The summand associated with π is exactly (91).

Proposition 13.4 (Chevalley identification). *For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is canonically the I -component of its chiral Chevalley coalgebra:*

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \cong \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}). \tag{95}$$

Under this identification the collision differential is the Chevalley coderivation induced by the chiral bracket.

Proof. The cofree commutative coalgebra is the sum over finite collections of primitive factors. Pulling its Ran component back to X^I records a surjection from I to the set of primitive factors. A surjection modulo relabeling of its target is a partition of I . The Chevalley coderivation merges two primitive factors by the bracket, hence agrees with the cover differential of the partition lattice. $\square$

13.5 The factorization coproduct

Suppose $I = I_1 \sqcup I_2$. Restriction to the open locus where the points of I_1 remain disjoint from the points of I_2 gives a coproduct

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \longrightarrow C_{I_1}^{\mathrm{coll}}(\mathfrak{g}) \boxtimes C_{I_2}^{\mathrm{coll}}(\mathfrak{g}). \tag{96}$$

A partition contributes to this restriction exactly when no block meets both I_1 and I_2 . The coproduct is therefore deconcatenation at the level of primitive blocks. Coassociativity follows from the associativity of disjoint union.

Proposition 13.5. *The family $\{C_I^{\mathrm{coll}}(\mathfrak{g})\}_I$, with the coproducts (96), is a conilpotent commutative factorization coalgebra. Its primitive part is $\mathfrak{g}[1]$.*

Conilpotence is measured by the number of blocks. Repeated reduced coproducts eventually reach singleton partitions. This filtration is intrinsic and requires no finite-dimensional dual.

13.6 Incidence geometry and Möbius inversion

The Euler characteristic of the collision complex is governed by the Möbius function of the partition lattice. For the interval from the discrete partition to the one-block partition,

$$\mu_{\text{Part}(I)}(\hat{0}, \hat{1}) = (-1)^{|I|-1}(|I| - 1)!.$$

This integer is also the top reduced Euler characteristic of the proper partition complex. It explains the factorial multiplicities that appear in Lie operad components and in the top cohomology of configuration spaces. The equality is combinatorial; the representations carried by the two sides require their respective coefficient systems.

The proper partition complex is homotopy equivalent to a wedge of $(|I| - 1)!$ spheres of dimension $|I| - 3$. Its top homology is the multilinear part of the Lie operad, up to the sign representation. This is the classical topological shadow of the Lie–commutative Koszul duality that becomes chiral on the Ran space.

13.7 Modules and marked points

Let M be a chiral module over $\mathfrak{g}$. Add a distinguished label $*$ that is never merged with another module label. A partition of $I \sqcup \{*\}$ with a distinguished block containing $*$ indexes a summand in the module collision complex. Merging an ordinary block with the distinguished block applies the module action; merging two ordinary blocks applies the bracket. The same length-two interval analysis proves that the square is the sum of the Lie Jacobiator and the module identity.

This pointed complex is the geometric origin of derived coinvariants and conformal blocks with insertions. It is also the correct place to compare BRST reduction, because the BRST differential acts on both the algebra and the marked module.

13.8 What the partition complex does not contain

The partition complex records which groups of points collide and in what incidence pattern. It does not record the relative shape of a collision. For three points on a complex line, the ratio of their relative positions survives after translation and dilation. That ratio belongs to the screen configuration space of the Fulton–MacPherson boundary. Differential forms on screens produce Arnold and Orlik–Solomon classes. They enter through the geometric resolution of the next section.

14 The Fulton–MacPherson forest resolution

14.1 Compactifying collisions

The configuration space $\text{Conf}_I(X)$ is the complement of the diagonals in X^I . Its Fulton–MacPherson compactification $\text{FM}_I(X)$ is obtained by successively blowing up partial diagonals in an order compatible with inclusion [40]. The result is smooth, and the boundary is a normal-crossing divisor. A boundary point records not only which labels collide but also the limiting relative configuration at every scale.

Boundary strata are indexed by nested collections of subsets, equivalently by rooted forests whose leaves are the labels in I . An internal vertex represents a cluster. An edge represents one scale of collapse. Roots represent the final clusters that remain distinct in X .

Let F be such a forest. The root components determine a partition $q(F)$ of I . The corresponding stratum maps to the partial diagonal $\Delta_{q(F)}(X^{q(F)})$. Its fiber is a product of screen configuration spaces. On a curve, a screen for a cluster S is the compactification of configurations of $|S|$ points on a one-dimensional tangent space modulo translation and dilation.

14.2 Tree categories over collision blocks

For a nonempty finite set B , let $\text{Tree}(B)$ be the category of rooted trees with leaves labelled by B and with no unary internal vertices. Morphisms contract internal edges. The corolla with leaf set B is terminal. Write

$$K(B) = N_*(N \text{Tree}(B)), \quad (97)$$

for the normalized chain complex of its nerve, placed in nonpositive cohomological degrees. The augmentation and the terminal-corolla inclusion are chain maps

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C}, \quad \mathbb{C} \xrightarrow{\iota_B} K(B), \quad \epsilon_B \iota_B = \text{id}. \quad (98)$$

Because $\text{Tree}(B)$ has a terminal object, ϵ_B is a quasi-isomorphism.

For a partition π of I , define

$$K(\pi) = \bigotimes_{B \in \pi}^{\text{unord}} K(B). \quad (99)$$

An element of $K(\pi)$ is a chain of contractions in a forest whose root components are the blocks of π . This is the incidence information carried by the face category of the Fulton–MacPherson boundary; it contains no cohomology of the screen spaces.

14.3 The corolla-normalized forest resolution

Set

$$W_I^{\text{FM}}(\mathfrak{g}) = \bigoplus_{\pi \in \text{Part}(I)} C_\pi(\mathfrak{g}) \otimes K(\pi). \quad (100)$$

Its differential has three terms. The internal term is induced by the differential of $\mathfrak{g}$. The vertical term is the normalized nerve differential on the factors $K(B)$. For a cover $\pi \triangleleft \rho$ merging blocks B, C , the horizontal term applies the suspended chiral bracket to the B, C coefficient factors and applies the chain map

$$K(B) \otimes K(C) \xrightarrow{\epsilon_B \otimes \epsilon_C} \mathbb{C} \xrightarrow{\iota_{B \cup C}} K(B \cup C) \quad (101)$$

while leaving the other tree factors unchanged. Thus a collision of two root components is represented by the terminal corolla of the merged block; all nonterminal tree data remains in the vertical resolution.

The maps (101) are symmetric and strictly associative because they factor through the augmentations. They commute with the vertical differential. Consequently the square of the horizontal differential is exactly the square of the partition collision differential, with a common terminal-corolla factor.

There is an augmentation

$$\epsilon_I : W_I^{\text{FM}}(\mathfrak{g}) \longrightarrow C_I^{\text{coll}}(\mathfrak{g}) \quad (102)$$

obtained by applying ϵ_B to every tree factor.

Theorem 14.1 (Forest resolution). *For every chiral Lie algebra $\mathfrak{g}$ and finite set I , the differential on (100) squares to zero and the augmentation (102) is a quasi-isomorphism. The construction is natural under relabelling and compatible with disjoint union.*

Proof. The vertical differential squares to zero. Its mixed commutator with the horizontal differential vanishes because the augmentations and corolla inclusions in (101) are chain maps. Strict associativity of the corolla-normalized merge identifies the horizontal square with the Jacobiator computed in Theorem 13.2; it therefore vanishes for a chiral Lie algebra.

Filter by the total nerve degree of the tree factors. The filtration is finite for fixed I . Its first spectral sequence computes the homology of every $K(B)$. Since each ϵ_B is a quasi-isomorphism, the first page is precisely $C_I^{\text{coll}}(\mathfrak{g})$, and the induced differential is its collision differential. The spectral-sequence comparison proves that ϵ_I is a quasi-isomorphism. Relabelling preserves terminal corollas, and disjoint union tensors the tree complexes, giving naturality and factorization compatibility. $\square$

Corollary 14.2. *The corolla-normalized forest complex resolves the chiral Chevalley coalgebra:*

$$W_I^{\text{FM}}(\mathfrak{g}) \xrightarrow{\sim} \text{coChev}_I^{\text{ch}}(\mathfrak{g}).$$

The word “geometric” refers to the tree categories labelling the faces of the compactification. It does not assert that the actual screen fibers are contractible or that their de Rham complexes are removed by the augmentation.

14.4 Screen spaces are genuine coefficients

Let $p_I : \text{FM}_I(X) \rightarrow X^I$ be the proper blowdown. The fiber over a point of a partial diagonal contains compactified screen configurations. For $|I| = 3$, the screen modulo translation and dilation has a nontrivial moduli coordinate. Its compactification is not a point. Therefore the derived pushforward of the constant sheaf on $\text{FM}_I(X)$ is not, in general, the partition incidence complex.

No-go theorem 14.3 (No contraction of screen cohomology). *The existence of a terminal corolla in the forest category does not imply that the proper geometric fibers of p_I are contractible. Consequently a quasi-isomorphism from a de Rham or constructible complex on $\text{FM}_I(X)$ to the collision complex requires a comparison map that accounts for screen cohomology.*

The distinction separates two independent structures:

incidence of collision strata	geometry inside a screen
partition lattice	configuration space
edge contraction	de Rham differential
chiral Jacobi	Arnold–Orlik–Solomon relations
Chevalley coderivation	propagator and period classes

14.5 The stratification spectral sequence

Filter $\mathrm{FM}_I(X)$ by the codimension of its boundary strata. For a constructible complex $\mathcal{F}$ that is locally constant along strata, the standard spectral sequence of a filtered space has first page

$$E_1^{p,q} = \bigoplus_{\mathrm{codim} S=p} H_c^{p+q}(S; \mathcal{F}|_S \otimes \mathrm{or}_S) \implies H_c^{p+q}(\mathrm{FM}_I(X); \mathcal{F}). \quad (103)$$

The differential is the signed sum of incidence maps between adjacent strata.

For coefficients pulled back from the root diagonal, a stratum splits locally into a base and a product of screens. The Kunneth theorem expresses its cohomology as

$$H_c^*(S; \mathcal{F}|_S) \cong H_c^*(\Delta_{q(F)}; \mathcal{F}_{q(F)}) \otimes \bigotimes_{v \in V_{\mathrm{int}}(F)} H_c^*(\mathrm{Screen}(v)). \quad (104)$$

After barycentric replacement of the face incidence category and corolla normalization, the degree-zero screen classes map to the forest incidence resolution. Positive-degree screen classes record the topology of relative configurations.

Theorem 14.4 (Incidence–screen separation). *For a coefficient system constant along Fulton–MacPherson strata, the boundary-codimension spectral sequence separates the collision incidence maps from the cohomology of screen configuration spaces. After replacing the face poset by its normalized nerve and applying the terminal-corolla contractions blockwise, the row obtained from degree-zero screen classes maps quasi-isomorphically to the forest resolution W_I^{FM} .*

This theorem is the precise meeting point of the chiral Chevalley complex and configuration-space forms. The two structures interact through the spectral-sequence differential but are not identified term by term.

14.6 Arnold forms on screens

On an affine coordinate chart, set

$$\eta_{ij} = d \log(z_i - z_j).$$

On $\mathrm{Conf}_3(\mathbb{C})$ the Arnold relation is

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0. \quad (105)$$

It follows by the elementary substitution $x = z_1 - z_2$, $y = z_2 - z_3$. The Orlik–Solomon algebra of the braid arrangement is generated by the classes of the η_{ij} with these three-term relations [4].

The Arnold relation expresses flatness of the universal logarithmic connection once the coefficients satisfy infinitesimal braid relations. It does not express the full chiral Jacobi identity. In the stratification spectral sequence, (105) belongs to the screen coefficient system; the chiral Jacobi identity belongs to the coefficient operation on the root diagonal. A square-zero total differential may use both, but each has its own source.

14.7 Real oriented compactification and Stokes’ theorem

For perturbative integrals one uses the real oriented blowup of the diagonals. The boundary hypersurfaces carry outward normal orientations. If α is a compactly supported form of top degree minus one,

Stokes' theorem gives

$$\int_{\mathrm{FM}_I^{\mathrm{or}}(M)} d\alpha = \sum_{H \subset \partial \mathrm{FM}_I^{\mathrm{or}}(M)} \int_H \alpha.$$

Applying Stokes twice yields cancellation of codimension-two corners because each corner occurs with opposite induced orientations from its two parent hypersurfaces. This is the analytic form of the incidence differential squaring to zero.

When forms carry algebraic coefficients, the total square contains three distinct terms: the de Rham square, the incidence square, and the coefficient Jacobiator. A correct proof specifies the bidegrees and the mixed commutators. The vanishing of boundary-of-boundary alone cannot create the coefficient identities.

14.8 Renormalization forests

The same forest combinatorics appears in the BPHZ forest formula. A divergent subgraph determines a collision subconfiguration; nested divergent subgraphs determine a forest; subtraction is performed recursively from smaller clusters to larger clusters. Fulton–MacPherson compactification turns this recursion into geometry. Counterterms are supported on boundary strata, and locality is the compatibility of their restrictions under further collisions.

This parallel is exact at the level of incidence categories. A particular quantum field theory supplies a propagator, a density, a gauge fixing, and a renormalization prescription. Those analytic data form the screen coefficient system. The universal forest resolution supplies the combinatorics on which they live.

15 Pole filtrations and normal geometry

15.1 Principal parts along a divisor

Let Y be a smooth variety and $D \subset Y$ a smooth divisor. For $m \geq 0$ there is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \longrightarrow \mathcal{O}_D((m+1)D) \longrightarrow 0. \quad (106)$$

Since $\mathcal{O}_D(D) \cong N_{D/Y}$, the associated graded of the pole filtration is

$$\mathrm{gr}_{m+1}^F \mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes(m+1)}. \quad (107)$$

If t is a local defining equation for D , the class of $t^{-(m+1)}$ is a local generator of this line. Under $t' = ut + O(t^2)$ it transforms by $u^{-(m+1)}$ modulo lower pole order. This is precisely the tensorial transformation law of the normal symbol.

For a normal-crossing divisor $D = \bigcup_{a \in A} D_a$, use the multi-filtration

$$F_m \mathcal{O}_Y(*D) = \mathcal{O}_Y \left(\sum_{a \in A} m_a D_a \right).$$

On a stratum $D_J = \bigcap_{a \in J} D_a$, the multi-associated graded is

$$\mathrm{gr}_m^F \mathcal{O}_Y(*D)|_{D_J} \cong \bigotimes_{a \in J} N_{D_a/Y}^{\otimes m_a} \otimes \mathcal{O}_{D_J}(*D_J^{\mathrm{res}}). \quad (108)$$

The residual divisor records further boundary components meeting the stratum.

15.2 The full operator product

Let D_{12} be the pair-collision divisor in $\mathrm{FM}_2(X) = X^2$. In a coordinate $t = z_1 - z_2$, the singular operator product is

$$\sum_{m \geq 1} A_m(z_2) t^{-m}.$$

The coefficient A_m belongs intrinsically to the m th normal symbol. For a vertex algebra, $A_m = a_{(m-1)}b$. Therefore

$$a_{(r)}b \longleftrightarrow \mathrm{gr}_{r+1}^F \mathcal{O}(*D_{12}) \cong N_{D_{12}/X^2}^{\otimes(r+1)}. \quad (109)$$

The correspondence is coordinate-free at the associated-graded level. A coordinate chooses a trivialization of the normal line and thereby names the coefficient as a mode.

The zeroth regular coefficient belongs to F_0 and is the normally ordered product; its higher regular Taylor coefficients are obtained from it by translation and normal differentiation. Thus the complete filtered chiral operation contains both the singular modes and the regular multiplication. Keeping only the associated graded forgets extensions between pole orders; these extensions encode the triangular mixing under coordinate changes.

15.3 Nested collisions

Let T be a rooted collision tree indexing a boundary stratum $D_T \subset \mathrm{FM}_I(X)$. Each internal edge e corresponds to one normal direction, hence to a line N_e . Locally the normal bundle splits

$$N_{D_T/\mathrm{FM}_I(X)} \cong \bigoplus_{e \in E_{\mathrm{int}}(T)} N_e. \quad (110)$$

The multi-associated graded of a meromorphic coefficient is therefore

$$\bigotimes_{e \in E_{\mathrm{int}}(T)} N_e^{\otimes m_e}. \quad (111)$$

Assigning a positive integer m_e to every internal edge is the geometric version of assigning a pole order to every intermediate OPE channel.

Tree substitution contracts a subtree and replaces it by one incoming branch. Over an arbitrary curve the root point and its tangent line must remain part of the colour: Fulton–MacPherson spaces are not, by themselves, a single uncoloured operad unless additional translation or framing data is supplied. The intrinsic object is therefore a coloured collision coefficient system over X , whose colours are root points together with their tangent lines.

Theorem 15.1 (Normal-cone collision operad). *The multi-normal symbols of nested collision strata*

form a coloured operad over X . For a collision tree T , its coefficient line is

$$\bigotimes_{e \in E_{\text{int}}(T)} N_e^{\otimes m_e},$$

with one positive pole order m_e on every internal edge. Substitution is induced by the canonical morphisms of Rees algebras for nested regular embeddings. It is associative and independent of local coordinates.

Proof. A nested collision stratum is an iterated regular embedding. The associated graded of its formal neighborhood is the symmetric algebra of the conormal bundle, and (110) identifies the independent conormal directions with internal edges. For a chain of nested regular embeddings, deformation to the normal cone is functorial: composing the corresponding Rees morphisms either before or after grouping a subtree gives the same morphism. Tensoring the resulting normal lines gives the stated substitution law and its associativity. The construction uses only ideals of diagonals and their normal bundles, hence is coordinate-free. $\square$

A strict filtered identification of formal neighborhoods requires choices of compatible local coordinates or tubular neighborhoods. Different choices act by filtration-positive automorphisms. The associated-graded coloured operad is canonical; the filtered operation is canonical as a chiral distribution, not as a chosen Laurent series.

15.4 The collision Rees space

The pole filtration can be assembled into a Rees algebra

$$\mathcal{R}_D(\mathcal{O}_Y(*D)) = \bigoplus_{m \geq 0} F_m \mathcal{O}_Y(*D) u^m. \quad (112)$$

For $u \neq 0$ its localization recovers the meromorphic coefficient algebra. At $u = 0$ it becomes the normal-symbol algebra. Locally, the deformation is the familiar equation

$$tv = u,$$

where v represents the inverse normal coordinate. The generic fiber contains the full pole; the special fiber records its normal cone.

For the full boundary of $\text{FM}_I(X)$ one obtains a multi-Rees space. Collision scales become independent deformation parameters. The special fiber is a union of normal-cone pieces indexed by forests. This geometry explains why classical limits are controlled by trees and why quantum corrections mix different pole orders.

15.5 The chiral and classical operads

The algebraic chiral operad is filtered by the order of poles along diagonals. Bakalov–De Sole–Heluani–Kac construct the corresponding classical operad and a canonical comparison from chiral symbols to classical operations [6, 7]. Geometrically, a chiral product passes to its normal symbols; when a filtration on the state space is compatible with the products, the regular multiplication becomes commutative and the leading singular symbols define the Poisson vertex bracket.

Theorem 15.2 (Chiral-to-classical symbol map). *On a translation-invariant formal disc, the pole filtration of the chiral operad has a canonical morphism of operads*

$$\mathrm{gr}^F \mathcal{P}^{\mathrm{ch}} \longrightarrow \mathcal{P}^{\mathrm{cl}}.$$

Under the normal-symbol identification (109), this is the symbol morphism of the geometric pole filtration. If a filtered vertex algebra is compatible with its filtration, the image of its chiral Maurer–Cartan operation is the Poisson vertex algebra operation on the associated graded.

The statement does not assert that every vertex algebra is a flat quantization of a prescribed Poisson vertex algebra, nor that the symbol map is an isomorphism without further hypotheses. The Li filtration is the standard compatible filtration.

15.6 Residues, jets, and differential operators

Higher coefficients may be extracted by pairing principal parts with jets. If t is a coordinate normal to D , then

$$A_m = \frac{1}{2\pi i} \oint t^{m-1} \left(\sum_{r \geq 1} A_r t^{-r} \right) dt. \quad (113)$$

The factor t^{m-1} is a choice of jet. Intrinsically, the pairing is between $N_D^{\otimes m}$ and $N_D^{\vee \otimes m}$. The contour integral is the analytic realization of this pairing. Ordinary residue is $m = 1$.

In a D -module, normal derivatives of delta distributions supply the dual description. The identity between principal parts and delta derivatives is a local Fourier transform in the normal direction. This is why the lambda bracket encodes all singular modes polynomially in λ .

15.7 Coordinate changes and the stress tensor

A projective connection is required to make the stress tensor transform tensorially. Under $z \mapsto w(z)$,

$$T(z) = \left(\frac{dw}{dz} \right)^2 T(w) + \frac{c}{12} \{w, z\}, \quad (114)$$

where $\{w, z\}$ is the Schwarzian derivative. The inhomogeneous term is not the failure of $d \log(z_1 - z_2)$ to be invariant. It is the central extension of the action of vector fields, detected by the fourth-order pole in the TT operator product.

The pole filtration makes the distinction visible. The central term belongs to the fourth normal symbol. The transformation law mixes it with lower terms through the choice of projective connection. The invariant object is the chiral operation of the Virasoro algebra together with its coordinate-change action.

15.8 Pole order is not homotopy depth

The maximal pole order of a chosen strong generating set is a local analytic invariant. The length of a minimal resolution, the nonvanishing range of transferred higher operations, and the genus at which a trace appears are homological or modular invariants. There is no general equality among them. A first-order system may have only simple poles and still have rich representation theory; a Virasoro field has a fourth-order self-OPE but its deformation complex is not determined by the number four.

This separation replaces scalar “depth classifications” by canonical filtrations: pole order, coradical degree, collision codimension, graph genus, and cohomological degree. Each filtration has a geometric origin and a convergent spectral sequence under its own boundedness hypotheses.

16 The formal disc and vertex algebras

16.1 Translation-equivariant chiral operations

Let V be a vector superspace with an even endomorphism ∂ . On the affine line, translation-invariant rational functions on configurations are generated by the differences $z_i - z_j$ and their inverses. The algebraic chiral operad has n -ary component

$$\mathcal{P}_V^{\text{ch}}(n) = \text{Hom}_{\mathcal{D}_n^T} \left(V^{\otimes n} \otimes \mathbb{C}[z_i - z_j, (z_i - z_j)^{-1}], V[\lambda_1, \dots, \lambda_n] / \langle \partial + \sum_i \lambda_i \rangle \right), \quad (115)$$

where $\mathcal{D}_n^T$ is the algebra of translation-invariant differential operators. The variables λ_i record normal derivatives. The relation in the target expresses simultaneous translation covariance.

The operad (115) is the formal-coordinate presentation of the Beilinson–Drinfeld chiral endomorphism operad. It contains the regular multiplication and all finite principal parts. Composition expands rational functions in the domains corresponding to nested collisions.

16.2 The universal Lie superalgebra of an operad

For a linear operad $\mathcal{P}$, set

$$W(\mathcal{P}) = \bigoplus_{k \geq -1} W_k(\mathcal{P}), \quad W_k(\mathcal{P}) = \mathcal{P}(k+1)^{\Sigma_{k+1}}.$$

Operadic insertion defines a pre-Lie product and hence a Lie superbracket. For $\mathcal{P} = \mathcal{P}_{\text{IV}}^{\text{ch}}$, write $W^{\text{ch}}(V)$. An odd binary operation is an element $X \in W_1^{\text{ch}}(\text{IV})$.

The following theorem is the algebraic selection principle.

Theorem 16.1 (Bakalov–De Sole–Heluani–Kac). *An odd element $X \in W_1^{\text{ch}}(\text{IV})$ satisfies*

$$[X, X] = 0 \quad (116)$$

if and only if X defines a nonunital vertex algebra structure on V with translation operator ∂ . The differential $\text{ad } X$ on $W^{\text{ch}}(V)$ is the vertex-algebra deformation differential.

The theorem includes all coefficients of the integral lambda bracket. Skewsymmetry, the normally ordered product, the noncommutative Wick formula, and the Borchers identity are different coefficient forms of the single equation (116) [6].

16.3 The coderivation identity

Let $\mathcal{C}$ be a conilpotent dg cooperad and let $\alpha : \mathcal{C} \rightarrow \mathcal{P}^{\text{ch}}$ be an operadic twisting morphism, for example the canonical twisting morphism from the bar cooperad. An operation X then determines, through α , a coderivation d_X of the cofree $\mathcal{C}$ -coalgebra $\text{cofree}_{\mathcal{C}}(sV)$. The commutator of coderivations corresponds to the convolution bracket.

Proposition 16.2 (Square of the vertex coderivation). *In every such bar-coalgebra model, an odd chiral operation X satisfies*

$$d_X^2 = \frac{1}{2}[d_X, d_X] = \frac{1}{2}d_{[X, X]}. \quad (117)$$

Consequently d_X is a differential exactly when the image of X is Maurer–Cartan in the chiral convolution Lie algebra; for the binary operation of Theorem 16.1, this is exactly the vertex-algebra condition.

Proof. Coderivations of a cofree conilpotent coalgebra are determined by their corestriction to co-generators. The corestriction of $[d_X, d_Y]$ is the signed sum of single insertions represented by the convolution pre-Lie product; antisymmetrizing gives $[X, Y]$. Setting $Y = X$ gives the formula. $\square$

Combining Theorems 15.1, 16.1 and 16.2 gives the geometric statement: the full pole-filtered collision operation produces a square-zero coderivation exactly when the vertex identities hold. The geometry represents the coefficients; the operadic bracket supplies the identity.

16.4 The Borchers identity

For homogeneous $a, b, c \in V$, the coefficient form of the vertex Jacobi identity is

$$\sum_{j \geq 0} (-1)^j \binom{n}{j} (a_{(m+n-j)} b_{(k+j)} c - (-1)^n (-1)^{p(a)p(b)} b_{(n+k-j)} a_{(m+j)} c) = \sum_{j \geq 0} \binom{m}{j} (a_{(n+j)} b)_{(m+k-j)} c. \quad (118)$$

The binomial coefficients arise from translating one formal variable relative to another. Equation (118) is therefore invisible to a complex that retains only a simple residue and discards jets. The normal-symbol filtration retains precisely the powers needed to recover the translated expansions.

At the level of distributions, the same identity is the equality of three expansions of a rational distribution on the complement of the three diagonals. The chiral operad records the expansions and their common meromorphic origin.

16.5 Vacuum and conformal structure

A vacuum vector $\mathbf{1}$ and the creation axiom extend the nonunital structure to a vertex algebra. A conformal vector ω supplies operators $L_n = \omega_{(n+1)}$ with Virasoro commutation relations and $L_{-1} = \partial$. The central charge is the coefficient of the fourth-order pole in

$$Y(\omega, z)Y(\omega, w) \sim \frac{c/2}{(z-w)^4} + \frac{2Y(\omega, w)}{(z-w)^2} + \frac{\partial_w Y(\omega, w)}{z-w}.$$

The vacuum, translation, and conformal vector are distinct structures. Chiral Koszul duality applies to the underlying chiral Lie object; conformal blocks and modular anomalies use the conformal structure.

16.6 The classical limit

Let $F_\bullet V$ be a filtration compatible with all vertex products, for example the Li filtration. The associated graded is commutative with a derivation and a lambda bracket. The regular normally

ordered product becomes the commutative product; the first nontrivial singular symbols become the Poisson vertex bracket.

Theorem 16.3 (Classical symbol operad). *There is a canonical morphism from the pole-associated graded of the chiral operad to the classical operad $\mathcal{P}^{\text{cl}}$. An odd Maurer–Cartan element in $W(\mathcal{P}_{\text{IV}}^{\text{cl}})$ is a Poisson vertex algebra structure. For a vertex algebra equipped with a compatible filtration, the image of the principal chiral symbol is the Maurer–Cartan element of the associated Poisson vertex algebra.*

The theorem makes quantization a lifting problem in the chiral deformation complex. Obstructions lie in the second cohomology of the classical differential; choices of lifts lie in the first. This is ordinary Maurer–Cartan obstruction theory applied to the filtered operad.

16.7 Deformations

Fix a vertex operation X . The complex

$$(W^{\text{ch}}(\Pi V), d_X = [X, -]) \quad (119)$$

controls formal deformations. If

$$X_{\hbar} = X + \hbar X_1 + \hbar^2 X_2 + \cdots,$$

then the coefficient of $\hbar$ in $[X_{\hbar}, X_{\hbar}] = 0$ says $d_X X_1 = 0$. Gauge-equivalent first-order deformations differ by a coboundary. The second-order obstruction is the class of $\frac{1}{2}[X_1, X_1]$ in H^2 .

This cohomology is the appropriate home of local deformation classes. It is not automatically the Hochschild cohomology of a mode algebra and not the cohomology of the moduli space of curves. Comparison maps may relate these theories in specific settings.

16.8 Homotopy chiral structures

Let $\mathcal{P}_{\infty} \rightarrow \mathcal{P}^{\text{ch}}$ be a cofibrant resolution and let $\mathcal{C}_{\infty}$ be a conilpotent cooperadic model carrying its canonical twisting morphism to $\mathcal{P}_{\infty}$. A homotopy chiral algebra is an algebra over $\mathcal{P}_{\infty}$, equivalently a Maurer–Cartan element in the completed convolution L_{∞} -algebra

$$\text{Hom}_{\Sigma}(\overline{\mathcal{C}_{\infty}}, \text{End}_V).$$

The higher operations are indexed by cells in a resolution of collision composition. Their identities are the components of the Maurer–Cartan equation. This is the correct setting for transferred structures and for BRST models before cohomology concentration.

A strict vertex algebra is a homotopy chiral algebra whose higher generators vanish in a chosen model. Homotopy transfer preserves the quasi-isomorphism type but need not preserve strictness. Therefore the nonvanishing of a transferred higher operation is a property of the contraction as well as of the quasi-isomorphism class; its cohomology class is the invariant obstruction.

16.9 The meaning of “chiral Koszul”

The adjective “chiral” refers to the monoidal category and the diagonal operation. The phrase “chiral Koszul duality” is reserved in this paper for the Chevalley–primitives equivalence on the Ran space. A

vertex algebra is a formal-coordinate manifestation of a chiral algebra. A quadratic presentation of a vertex algebra may admit a smaller quadratic dual, but that is an additional Koszul property, not the definition of the subject.

17 The chiral Chevalley coalgebra on the Ran space

17.1 The Ran space

The Ran space of X is the prestack of nonempty finite subsets of X :

$$\mathrm{Ran}(X) = \operatorname{colim}_{I \in \mathrm{Fin}^{\mathrm{surj}, \mathrm{op}}} X^I.$$

A point of X^I maps to the subset of its coordinate values; coincident coordinates are allowed in the colimit. The transition maps are partial diagonals. Thus the Ran space remembers every finite collision simultaneously.

The category $D(\mathrm{Ran} X)$ carries two monoidal structures. The star tensor product is induced by union of finite subsets without a disjointness condition. The chiral tensor product is induced by union restricted to disjoint subsets. Chiral Lie algebras are Lie algebra objects for the chiral tensor product. Factorization objects are those whose restriction to disjoint configurations is the tensor product of their one-point pieces.

The Ran space is not a finite-dimensional variety, but its category of D -modules is controlled by the Cartesian powers X^I and their diagonal transition maps. Every formula in the preceding sections is therefore a labelled chart of a Ran object.

17.2 Chevalley chains

Let $\mathfrak{g}$ be a chiral Lie algebra. Its chiral Chevalley chains are the cofree commutative coalgebra on $\mathfrak{g}[1]$ with the coderivation induced by the chiral bracket:

$$\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) = \left(\mathrm{coSym}^{\mathrm{ch}}(\mathfrak{g}[1]), d_{\mathfrak{g}} + d_{\mu} \right). \quad (120)$$

Pullback to X^I gives the partition decomposition of [Theorem 13.4](#). The factorization coproduct is induced by the decomposition of a finite set into disjoint subsets.

The primitive filtration is the filtration by the number of factors. Its associated graded is the cofree commutative coalgebra with only the internal differential. This filtration is complete in the conilpotent direction: every element has finite primitive length in a labelled finite component.

Proposition 17.1 (Geometric realization of Chevalley chains). *The forest incidence objects $W_I^{\mathrm{FM}}(\mathfrak{g})$ assemble, under disjoint union of labels and diagonal pullback, into a simplicial resolution of $\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})$ on $\mathrm{Ran}(X)$.*

Proof. Let $\iota_I : C_I^{\mathrm{coll}}(\mathfrak{g}) \rightarrow W_I^{\mathrm{FM}}(\mathfrak{g})$ insert the terminal corolla in every block. Then $\epsilon_I \iota_I = \mathrm{id}$. For a surjection of label sets $\phi : I \twoheadrightarrow J$, let T_{ϕ} be the canonical Ran transition on the partition collision complexes and define the transition on the resolution by

$$\tilde{T}_{\phi} = \iota_J T_{\phi} \epsilon_I.$$

Because $\epsilon_I = \text{id}$, the maps $\tilde{T}_\phi$ are strictly functorial under composition, and ϵ_I is a natural transformation. Disjoint union tensors the tree complexes and preserves terminal corollas, so these transitions are compatible with the factorization coproduct. They therefore assemble the labelled resolutions over the colimit presentation of $\text{Ran}(X)$. $\square$

17.3 Primitives

Let C be a coaugmented conilpotent commutative coalgebra in the chiral monoidal category. The primitive functor is defined invariantly as the right adjoint to the chiral Chevalley functor. With the Chevalley convention of (120), its value is shifted by $[-1]$ to produce the corresponding chiral Lie object.

In a strict conilpotent dg-coalgebra model, let

$$\overline{C} = \text{Cofib}(\mathbf{1} \longrightarrow C)$$

be the coaugmentation coideal and let $\overline{\Delta} : \overline{C} \rightarrow \overline{C} \otimes^{\text{ch}} \overline{C}$ be the reduced coproduct. When strict primitives compute the derived right adjoint—in particular for cofree conilpotent models and their fibrant replacements—one may represent the primitive complex by

$$\text{Prim}(C) \simeq \text{Fib}\left(\overline{C} \xrightarrow{\overline{\Delta}} \overline{C} \otimes^{\text{ch}} \overline{C}\right). \quad (121)$$

For a cofree conilpotent coalgebra this complex is the cogenerator complex.

The Chevalley and primitive constructions form an adjunction

$$\text{coChev}^{\text{ch}} : \text{Lie}_{\text{ch}}(X) \rightleftarrows \text{Coalg}_{\text{Com}}^{\text{ch}, \text{conil}}(\text{Ran } X) : \text{Prim}[-1]. \quad (122)$$

The shift depends on the Chevalley convention. The substantive point is that the bracket and the reduced coproduct are Koszul-dual operations.

17.4 Francis–Gaitsgory duality

The chiral monoidal category on the Ran space is pro-nilpotent with respect to cardinality. This property removes the convergence obstruction that ordinary Lie–commutative Koszul duality encounters in an arbitrary monoidal category.

Theorem 17.2 (Francis–Gaitsgory chiral Koszul duality). *In the factorization and conilpotent subcategories of the pro-nilpotent chiral monoidal category of D -modules on $\text{Ran}(X)$ considered by Francis and Gaitsgory, the chiral Chevalley functor and the primitive functor are inverse equivalences between chiral Lie algebras and conilpotent factorization coalgebras:*

$$\text{Lie}_{\text{ch}}(X) \simeq \text{CoFact}_{\text{conil}}(\text{Ran } X). \quad (123)$$

The unit and counit of the adjunction are equivalences.

This theorem is the invariant core of chiral Koszul duality [38]. It is stronger than the statement that a quadratic algebra is Koszul. No presentation is chosen and no linear dual is taken.

17.5 Chiral homology

The chiral homology of $\mathfrak{g}$ on a complete curve X is the de Rham cohomology of its factorization coalgebra:

$$\int_X^{\text{ch}} \mathfrak{g} := R\Gamma_{\text{dR}}(\text{Ran } X, \text{coChev}^{\text{ch}}(\mathfrak{g})). \quad (124)$$

The cardinality filtration produces a spectral sequence whose first page is a sum of cohomologies of configuration spaces with coefficients in external tensor powers of $\mathfrak{g}$. The collision differential is induced by the chiral bracket.

For an ordinary Lie algebra $\mathfrak{g}_0$ placed constantly on a point, this reduces to the usual Chevalley complex. For a current chiral algebra on a curve, it becomes a global version of Lie algebra homology with local singularities. For modules inserted at marked points, the pointed collision complex computes derived coinvariants.

17.6 The factorization envelope

Beilinson and Drinfeld associate to a chiral Lie algebra its chiral enveloping algebra $U^{\text{ch}}(\mathfrak{g})$. Its factorization structure is generated by the same local operation μ , but the output is associative rather than coalgebraic. The Chevalley coalgebra and the enveloping factorization algebra play different roles:

$$\begin{aligned} \text{coChev}^{\text{ch}}(\mathfrak{g}) & \text{ resolves primitives and homology,} \\ U^{\text{ch}}(\mathfrak{g}) & \text{ acts on modules and defines centers.} \end{aligned}$$

The Poincaré–Birkhoff–Witt theorem identifies the associated graded of $U^{\text{ch}}(\mathfrak{g})$ with the symmetric chiral algebra under the standard filtration. This is a theorem about the enveloping construction, not an assertion that every vertex algebra is a universal enveloping algebra.

17.7 Analytic realization

Over $\mathbb{C}$, regular holonomic D -modules correspond under Riemann–Hilbert to constructible sheaves. The chiral operation becomes a morphism of constructible objects with singular support along diagonals. On the open configuration space, horizontal sections form a local system. The difference between the algebraic and analytic descriptions is then the difference between a regular singular connection and its monodromy representation.

This passage explains the ordered theory. The Ran coalgebra retains symmetric factorization. Pulling its analytic realization to labelled configurations and taking horizontal sections retains paths and braid monodromy. The two constructions are related but not identical: de Rham cohomology forgets a chosen basis of horizontal solutions, while monodromy acts on it.

17.8 Completion and cardinality

The Ran category is pro-nilpotent by cardinality, but additional completions may enter from conformal weight, pole order, or infinite-dimensional state spaces. These are independent filtrations. If C_N is an inverse system of finite-window complexes with surjective transition maps on cohomology, then the Milnor sequence

$$0 \longrightarrow \varprojlim^1 H^{m-1}(C_N) \longrightarrow H^m(\varprojlim C_N) \longrightarrow \varprojlim H^m(C_N) \longrightarrow 0 \quad (125)$$

shows that completion is exact when the derived limit vanishes. This is the precise content of a Mittag-Leffler argument.

No unrestricted completion theorem is needed for [Theorem 17.2](#). Completion enters when one replaces abstract Ran objects by explicit infinite products of modes or conformal weights.

17.9 The Beilinson hierarchy

The objects generated by one chiral operation form a hierarchy:

$$\mathfrak{g} \longmapsto \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \int_X^{\mathrm{ch}} \mathfrak{g}, \quad \mathfrak{g} \longmapsto U^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \mathrm{Mod}(U^{\mathrm{ch}}(\mathfrak{g})). \quad (126)$$

The first branch is coalgebraic and homological. The second is algebraic and representation-theoretic. Centers, traces, and conformal blocks are built from the second branch; primitives and factorization homology are built from the first. Chiral Koszul duality connects the primitive object to its factorization coalgebra without collapsing the two branches.

18 Chiral Koszul duality, bar–cobar, and presentations

18.1 Three meanings of Koszul duality

The phrase “Koszul duality” is used for three related constructions.

The first is the Lie–commutative equivalence of [Theorem 17.2](#). This is chiral Koszul duality in the strict sense of this paper.

The second is bar–cobar reconstruction for augmented associative or operadic algebras. An augmented algebra A has a conilpotent bar coalgebra $B(A)$, and a conilpotent coalgebra C has a cobar algebra $\Omega(C)$. Under the usual conilpotence and completeness conditions, the unit and counit are equivalences.

The third is quadratic Koszul duality. A quadratic presentation $A = T(V)/(R)$ determines a smaller coalgebra $A^\natural$ generated by sV with corelations s^2R . The algebra is Koszul when the canonical map $A^\natural \rightarrow B(A)$ is a quasi-isomorphism.

The first construction is presentation-free. The second is a universal resolution. The third is a property of a presentation. They coincide only when comparison maps have been constructed.

18.2 Twisting morphisms

Let C be a conilpotent dg coalgebra and A an augmented dg algebra in a complete symmetric monoidal category. A twisting morphism is a degree-one map $\tau : C \rightarrow A$ satisfying

$$d\tau + \mu_A(\tau \otimes \tau)\Delta_C = 0. \quad (127)$$

This is a Maurer–Cartan equation in the convolution algebra $\mathrm{Hom}(C, A)$. It determines a coalgebra map $C \rightarrow B(A)$ and an algebra map $\Omega(C) \rightarrow A$.

Theorem 18.1 (Fundamental theorem of twisting morphisms). *Assume that the bar and cobar constructions are defined and that the coradical filtrations converge. For a twisting morphism $\tau : C \rightarrow A$, the following are equivalent:*

- (i) *the map $C \rightarrow B(A)$ is a quasi-isomorphism;*
- (ii) *the map $\Omega(C) \rightarrow A$ is a quasi-isomorphism;*
- (iii) *the left twisted tensor product $A \otimes_{\tau} C$ is acyclic;*
- (iv) *the right twisted tensor product $C \otimes_{\tau} A$ is acyclic.*

The theorem is classical over a field and extends to filtered chiral settings when the monoidal products preserve the relevant limits and the comparison spectral sequences converge [55]. Every hypothesis in a chiral application concerns the ambient category, not the formal equation (127).

18.3 Quadratic recognition

Let $A = T(V)/(R)$ be a connected weight-graded quadratic algebra. Its quadratic coalgebra A^i comes with a canonical twisting morphism. The usual Koszul criterion is diagonal concentration of bar homology.

Theorem 18.2 (Quadratic Koszul criterion). *For a connected weight-graded quadratic algebra A , the following are equivalent:*

- (i) *$A^i \rightarrow B(A)$ is a quasi-isomorphism;*
- (ii) *$\Omega(A^i) \rightarrow A$ is a quasi-isomorphism;*
- (iii) *$\mathrm{Tor}_{p,q}^A(\mathbb{C}, \mathbb{C}) = 0$ for $p \neq q$;*
- (iv) *the quadratic twisted tensor complexes are acyclic.*

For a chiral algebra presented by generators and relations, the same criterion applies in each finite window once the free chiral algebra, the quadratic relation object, and the filtered comparison have been defined. It does not imply that universal affine, Virasoro, lattice, and W -algebras are all Koszul. Each family requires its own presentation and spectral-sequence proof.

18.4 PBW deformations

A PBW theorem identifies the associated graded of a filtered algebra with a prescribed graded algebra. By itself, this does not identify the bar complex of the filtered algebra with a quadratic dual: lower-order linear and constant terms alter the Koszul differential and may introduce curvature. The invariant comparison statement is the following filtered lemma.

Criterion 18.3 (Filtered Koszul comparison). *Let $q : C \rightarrow B(A)$ be a morphism from a filtered conilpotent coalgebra to the bar coalgebra of a filtered augmented algebra. Assume that*

- (a) *the filtrations on C and $B(A)$ are exhaustive, separated, complete, and preserved by q ;*
- (b) *the induced map $\mathrm{gr} q : \mathrm{gr} C \rightarrow B(\mathrm{gr} A)$ is a quasi-isomorphism;*
- (c) *the two comparison spectral sequences converge strongly.*

Then q is a quasi-isomorphism.

Proof. The cone of q inherits a complete exhaustive filtration. Its associated graded is the cone of $\text{gr } q$ and is acyclic by hypothesis. Strong convergence therefore gives zero cohomology for the filtered cone, so q is a quasi-isomorphism. $\square$

For a quadratic-linear-constant PBW deformation of a quadratic Koszul algebra, the appropriate C is the nonhomogeneous or curved Koszul coalgebra. The PBW compatibility identities ensure that its coderivation squares to the prescribed curvature, and the associated-graded comparison is the ordinary quadratic Koszul resolution. Thus PBW, classical Koszulness, and the curved compatibility identities together imply the filtered comparison. PBW alone does not.

In a vertex algebra, an application must additionally control derivatives, infinitely many modes, and the completion in conformal weight or pole order. No family is declared Koszul merely from the existence of a PBW spanning set.

18.5 Verdier duality and a dual algebra

The chiral Chevalley coalgebra is not a linear dual. Nevertheless, if its labelled components are Verdier dualizable, duality exchanges coalgebra maps with algebra maps. Let $\mathbb{D}_{\text{Ran}}$ denote Verdier duality on an appropriate constructible or holonomic subcategory.

Proposition 18.4 (Verdier-dual algebra). *Let C be a factorization coalgebra whose finite labelled components are Verdier dualizable, and suppose Verdier duality commutes with the factorization restriction maps. Then*

$$C^\vee := \mathbb{D}_{\text{Ran}} C$$

is a factorization algebra. If biduality holds componentwise and is compatible with factorization, then $C \rightarrow \mathbb{D}_{\text{Ran}} \mathbb{D}_{\text{Ran}} C$ is an equivalence of factorization coalgebras.

Applied to $C = \text{coChev}^{\text{ch}}(\mathfrak{g})$, this produces a Verdier-dual factorization algebra. Identifying it with a familiar vertex algebra, a reflected level, or a quantum group is a separate recognition theorem.

18.6 Completion

Let $F^1 A$ be the augmentation ideal. The completed bar and cobar constructions use products over tensor length rather than direct sums. Their differentials converge when they raise the filtration. The completed counit

$$\widehat{\Omega} \widehat{B}(A) \longrightarrow A$$

is an equivalence for complete augmented algebras in the standard pronilpotent setting.

Curved algebras require another distinction. If a curvature element m_0 is present, the unary operation satisfies $m_1^2 = [m_0, -]$ rather than zero. A central scalar has zero commutator and therefore cannot be the same datum as a nonzero inner curvature. Curved bar–cobar duality belongs to the coderived and contraderived categories of Positselski [60]. It does not alter the square-zero chiral differential of an honest vertex algebra.

18.7 Koszul dual modules

A twisting morphism $\tau : C \rightarrow A$ defines adjoint functors between A -modules and C -comodules. Under the hypotheses of Theorem 18.1, these functors give an equivalence on the appropriate derived

categories. For a quadratic Koszul algebra, the resolution is linear and yields the familiar exchange of Ext and Tor.

In the chiral setting, modules are factorization modules at marked points. The pointed collision complex is the bar resolution. A module-level Koszul duality therefore requires the algebra comparison and the compatibility of marked-point factorization. Fusion products and conformal blocks are preserved only when a braided tensor equivalence of module categories has been established.

18.8 Maurer–Cartan moduli

Twisting morphisms form a Maurer–Cartan space in a convolution dg Lie or L_∞ -algebra. Gauge equivalence is homotopy of twisting data. The tangent complex at τ is the twisted convolution complex

$$(\mathrm{Hom}(C, A), d + [\tau, -]).$$

Its first cohomology contains infinitesimal deformations and its second contains obstructions.

This is the natural moduli object associated with Koszul presentations. Koszulness itself is the open condition that the comparison cone be acyclic in a finite perfect family; globally it is a derived condition on the vanishing of that cone. There is no canonical finite list of scalar invariants that classifies the bar coalgebra.

18.9 Exact limits of the notion

The following implications do not hold without additional hypotheses:

$$\begin{aligned} \text{PBW presentation} &\not\Rightarrow \text{Koszul}, & C_2\text{-cofinite} &\not\Rightarrow \text{Koszul}, \\ \text{Verdier dualizable} &\not\Rightarrow \text{self-dual}, & \text{braided monodromy} &\not\Rightarrow \text{specified Yangian}, \\ & & \text{equal central charges} &\not\Rightarrow \text{equivalent chiral algebras}. \end{aligned} \tag{128}$$

Each failed implication confuses a coarse shadow with a structured object. The correct replacements are comparison maps and recognition theorems.

19 Duality, Hochschild theory, and the derived center

19.1 The center requires an associative object

A chiral Lie algebra has a bracket, not a bimodule category. The expression $\mathrm{RHom}_{A-A}(A, A)$ is therefore undefined when A denotes only a chiral Lie object. To form a center, one first passes to an associative chiral or factorization algebra, such as the chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$.

Definition 19.1 (Derived chiral center). For a chiral Lie algebra $\mathfrak{g}$ for which the chiral enveloping factorization algebra is defined, set

$$Z_{\mathrm{ch}}^{\mathrm{der}}(\mathfrak{g}) := \mathrm{HH}_{\mathrm{Fact}}^*(U^{\mathrm{ch}}(\mathfrak{g}), U^{\mathrm{ch}}(\mathfrak{g})). \tag{129}$$

For an associative factorization algebra A , write $Z_{\mathrm{ch}}^{\mathrm{der}}(A) = \mathrm{HH}_{\mathrm{Fact}}^*(A, A)$.

Hochschild cochains are derived endomorphisms of the diagonal bimodule. The bar resolution supplies a concrete model. In the chiral setting the resolution is supported on configurations with

an incoming and an outgoing boundary component, and insertion of configurations produces brace operations.

19.2 The E_2 structure

The higher Deligne theorem gives Hochschild cochains of an E_1 algebra an E_2 structure. The same statement holds for associative factorization algebras in a suitable symmetric monoidal category. The product is cup product; the shifted Lie bracket is the Gerstenhaber bracket; higher braces encode homotopies.

Theorem 19.2 (Derived center structure). *Let A be an associative factorization algebra whose bimodule category admits the diagonal bar resolution. Then $Z_{\text{ch}}^{\text{der}}(A)$ is an E_2 algebra. Its action on A is universal among homotopy-coherent central actions.*

The universal property is algebraic. An object B acting on A through the Swiss-cheese operad determines a morphism of E_2 algebras

$$B \longrightarrow Z_{\text{ch}}^{\text{der}}(A). \quad (130)$$

The map sends a closed operation to the family of open operations obtained by inserting it next to boundary inputs.

19.3 The center and a physical bulk

Let Obs_{bulk} be the factorization algebra of local observables of a field theory with boundary algebra A . The bulk-to-boundary operator product defines a map

$$\text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ch}}^{\text{der}}(A). \quad (131)$$

If the map is a quasi-isomorphism, the bulk local observables are reconstructed from the boundary algebra. If a decoupled bulk sector acts trivially on the boundary, it lies in the kernel of every boundary measurement and prevents (131) from being an equivalence.

No-go theorem 19.3 (Boundary data do not determine a decoupled bulk). *A boundary chiral algebra and all of its algebraic centers are unchanged by tensoring the bulk theory with a decoupled theory whose boundary action is trivial. Therefore no theorem can identify an arbitrary physical bulk with the derived center of its boundary without a nondegeneracy or generation hypothesis.*

This statement is the exact mathematical limit of holographic reconstruction from local boundary operations.

19.4 The annulus and Hochschild homology

Factorization homology of the circle with coefficients in an E_1 algebra is Hochschild homology:

$$\int_{S^1} A \simeq A \otimes_{A \otimes A^{\text{op}}} A. \quad (132)$$

For a boundary theory, the annulus is obtained by thickening the circle. Its state space is therefore a trace of the boundary algebra. The pairing between Hochschild cochains and Hochschild chains is the algebraic form of inserting a bulk local operator into an annulus amplitude.

A cyclic trace on A thickens planar trees to ribbon graphs. This is the point at which genus enters. Without a cyclic pairing, the derived center has an E_2 structure but no canonical loop contraction.

19.5 Morita invariance

Hochschild cochains and chains are Morita invariant. If A and B are derived Morita equivalent associative factorization algebras, then

$$\mathrm{HH}_{\mathrm{Fact}}^*(A, A) \simeq \mathrm{HH}_{\mathrm{Fact}}^*(B, B)$$

as E_2 algebras, and their Hochschild homologies agree.

A Koszul duality between module categories may therefore induce an equivalence of centers. The implication runs through Morita equivalence, not through a scalar complementarity rule. A coalgebraic Koszul dual without a module-category equivalence does not by itself identify centers.

19.6 A small convolution model

Assume A is augmented, complete, and Koszul with a perfect bar coalgebra $B(A)$. The diagonal bar resolution identifies Hochschild cochains with a twisted convolution complex

$$Z_{\mathrm{ch}}^{\mathrm{der}}(A) \simeq \mathrm{Hom}^\tau(B(A), A), \quad (133)$$

where the differential is

$$df = d_A f - (-1)^{|f|} f d_B + [\tau, f].$$

The braces insert one coalgebra input into another before applying f . This model is finite and explicit when the bar coalgebra is locally finite.

The formula (133) explains the relation between deformation theory and the center. A first-order associative deformation is a degree-two Hochschild cocycle; its obstruction is the Gerstenhaber square. The chiral deformation complex of Equation (119) and the factorization Hochschild complex may be compared, but they govern different structures: one deforms the chiral operation, the other the associative factorization envelope.

19.7 Verdier duality and centers

Verdier duality exchanges $!$ - and $*$ -extensions across diagonals. For a perfect factorization algebra A , duality produces an opposite or dual factorization algebra. The induced map on bimodule categories reverses left and right actions. Consequently the Hochschild complex of the dual is the dual of the Hochschild chain complex of A under a Calabi–Yau pairing.

A degree- d Calabi–Yau structure on A gives a nondegenerate pairing

$$\mathrm{HH}^i(A, A) \otimes \mathrm{HH}_{d-i}(A) \longrightarrow \mathbb{C}.$$

The degree is determined by the cyclic structure, not by the dimension of a moduli space chosen later. This pairing is the algebraic origin of BV operators on Hochschild cohomology.

19.8 Topological enhancement

The derived centre is E_2 in the transverse directions. A coherent null-homotopy of one further real translation gives an E_3 enhancement. A coherent topologization of the whole complex curve supplies two additional real topological directions and gives an E_4 algebra, as in Proposition 9.8. A holomorphic centre is not topological merely because it is a centre.

In three-dimensional Chern–Simons theory, local bulk observables are E_3 because the bulk has three topological directions. Comparison with a boundary centre therefore uses the appropriate intermediate topologization of the boundary object. In a four-real-dimensional holomorphic–topological theory whose holomorphic direction is also topologized, the corresponding local structure is E_4 . In either dimension, proving that the bulk-to-centre map is an equivalence is a field-theoretic calculation.

19.9 Centers at critical level

For the universal affine vertex algebra at the critical level, the ordinary vertex-algebra center is large: the Feigin–Frenkel theorem identifies it with functions on opers for the Langlands dual group on the formal disc [34]. This center is a commutative chiral algebra inside the affine algebra. It is not the same object as factorization Hochschild cochains, although the latter receives operations from it.

The distinction is visible in degrees. The vertex center consists of fields commuting with all fields under nonnegative modes. The derived center includes higher self-extensions of the diagonal bimodule. At noncritical level the ordinary center may be scalar while the derived center remains nontrivial.

Part III

Exchange, reduction, and quantum symmetry

20 Labelled chiral configurations and monodromy

20.1 Exchange geometry is not the ordered line

The order in an E_1 algebra comes from chambers of configurations on a real line. The monodromy in this section comes instead from moving labelled chiral insertion points in the complex curve, or from an E_2 transverse enhancement. Nothing below is implied by the bare E_1 multiplication of an ordered chiral algebra.

20.2 Labels retain paths

The labelled configuration space of n points is

$$\mathrm{Conf}_n(X) = \{(x_1, \dots, x_n) \in X^n \mid x_i \neq x_j \text{ for } i \neq j\}.$$

The symmetric group acts freely, and the quotient is the unordered configuration space

$$\mathrm{UConf}_n(X) = \mathrm{Conf}_n(X)/\Sigma_n.$$

For $X = \mathbb{C}$,

$$\pi_1(\mathrm{Conf}_n(\mathbb{C})) = P_n, \quad \pi_1(\mathrm{UConf}_n(\mathbb{C})) = B_n,$$

the pure braid and braid groups. A local system on the labelled space therefore retains pure braid monodromy; an equivariant local system descends to the unordered space and retains full braid monodromy.

A symmetric factorization coalgebra is not obtained by forgetting labels in the naive set-theoretic sense. It carries coherent descent data under all relabelings and diagonals. Labelled analytic continuation is extra information: it is a representation of the fundamental groupoid of the configuration space.

20.3 Derived descent

Let M be a chain complex with an action of a finite group G . The homotopy coinvariants are

$$M_{hG} = M \otimes_{\mathbb{C}[G]}^L \mathbb{C}.$$

The augmentation ideal $I_G = \ker(\mathbb{C}[G] \rightarrow \mathbb{C})$ fits into a distinguished triangle

$$M \otimes_{\mathbb{C}[G]}^L I_G \longrightarrow M \longrightarrow M_{hG}. \quad (134)$$

Thus the homotopy fiber of descent is canonical. In characteristic zero and for finite G , the group algebra is semisimple, so derived coinvariants agree with ordinary coinvariants. The fiber still records all nontrivial isotypic components.

For a braid-group local system, the relevant descent is not tensoring over $\mathbb{C}[\Sigma_n]$ alone. One must retain the monodromy representation and its equivariance under the extension

$$1 \longrightarrow P_n \longrightarrow B_n \longrightarrow \Sigma_n \longrightarrow 1.$$

Discarding the pure braid action loses analytic continuation even when the underlying vector space is unchanged.

20.4 The descent fiber and collision filtrations

For a coefficient system M_n carrying a Σ_n -action, define the descent fiber

$$\mathcal{C}_n(M) = \text{Fib}(M_n \rightarrow M_{n,h\Sigma_n}). \quad (135)$$

By (134), it is canonically $M_n \otimes_{\mathbb{C}[\Sigma_n]}^L I_{\Sigma_n}$. A universal chain model is obtained from the ordinary group bar resolution of the trivial $\mathbb{C}[\Sigma_n]$ -module. In characteristic zero the group algebra is semisimple, so this derived fiber is represented by the direct sum of the nontrivial isotypic components.

The proper partition complex is a different universal object. It is the incidence and Koszul complex of collision partitions, and its top homology is the Lie representation. It does not resolve the descent fiber of an arbitrary Σ_n -module. When M_n carries a functorial filtration indexed by collision partitions, an explicit comparison morphism from the partition-incidence complex to the group-bar model may filter $\mathcal{C}_n(M)$ by binary collisions, ternary collision coherence, and higher strata. Such a filtration belongs to the coefficient system and must be constructed; it is not automatic from the group action.

The descent fiber need not contain an r -matrix. It contains whatever nontrivial isotypic and braid data the coefficient system actually carries. An r -matrix appears when the first-order monodromy of a flat connection has the form required by the classical Yang–Baxter equation.

20.5 Dual local systems

Let M be an oriented manifold and $\mathcal{L}$ a finite-rank local system with monodromy $\rho : \pi_1(M) \rightarrow GL(V)$. Its Verdier dual is

$$\mathbb{D}\mathcal{L} \cong \mathcal{L}^\vee[\dim M],$$

up to the orientation line. The monodromy of the dual local system is

$$\rho^\vee(\gamma) = \rho(\gamma^{-1})^t. \quad (136)$$

The inverse occurs because parallel transport on the dual is contragredient; the transpose is the linear dual.

Corollary 20.1. *Verdier duality sends a braid operator R to the inverse transpose R^{-t} . If the representation is identified with its dual by a nondegenerate invariant pairing, this becomes inverse braiding.*

The further notation $q \mapsto q^{-1}$ is justified only when a family of braid operators has been identified with a quantum-group family parametrized by q .

20.6 Associators

The monodromy of a flat connection on the configuration space of three points depends on comparing asymptotic zones. A Drinfeld associator is a group-like formal series satisfying the pentagon and hexagon equations. Associators form a torsor under the Grothendieck–Teichmüller group GRT_1 after the usual normalization [30, 77].

An arbitrary linear splitting of a symmetrization map is not an associator. The pentagon equation is a nonlinear condition in arity four; the hexagon relates the associator to the braiding. The torsor arises only inside the solution space of these equations.

In the de Rham realization, the Knizhnik–Zamolodchikov associator is the regularized parallel transport between tangential base points at 0 and 1 on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its coefficients are multiple zeta values. This arithmetic belongs to chiral configuration-space monodromy, not to a scalar obtained by averaging the underlying tensor space.

20.7 Tannaka reconstruction

Let $\mathcal{C}$ be a rigid braided tensor category and let $F : \mathcal{C} \rightarrow \text{Vect}$ be a faithful exact fibre functor. The coend

$$H = \int^{M \in \mathcal{C}} F(M)^\vee \otimes F(M) \quad (137)$$

reconstructs a bialgebra object when F is strong monoidal; rigidity supplies an antipode when the coend belongs to the finite or continuous-dual class. If F is only quasi-monoidal, the associativity constraint produces a coassociator and H is a coquasi-bialgebra. The braiding produces a coquasitriangular structure. Under the relevant finite-dimensional or continuous duality, one obtains a quasitriangular Hopf or quasi-Hopf algebra, respectively.

Theorem 20.2 (Ordered Tannaka principle). *A rigid tensor-compatible monodromy category with a faithful strong monoidal fibre functor reconstructs a coquasitriangular bialgebra, and a Hopf algebra*

whenever the antipode is defined in the chosen finite or continuous-dual class. With a faithful quasi-monoidal fibre functor, it reconstructs a coquasitriangular coquasi-bialgebra. A quasi-Hopf presentation becomes a strict Hopf presentation only after a twist that trivializes the coassociator.

The theorem is universal. Identifying the reconstructed object with $U_q(\mathfrak{g})$, a Yangian, an elliptic quantum group, or a Hall algebra is a recognition theorem using the specific connection and representation category.

20.8 Surface braid groups

For a curve of genus $g > 0$, $\pi_1(\text{Conf}_n(X))$ is a surface pure braid group. Its generators include collisions and transport around the A - and B -cycles of the curve. Flat connections such as the KZB connection represent these generators by operators satisfying elliptic braid relations.

The elliptic theory is not determined by the genus-zero associator. Quasi-periodicity, the choice of a dynamical parameter, and the mapping-class-group action supply additional data. At higher genus, conformal-block monodromy is a representation of the mapping class group of the punctured surface, not merely an iteration of B_n .

20.9 What symmetrization preserves

Symmetrization preserves invariant traces, dimensions of invariant subspaces, and any operation that factors through the quotient. It may preserve nontrivial periods when those periods lie in the invariant part of the local system. Therefore it is false that symmetrization erases all arithmetic or all monodromy. The exact statement is that it kills the non-invariant quotient measured by (135).

For conformal blocks, passing from vectors to their dimension preserves the Verlinde number and loses the mapping-class-group representation. Passing from a braided category to its Grothendieck ring preserves fusion coefficients and loses the associator and twists. These are concrete instances of the same descent.

21 KZ geometry, quantum groups, Gaudin systems, and lines

21.1 The logarithmic connection

Let $\mathfrak{g}$ be a finite-dimensional complex simple Lie algebra with invariant tensor $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$. On $V_1 \otimes \cdots \otimes V_n$, define

$$\nabla_{\text{KZ}} = d - \hbar \sum_{i < j} \Omega_{ij} d \log(z_i - z_j), \quad \hbar = \frac{1}{k + h^\vee}. \quad (138)$$

The parameter depends on the normalization of the invariant form; the shifted level is invariantly determined by the Sugawara construction.

The infinitesimal braid relations are

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{if } \{i, j\} \cap \{k, l\} = \emptyset, \quad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0 \quad (139)$$

for distinct i, j, k .

Theorem 21.1 (Flatness of the KZ connection). *The connection (138) is flat.*

Proof. The logarithmic forms are closed, so for the convention $\nabla = d - A$ the curvature is $A \wedge A$. Terms with disjoint index pairs vanish by the first relation in (139). For every triple i, j, k , the remaining terms factor as the Arnold relation

$$d \log z_{ij} \wedge d \log z_{jk} + d \log z_{jk} \wedge d \log z_{ki} + d \log z_{ki} \wedge d \log z_{ij} = 0$$

times the second infinitesimal braid relation. Hence the curvature vanishes. $\square$

This proof displays the exact division of labor. The differential forms supply the Arnold identity. The coefficient tensor supplies the infinitesimal braid identity. Neither identity alone implies flatness.

21.2 From affine currents to KZ

For the affine vertex algebra $V_k(\mathfrak{g})$, current fields satisfy

$$J^a(z)J^b(w) \sim \frac{k(a, b)}{(z - w)^2} + \frac{J^{[a, b]}(w)}{z - w}. \quad (140)$$

Ward identities for primary insertions convert the simple-pole action into the differential equation (138); the double pole and the Sugawara stress tensor produce the shift by $\hbar^\vee$. The KZ connection is therefore a representation-theoretic shadow of the affine chiral operation, not the bar differential of the entire vertex algebra.

21.3 Drinfeld–Kohno

Let $q = \exp(\pi i \hbar)$ in the standard convention. The Drinfeld–Kohno theorem identifies the braided tensor category defined by KZ monodromy with the category of finite-dimensional modules over the Drinfeld–Jimbo quantum group $U_q(\mathfrak{g})$, with an associator determined by KZ parallel transport [51, 30, 50].

Theorem 21.2 (Drinfeld–Kohno). *For generic $\hbar$, the braid-group representations obtained from the monodromy of the KZ connection are naturally equivalent, as braided tensor data, to those obtained from the universal R -matrix of $U_q(\mathfrak{g})$ with $q = e^{\pi i \hbar}$.*

The theorem is the rigorous statement that a quantum group is present in the ordered affine theory. It includes the associator. It does not identify the kernel of symmetrization with a Hopf algebra.

21.4 The classical r -matrix

The one-form in (138) has coefficient

$$r_{ij}(z_i - z_j) = \hbar \frac{\Omega_{ij}}{z_i - z_j}. \quad (141)$$

The flatness equation at first order in $\hbar$ is the classical Yang–Baxter equation. The monodromy is a path-ordered exponential of the connection; its semiclassical expansion begins with the classical r -matrix.

The same rational kernel appears in several constructions once their normalizations are matched: the KZ connection, the rational Gaudin model, and the line-operator OPE of four-dimensional Chern–

Simons theory. The equality is a theorem only after the coefficient spaces and coupling constants have been identified.

21.5 Gaudin Hamiltonians

Fix distinct points $z_1, \dots, z_n$. The Gaudin Hamiltonians are

$$H_i = \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}. \quad (142)$$

The infinitesimal braid relations imply

$$[H_i, H_j] = 0.$$

They are the residues of the KZ connection in the z_i variable. Their commutativity is therefore another coefficient of flatness.

For highest-weight modules, the Bethe ansatz constructs simultaneous eigenvectors from critical points of a master function. In the simplest $\mathfrak{sl}_2$ case, Bethe roots t_a satisfy

$$\sum_{i=1}^n \frac{\lambda_i}{t_a - z_i} - \sum_{b \neq a} \frac{2}{t_a - t_b} = 0. \quad (143)$$

The Schechtman–Varchenko hypergeometric solutions of the KZ equations relate these critical points to horizontal sections. Thus KZ monodromy and the Gaudin spectrum are two readings of the same flat connection.

21.6 Four-dimensional Chern–Simons theory

Costello–Witten–Yamazaki construct a four-dimensional gauge theory that is topological in two directions and holomorphic in one complex direction. Wilson lines at distinct holomorphic positions have an operator product governed by a spectral-parameter R -matrix. In the rational case, the algebra of line operators is a Yangian; trigonometric and elliptic choices produce quantum loop and elliptic structures [19, 20].

The spectral parameter is geometric: it is the difference of holomorphic positions of the lines. The Yang–Baxter equation follows from isotopy of three lines. The one-loop computation fixes the first correction to the R -matrix. This is a physical construction of ordered monodromy, not a consequence of chiral Koszul duality alone.

21.7 Yangians and recognition

A Yangian $Y_h(\mathfrak{g})$ is a filtered deformation of $U(\mathfrak{g}[t])$ with a rational R -matrix on suitable representations. To recognize a reconstructed quasi-bialgebra as a Yangian one needs, at minimum, a PBW theorem, evaluation representations, the rational difference property, and a proof that no additional relations occur.

Criterion 21.3 (Yangian recognition). *Let H be a filtered quasi-bialgebra reconstructed from an ordered tensor category. If*

- (i) $\text{gr } H \cong U(\mathfrak{g}[t])$ as a bialgebra;

- (ii) *the fundamental evaluation modules are faithful on the completed algebra;*
- (iii) *their R -matrices agree with the rational Yangian R -matrices;*
- (iv) *the defining RTT or Drinfeld relations generate the full kernel of the representation map;*

then H is isomorphic, up to the associator twist already present in the tensor functor, to the corresponding completed Yangian.

The criterion states exactly what an ordered chiral construction must prove before the word “Yangian” is justified.

21.8 Elliptic transport

On an elliptic curve, the KZB connection replaces $d\log(z_i - z_j)$ by elliptic kernels with prescribed quasi-periodicity. Its flatness uses Fay identities and dynamical correction terms. The monodromy represents the elliptic braid group and the modular group. Depending on the setting, reconstruction leads to elliptic or dynamical quantum groups.

No universal formula sends genus-zero Yangians to genus-one quantum groups. Rational, trigonometric, and elliptic structures are different degenerations of connections with different base geometry. Their relation is controlled by limits of the curve and of the coefficient category.

21.9 Knots and ribbon traces

A ribbon category assigns invariants to framed links by composing braidings, duality maps, and twists. The Reshetikhin–Turaev construction applied to $U_q(\mathfrak{g})$ produces quantum link invariants [63]. Chern–Simons theory supplies the physical path-integral interpretation. The KZ connection supplies a monodromy realization of the same braided tensor category.

The closure of a braid uses a categorical trace. This is the precise point at which ordered monodromy becomes a scalar knot invariant. The scalar does not retain the individual braid representation, just as a conformal-block dimension does not retain mapping-class-group monodromy.

21.10 Duality of monodromy

By (136), dualizing the KZ local system sends the monodromy to its inverse transpose. Under the invariant form on representations and the Drinfeld–Kohno equivalence, this agrees with the dual quantum-group representation and inverse braiding. A level reflection such as $k \mapsto -k - 2h^\vee$ has the same effect on the parameter $\hbar = (k + h^\vee)^{-1}$, namely $\hbar \mapsto -\hbar$, hence $q \mapsto q^{-1}$. This is a statement about the KZ family. It is not a universal formula for every chiral Koszul dual.

22 Quantum line defects and meromorphic tensor categories

22.1 Fusion before exchange

Let $A \in \text{Alg}_{E_1}(\text{FactD}(X))$. Its modules can be placed in a fixed order along the transverse line. Relative tensor product gives fusion:

$$M \star_A N := M \otimes_A^L N.$$

Associativity follows from excision for three consecutive intervals. There is no canonical map $M \star_A N \rightarrow N \star_A M$ because an isotopy in one dimension cannot exchange the intervals without a collision.

No-go theorem 22.1 (Order does not imply an R -matrix). *The forgetful functor*

$$\mathrm{Alg}_{E_2}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$$

has no natural section and is not essentially surjective on discrete algebra objects. Consequently an ordered chiral algebra has no functorially determined braiding, classical r -matrix, or quantum group.

Proof. The multiplication on the homology of an E_2 algebra is graded commutative. A noncommutative algebra concentrated in cohomological degree zero, such as $M_2(\mathbb{C})$ or a free associative algebra on two generators, therefore cannot be the underlying discrete E_1 algebra of an E_2 algebra with the same product. A natural section would provide such a lift and is impossible. Geometrically, the space of two ordered intervals in an interval has two disconnected order components and contains no exchange path; the path exists only after a second transverse direction is supplied. $\square$

The theorem fixes the location of quantum symmetry. Fusion belongs to the ordered chiral algebra. Exchange belongs to an E_2 enhancement, to the Hochschild centre, or to a separately supplied meromorphic local system.

22.2 The meromorphic line category

Let $\mathcal{L}$ be a category of line defects varying over the spectral curve X . For distinct spectral points, fusion is defined and is compatible with factorization. An exchange structure consists of isomorphisms

$$R_{L,M}(x,y) : L_x \star M_y \longrightarrow M_y \star L_x, \quad x \neq y,$$

which are horizontal for the D -module connection and satisfy the braid relation on $\mathrm{Conf}_3(X)$.

Definition 22.2 (Meromorphic tensor category). A meromorphic tensor category on X consists of a factorization category $\mathcal{L}$ on the ordered configuration spaces of X , an associative fusion product on disjoint ordered blocks, and a braid-group descent datum on the locus of distinct points. The descent datum is meromorphic along the collision divisors and satisfies the pentagon and hexagon identities.

The associator compares the three asymptotic regions of four ordered insertions. The braiding compares two chambers after a transverse exchange. The pole of the exchange operator is a chiral singularity; its homotopy class is topological.

Theorem 22.3 (Line-category recognition). *Let $\mathcal{T}$ be a renormalized quantum field theory on $X \times \mathbb{R}^2$ that is holomorphic on X and topological on $\mathbb{R}^2$. Assume that the fusion and exchange Ward identities are anomaly-free on a chosen class of line defects, that their morphism complexes are perfect over the meromorphic functions on X , that the line OPE extends meromorphically across the collision divisors, and that the class is closed under fusion and duality. Then these line defects form a rigid meromorphic tensor category in the sense of Theorem 22.2.*

Proof. Fusion is factorization in one transverse direction. Associativity is isotopy of three parallel lines. The exchange of two lines is a path in the configuration space of two points in the transverse plane. Isotopy of three exchange paths gives the braid relation. Isotopy of decompositions of four

lines gives the pentagon; compatibility of exchange with fusion gives the hexagons. Holomorphic dependence on spectral position and locality of counterterms make the structure meromorphic along diagonals. Dual line orientations give categorical duals under the perfectness hypothesis. $\square$

22.3 The centre and categorical exchange

The internal Hochschild centre $Z_{\text{ord, ch}}(A)$ is an E_2 algebra in $\text{FactD}(X)$. This means that its local multiplication carries the homotopy-coherent exchange of little two-disks. It does not mean that its ordinary module category is braided: modules over an E_2 algebra are naturally E_1 -monoidal. A braided line category is instead supplied by the Drinfeld centre of a monoidal line category, or by modules over an E_3 enhancement.

Corollary 22.4 (Exchange through a complete centre). *Suppose the bulk-to-boundary map*

$$\beta\tau : \text{Obs}_{\text{bulk}} \longrightarrow Z_{\text{ord, ch}}(A)$$

is an equivalence of local E_2 algebras. Suppose in addition that a monoidal-affineness theorem identifies the selected bulk line category with the Drinfeld centre $Z(\text{Bimod}_A)$, and identifies this categorical centre with modules over an E_3 topological enhancement of $Z_{\text{ord, ch}}(A)$. Then the corresponding perfect line categories are equivalent as braided monoidal categories.

The extra hypotheses are necessary. Equality of local bulk observables does not by itself classify extended line defects, and an E_2 algebra alone does not make its module category braided.

22.4 Tannaka reconstruction and its exact output

Choose a faithful exact meromorphic fibre functor

$$F : \mathcal{L} \longrightarrow \text{Vect}_K, \quad K = \mathbb{C}(X).$$

The coend

$$H_{\mathcal{L}} = \int^{L \in \mathcal{L}} F(L)^\vee \otimes_K F(L)$$

represents tensor automorphisms of F . The tensor product, associator, and braiding produce respectively a coproduct, coassociator, and universal R -form.

Theorem 22.5 (Quantum symmetry of line defects). *If F is strong monoidal, $H_{\mathcal{L}}$ is a coquasitriangular bialgebra over K , and rigidity supplies a Hopf antipode in the finite or continuous-dual class. If F is only quasi-monoidal, $H_{\mathcal{L}}$ is a coquasitriangular coquasi-bialgebra whose coassociator is the transported tensor constraint. When the relevant dual exists, the dual objects are quasitriangular Hopf or quasi-Hopf algebras, respectively. The construction is invariant under braided tensor equivalence compatible with the chosen fibre functor.*

The theorem reconstructs quantum symmetry without naming a presentation. The names Yangian, quantum loop algebra, elliptic quantum group, or Hall algebra require additional relations and a PBW theorem.

22.5 Rational gauge theory and the Yangian

In four-dimensional Chern–Simons theory on $\Sigma \times X$, where the theory is topological on Σ and holomorphic on X , parallel Wilson lines form a meromorphic tensor category. In the rational theory $X = \mathbb{C}$, perturbation theory produces

$$R_{V,W}(z) = 1 + \hbar \frac{\Omega_{V,W}}{z} + O(\hbar^2). \quad (144)$$

The one-loop anomaly of a line is cancelled by deforming the classical current algebra to the Yangian. The resulting line OPE is the RTT or Drinfeld coproduct of $Y_{\hbar}(\mathfrak{g})$.

Theorem 22.6 (Rational Wilson-line algebra). *For the perturbative Wilson-line categories constructed in anomaly-free rational four-dimensional Chern–Simons theory by Costello–Witten–Yamazaki, the algebra of line operators is the completed Yangian deformation of $U(\mathfrak{g}[z])$ on the stated class of representations; exchange is given by the corresponding rational universal R -matrix. The spectral parameter is the difference of the holomorphic positions. The assertion belongs to the completed formal $\hbar$ -adic category in which the line anomaly and its cancellation are defined [19, 20].*

The theorem is a field-theoretic recognition theorem. It uses the gauge-theory Ward identities and anomaly calculation, not merely the existence of an ordered bar construction.

22.6 Trigonometric and elliptic geometries

Changing the holomorphic curve changes the meromorphic tensor category. For $X = \mathbb{C}^\times$, the trigonometric kernel and its periodicity lead to a quantum loop algebra. For an elliptic curve, quasi-periodicity and a dynamical parameter lead to elliptic or dynamical quantum groups. These structures are not obtained by exponentiating a universal scalar. They are reconstructed from different meromorphic connections.

Principle 22.7 (The spectral curve selects the quantum group). *For the standard rational, trigonometric, and elliptic meromorphic line categories, Tannaka reconstruction produces the corresponding rational, quantum-loop, and elliptic or dynamical quantum symmetries after their presentation-recognition theorems are applied. A degeneration of the spectral curve induces a degeneration of the exchange kernel only when the line category, fibre functor, and meromorphic connection extend flatly over the degenerating family.*

22.7 Shifted Yangians and boundary conditions

A line defect may end on a boundary or meet an 't Hooft defect. Magnetic charge changes the allowed pole orders of the Drinfeld currents. Algebraically this produces shifted Yangians. Geometrically the shift is a modification of the factorization module at a marked transverse boundary condition.

Let μ be a coweight. A shifted object is represented by a pointed module bar

$$\text{Bar}_X^{\text{ord}}(M_-, A, M_+) = M_- \otimes_A^L M_+,$$

where the endpoint modules encode the two boundary charges. The unshifted algebra is recovered when both modules are the augmentation boundary.

Criterion 22.8 (Shifted recognition). *A line algebra is identified with a shifted Yangian only after the following data agree: the coweight filtration of currents, the shifted coproduct, the PBW associated graded, the fundamental line modules, and the rational exchange kernel. Agreement of the unshifted R -matrix alone does not determine the shift.*

22.8 KZ and gauge-theory exchange

For affine conformal blocks, the KZ connection gives braid monodromy with coupling $(k + h^\vee)^{-1}$. For four-dimensional Chern–Simons lines, the gauge coupling fixes the rational R -matrix. These two constructions are related through the Drinfeld–Kohno theorem after the representation categories and normalizations have been identified. The equality of kernels is not formal: the affine theory is a two-dimensional chiral theory, whereas the gauge theory is a four-dimensional holomorphic–topological theory.

Theorem 22.9 (Comparison through the line category). *Let $\mathcal{L}_{\text{KZ}}$ be a rigid braided tensor subcategory of the Drinfeld–KZ category at a nonresonant shifted level, and let $\mathcal{L}_{\text{gauge}}$ be a rigid Wilson-line subcategory of a rational gauge theory. A braided tensor equivalence matching tensor generators and the first-order Casimir kernel identifies their braid monodromies. Under compatible Tannaka fibre functors, both reconstruct the same quasitriangular Hopf or quasi-Hopf object up to the transported associator twist.*

Proof. A tensor equivalence preserves the coend reconstruction. Matching the Casimir coefficient fixes the formal deformation parameter. The braid operator on tensor generators then agrees, and rigidity extends the agreement to the generated category. The associator records the difference between strict and KZ tensor structures. $\square$

The theorem isolates the actual comparison datum. Without the tensor equivalence, similarity of rational kernels is evidence rather than identity.

22.9 Koszul duality of line couplings

Let A be the algebra of local observables near a line and let $\varepsilon : A \rightarrow \mathbf{1}$ describe the distinguished trivial line. Formal deformations supported on that line are governed by the derived deformation theory of the augmentation module. Its endomorphism algebra is the Koszul dual

$$A^\dagger = \text{RHom}_A(\mathbf{1}, \mathbf{1}).$$

Theorem 22.10 (Koszul duality for the augmentation-generated line sector). *Let A be a complete augmented E_1 algebra of local observables in the formal neighbourhood of a line defect, and put $A^\dagger = \text{RHom}_A(\mathbf{1}, \mathbf{1})$. The adjunction of Theorem 7.5 identifies the thick, or presentably completed, sector of A -modules generated by the augmentation with the corresponding sector of $A^\dagger$ -modules. The bar complex $\mathbf{1} \otimes_A^L \mathbf{1}$ computes the derived matrix coefficients of the augmentation line.*

Proof. The augmentation module is the distinguished line. Its derived endomorphisms form $A^\dagger$. The endomorphism adjunction is an equivalence on the generator and therefore on its thick closure; compact generation extends it to the stated presentable completion. The two-sided bar computes the derived tensor product of the two augmentation modules. $\square$

This theorem is the precise common content of Koszul duality in defect physics. Identifying $A^!$ with a named Yangian is a further calculation.

22.10 Transfer matrices and scalar observables

Let V be an auxiliary object of a rigid line category and let $M_1, \dots, M_n$ be quantum spaces. The monodromy operator

$$T_V(z) = R_{V, M_n}(z - z_n) \cdots R_{V, M_1}(z - z_1)$$

obeys the RTT relation. Taking the categorical trace over V gives commuting transfer matrices:

$$[t_V(z), t_W(w)] = 0.$$

The proof is the braid relation followed by cyclicity of the trace.

This passage loses information. The transfer matrix is a scalar or endomorphism-valued trace of ordered exchange. Bethe equations and Baxter relations describe its spectrum in selected representations. They are not universal consequences of an E_1 chiral algebra; they follow from the particular reconstructed quantum group and its representation theory.

22.11 Knots and framing

Closing a braid requires duals and a ribbon twist. The Reshetikhin–Turaev invariant is the categorical trace of the closed braid. In Chern–Simons theory the twist is the framing anomaly. Thus three structures enter in order:

$$\text{fusion} \longrightarrow \text{braiding} \longrightarrow \text{ribbon trace}.$$

They are governed respectively by E_1 , E_2 , and cyclic or framed data. The distinction is the categorical form of the distinction between an ordered word, an exchange path, and a closed world-line.

23 BRST descent and W -algebras

23.1 A differential compatible with chirality

Let (C, Q) be a cochain complex equipped with a chiral operation X . The statement that Q is a derivation of the chiral structure is

$$[Q, X] = 0 \tag{145}$$

in the chiral deformation Lie algebra. If $Q^2 = 0$ and $[X, X] = 0$, then

$$[Q + X, Q + X] = 0.$$

Thus the BRST differential and the chiral operation form one Maurer–Cartan element with unary and binary parts.

Proposition 23.1 (BRST descent of chiral operations). *If Q is a square-zero derivation of a chiral algebra C , then $H_Q(C)$ inherits a chiral algebra structure. The induced operation is independent of the choice of cocycle representatives.*

Proof. Equation (145) says that the operation sends cocycles to cocycles and that changing one input by a boundary changes the output by a boundary. The chiral Jacobi identity descends because it holds before taking cohomology. $\square$

The proposition applies equally to vertex algebras in a formal coordinate. It is the algebraic core of Hamiltonian reduction.

23.2 The BRST–collision bicomplex

The collision complex of C carries the internal differential induced by Q and the collision differential induced by X . They anticommute by (145). Hence

$$(C_I^{\text{coll}}(C), Q + d_{\text{coll}}) \quad (146)$$

is a bicomplex.

Filter by BRST degree. The first spectral sequence has

$$E_1 \cong C_I^{\text{coll}}(H_Q(C)).$$

Filter by collision degree. The other spectral sequence computes BRST cohomology of the collision resolution.

Theorem 23.2 (Bar–BRST comparison). *Assume that the tensor products and sums forming the coefficient objects are Q -acyclic outside degree zero, and that the total bicomplex (146) is bounded below in one grading and complete with respect to an exhaustive filtration in the other. Then the canonical map of collision complexes*

$$C_I^{\text{coll}}(H_Q^0(C)) \longrightarrow H_Q^0(C_I^{\text{coll}}(C), Q) \quad (147)$$

is an isomorphism, while $H_Q^j(C_I^{\text{coll}}(C), Q) = 0$ for $j \neq 0$. Consequently the total complex is quasi-isomorphic to $C_I^{\text{coll}}(H_Q^0(C))$ with its induced collision differential.

Proof. Taking vertical Q -cohomology commutes with the coefficient tensor products under the stated acyclicity hypothesis, which proves (147) and the higher vanishing. The BRST-first spectral sequence therefore has one nonzero row. Boundedness and completeness identify its abutment with the total cohomology, giving the final quasi-isomorphism. $\square$

The theorem is exact and strong, but its concentration hypothesis is substantive. Without it, the result is a derived BRST object rather than the ordinary bar complex of degree-zero cohomology.

23.3 Quantum Drinfeld–Sokolov reduction

Let $\mathfrak{g}$ be a simple Lie algebra and $f \in \mathfrak{g}$ nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from the affine vertex algebra $V_k(\mathfrak{g})$, charged ghosts, neutral fields when required, and a differential determined by f . The affine W -algebra is

$$\mathcal{W}^k(\mathfrak{g}, f) = H_{\text{BRST}}^0(C_{\text{DS}}^k(\mathfrak{g}, f)). \quad (148)$$

For the universal W -algebra, the BRST cohomology is concentrated in degree zero in the standard construction; the resulting vertex algebra has an associated graded described by functions on an arc space of the Slodowy slice [2, 3].

The chiral operation on the BRST complex descends by Theorem 23.1. The complicated nonlinear OPEs of W -generators are therefore transferred collision operations. Their composite fields are not exceptions to chiral geometry; they are the output of homological reduction.

23.4 Homotopy transfer

Choose a contraction

$$(H, 0) \xrightleftharpoons[p]{i} (C, Q), \quad Qi = 0, \quad pQ = 0, \quad Qh + hQ = \text{id} - ip.$$

Homotopy transfer expresses the induced higher operations on H as sums over rooted trees. Vertices carry the original chiral operation, internal edges carry h , leaves carry i , and the root carries p .

If the transferred structure is strict, the tree sums beyond the binary operation cancel. If it is not strict, H carries a homotopy chiral structure. The cohomology classes of the higher operations measure the obstruction to strictification; their raw coefficients depend on the contraction.

In Drinfeld–Sokolov reduction, the cohomology is known independently to carry a strict vertex-algebra structure. This fact alone does not trivialize the higher operations produced by an arbitrary transfer. If the contraction extends to a $\mathcal{P}_\infty$ quasi-isomorphism from the transferred homotopy chiral structure to the strict BRST-cohomology model, then the two Maurer–Cartan elements are gauge equivalent and the transferred higher operations are removable in that model. Without such a comparison, they remain legitimate transfer data.

23.5 Feigin–Frenkel duality for principal W -algebras

For principal nilpotent f , the Feigin–Frenkel duality relates the principal W -algebras of Langlands dual Lie algebras. With lacing number $r^\vee$, the levels k and ℓ satisfy

$$r^\vee(k + h^\vee)(\ell + {}^L h^\vee) = 1. \quad (149)$$

Under this relation there is an isomorphism

$$\mathcal{W}^k(\mathfrak{g}) \cong \mathcal{W}^\ell({}^L \mathfrak{g}). \quad (150)$$

This is the standard W -algebra duality. It is not the affine reflection $k \mapsto -k - 2h^\vee$.

The duality may be constructed through screening operators or free-field realizations. Its relation to a bar coalgebra requires a separate comparison between the BRST resolution and the chiral Chevalley construction.

23.6 Screening operators

A free-field realization embeds a W -algebra into an intersection of kernels of screening operators. The screenings are contour integrals of vertex operators and commute with the W -algebra up to total derivatives. Their kernels form a complex whose cohomology realizes the reduced algebra.

Geometrically, a screening contour is a cycle in a configuration space. Its monodromy and intersection pairing depend on the charges of the free fields. Multiple screenings produce hypergeometric local systems and quantum-group actions. This is an ordered configuration-space realization of BRST reduction.

23.7 Critical level

At $k = -h^\vee$, the Sugawara stress tensor is undefined because its denominator vanishes. On the formal disc D , the center of the critical vacuum vertex algebra is

$$Z(V_{-h^\vee}(\mathfrak{g})) \cong \mathcal{O}(\mathrm{Op}_{LG}(D)). \quad (151)$$

The center of the completed critical enveloping algebra over the punctured disc is described instead by functions on opers on $D^\times$, with the appropriate completion and central-character formulation [34, 36]. These are related but distinct statements. The vertex center is a commutative chiral algebra and underlies the local geometric Langlands construction.

The critical level is singular for KZ because $\hbar = (k + h^\vee)^{-1}$ diverges. It is not a limit in which q becomes one. The affine chiral algebra remains well defined, but its conformal structure and monodromy description change character.

23.8 What BRST reduction preserves

BRST reduction preserves any structure represented by a chain map commuting with Q . It preserves the chiral operation, module actions, and pairings that satisfy this condition. It does not automatically preserve:

- a chosen quadratic presentation;
- Koszulness of that presentation;
- a particular braid local system;
- a Verdier-dual identification;
- positive-genus sewing before compatibility with Q is proved.

The precise theorem is [Theorem 23.2](#). Every stronger statement is a comparison theorem for the corresponding additional structure.

23.9 The physical meaning

Classical Hamiltonian reduction imposes constraints and divides by gauge transformations. BRST replaces this quotient by a cochain complex whose differential combines constraints and ghosts. Quantum Drinfeld–Sokolov reduction is the chiral version. The Virasoro or W -algebra is the algebra of gauge-invariant boundary observables after reduction.

From the field-theoretic viewpoint, the ghost central charge is part of the anomaly calculation. BRST nilpotence requires cancellation of the total anomaly in a coupled theory. This is an equality of central extensions in the BRST complex, not a universal formula for a Koszul dual.

Part IV

Two genera and two master equations

24 Two genera and the two-edge master equation

24.1 Two loop-forming operations

Let A be an ordered chiral algebra. A cyclic trace on its transverse E_1 multiplication closes an interval into a circle and produces Hochschild chains, annuli, ribbon graphs, and open topological surfaces. A modular sewing structure on the chiral coefficient glues marked points of algebraic curves by plumbing and produces conformal blocks. Both operations contract pairs of flags, but their moduli spaces and anomaly lines are different.

Definition 24.1 (Transverse Calabi–Yau structure). A transverse Calabi–Yau structure of degree $d_\perp$ is a trace

$$\mathrm{tr}_\perp : \mathrm{HH}_0^{\mathrm{FactD}(X)}(A) \longrightarrow \mathbf{1}[-d_\perp]$$

whose pairing $\langle a, b \rangle_\perp = \mathrm{tr}_\perp(ab)$ is perfect and cyclic.

Definition 24.2 (Chiral modular sewing structure). A chiral modular sewing structure of degree d_X consists of perfect pairings on the relevant insertion or module objects and operations assigned to stable pointed complex curves, compatible with separating and nonseparating clutching, the factorisation D -module maps, and the determinant twists.

Neither definition implies the other. The first is an open two-dimensional topological structure internal to the chiral coefficient. The second varies the complex curve.

24.2 Bicoloured stable graphs

A bicoloured stable graph Γ has two sets of internal edges,

$$E(\Gamma) = E_X(\Gamma) \sqcup E_\perp(\Gamma).$$

An X -edge represents a chiral plumbing contraction. A $\perp$ -edge represents a transverse cyclic contraction. Vertices carry the operation appropriate to the incident flags; transverse flags retain their cyclic order. The orientation line is

$$\mathrm{or}(\Gamma) = \det(E_X(\Gamma)) \otimes \det(E_\perp(\Gamma)). \quad (152)$$

Contracting a loop of colour X raises the chiral genus g by one; contracting a loop of colour $\perp$ raises the ribbon genus h by one.

Let $\mathbf{G}^{\mathrm{bi}}$ be the groupoid of these graphs with isomorphisms preserving the edge colour and cyclic orders. The free completed graph object generated by the two sewing systems is

$$\mathbb{F}^{\mathrm{bi}}(A) = \prod_{[\Gamma] \in \mathbf{G}^{\mathrm{bi}}} (\mathrm{or}(\Gamma) \otimes A(\Gamma))_{\mathrm{Aut}(\Gamma)}. \quad (153)$$

This definition does not presuppose a tensor product of two modular operads. It directly records the two kinds of edge and their common vertex operations.

24.3 The two boundary operators

Let ∂_X be the signed sum of contractions of one X -edge and $\partial_\perp$ the signed sum of contractions of one $\perp$ -edge.

Proposition 24.3 (Independent graph boundaries). *With the orientation convention (152),*

$$\partial_X^2 = 0, \quad \partial_\perp^2 = 0, \quad \partial_X \partial_\perp + \partial_\perp \partial_X = 0.$$

Hence $\partial_X + \partial_\perp$ is square-zero.

Proof. For two edges of the same colour, the two contraction orders induce opposite orientations in the determinant of that edge set. For edges of different colours, the final graph is the same in either order, while exchanging the two odd determinant factors in (152) changes the sign. $\square$

The proposition is a graph-theoretic identity. A geometric realization must additionally show that the chosen ribbon and Deligne–Mumford boundary maps represent these edge contractions.

24.4 The bicoloured Feynman transform

Let $\mathcal{P}_X$ be the modular or cyclic object governing chiral sewing, and let $\mathcal{P}_\perp$ govern transverse cyclic sewing. Assume that their endomorphism actions on A obey the interchange identities on common flags. Define the *two-edge Feynman transform* to be the free completed bicoloured graph object with differential

$$d_{\mathbb{F}} = d_{\text{int}} + \partial_X + \partial_\perp.$$

By Proposition 24.3, this is a differential.

The corresponding completed convolution complex has components

$$\mathfrak{g}_A^{\text{bi}} = \prod_{g,h,n} \text{Hom}_{\Sigma_n} \left(\mathcal{P}^{\text{bi}}(g, h, n), \text{End}_A(g, h, n) \right), \quad (154)$$

completed by the number of internal edges and vertices. There are brackets $[-, -]_X$ and $[-, -]_\perp$ from joining two vertices with an edge of the indicated colour, and loop operators Δ_X and $\Delta_\perp$ from joining two flags of one vertex.

Theorem 24.4 (Two-edge master equation). *A morphism from the two-edge Feynman transform to the cyclic endomorphism object of A is equivalent to an element*

$$\Theta_A^{\text{bi}} \in \mathfrak{g}_A^{\text{bi}}[[\hbar_X, \hbar_\perp]]$$

satisfying

$$d\Theta_A^{\text{bi}} + \frac{1}{2}[\Theta_A^{\text{bi}}, \Theta_A^{\text{bi}}]_X + \frac{1}{2}[\Theta_A^{\text{bi}}, \Theta_A^{\text{bi}}]_\perp + \hbar_X \Delta_X \Theta_A^{\text{bi}} + \hbar_\perp \Delta_\perp \Theta_A^{\text{bi}} = 0. \quad (155)$$

Proof. The free transform is generated by corollas, and its differential is the sum of the internal differential and all one-edge contractions. Evaluating a morphism on the generators gives Θ_A^{bi} . Compatibility with $d_{\mathbb{F}}$ is exactly (155). Conversely, the equation extends the assignment on corollas uniquely to a chain morphism on the free completed graph object. $\square$

The hypotheses include both sewing systems. The theorem identifies their common master equation; no sewing operation is inferred from a bare ordered chiral algebra.

24.5 The two loop operators

On the suspended convolution complex, both loop contractions are odd.

Proposition 24.5 (Supercommutation of loop contractions). *With the determinant convention (152),*

$$\Delta_X \Delta_{\perp} + \Delta_{\perp} \Delta_X = 0.$$

After passing to unsuspended scalar amplitudes, the induced even contractions commute.

At bidegree $(g, h) = (0, 1)$, one sees the annulus and the Hochschild trace. At $(1, 0)$, one sees genus-one chiral sewing and its projective anomaly. At $(1, 1)$, both loop operations occur; the mixed codimension-two identity is their supercommutation.

24.6 Geometric realization

A geometric realization of the $\perp$ -edges uses ribbon surfaces or cyclically ordered boundary configurations. A realization of the X -edges uses stable curves and clutching morphisms. Compatibility requires a Cartesian or correspondent model in which performing one degeneration does not alter the local definition of the other except through the prescribed interchange map.

Criterion 24.6 (Geometric two-edge realization). *A pair of geometric sewing theories realizes the two-edge master equation when:*

- (i) *every codimension-one boundary of either type has a coefficient contraction;*
- (ii) *same-colour codimension-two boundaries satisfy the usual modular or ribbon boundary identity;*
- (iii) *mixed codimension-two boundaries form Cartesian comparison squares and carry the product orientation (152);*
- (iv) *the coefficient contractions commute over those squares;*
- (v) *the graph filtration is complete.*

Under these conditions, the geometric boundary differential maps to $d_{\mathbb{F}}$ and Theorem 24.4 applies.

24.7 Open topological field theory and chiral conformal theory

A smooth proper Calabi–Yau E_1 algebra defines an oriented open topological field theory; its value on the circle is Hochschild homology,

$$\int_{S^1} A \simeq \mathrm{HH}_{\bullet}(A).$$

Ribbon graphs compute its amplitudes. This is the $g = 0$ row with arbitrary h .

A chiral conformal theory assigns blocks to complex curves and varies with complex structure through a projectively flat connection. This is the $h = 0$ column with arbitrary g .

No-go theorem 24.7 (No universal equality of the genera). *There is no natural identity $g = h$. A cyclic associative algebra may have nontrivial ribbon amplitudes without a family of algebraic curves, while a chiral conformal theory may have higher-genus blocks without being determined by one finite-dimensional cyclic E_1 algebra.*

A string duality may relate the two expansions, but that relation is a theorem of the specific dynamics.

24.8 The double expansion

For a matrix realization, a connected ribbon graph is weighted by a power of N determined by h and the number of boundaries. Chiral sewing is weighted by a string coupling determined by g . When both exist, the connected potential has the form

$$\mathcal{F}_A(N, g_s) = \sum_{g, h, b} N^{2-2h-b} g_s^{2g-2} \mathcal{F}_{g, h, b}(A). \quad (156)$$

The coefficients are traces of the bidegree components of Θ_A^{bi} . Keeping the two parameters separate is the exact algebraic expression of the distinction between matrix-planar and worldsheet-genus expansions.

24.9 Duality of the two-edge theory

Theorem 24.8 (Duality of bicoloured amplitudes). *Suppose:*

- (i) A and $A^!$ are joined by an invertible bimodule Morita context on the selected complete perfect sector;
- (ii) their transverse Calabi–Yau traces correspond;
- (iii) their chiral sewing systems are identified by a tensor equivalence preserving projective connections, pairings, and determinant twists.

Then the equivalence induces an isomorphism of two-edge convolution complexes carrying Θ_A^{bi} to $\Theta_{A^!}^{\text{bi}}$.

Proof. Morita invariance identifies Hochschild chains, ribbon contractions, and $\Delta_\perp$. The sewing equivalence identifies the chiral contractions and Δ_X . Preservation of determinant lines identifies the orientation signs. Hence every term of (155) is intertwined. $\square$

A numerical complementarity statement follows only after a trace has been applied to this structured equivalence.

24.10 Physical meaning

The operator $\Delta_\perp$ contracts two open or line-defect states and creates a loop of the transverse topological theory. The operator Δ_X contracts two chiral insertions through a plumbing fixture and creates a handle on the complex curve. Their supercommutation is locality for two independent degenerations. The universal object is therefore not a scalar curvature but a bicoloured graph operation that remembers which edge was contracted and on which moduli space the contraction lives.

25 Cyclicity, stable graphs, and the master equation

25.1 Why genus requires a pairing

A genus-zero operation has outputs and inputs. To create a loop, one must identify two flags. Such an identification requires a pairing. Let V be a complex with a nondegenerate graded-symmetric pairing

$$\langle -, - \rangle : V \otimes V \longrightarrow \mathbb{C}[d] \quad (157)$$

that is invariant under the genus-zero operations. The pairing converts an output into an input and permits contraction along internal edges.

Without (157), there is no canonical nonseparating sewing. A chiral algebra alone therefore does not determine a modular object. A cyclic chiral algebra, a conformal-block pairing, or a field-theoretic propagator supplies the additional datum.

25.2 Modular operads

A modular operad has operations indexed by genus g and a finite set of legs. It allows two kinds of composition: joining a leg of one operation to a leg of another, and joining two legs of the same operation. The first adds genera; the second raises genus by one. Stable graphs encode iterated compositions [43].

Let $\mathcal{P}$ be a dg modular cooperad and let $\text{End}_V^{\text{mod}}$ be the endomorphism modular operad determined by the pairing (157). The completed convolution space

$$\mathfrak{g}_{\mathcal{P},V} = \prod_{g,n} \text{Hom}_{\Sigma_n}(\mathcal{P}(g,n), \text{End}_V^{\text{mod}}(g,n)) \quad (158)$$

contains a differential, a bracket joining two vertices, and a loop operator Δ joining two legs of one vertex.

25.3 The Feynman transform

The Feynman transform $F\mathcal{P}$ is the free modular operad on the shifted dual of $\mathcal{P}$, with the standard determinant twists, and with differential given by the internal differential and the signed sum over edge contractions. A morphism $F\mathcal{P} \rightarrow \text{End}_V^{\text{mod}}$ is determined by its values on generators. Compatibility with the differential is the master equation.

Theorem 25.1 (Feynman transform and master equation). *Giving V the structure of an algebra over the Feynman transform $F\mathcal{P}$ is equivalent to giving an element $\Theta \in \mathfrak{g}_{\mathcal{P},V}$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0. \quad (159)$$

Proof. The free modular operad is generated by corollas. The internal part of its differential gives $d\Theta$. Contracting an edge between two vertices gives the bracket term, with the factor $1/2$ accounting for interchange of the vertices. Contracting a loop at one vertex gives $\Delta\Theta$. Equality with the endomorphism differential is exactly (159). $\square$

The theorem is formal and unconditional. Its hypothesis is the existence of a morphism from the Feynman transform, equivalently the operations that Θ is supposed to encode. Boundary-of-boundary proves the equation for constructed sewing maps; it does not construct the maps.

25.4 Stable curves

The boundary strata of $\overline{\mathcal{M}}_{g,n}$ are indexed by stable graphs. Separating nodes correspond to edges joining distinct vertices. Nonseparating nodes correspond to loops. Clutching morphisms realize graph contraction geometrically.

Borel–Moore chains on the clutching spaces, with the usual determinant twists, form a modular operad under proper pushforward along gluing. After aritywise finite duality, or in a specified completed continuous dual, the corresponding cochains form a modular cooperad. If a system of multilinear maps on V is compatible with every clutching morphism, it defines an algebra over the appropriate Feynman transform. The master equation then states that the two ways of reaching every codimension-two boundary stratum cancel with orientation signs.

Principle 25.2 (Sewing principle). *The topology of $\overline{\mathcal{M}}_{g,n}$ supplies the relations among sewing maps. The coefficient theory must supply the sewing maps themselves.*

25.5 Graph filtration and obstruction theory

Filter $\mathfrak{g}_{\mathcal{P},V}$ by the number of internal edges, or more generally by stable-graph complexity. The filtration is complete and compatible with the bracket. Suppose a genus-zero Maurer–Cartan element Θ_0 is fixed. A partial lift

$$\Theta_{\leq r} = \Theta_0 + \Theta_1 + \cdots + \Theta_r$$

has curvature in filtration $r+1$. Its leading term is a cocycle for the twisted differential $d_{\Theta_0} = d + [\Theta_0, -]$.

Theorem 25.3 (Filtered obstruction theorem). *The obstruction to extending $\Theta_{\leq r}$ to filtration $r+1$ is a canonical class*

$$\mathfrak{o}_{r+1}(\Theta_{\leq r}) \in H^2\left(\mathrm{gr}_F^{r+1} \mathfrak{g}_{\mathcal{P},V}, d_{\Theta_0}\right). \quad (160)$$

It vanishes exactly when an extension exists. When it vanishes, the set of extensions modulo gauge transformations trivial in lower filtration is a torsor for the corresponding H^1 .

Proof. This is the standard obstruction theory of a complete filtered Maurer–Cartan equation. The Bianchi identity implies that the leading curvature is closed. Changing a lift alters it by a coboundary. Solving the next equation is equivalent to choosing a primitive; two primitives differ by a closed degree-one element. $\square$

This graph obstruction tower is invariant. Restricting to an arbitrary primary line need not preserve the bracket and does not produce an invariant tower unless the restriction is a morphism of deformation complexes.

25.6 Connected graph cumulants

Suppose a cyclic trace assigns a scalar weight $W(\Gamma)$ to every stable graph and the graph sum converges formally in $\hbar$:

$$Z = 1 + \sum_{\Gamma} \frac{\hbar^{g(\Gamma)}}{|\mathrm{Aut} \Gamma|} W(\Gamma). \quad (161)$$

The exponential formula for combinatorial species gives

$$\log Z = \sum_{\Gamma \text{ connected}} \frac{\hbar^{g(\Gamma)}}{|\mathrm{Aut} \Gamma|} W(\Gamma). \quad (162)$$

Thus connected amplitudes are cumulants of the full partition function. This is a canonical scalar projection once the trace is fixed. It does not classify the underlying chiral algebra.

25.7 Graphwise duality

Let V and $V^\vee$ be paired cyclic objects. Transposing every vertex operation and using the inverse pairing on every internal edge sends a graph weight for V to a graph weight for $V^\vee$. Orientation and determinant lines contribute the standard signs.

Proposition 25.4 (Graphwise transpose). *Assume that the coefficient systems are aritywise perfect, that continuous duality commutes with the stable-graph completion, and that the cyclic pairing identifies their endomorphism modular operads. Then transposition at every vertex and contraction with the inverse pairing on every edge induce a filtered isomorphism between the two modular convolution complexes. It intertwines their master equations and sends connected graph weights to their transposes.*

Any scalar complementarity formula is a trace of this graphwise statement and depends on the chosen duality and trace. There is no universal scalar conductor shared by levels, central charges, and Hodge classes.

25.8 The Batalin–Vilkovisky reading

In the BV formalism, fields form a shifted symplectic space, the bracket is induced by the symplectic pairing, and Δ is the second-order operator corresponding to contraction of a loop. The quantum action S satisfies

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \quad (163)$$

which expands to

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0.$$

This is the same algebraic shape as (159). Equality of the two equations for a particular theory requires a map from the renormalized BV complex to the modular endomorphism complex. Compactified configuration spaces supply that map in perturbative models where the Feynman weights extend to the boundary.

25.9 Anomalies

If the quantum master equation fails, the failure is a degree-one local cocycle, the anomaly. Its cohomology class is independent of the choice of counterterm. An anomaly can be cancelled by adding a local counterterm exactly when the class vanishes. A projective connection on conformal blocks is a different but related manifestation: the curvature is central rather than zero. Tensoring by a line with opposite curvature converts projective flatness into flatness.

The next section treats this anomaly line geometrically on the moduli space of curves.

26 Conformal blocks, sewing, and the anomaly line

26.1 Coinvariants and blocks

Let V be a conformal vertex algebra and let $M_1, \dots, M_n$ be modules inserted at marked points of a smooth curve C . The global chiral algebra acts on $M_1 \otimes \dots \otimes M_n$. The coinvariants are the quotient by this action; conformal blocks are their duals. Varying the pointed curve produces sheaves on moduli.

Damiolini–Gibney–Tarasca construct sheaves of coinvariants for vertex algebras on moduli of pointed stable curves and prove factorization and vector-bundle results under finiteness and semisimplicity hypotheses [22, 24]. For strongly rational vertex operator algebras, Damiolini–Woitke prove that the conformal blocks define a modular functor [25].

The modular functor is the positive-genus coefficient system required by Theorem 25.2. Its genus-zero restriction is built from intertwining operators and the vertex tensor category. Its sewing maps are convergent analytic operations and satisfy the mapping-class-group relations.

26.2 Projective flatness

The sheaf of conformal blocks carries a projectively flat connection. Infinitesimal changes of complex structure act through the stress tensor. The central term in the Virasoro algebra makes the action projective. Algebraically, the sheaf is a module for an Atiyah algebra rather than for the untwisted tangent sheaf.

Let $\lambda = c_1(\mathbb{E})$ be the Hodge class and ψ_i the cotangent-line classes at the markings. If a_i is the conformal weight of M_i , the interior projective class has the form

$$\mathrm{rk}(\mathbb{V}) \left(\frac{c}{2} \lambda + \sum_i a_i \psi_i \right). \quad (164)$$

On the compactification, boundary terms are determined by factorization through intermediate modules. Their coefficients depend on fusion rules and conformal weights [22, 23].

Theorem 26.1 (Chern-class form of the anomaly). *For a conformal-block bundle satisfying the factorization hypotheses, its first Chern class is the sum of the interior class (164) and explicit boundary divisor contributions obtained by sewing through simple modules. In particular, the Hodge coefficient is $\mathrm{rk}(\mathbb{V})c/2$, while the full class is not determined by c alone.*

This theorem separates the local central charge from the global geometry. Two theories with the same central charge may have different ranks, conformal weights, fusion rules, and boundary classes.

26.3 The anomaly line

A projectively flat bundle E has curvature

$$\nabla_E^2 = \Omega \mathrm{id}_E$$

for a scalar two-form Ω . Let L be a line bundle with connection satisfying $\nabla_L^2 = -\Omega$. Then

$$\nabla_{E \otimes L}^2 = 0.$$

If only a rational multiple of the required line exists, one works with a twisted D -module or on the corresponding gerbe.

Proposition 26.2 (Anomaly cancellation by a line). *Tensoring a projectively flat conformal-block system by a line or twisted line of opposite Atiyah class produces a flat system. The construction is compatible with sewing when the anomaly lines satisfy the clutching multiplicativity dictated by the Hodge and cotangent-line bundles.*

The proposition does not change the vertex algebra differential. It changes the connection on the family of blocks over moduli.

26.4 Sewing and the master equation

Choose a perfect pairing between a module and its contragredient. Sewing two marked points inserts the inverse pairing and sums over a homogeneous basis. In a local sewing parameter q , the operation contains the propagator

$$q^{L_0 - c/24}.$$

Convergence is analytic; factorization identifies the limit at $q = 0$ with the direct sum over intermediate modules.

The modular-functor sewing maps define an algebra over the chains of the surface operad. After the anomaly twist of [Theorem 26.2](#), they determine a solution of the master equation in the modular convolution algebra. This conclusion follows from [Theorem 25.1](#); no additional universal scalar is introduced.

26.5 Genus one

At genus one, Zhu's theorem gives modular transformation laws for trace functions of rational vertex operator algebras [\[78\]](#). For a holomorphic VOA,

$$Z_V(\tau) = \text{Tr}_V q^{L_0 - c/24}$$

is a modular function with the appropriate multiplier. For a rational VOA, the vector of irreducible characters transforms under a finite-dimensional representation of $SL_2(\mathbb{Z})$.

The projective anomaly appears as the factor $q^{-c/24}$ and the multiplier system. The modular S -matrix and fusion coefficients are related by the Verlinde formula when the representation category is modular [\[47, 73\]](#). The dimension of a conformal-block space is a trace of the modular functor; the $SL_2(\mathbb{Z})$ representation is strictly richer.

26.6 Prime forms and analytic kernels

On a compact Riemann surface, the prime form $E(z, w)$ is a canonical section, after the choice of a theta characteristic, that vanishes simply on the diagonal. The differential $d_z \log E(z, w)$ has the correct local simple pole but is quasi-periodic. Adding a harmonic correction produces a single-valued Green kernel. Fay's identities control products of prime forms and Szego kernels [\[32\]](#).

These kernels are analytic representatives of propagators and connections. Their quasi-periodicity produces monodromy and projective curvature. They do not define a universal fiberwise differential whose square is $c_1(\mathbb{E})$. The base and fiber degrees must remain distinct.

26.7 Free bosons and determinant lines

For a rank- r free boson, the oscillator partition function is

$$\eta(\tau)^{-r}.$$

The factor η^{-r} is the regularized determinant of the chiral Laplacian, while the nonchiral partition function includes $(\operatorname{Im} \tau)^{-r/2} |\eta(\tau)|^{-2r}$. On higher-genus surfaces, the determinant line of the $\bar{\partial}$ operator carries the Quillen metric and curvature given by the family index theorem.

This is the precise meeting point of Polyakov's conformal anomaly and the geometry of determinant lines. The coefficient is the central charge of the free fields. Interacting theories add non-Gaussian conformal blocks; they are not determined by the determinant line.

26.8 Lattice theories and Siegel theta series

For an even positive-definite lattice L of rank r , the genus-one character of the lattice VOA is

$$Z_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^r}. \quad (165)$$

At higher genus the zero modes contribute the Siegel theta series $\Theta_L^{(g)}(\Omega)$, while oscillators contribute determinant factors. Equal genus-one theta series need not imply equal lattices or equal VOAs.

For the two even unimodular lattices of rank sixteen, the degree- g Siegel theta series agree for $g \leq 3$. Their degree-four difference is proportional to the Schottky form and vanishes on the Jacobian locus. Hence genus-two does not separate the corresponding lattice theories. This example shows why every arithmetic claim must specify the genus and the moduli locus.

26.9 Genus two and beyond

Genus two introduces a three-dimensional Siegel upper half-space, separating and nonseparating boundary divisors, and stable graphs with loops. The period matrix and the off-diagonal sewing parameter couple channels that are independent at genus one. For rational VOAs, the modular functor gives finite-dimensional blocks with mapping-class-group monodromy. For lattice VOAs, Siegel theta series provide explicit sections.

No universal genus expansion follows from central charge alone. The stable-graph sum depends on all vertex and propagator data. The central charge controls the anomaly line; fusion and OPE coefficients control the non-scalar graph weights.

27 Anomalies, curvature, and invertible field theories

27.1 An anomaly has a domain

The word *anomaly* denotes an obstruction to making a classical symmetry act strictly after quantization. Its mathematical type depends on the symmetry and on the object on which the symmetry acts. The following classes occur in the present subject:

central extension of a local operator algebra,	$[\omega_{\text{loc}}] \in H_{\text{Lie}}^2(\mathfrak{s}; \mathbb{C}),$
projective curvature of a family of states,	$[\Omega_{\text{fam}}] \in H^2(B; \mathbb{C}),$
BV quantization obstruction,	$[\mathfrak{A}_{\text{BV}}] \in H_{\text{loc}}^1(\text{Obs}^{\text{cl}}),$
BRST obstruction,	$[Q^2] \in H^2(\text{End } C),$
framing anomaly,	$\alpha_{\text{fr}} \in \pi_0 \operatorname{InvFT}_{d+1}^{\text{fr}},$
curvature of a curved Koszul object,	$m_0 \in C^2.$

The symbols lie in different complexes. A theorem can identify their images under transgression or a comparison map. No scalar is their common definition.

27.2 Local central extensions

The affine current algebra has central term

$$J^a(z)J^b(w) \sim \frac{k(a, b)}{(z - w)^2} + \frac{J^{[a, b]}(w)}{z - w}.$$

The Virasoro algebra has central term

$$T(z)T(w) \sim \frac{c/2}{(z - w)^4} + \frac{2T(w)}{(z - w)^2} + \frac{\partial T(w)}{z - w}.$$

The parameters k and c classify different central extensions. The Sugawara construction relates them for a chosen affine theory:

$$c_{\text{Sug}} = \frac{k \dim \mathfrak{g}}{k + h^\vee}.$$

This rational function is not the affine level and is not defined at the critical level.

A central extension is part of the chiral operation. It does not prevent the chiral or ordered bar differential from squaring to zero. Square-zero follows from the complete algebra identities, including invariance of the cocycle.

Proposition 27.1 (Central terms and the ordered bar). *Let A be an augmented dg E_1 algebra in $\text{FactD}(X)$ whose chiral coefficient carries a central extension. The ordered bar coderivation satisfies $b^2 = 0$. The central term contributes to b through the multiplication of A or through its coefficient differential; it is not a scalar curvature of the simplicial bar.*

Proof. The simplicial identities of the bar construction use only the associativity and chain-map property of multiplication. A central extension satisfying the algebra identities is already included in that multiplication. Hence the standard proof of $b^2 = 0$ is unchanged. $\square$

27.3 Projective curvature over moduli

Let $\mathbb{V} \rightarrow B$ be a bundle of conformal blocks. A projectively flat connection satisfies

$$\nabla_{\mathbb{V}}^2 = \Omega \text{id}_{\mathbb{V}}.$$

The class $[\Omega]$ is the curvature of a line-valued central extension of the tangent action on B . For pointed stable curves, the interior contribution to the first Chern class is

$$\text{rk}(\mathbb{V}) \left(\frac{c}{2} \lambda + \sum_i a_i \psi_i \right),$$

and factorization supplies boundary terms. The central charge controls one coefficient of this class; the conformal weights and fusion channels control the rest.

Theorem 27.2 (Type of the conformal anomaly). *The conformal anomaly of a family of chiral blocks is the central extension encoded by the Atiyah algebra of the block bundle. After a projective connection*

is chosen, the same class is represented by its scalar curvature. It acts in the moduli direction and is not the square of the internal differential on an individual block complex.

Proof. The internal differential is defined before the base curve varies and has cohomological degree one in the fibre. The projective curvature is a two-form on the base acting by a scalar endomorphism of the fibre. They belong to the vertical and horizontal parts of the superconnection. The family Ward identity identifies the horizontal curvature with the central extension of vector fields represented by the stress tensor. $\square$

27.4 The superconnection equation

The correct object combining fibre and base is a superconnection

$$\mathbb{D} = d_{\text{fb}} + \nabla + \mathbb{A}_2 + \mathbb{A}_3 + \cdots.$$

Its square has total degree two. Flatness means $\mathbb{D}^2 = 0$; projective flatness means $\mathbb{D}^2 = \Omega \text{id}$.

Proposition 27.3 (Cancellation by an anomaly line). *If a line bundle L with connection satisfies $\nabla_L^2 = -\Omega$, then the tensor-product superconnection on $\mathbb{V} \otimes L$ is flat. If the class of Ω is only rationally integral, the same statement holds in the corresponding gerbe of twisted bundles.*

The line L must exist. Writing a formal symbol L^{-1} does not construct a root of a determinant line.

27.5 Polyakov's Weyl anomaly

Let $g = e^{2\sigma}\hat{g}$ be two metrics in the same conformal class on a closed surface. For a two-dimensional conformal field theory of central charge c , the change of the renormalized effective action has the local form

$$W[e^{2\sigma}\hat{g}] - W[\hat{g}] = \frac{c}{24\pi} \int (|\nabla_{\hat{g}}\sigma|^2 + R_{\hat{g}}\sigma) \, d\text{vol}_{\hat{g}}, \quad (166)$$

up to convention and local topological terms. Differentiating gives the trace anomaly

$$\langle T^\mu_\mu \rangle = -\frac{c}{24\pi} R.$$

The family index theorem identifies the curvature of the corresponding determinant line over moduli. This is the analytic origin of the $c\lambda/2$ term in the chiral block connection.

The Polyakov action concerns the response to a Weyl transformation. The ordered E_1 multiplication concerns fusion in a transverse line. Their relation passes through the stress tensor and the determinant line, not through the associative bar differential.

27.6 BRST anomaly

A BRST operator Q is required to satisfy $Q^2 = 0$. In the bosonic string, the total BRST charge squares to a multiple of the total central-charge defect and the intercept defect. Matter and ghosts cancel the central extension when

$$c_{\text{matter}} + c_{bc} = 0, \quad c_{bc} = -26.$$

This is additivity under tensor product of chiral theories.

Theorem 27.4 (BRST cancellation is tensorial). *Let V and G be conformal complexes with commuting BRST data. The BRST operator on $V \otimes G$ is $Q_V \otimes 1 + 1 \otimes Q_G$. Its square is the sum of the two anomaly endomorphisms. Hence nilpotence is cancellation of anomaly classes under tensor product. It is not a general identity between V and a Koszul dual of V .*

27.7 BV anomaly

For a classical interaction I , define the renormalized quantum master defect

$$\mathfrak{A}(I_h) = QI_h + \frac{1}{2}\{I_h, I_h\} + \hbar\Delta I_h.$$

At the first unsolved order in $\hbar$, $\mathfrak{A}$ is closed for the classical BV differential. Its cohomology class is independent of counterterm representatives. A quantization exists to the next order precisely when this class vanishes.

Proposition 27.5 (Locality of the BV anomaly). *The BV anomaly is represented by a local functional. In a configuration-space renormalisation scheme, its graph representatives are singular near partial diagonals before extension; after Fulton–MacPherson or Axelrod–Singer compactification, the obstruction is represented by the uncanceled boundary terms of the renormalised graph weights together with local counterterm faces.*

This is where Fulton–MacPherson or Axelrod–Singer geometry enters the quantum theory. The coefficient system consists of propagators and graph weights, not only the incidence complex of collisions.

27.8 Framing anomaly

Chern–Simons theory is invariant under orientation-preserving diffeomorphisms only after a choice of framing or a compensating gravitational Chern–Simons term. The framing change multiplies the partition function and Wilson-line observables by phases determined by the central charge of the associated modular tensor category. This anomaly lives in three-dimensional geometry.

The ribbon twist of the line category records the same phase on framed knots. Its relation to the two-dimensional WZW central charge follows from the Chern–Simons/WZW correspondence. It is not determined by an arbitrary E_1 chiral algebra.

27.9 Curved Koszul duality

A curved dg algebra or coalgebra has a curvature element m_0 satisfying

$$d^2(x) = [m_0, x]$$

in the associative case, with the corresponding A_∞ identities in the homotopy case. If m_0 is a central scalar, the commutator vanishes. Such a scalar may still change the Maurer–Cartan equation, but it does not make d^2 nonzero.

Curvature arises naturally when an inhomogeneous relation has a constant term. For example, quadratic-linear-constant Koszul duality places the constant term in m_0 . This algebraic curvature is not $c_1(\mathbb{E})$ unless a family comparison sends it to that class.

No-go theorem 27.6 (No universal anomaly scalar). *There is no functorial scalar $\kappa(A)$ that simultaneously determines the affine level, the Virasoro central charge, the projective Chern class of all conformal-block bundles, the BRST anomaly, the framing anomaly, and the curvature of every Koszul dual.*

Proof. The affine level and Virasoro central charge vary independently before a stress tensor is chosen. Conformal-block Chern classes also depend on module weights and boundary fusion coefficients. BRST anomalies depend on the ghost complex; framing anomalies depend on a three-dimensional ribbon structure; curved Koszul terms depend on the chosen inhomogeneous presentation. Since these data can be varied independently while holding the others fixed, no such scalar is functorial. $\square$

27.10 Anomalies as invertible theories

For local anomalies admitting the standard anomaly-field-theory description, the anomaly is represented by an invertible field theory one dimension higher. The anomalous theory is relative to its state lines, and cancellation is a trivialisation of the tensor product anomaly.

For chiral conformal blocks, the invertible theory is represented by a determinant or Hodge line. For fermions it is the determinant line of the Dirac operator. For Chern–Simons theory it includes the framing line. This viewpoint explains additivity: tensoring theories tensors anomaly lines and adds their characteristic classes.

Principle 27.7 (Anomaly comparison). *Two anomalies are the same only when a morphism of invertible theories identifies their state lines, connections, and gluing laws. Equality of one numerical coefficient is a consequence of such an identification, not its substitute.*

27.11 The anomaly square

The complete theory contains four compatible arrows:

$$\begin{array}{ccc}
 \text{local central extension} & \xrightarrow{\text{Ward identity}} & \text{Atiyah algebra of blocks} \\
 \text{BV descent} \downarrow & & \downarrow c_1 \\
 \text{local BV anomaly} & \xrightarrow{\text{family index}} & \text{curvature of determinant line.}
 \end{array} \tag{167}$$

Each arrow is constructed in a specified field theory. For free chiral fields and WZW models, the standard Ward-identity and family-index calculations construct the arrows of this square. For a general ordered chiral algebra, the upper-left corner exists while the other corners require a stress tensor, a family of states, and a BV realization.

The square is the precise form of the unity sought by the subject. It gives one language without declaring unlike objects equal.

Part V

Exact mathematics and exact physics

28 Exact examples

28.1 Heisenberg

Let $\mathcal{H}_k$ be generated by a current J with

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \quad (168)$$

The chiral operation has no simple-pole bracket. Its singular part lies in the second normal symbol of the diagonal. The corresponding ordinary Lie conformal bracket is

$$[J_\lambda J] = k\lambda.$$

The Jacobi identity is immediate because the bracket is central.

The chiral Chevalley coalgebra is cofree on the abelian chiral Lie algebra with the central extension built into the operation. Ordered monodromy is scalar. For vacuum and Fock-module insertions on a compact curve, the conformal blocks admit Fock-space descriptions and their partition functions are determinant expressions. The rank-one conformal vector has central charge one, independent of the nonzero normalization k after rescaling the current. This already separates level from central charge.

28.2 Affine Kac–Moody

For $V_k(\mathfrak{g})$ the current OPE is (140). The simple pole is the Lie bracket; the double pole is the invariant bilinear form. The chiral Jacobi identity splits into the ordinary Jacobi identity and invariance of the form:

$$([a, b], c) = (a, [b, c]).$$

The Chevalley differential formed from the simple-pole bracket squares to zero by itself. The level term is not required to repair Jacobi. It supplies a central extension and enters conformal and representation-theoretic constructions.

At noncritical level the Sugawara vector has central charge

$$c_{\text{aff}} = \frac{k \dim \mathfrak{g}}{k + h^\vee}. \quad (169)$$

The KZ connection and its quantum-group monodromy are described in Section 21. At positive integral level, the integrable modules of the simple affine VOA $L_k(\mathfrak{g})$ form a modular tensor category, and their conformal blocks satisfy the Verlinde formula.

At level one for simply laced $\mathfrak{g}$, the Frenkel–Kac lattice realization expresses the affine currents inside a lattice VOA [35]. The nonabelian simple-pole bracket remains present; it is not killed by the realization.

28.3 Virasoro

The Virasoro vertex algebra is generated by T with

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \quad (170)$$

The three singular terms lie in the fourth, second, and first normal symbols. Their relation is fixed by conformal covariance. The central charge c is the coefficient of the fourth-order symbol and the central extension of the vector-field algebra.

The vacuum Verma module has a level-four quasi-primary

$$\Lambda =: TT : - \frac{3}{10} \partial^2 T$$

whose norm is proportional to $c(5c + 22)$. The zeros at $c = 0$ and $c = -22/5$ signal null vectors. This Gram determinant is representation-theoretic data. It does not define a universal quartic obstruction tower.

Conformal blocks carry a projective connection with Hodge coefficient $c/2$. At $c = 26$, coupling to the (b, c) ghost system of weights $(2, -1)$ cancels the total central charge. At $c = 13$ no universal self-duality theorem for Virasoro algebras follows; 13 is merely the midpoint of the reflection $c \mapsto 26 - c$ used in bosonic string bookkeeping.

28.4 Free fermions

A complex free fermion has fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w}.$$

The chiral operation lies in the first normal symbol. Wick's theorem determines all higher correlators as determinants. The representation category, spin structures, and modular transformation law depend on whether one uses a real, complex, or lattice extension.

Boson-fermion correspondence identifies the charged free fermion VOA with a rank-one lattice VOA. This is an isomorphism of vertex algebras after the lattice extension; it is not the statement that the Heisenberg and fermion chiral Lie algebras are linear Koszul duals.

28.5 The $\beta\gamma$ and bc systems

A first-order bosonic system has

$$\beta(z)\gamma(w) \sim \frac{1}{z-w},$$

while a fermionic bc system has the same singular kernel with odd parity. Their oscillator algebras are Weyl and Clifford algebras. For fields of weights $(\lambda, 1 - \lambda)$, the central charges are

$$c_{\beta\gamma} = 2(6\lambda^2 - 6\lambda + 1), \quad c_{bc} = -2(6\lambda^2 - 6\lambda + 1). \quad (171)$$

The opposite signs reflect statistics. The determinant lines cancel when the systems are paired with identical weights.

The simple pole does not imply that the full algebra has only one homotopical layer. Composite fields and modules may generate nontrivial higher operations. Pole order and deformation cohomology are independent.

28.6 Lattice vertex algebras

Let L be an even positive-definite lattice. The lattice VOA V_L has central charge $\text{rk } L$. Its fields include Heisenberg currents and exponentials e^α . Their operator products contain powers

$$(z - w)^{(\alpha, \beta)}$$

and a cocycle $\epsilon(\alpha, \beta)$. Locality requires the commutator of the cocycle to match the parity determined by the lattice form.

If L is even unimodular, V_L is holomorphic. The character is (165). Inverting the cocycle gives the contragredient braiding; a symmetric cocycle may make the theory self-contragredient. This is a statement about the lattice extension, not about all chiral Koszul duals.

28.7 Leech and Moonshine

The Leech lattice VOA V_Λ and the Moonshine VOA $V^\natural$ both have central charge 24. Their weight-one spaces differ:

$$(V_\Lambda)_1 \cong \mathbb{C}^{24}, \quad (V^\natural)_1 = 0.$$

Their characters also differ in the constant term after the vacuum normalization. Thus central charge does not classify holomorphic VOAs, their chiral operations, or their factorization coalgebras.

The Monster action on $V^\natural$ produces McKay–Thompson series. These are traces of automorphisms on the state space. They are arithmetic shadows of the VOA, not coefficients of a universal modular Koszul invariant.

28.8 Principal W -algebras

The principal W -algebra $\mathcal{W}^k(\mathfrak{g})$ is obtained by BRST reduction. At generic level its standard strong generators have conformal weights equal to the exponents of $\mathfrak{g}$ plus one; special levels may impose decoupling relations or collapse generators. Their OPEs are nonlinear and contain normally ordered composites. The full chiral operation nevertheless satisfies the same Maurer–Cartan equation (116).

For $\mathfrak{sl}_2$, the principal W -algebra is Virasoro. For $\mathfrak{sl}_3$, it is generated by T and a weight-three field W ; the WW OPE contains T , its derivatives, and the quasi-primary Λ . These terms are fixed by Jacobi and conformal covariance. No single scalar extracted from the WW OPE determines the algebra.

Feigin–Frenkel duality uses (149). The central charges of dual pairs agree because the vertex algebras are isomorphic. This duality is distinct from KZ inverse braiding.

28.9 Critical affine algebras

At $k = -h^\vee$, the affine OPE remains (140), but the Sugawara conformal vector ceases to exist. The Feigin–Frenkel center becomes large and is identified with ops. KZ equations in their usual WZW

normalization become singular. The critical level is therefore a change of structure, not an uncurved point of a universal bar differential.

28.10 A comparison table

family	highest singular symbol	ordered structure	positive-genus structure
Heisenberg	second normal symbol k	scalar monodromy	determinant line and Fock sewing
affine $V_k(\mathfrak{g})$	bracket plus second symbol	KZ braid monodromy	WZW conformal blocks
Virasoro	fourth symbol $c/2$	conformal-block monodromy	projective anomaly $c\lambda/2$ plus boundary data
free first-order	first symbol	Weyl/Clifford statistics	determinant lines
lattice V_L	lattice powers and cocycle	lattice braiding	Siegel theta series
principal W	nonlinear finite OPEs	screening monodromy	W -conformal blocks when finite
critical affine	bracket and critical central extension	oper geometry replaces KZ	Feigin–Frenkel center

The table lists structures, not a classification by one number. Each row realizes the diagonal principle in a different representation category.

29 Strictly noncommutative chiral algebras

29.1 Free ordered chiral algebras

Let $M \in \text{FactD}(X)$. The free augmented ordered chiral algebra is

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus M^{\otimes 3} \oplus \cdots, \quad (172)$$

with concatenation. The tensor products are taken in $\text{FactD}(X)$, so each word retains the full chiral coefficient system of its letters.

Proposition 29.1 (Bar homology of the free algebra). *Assume the tensor products in (172) are exact. The reduced bar of $T_X(M)$ contracts onto $M[1]$. Equivalently,*

$$\mathbf{1} \otimes_{T_X(M)}^L \mathbf{1} \simeq \mathbf{1} \oplus M[1]$$

as a conilpotent coalgebra, with zero reduced coproduct on $M[1]$.

Proof. The word-length decomposition gives a fibre sequence of left $T_X(M)$ -modules

$$T_X(M) \otimes M \xrightarrow{\text{mult}} T_X(M) \longrightarrow \mathbf{1}.$$

On the summand of words of positive length, multiplication is the canonical identification obtained by separating the last letter; the cofiber is the empty word. Applying $\mathbf{1} \otimes_{T_X(M)}^L -$ leaves the two-term complex $M \rightarrow \mathbf{1}$ with zero reduced differential, hence the class $M[1]$. Equivalently, the standard extra degeneracy of the normalized bar contracts every bar word having an internal cut and leaves the one-letter cogenerator. $\square$

Thus the Koszul dual algebra of a free associative object is the square-zero extension

$$T_X(M)^! \simeq \mathbf{1} \oplus M^\vee[-1]$$

when M is dualizable. Noncommutative freedom on one side becomes a first-order infinitesimal object on the other.

29.2 Square-zero chiral algebras

Conversely, let

$$A = \mathbf{1} \oplus M, \quad M^2 = 0.$$

The bar multiplication differential vanishes, and

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) = \bigoplus_{r \geq 0} M[1]^{\otimes r}$$

with deconcatenation. Its continuous dual is the completed tensor algebra

$$A^! \simeq \widehat{T}_X(M^\vee[-1]).$$

This pair is the local model of an infinitesimal boundary coupling and its algebra of iterated defect insertions.

29.3 Quiver chiral algebras

Let Q be a finite quiver, let

$$S = \bigoplus_{i \in Q_0} \mathbf{1}e_i,$$

and assign an object $M_a \in \mathrm{FactD}(X)$ to every arrow $a : i \rightarrow j$. Put

$$M = \bigoplus_{a \in Q_1} e_{t(a)} M_a e_{s(a)}.$$

The relative tensor algebra

$$A_Q = T_S(M) \tag{173}$$

is an E_1 algebra in factorization D -modules. Multiplication concatenates composable arrows and is zero otherwise. The augmentation $A_Q \rightarrow S$ forgets nontrivial paths.

The relative bar

$$B_S(A_Q) = S \otimes_{A_Q}^L S$$

retains the ordered path structure. If Q has no relations, it contracts to $S \oplus M[1]$. If relations $R \subset M \otimes_S M$ are imposed, the quadratic coalgebra begins

$$S \oplus M[1] \oplus R[2] \oplus \cdots.$$

Example 29.2 (Two boundary conditions). For the quiver $1 \rightleftarrows_u^v 2$, the objects M_u and M_v are boundary-changing fields. The composites uv and vu are endomorphisms of different boundaries and need not agree. Their chiral insertion points may collide on X , while the order of composition is fixed

by the transverse interval. This is strict noncommutative chirality in its smallest form.

A cyclic potential W determines a cyclic functional on the completed path algebra. After choosing the standard Ginzburg or cyclic A_∞ construction, its cyclic derivatives enter the differential and are the classical equations of motion of the corresponding open-string field theory. The ribbon genus of Section 24 is then the genus of the resulting open-string Feynman surface.

29.4 Matrix and Chan–Paton extensions

Let V be a commutative algebra object in $\text{FactD}(X)$ and let E be a finite-dimensional vector space. The matrix extension

$$A = \text{End}(E) \otimes V \quad (174)$$

is an E_1 algebra in $\text{FactD}(X)$. Its multiplication is matrix multiplication times the chiral factorization product. The OPE coefficients are $\text{End}(E)$ -valued and preserve their order.

The algebra (174) is Morita equivalent to V . Hence

$$Z_{\text{ord, ch}}(A) \simeq Z_{\text{ord, ch}}(V)$$

and their perfect module categories have equivalent centres. The open amplitudes differ by Chan–Paton traces, producing powers of $\dim E$ on ribbon graphs.

Proposition 29.3 (Chan–Paton weight). *For a connected ribbon graph Γ contributing to a cyclic amplitude of (174), contraction of matrix indices contributes*

$$(\dim E)^{F(\Gamma)},$$

where $F(\Gamma)$ is the number of index faces. For a closed ribbon surface of genus h with b boundary components, this is $(\dim E)^{2-2h-b}$ after the standard normalization of vertices and propagators.

This is the exact algebraic origin of the 't Hooft expansion. It belongs to the transverse ribbon genus and is independent of the genus of the chiral curve.

29.5 Universal enveloping ordered chiral algebras

Let $\mathfrak{g}$ be a Lie algebra object in $\text{FactD}(X)$. Its universal enveloping algebra

$$U_X(\mathfrak{g}) \in \text{Alg}_{E_1}(\text{FactD}(X))$$

is defined by the usual coequalizer of the tensor algebra by

$$x \otimes y - (-1)^{|x||y|} y \otimes x - [x, y].$$

The Poincaré–Birkhoff–Witt filtration has

$$\text{gr } U_X(\mathfrak{g}) \simeq \text{Sym}_X(\mathfrak{g})$$

under the standard flatness hypotheses.

Theorem 29.4 (Associative bar and internal Chevalley chains). *For a flat Lie object $\mathfrak{g}$ in $\text{FactD}(X)$, the canonical twisting morphism induces a quasi-isomorphism*

$$\text{Chev}_X(\mathfrak{g}) \xrightarrow{\sim} \text{Bar}_X^{\text{ord}}(U_X(\mathfrak{g})).$$

Proof. Filter both sides by symmetric degree. The associated graded map is the ordinary Koszul resolution of $\text{Sym}_X(\mathfrak{g})$ by the exterior coalgebra on $\mathfrak{g}[1]$. PBW identifies the associated graded of the enveloping algebra. Completeness and exactness of the filtration lift the associated-graded quasi-isomorphism. $\square$

This theorem is the internal Lie–associative Koszul comparison in the pointwise coefficient category. It is distinct from the Beilinson–Drinfeld chiral Chevalley construction on the Ran space. The internal Lie bracket maps to the commutator in its enveloping algebra, and internal Lie chains compute the ordered bar of that enveloping algebra.

29.6 Affine currents as chiral coefficients

The affine current algebra is a Beilinson–Drinfeld chiral Lie algebra: its simple pole is the Lie bracket and its double pole is the invariant cocycle. Its chiral enveloping construction produces the universal affine chiral coefficient. This construction takes place in the chiral pseudo-tensor category and is distinct from the internal enveloping algebra of Theorem 29.4, which begins with a Lie object for the pointwise tensor product.

To obtain a strictly noncommutative ordered chiral algebra from affine data one must supply a transverse associative object: for example an endomorphism algebra of affine boundary conditions, a quiver of affine modules, or a free tensor algebra generated by the affine factorisation coefficient. The resulting internal bar resolves that transverse multiplication. The chiral Chevalley complex resolves the affine chiral Lie operation. A comparison between them is an enveloping or Morita theorem and is never automatic. In either construction the level is not required to repair Jacobi; invariance of the cocycle is the identity that makes the central extension chiral-Lie.

29.7 Weyl and Clifford chiral algebras

Let M carry a perfect pairing $\omega : M \otimes M \rightarrow \mathbf{1}[d]$. The tensor algebra modulo

$$xy - (-1)^{|x||y|}yx = \hbar\omega(x, y) \tag{175}$$

produces a Weyl or Clifford object according to parity. In the chiral category, M may be a factorization D -module and ω may contain a higher normal symbol along the diagonal.

The relation is quadratic-linear-constant. Its Koszul dual is curved: the constant term becomes the curvature m_0 of the dual coalgebra. The curvature belongs to the algebraic presentation. It is not a Hodge class on moduli.

Example 29.5 (The $\beta\gamma$ system). The chiral Weyl algebra generated by β and γ has singular relation

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}.$$

With matrix Chan–Paton factors, the ordered product is matrix multiplication while the singular coefficient is the Weyl pairing. The two operations are visibly independent.

Example 29.6 (The bc system). The odd analogue has Clifford relation. Its determinant line contributes the ghost conformal anomaly. The Weyl–Clifford Koszul exchange is a statement about quadratic relations; the cancellation of their conformal anomalies is a separate determinant-line calculation.

29.8 Chiral differential operators

For a smooth variety Y , a sheaf of chiral differential operators exists after its obstruction gerbe has been trivialized. It supplies a chiral factorisation coefficient whose local fields include functions and vector fields on Y . The vertex or chiral operation of that coefficient is not, by itself, the transverse E_1 multiplication of this paper.

Strict ordered examples are obtained from associative endomorphism objects in a category of chiral-differential-operator modules, from quiver endomorphisms of boundary conditions, or from matrix and tensor-algebra extensions of a dualizable coefficient. When Y carries a group action, BRST or quantum Hamiltonian reduction acts on such an ordered endomorphism algebra provided the BRST operator is a derivation of the transverse product. The resulting comparison is governed by Section 23.

29.9 Boundary endomorphism algebras

Let $\mathcal{B}$ be a factorization category of boundary conditions on X . For objects b_i , set

$$A_{ij} = \mathrm{RHom}_{\mathcal{B}}(b_i, b_j).$$

Composition gives

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

The direct sum

$$A_{\mathcal{B}} = \bigoplus_{i,j} A_{ij}$$

is an algebra over the semisimple idempotent object $S = \oplus_i \mathbf{1}e_i$. Its relative bar computes chains of boundary changes. This is the categorical form of the quiver example and the native algebra of open strings.

Theorem 29.7 (Morita reconstruction from a boundary generator). *Let $b = \oplus_i b_i$ be a compact generator of a stable factorization category $\mathcal{B}$. Then*

$$A_b = \mathrm{RHom}_{\mathcal{B}}(b, b)$$

is an E_1 algebra in $\mathrm{FactD}(X)$ and

$$\mathcal{B} \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

under the usual compact-generation hypotheses.

The theorem explains why noncommutative chiral algebras are primary in boundary physics. The category of boundary conditions comes first; its endomorphism algebra is a coordinate chart on that category.

29.10 Factorization quantum groups

A factorization quantum group is not merely an E_1 algebra. It combines an algebra object, a compatible coalgebra or tensor-category structure, and meromorphic exchange. In the ordered language these pieces occupy different levels:

E_1 multiplication	: fusion,
factorization coproduct	: splitting of lines or representations,
E_2 enhancement	: exchange,
ribbon trace	: closure and link invariants.

A Yangian possesses all these structures in a compatible completed sense. A generic ordered chiral algebra possesses only the first.

29.11 A noncommutative Koszul pair

Let V be dualizable and let $R \subset V \otimes V$ be a factorization subobject. Define

$$A = T_X(V)/(R), \quad A^{!,\text{quad}} = T_X(V^\vee[-1])/(R^\perp).$$

The canonical twisting morphism $A^i \rightarrow A$ gives the Koszul complex

$$K(A) = A \otimes A^i.$$

Theorem 29.8 (Quadratic recognition in the ordered chiral category). *Assume positive weight, degreewise dualizability, and strong convergence of the weight filtration. The following are equivalent:*

- (i) $A^i \rightarrow \text{Bar}_X^{\text{ord}}(A)$ is a quasi-isomorphism;
- (ii) $\text{Cobar}_X(A^i) \rightarrow A$ is a quasi-isomorphism;
- (iii) the left and right quadratic Koszul complexes are acyclic off weight zero;
- (iv) $\text{Tor}^A(\mathbf{1}, \mathbf{1})$ is concentrated on the quadratic diagonal.

The proof is the fundamental theorem of twisting morphisms internal to $\text{FactD}(X)$. The chiral coefficient changes the tensor products and support conditions, but not the formal equivalence of the four tests. When these conditions hold and finite duals exist, $A^{!,\text{quad}}$ identifies with the boundary endomorphism algebra $\text{RHom}_A(\mathbf{1}, \mathbf{1})$; before that comparison the two symbols denote different constructions.

29.12 What these examples establish

The examples exhibit three independent kinds of noncommutativity.

Quiver and matrix algebras have noncommutative transverse multiplication even when their chiral coefficient is free or commutative.

Internal enveloping algebras have noncommutative transverse multiplication generated by an internal Lie object. Affine current algebras supply chiral coefficients; a strictly ordered affine example requires the additional boundary, quiver, tensor, or endomorphism multiplication described above.

Yangians and line categories add meromorphic braid exchange to noncommutative fusion.

The ordered chiral language accommodates all three without identifying them. That separation is the condition under which their Koszul dualities can be stated correctly.

30 Arithmetic traces and automorphic forms

30.1 Arithmetic enters through periods and traces

The chiral operation is defined over an algebraic or analytic coefficient field. Arithmetic appears when one takes periods of configuration spaces, traces of monodromy, dimensions of conformal blocks, theta sums over lattices, or regularized determinants. These operations are functorial shadows. Their arithmetic is controlled by the geometry of the chosen trace.

There are two basic sources. Ordered iterated integrals produce multiple zeta values and their elliptic analogues. Symmetric and modular traces produce modular forms, theta series, and characteristic numbers. The two sources can interact, but neither is determined by the other.

30.2 The KZ associator

The regularized holonomy of the KZ connection on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ is the Drinfeld associator. Its coefficients are multiple zeta values. The weight filtration is the length of iterated integration. The pentagon equation expresses compatibility around the five degenerations forming the boundary of a real associahedral pentagon inside $\overline{\mathcal{M}}_{0,5}$; the complex Deligne–Mumford boundary itself has ten irreducible divisors.

This is a direct arithmetic consequence of ordered configuration geometry. Passing to the Grothendieck ring of the representation category discards the associator but retains fusion multiplicities. Thus ordered and symmetric traces have visibly different arithmetic content.

30.3 Verlinde numbers

For a modular tensor category with simple objects indexed by λ and modular S -matrix, the dimension of the genus- g conformal-block space is

$$\dim \mathbb{V}_g(\lambda_1, \dots, \lambda_n) = \sum_{\mu} (S_{0\mu})^{2-2g-n} \prod_{i=1}^n S_{\lambda_i \mu}. \quad (176)$$

For $\widehat{\mathfrak{sl}}_2$ at level k and no insertions,

$$\dim \mathbb{V}_g = \left(\frac{k+2}{2} \right)^{g-1} \sum_{j=1}^{k+1} \sin \left(\frac{\pi j}{k+2} \right)^{2-2g}. \quad (177)$$

The cosecant power sums are polynomials in $k+2$ with Bernoulli-number coefficients. This arithmetic follows from the modular tensor category and the geometry of flat connections. It is not a bar-cohomology dimension formula.

30.4 Hodge integrals

Faber and Pandharipande evaluate the one-point Hodge integrals

$$\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g. \quad (178)$$

Their generating function is

$$1 + \sum_{g \geq 1} \left(\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g \right) t^{2g} = \frac{t/2}{\sin(t/2)}. \quad (179)$$

This identity is a theorem of the tautological geometry of moduli [33]. It contributes to any field theory whose stable-graph trace reduces to these Hodge integrals. It is not the universal genus expansion of every chiral algebra.

30.5 Theta series

For an even lattice L , the degree- g Siegel theta series is

$$\Theta_L^{(g)}(\Omega) = \sum_{x_1, \dots, x_g \in L} \exp \left(\pi i \sum_{a,b} (x_a, x_b) \Omega_{ab} \right). \quad (180)$$

It is a Siegel modular form when L is even unimodular. The genus- g partition function of a lattice theory contains (180) multiplied by an oscillator determinant.

Theta series are traces over zero modes. They forget the cocycle and many OPE coefficients. Nonisometric lattices may have equal theta series in low genus. The Schottky form illustrates the first place at which the two rank-sixteen even-unimodular lattice theta series differ on the full Siegel space, while their restrictions to the genus-four Jacobian locus agree [59, 66].

30.6 Symmetric products and the DMVV formula

Let $\phi_M(\tau, z)$ be the elliptic genus of a compact Calabi–Yau manifold M . Dijkgraaf–Moore–Verlinde–Verlinde prove the product formula

$$\sum_{N \geq 0} p^N \phi_{\text{Sym}^N M}(\tau, z) = \prod_{n > 0, m \geq 0, \ell \in \mathbb{Z}} \left(1 - p^n q^m y^\ell \right)^{-c(nm, \ell)}, \quad (181)$$

where $c(m, \ell)$ are the Fourier coefficients of ϕ_M [26]. The logarithm of the product is a sum of Hecke transforms. This is the plethystic exponential of connected single-particle data.

The formula has the same combinatorial exponential structure as a cofree commutative coalgebra and as the connected-graph identity (162). The equality of combinatorial forms does not identify the DMVV state space with a chiral Chevalley coalgebra. Such an identification requires a functor from the sigma-model or BPS category to chiral factorization data.

30.7 Borchers products

A weak Jacobi form with integral Fourier coefficients can produce an automorphic product whose exponents are those coefficients. Gritsenko and Nikulin use such products to construct automorphic corrections of Lorentzian Kac–Moody algebras [44]. The Igusa cusp form Φ_{10} is the multiplicative lift relevant to the K3 elliptic genus, and the DMVV formula is naturally expressed using $1/\Phi_{10}$.

The weight of a Borchers product is half the constant coefficient of its input in the standard normalization. This weight is an automorphic invariant. It is not the central charge, the rank of a lattice, or a universal Koszul conductor.

30.8 K3 and geometric quantum groups

Maulik and Okounkov construct Yangians from the geometry of Nakajima quiver varieties using stable envelopes and R -matrices [56]. The cohomology of Hilbert schemes of points on a surface carries Heisenberg and, in suitable settings, larger quantum-group actions. Braverman–Finkelberg–Nakajima construct Coulomb-branch algebras that include shifted Yangians [10].

These are exact geometric constructions of quantum groups. A K3 surface also supplies a lattice of signature $(4, 20)$, elliptic genera, and Hilbert-scheme Fock spaces. None of these facts alone identifies a K3 Hall algebra with the ordered Koszul dual of a particular chiral algebra. The identification would require a functor preserving products, coproducts, pairings, completions, and R -matrices.

30.9 Moonshine

Moonshine associates modular functions to traces of Monster elements on $V^\natural$. Borchers’ proof uses a generalized Kac–Moody algebra and an automorphic denominator identity. The chiral algebra supplies the graded representation and its vertex operators; the denominator identity supplies an automorphic trace.

Chiral Koszul duality applies to the local chiral operation of $V^\natural$. Moonshine is additional equivariant and arithmetic structure. A trace remembers the conjugacy class and graded multiplicities but not the complete OPE.

30.10 Genus one and genus two

Genus one detects characters, modular transformations, and one-variable theta series. Genus two detects interactions between two cycles and introduces Siegel modular forms. Some distinctions first appear only on the full Siegel space and disappear on the Jacobian locus. Therefore the phrase “genus two detects” has mathematical content only after the function, the ambient moduli space, and the restriction locus have been named.

The same precision applies to L -functions. A modular form determines an L -function; a graph amplitude equals a special value only after an integral representation or a Rankin–Selberg unfolding has been proved. No operator-product coefficient is automatically an L -value.

30.11 Arithmetic duality

Dualizing a local system sends its period matrix to the contragredient one. Poincaré duality pairs homology and cohomology periods. Functional equations of L -functions reflect duality of motives. These are distinct dualities, connected only when the relevant realization functors commute.

The common language is a diagram of realizations:

algebraic coefficient object $\longrightarrow$ de Rham local system $\longrightarrow$ Betti monodromy $\longrightarrow$ period or trace.

Chiral Koszul duality acts at the first stage. Arithmetic enters at the final stages.

31 Quantum field theory in the same language

31.1 Observables as a factorization algebra

A perturbative quantum field theory assigns a cochain complex of observables to each open set. Observables with disjoint supports multiply. Locality and descent make these complexes a factorization algebra [14, 15]. The differential combines the equations of motion, gauge symmetry, and quantum corrections.

A holomorphic theory on a complex curve produces a holomorphic factorization algebra. Under suitable regularity, its local observables determine a vertex algebra or chiral algebra. A topological–holomorphic theory on $M \times X$ may produce a chiral algebra on a boundary parallel to X . The OPE is the short-distance expansion of the factorization product.

31.2 Renormalization and compactified configurations

A Feynman graph with vertices at points of a manifold defines a differential form or distribution on the configuration space of distinct vertices. Divergences occur at collision diagonals. Compactifying the configuration space resolves these diagonals into boundary faces indexed by forests. Renormalization extends the weight to the boundary by adding local counterterms.

The forest formula and the Fulton–MacPherson forest resolution have the same incidence category. The coefficient systems differ: a Feynman weight contains propagators, gauge-fixing data, and densities; the chiral Chevalley complex contains the diagonal operation. A comparison theorem identifies them when the boundary asymptotics of the propagator reproduce the chiral operation.

31.3 Classical and quantum master equations

Let $\mathcal{E}$ be the BV complex of fields with odd Poisson bracket $\{-, -\}$. A classical action satisfies

$$QS + \frac{1}{2}\{S, S\} = 0.$$

A quantization satisfies

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0$$

after renormalization. The anomaly is the cohomology class of the failure of this equation.

The modular equation (159) has the same form because both are generated by graph contractions. The identification is exact when the Feynman rules define an algebra over the same modular cooperad. The boundary strata of compactified configurations then prove the master equation by Stokes' theorem.

31.4 Chern–Simons theory and affine currents

Chern–Simons theory on a three-manifold with the standard chiral boundary condition induces a chiral Wess–Zumino–Witten boundary theory. The boundary currents form an affine Kac–Moody algebra. In the abelian theory, the boundary current OPE is proportional to the level and is computed directly from the Green kernel. In the nonabelian theory, one-loop effects produce the familiar shifted coupling in KZ and surgery formulas.

Gwilliam and Williams construct a one-loop exact quantization of a mixed holomorphic–topological presentation of Chern–Simons theory and analyze its boundary behavior [45]. The factorization algebra of boundary observables is the analytic counterpart of the affine chiral algebra.

Wilson lines form a braided tensor category. Their exchange is described by quantum-group R -matrices; their closure produces knot invariants. This is the physical realization of Theorems 20.2 and 21.2.

31.5 Four-dimensional Chern–Simons and integrability

In four-dimensional Chern–Simons theory, line operators are topological along a surface and holomorphic in a spectral parameter. Their OPE produces rational, trigonometric, or elliptic R -matrices depending on the curve. The Yang–Baxter equation follows from moving three lines without crossing a singular configuration. Transfer matrices commute because separated line networks are isotopic.

The Gaudin model arises from the semiclassical rational connection. Spin chains arise from choosing representations on parallel lines. Baxter relations are exact sequences or functional relations in the representation category. These structures are ordered because the sequence and positions of lines matter.

31.6 BRST, ghosts, and strings

Gauge fixing introduces ghosts and a BRST differential. The total central charge of matter and ghosts controls the conformal anomaly of the worldsheet theory. For the bosonic string, the (b, c) ghost system has central charge -26 ; BRST nilpotence requires matter central charge 26 together with the usual normal-ordering or intercept condition in the state-space formulation.

Polyakov’s path integral produces determinant lines over moduli. The Weyl anomaly is the curvature of these determinant lines. The Belavin–Knizhnik theorem and the theory of Quillen metrics make this relation geometric. The cancellation of matter and ghost determinant lines is the global form of the central-charge condition.

This anomaly cancellation is not a statement that the matter VOA is Koszul dual to the ghost VOA. It is a tensor-product cancellation in the conformal-block and determinant-line theory.

31.7 Brown–Henneaux and Cardy

Three-dimensional gravity with negative cosmological constant has asymptotic Virasoro symmetry with

$$c = \frac{3\ell}{2G} \tag{182}$$

by the Brown–Henneaux theorem [12]. In a unitary modular-invariant two-dimensional CFT with a sparse light spectrum and vacuum dominance, the Cardy formula gives the high-energy density of

states. Substituting (182) reproduces the Bekenstein–Hawking entropy of the BTZ black hole in the semiclassical regime.

The rigorous algebraic input is the boundary Virasoro algebra and modular covariance of its partition function. A derived center of the Virasoro algebra is not, by itself, a nonperturbative theory of three-dimensional gravity.

31.8 Entanglement in two-dimensional conformal field theory

For the vacuum of a unitary CFT on the line, the entanglement entropy of an interval of length L with ultraviolet cutoff ϵ is

$$S_A = \frac{c + \bar{c}}{6} \log \frac{L}{\epsilon} + \text{const.} \quad (183)$$

The replica derivation uses twist fields and conformal maps. The coefficient is fixed by the left- and right-moving stress-tensor anomalies; in a parity-invariant theory with $c = \bar{c}$ it reduces to $c/3$.

Equation (183) is a trace statement. It does not determine the OPE, braid monodromy, or representation category. Two CFTs with the same c may have different entanglement spectra and operator algebras.

31.9 Bulk–boundary maps

A boundary condition for a factorization algebra of bulk observables defines an action on boundary observables. By the universal property of the derived center, there is a map

$$\text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ch}}^{\text{der}}(A).$$

Costello and Gaiotto use this structure in twisted holography, where protected bulk sectors and boundary vertex algebras can sometimes be compared exactly [16]. The comparison is a theorem of the particular twisted theory, not a universal identity for all chiral algebras.

A decoupled bulk factor remains invisible to the boundary. Conversely, a boundary algebra may admit several bulk completions with different global observables. The center classifies central actions, not all global physics.

31.10 Celestial chiral algebras

In celestial and twistor constructions, scattering amplitudes can be encoded by correlators of chiral algebras. Costello–Paquette and collaborators construct examples in which gauge-theory amplitudes arise from current-algebra correlators and their deformations [17]. Associativity of the celestial OPE is then a nontrivial constraint on loop corrections.

The precise statement depends on the background geometry, asymptotic states, and the map from scattering observables to chiral fields. Soft theorems provide currents in the chiral algebra, but no universal sequence of OPE coefficients equals the gravitational soft expansion across all theories.

31.11 Calabi–Yau categories and topological strings

A Calabi–Yau A_{∞} category has a cyclic pairing. Costello’s construction associates to suitable Calabi–Yau categories a two-dimensional topological conformal field theory whose closed states are Hochschild chains. The Connes operator and the cyclic pairing supply the BV structure.

A holomorphic field theory on a Calabi–Yau variety may produce factorization algebras and BPS operator algebras. Cohomological Hall algebras, shifted Yangians, and Borchers products arise in specific geometries. Passing from such a category to a chiral algebra on a curve requires an explicit compactification, reduction, or boundary functor. The dimension shift and cyclic pairing alone do not define the functor.

31.12 The physical comparison diagram

The exact relation between the mathematical and physical objects is a commutative diagram of maps, not an equality of names:

$$\begin{array}{ccc}
 \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) & \longrightarrow & \text{algebraic factorization coalgebra} \\
 \downarrow & & \downarrow \\
 \text{renormalized collision complex} & \longrightarrow & \mathrm{Obs}_{\partial}
 \end{array} \tag{184}$$

The bulk-to-boundary action has the separately typed form

$$\mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ch}}^{\mathrm{der}}(\mathrm{Obs}_{\partial}). \tag{185}$$

The vertical maps are Riemann–Hilbert, analytification, or explicit field-theoretic comparison maps. When they are quasi-isomorphisms, the mathematical and physical master equations are the same equation in two models.

32 The physical derivation of ordered chirality

32.1 A local experiment

Consider a quantum field theory on

$$X \times \mathbb{R}_t,$$

holomorphic along the complex curve X and topological along the oriented coordinate t . Let $\mathcal{O}_i(z_i, t_i)$ be local observables with

$$t_1 < t_2 < \cdots < t_n.$$

Topological invariance permits the t_i to move without changing a correlator, provided their order does not change. Holomorphy permits the z_i to move except at collisions. Hence a correlator is locally constant in the ordered chamber of the t_i and meromorphic or distributional in the z_i .

The short-distance product has the form

$$\mathcal{O}_1(z_1, t_1) \cdots \mathcal{O}_n(z_n, t_n) \sim \sum_{\alpha} C_{\alpha}(z_1, \dots, z_n) \mathcal{O}_{\alpha}(z, t), \tag{186}$$

where the coefficient C_{α} is a chiral distribution and the multiplication order is the order of the t_i . Formula (186) is the physical content of an E_1 algebra in factorization D -modules.

Principle 32.1 (Ordered OPE principle). *Holomorphic collision determines the coefficient distribution. Topological ordering determines the associative word. The full product is their interchange law.*

This principle is more primitive than a vertex-algebra expansion. A local coordinate on X converts the coefficient distribution into modes. Choosing an ordering prescription on $\mathbb{R}$ converts the topological factor into a time-ordered product.

32.2 Topological descent and the E_1 product

Let Q be the cohomological supercharge of the twist. Topological invariance in the real direction means that the translation generator is Q -exact:

$$P_t = [Q, G_t].$$

For a Q -closed observable $\mathcal{O}$,

$$\partial_t \mathcal{O} = [Q, G_t \mathcal{O}].$$

The difference between insertions at two values of t is therefore Q -exact. Ordered products are independent of the positions on cohomology.

The identity $P_t = [Q, G_t]$ is the first descent equation. A chain-level E_1 structure requires the coherent extension of this identity to families of interval embeddings: the descendant operators must satisfy the Cartan homotopy relations and their higher compatibility on compactified configuration cells. Integrating the coherent descendants gives the A_∞ operations. The boundary of an associahedron is the sum of its parenthesized degenerations, so Stokes' theorem gives the Stasheff identities.

Theorem 32.2 (Descent produces ordered multiplication). *Let a factorization algebra of observables on $X \times \mathbb{R}$ be holomorphic in X and equipped with coherent topological descent in the real direction, so that its restriction to intervals is a locally constant factorization algebra. Then restriction to product disks determines an E_1 algebra in holomorphic factorization observables on X . A local operator G_t with $P_t = [Q, G_t]$ exhibits the first coherence; when it extends to the stated descent datum, the product on Q -cohomology is independent of the transverse positions.*

The theorem is the physical proof of Theorem 10.2 for a cohomological field theory.

32.3 Why the OPE is a D -module

The coefficient in (186) is not a function at the collision locus. Ward identities imply differential equations in the insertion points. Contact terms are distributions supported on diagonals. A right D -module records both the differential equations away from collisions and the allowed supported terms at collisions.

For a conserved holomorphic current,

$$\bar{\partial} J = 0$$

away from other insertions. In a correlator,

$$\bar{\partial}_z \frac{1}{z - w} = \pi \delta^{(2)}(z - w)$$

in the distributional normalization. Thus the singular OPE and the contact Ward identity are the same distribution seen before and after applying $\bar{\partial}$. The D -module language retains this equality without choosing a Laurent contour at every step.

32.4 Boundary conditions and open strings

Let the theory occupy $X \times \mathbb{R}_{\geq 0}$ and let b be a boundary condition at $t = 0$. Boundary operators form the endomorphism algebra of b . If several boundary conditions b_i are present, a boundary-changing field lies in

$$A_{ij} = \mathrm{RHom}(b_i, b_j).$$

Composition has the order

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

This is the open-string product. The insertion point on X supplies its chiral dependence.

The strip with boundary conditions b_i and b_j has state complex A_{ij} . Gluing strips is composition. Closing a strip into an annulus gives Hochschild homology when a cyclic trace and the required dualizability are present. Thickening the Feynman graphs of a cyclic open-string interaction then gives ribbon surfaces. These hypotheses are exactly the transverse modular data of Section 24.

32.5 The interval bar as a propagator-independent amplitude

Place the augmentation boundary at the two ends of an interval. Factorization homology gives

$$\int_{([0,1], \partial)} A \simeq \mathbf{1} \otimes_A^L \mathbf{1}.$$

A cellular decomposition of the interval yields the bar complex. Its differential composes adjacent operator insertions. This construction uses no gauge-fixing propagator and no logarithmic form. It is topological and exact.

A Feynman expansion is another model of the same derived tensor product when the field theory admits an action and a gauge fixing. The comparison map sends a bar word to the corresponding time-ordered amplitude. Its proof is a homological perturbation or factorization-excision theorem, not an identification of the bar differential with a residue.

32.6 A second topological direction and exchange

Now place the theory on $X \times \mathbb{R}^2$. Two parallel lines can move around one another. The exchange path defines

$$R_{12}(z_1, z_2) : L_{z_1} \otimes L_{z_2} \longrightarrow L_{z_2} \otimes L_{z_1}.$$

Three-line isotopy gives the Yang–Baxter equation. The exchange depends meromorphically on $z_1 - z_2$ because a singularity occurs when the spectral points collide.

This derivation explains the codimension hierarchy:

$$\begin{aligned} \mathbb{R}^1 & : E_1 \text{ fusion,} \\ \mathbb{R}^2 & : E_2 \text{ braid exchange,} \\ \mathbb{R}^3 & : E_3 \text{ higher linking operations.} \end{aligned}$$

The chiral curve is a coefficient direction and does not add to the little-disks index until its holomorphic translations become topological.

32.7 Four-dimensional Chern–Simons theory

The action

$$S[A] = \frac{1}{2\pi\hbar} \int_{\Sigma \times X} \omega \wedge \left(\frac{1}{2}(A, dA) + \frac{1}{6}(A, [A, A]) \right)$$

is topological on the oriented surface Σ and holomorphic on the curve X . A Wilson line lies on a curve in Σ and at a point of X . The point of X is its spectral parameter.

At tree level, exchange of two Wilson lines gives the Casimir kernel

$$r(z) = \frac{\Omega}{z}.$$

The configuration of three lines has two isotopic resolutions, which gives the classical Yang–Baxter equation. Quantization replaces r by an R -matrix. The one-loop anomaly of line gauge invariance forces the Yangian deformation in the rational theory. Thus the Yangian is not inferred from order; in the formal perturbative rational theory it is selected by the gauge-theory anomaly and Ward identities [19, 20].

For $X = \mathbb{C}^\times$ and for elliptic X , the propagator has trigonometric and elliptic periodicity. The quantum symmetry changes accordingly. The spectral curve is part of the physics.

32.8 Three-dimensional gauge theory and boundary chiral algebra

A holomorphic twist of a three-dimensional supersymmetric theory on $X \times \mathbb{R}_{\geq 0}$ may admit a boundary condition preserving the holomorphic supercharge. Boundary local operators form a chiral algebra on X . Boundary line junctions carry an ordered product inherited from the normal direction. Bulk lines act by modules.

In a Lagrangian theory, the boundary algebra is computed from boundary fields and BRST reduction. Gauge anomalies must cancel for the boundary condition to exist quantum mechanically. Different boundary conditions can give different E_1 chiral algebras with equivalent central bulk actions. This is the physical origin of Morita and Koszul dual boundary descriptions.

32.9 Drinfeld–Sokolov reduction as gauge fixing

Start from affine currents and couple them to ghosts imposing a moment-map constraint. The BRST charge is

$$Q = \oint \left(\sum_a c^a (J_a - \chi_a) - \frac{1}{2} \sum_{a,b,c} f_{ab}^c : c^a c^b b_c : + \dots \right).$$

The equation $Q^2 = 0$ follows from the current algebra, invariance of the level cocycle, and cancellation of the gauge anomaly. The reduced chiral algebra is $H_Q(A)$.

Because Q is a derivation of the transverse product, it acts on the ordered bar. The total differential is

$$D = b_A + Q_{\text{bar}}.$$

The spectral sequence obtained by first taking BRST cohomology compares the bar of the reduced algebra with the reduction of the bar. Degeneration is a theorem of the particular good grading and filtration. The higher W -operations are the homotopy-transferred remnants of affine multiplication and ghost contraction.

32.10 Polyakov's determinant and the chiral anomaly

For a free scalar on a Riemann surface,

$$Z[g] = (\det' \Delta_g)^{-1/2}$$

up to zero-mode volume. Under $g \mapsto e^{2\sigma}g$, zeta regularization gives the Polyakov action

$$\log \frac{Z[e^{2\sigma}g]}{Z[g]} = \frac{1}{24\pi} \int (|\nabla \sigma|^2 + R\sigma) d\text{vol}.$$

For a theory of central charge c , the right side is multiplied by c . The same coefficient appears in the fourth-order pole of the stress tensor because both measure the central extension of local conformal transformations.

On a family of curves, the determinant forms a line bundle with Quillen connection. Its curvature is computed by the family index theorem. This is the exact bridge from the local Virasoro cocycle to the Hodge class. It does not identify the central charge with affine level, ribbon genus, or a Koszul curvature.

32.11 Worldsheet sewing

Choose local coordinates $zw = q$ around two punctures. Sewing inserts

$$q^{L_0 - c/24}$$

between dual states. The factor $q^{-c/24}$ is the vacuum energy on the cylinder. The sum over a homogeneous basis is the inverse two-point pairing. Convergence for $|q| < 1$ is the analytic sewing theorem.

A nonseparating sewing increases the genus of the chiral curve. An annular trace of the transverse E_1 algebra increases ribbon genus. The two operations can occur in the same amplitude, giving the bidegree of Section 24.

32.12 The Chern–Simons/WZW comparison theorem and its hypotheses

For compact gauge group at integral level, and after the standard choice of polarization and framing data, the Chern–Simons/WZW and Reshetikhin–Turaev constructions give the fundamental example in which the quantized surface state spaces, boundary current algebra, Wilson-line labels, conformal blocks, and ribbon-category invariants are connected by comparison theorems.

The boundary current algebra is affine Kac–Moody. Canonical quantization on a surface gives the WZW conformal-block space. Wilson lines are labelled by admissible representations. The Chern–Simons/WZW and Drinfeld–Kohno comparison theorems identify their exchange with the corresponding quantum-group braiding. Closing lines with the ribbon trace gives link invariants. Under these comparison theorems, the framing dependence of the three-dimensional theory matches the ribbon twist and the projective anomaly data of the WZW conformal blocks.

Each statement has a different mathematical carrier:

current OPE	: chiral D -module,
boundary composition	: E_1 algebra,
Wilson exchange	: E_2 line category,
surface states	: conformal blocks,
framing phase	: invertible anomaly theory.

The strength of the example lies in the comparison maps between these carriers, not in collapsing them into one object.

32.13 Large N and the double genus

For N coincident branes, boundary operators acquire matrix indices. A connected ribbon graph contributes N^{2-2h-b} . Independently, a closed-string worldsheet of genus g contributes g_s^{2g-2} . Open–closed duality can reorganize the first expansion into the second after summing boundaries and applying a dynamical transition.

Before that transition, the amplitude is naturally

$$\sum_{g,h,b} g_s^{2g-2} N^{2-2h-b} \mathcal{F}_{g,h,b}.$$

The bimodular formalism is the algebraic form of this double expansion. The relation between g_s and N is a physical duality parameter, not an identity in operad theory.

32.14 Holographic reconstruction

Let A be the ordered chiral algebra of a boundary condition. Bulk observables act centrally on A , giving

$$\beta : \text{Obs}_{\text{bulk}} \longrightarrow Z_{\text{ord, ch}}(A).$$

If the boundary condition is complete, β is a quasi-isomorphism. Then the protected bulk local algebra is reconstructed from the boundary. Two boundary descriptions have the same centre when an explicit Morita context identifies their bimodule categories; a Koszul equivalence of augmentation-generated sectors does not by itself imply this stronger statement.

This is the precise algebraic statement behind twisted holography. It reconstructs a protected local sector. A metric, Hilbert space, nonperturbative spectrum, and integration cycle require additional physical structure.

32.15 The field-theoretic meaning of the dual

The augmentation $A \rightarrow \mathbf{1}$ is a distinguished boundary or defect. Its derived endomorphism algebra

$$A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1})$$

acts on deformations of that defect. The bar complex

$$\mathbf{1} \otimes_A^L \mathbf{1}$$

is the complex of interval amplitudes between two copies of the defect. This is why Koszul duality repeatedly appears in line-operator physics: it exchanges local bulk couplings near the defect with operators living on the defect.

The dual is a completion near the chosen augmentation. It need not recover sectors disconnected from that boundary condition. The physical meaning of double duality is therefore completion of the local theory around the chosen vacuum.

32.16 The physical theorem in one sentence

A holomorphic–topological quantum field theory whose protected observables are product-generated, algebraizable as factorisable D -modules, locally constant in the ordered direction, and anomaly-free produces an ordered chiral algebra: collision on the complex curve gives the chiral coefficient and ordered fusion gives the E_1 product. A second topological direction supplies exchange, a cyclic trace supplies ribbon loops, variation of the curve supplies conformal sewing, and each anomaly is the invertible theory obstructing the corresponding gluing.

33 Higher-dimensional sources and reduction to a curve

33.1 Pushforward is the source of chirality

Let Y be a compact framed d -manifold and let $p : Y \times X \times \mathbb{R} \rightarrow X \times \mathbb{R}$. Suppose $\mathcal{F}$ is a factorization algebra that is locally constant in the Y and $\mathbb{R}$ directions and holomorphic in X . Define

$$(p_*^{\text{fact}} \mathcal{F})(U \times I) := \int_{Y \times U \times I} \mathcal{F}. \quad (187)$$

Excision gives factorization maps on the target.

Theorem 33.1 (Topological pushforward to ordered chirality). *Assume the local constancy, dualisability, and excision hypotheses required for factorisation homology, and assume that the pushforward is product-generated and its holomorphic coefficient admits the factorisable D -module realisation of Theorem 10.1. Then (187) is a rectangular holomorphic–topological factorisation algebra on $X \times \mathbb{R}$ and determines an ordered chiral algebra in $\text{Alg}_{E_1}(\text{FactD}(X))$.*

Proof. Fubini for factorization homology identifies integration over disjoint product opens with iterated integration over Y and over the target open. Local constancy in Y and $\mathbb{R}$ is preserved by the pushforward. Holomorphic differential equations in X commute with integration in the topological directions. Weiss descent follows from excision. Apply Theorem 10.2. $\square$

The theorem gives the invariant shape of the higher-dimensional constructions treated here under these hypotheses. The source is a factorization algebra; the reduction is factorization homology; the output is an E_1 algebra in chiral coefficients.

33.2 Dimensional bookkeeping

If a theory is locally constant in m transverse directions, its local operators carry E_m structure. Integrating over a framed r -manifold with $r < m$ leaves E_{m-r} structure. A holomorphic curve remains

a coefficient direction. Thus a theory on

$$Y^r \times X \times \mathbb{R}^{m-r}$$

can reduce to an E_{m-r} algebra in $\text{FactD}(X)$.

This formula replaces a universal “chirality ladder.” The output depends on the actual number of topological directions and on the tangential structure used in the pushforward.

33.3 Calabi–Yau categories

A smooth proper Calabi–Yau A_∞ category $\mathcal{C}$ has a nondegenerate cyclic trace. Its open sector is the family of morphism complexes $\text{RHom}(E, F)$ with A_∞ composition. Its closed state complex is Hochschild homology $\text{HH}_\bullet(\mathcal{C})$. The Connes operator and the cyclic pairing supply the circle and surface operations of the associated topological conformal field theory.

These data are topological. To obtain a chiral algebra on X , one needs a family $\mathcal{C}_x$ varying holomorphically with $x \in X$, or a holomorphic field theory whose boundary category is $\mathcal{C}$. The ordered chiral algebra is then the endomorphism object of a chosen boundary generator, or the pushforward (187).

Theorem 33.2 (Categorical source theorem). *Let $\mathcal{C}$ be a factorization category over X with a compact generator b and with an ordered topological composition direction. Then*

$$A_b = \text{RHom}_{\mathcal{C}}(b, b)$$

is an ordered chiral algebra and

$$\mathcal{C} \simeq \text{Mod}_{A_b}(\text{FactD}(X))$$

under compact Morita hypotheses. If $\mathcal{C}$ is Calabi–Yau, the induced trace on A_b supplies the transverse cyclic structure of Theorem 24.1.

This theorem is the correct passage from a Calabi–Yau category to noncommutative chirality. It requires a factorization family over the curve; the Calabi–Yau trace alone does not create one.

33.4 Naturality of obstruction classes

Let $\mathfrak{g}$ and $\mathfrak{h}$ be complete filtered L_∞ algebras controlling two deformation problems, and let

$$F : \mathfrak{g} \longrightarrow \mathfrak{h}$$

be a filtered L_∞ morphism. It sends Maurer–Cartan elements to Maurer–Cartan elements and gauge equivalences to gauge equivalences.

Proposition 33.3 (Transgression of obstructions). *Let $\alpha_{\leq r}$ be a Maurer–Cartan solution modulo filtration $F^{r+1}\mathfrak{g}$, with obstruction*

$$o_{r+1}(\alpha) \in H^2(\text{gr}^{r+1}\mathfrak{g}, d_\alpha).$$

Then the obstruction to extending $F_(\alpha_{\leq r})$ is the image*

$$F_*o_{r+1}(\alpha) \in H^2(\text{gr}^{r+1}\mathfrak{h}, d_{F_*(\alpha)}).$$

The proof is the order- $r + 1$ component of the L_∞ identities. This proposition is the rigorous content of every assertion that a Calabi–Yau, Hall, or gauge-theoretic obstruction becomes a chiral obstruction. The essential construction is the morphism F .

33.5 Cohomological Hall algebras

Let $\mathcal{M}$ be a moduli stack of objects in a suitable abelian or derived category. Correspondences of short exact sequences define a convolution product on an appropriate homology theory of $\mathcal{M}$. The resulting cohomological Hall algebra is an E_1 algebra. Its multiplication is noncommutative because extension flags are ordered.

A coproduct requires additional geometry, typically localization, a stability condition, or a choice of chamber. A Drinfeld double requires a nondegenerate Hopf pairing. An R -matrix requires a stable-envelope or line-exchange construction. These are separate structures.

Criterion 33.4 (From a Hall algebra to a quantum group). *A Hall algebra determines a quasitriangular quantum group only after the construction of:*

- (i) *a compatible completed coproduct;*
- (ii) *a nondegenerate pairing and a double;*
- (iii) *a PBW theorem and a presentation of the relation ideal;*
- (iv) *a universal or categorical R -matrix;*
- (v) *convergence of the completion on the chosen representation category.*

The ordered chiral pushforward of a Hall theory retains the E_1 multiplication. The remaining items must survive the pushforward to identify its line category with a quantum group.

33.6 Coulomb branches and shifted Yangians

For three-dimensional $\mathcal{N} = 4$ gauge theories, the Braverman–Finkelberg–Nakajima construction defines Coulomb-branch algebras by convolution on affine-Grassmannian spaces. In type A quiver examples, quantized Coulomb branches are truncated shifted Yangians. The coweight shift records magnetic or boundary data.

In the holomorphic–topological twist, these algebras act on line defects. A boundary reduction to X gives a factorization family of modules with spectral parameter. The ordered chiral algebra records fusion of these lines. The BFN identification supplies the named shifted Yangian after the geometric convolution algebra and the Drinfeld presentation have been compared.

33.7 K3 geometry

Let S be a K3 surface. Its Mukai lattice

$$\tilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

has signature $(4, 20)$. Nakajima correspondences give a Heisenberg action on

$$\bigoplus_{n \geq 0} H^*(S^{[n]}).$$

The elliptic genera of the symmetric products satisfy the Dijkgraaf–Moore–Verlinde–Verlinde product formula. The reciprocal Igusa cusp form encodes the second-quantized K3 elliptic genus.

These are exact statements about cohomology, modular forms, and second quantization. They do not by themselves define an ordered chiral algebra or its Koszul dual.

Principle 33.5 (K3 separation principle). *The Mukai lattice, Nakajima Heisenberg action, Hall multiplication, elliptic genus, Siegel modular form, and a possible line-defect quantum group are distinct realizations of K3 geometry. A common chiral object exists only after explicit functors identify their operations and pairings.*

A valid construction may begin with a K3-associated field theory on $S \times X \times \mathbb{R}$, integrate over S or a cycle in S , and apply Theorem 33.1. Its comparison with a Hall algebra or a Borchers product is then a theorem about that pushforward.

33.8 Borchers products and scalar traces

A Borchers lift takes a suitable vector-valued modular or Jacobi form to an automorphic product. The weight is determined by the constant term of the input. The product can be interpreted as a denominator identity of a generalized Kac–Moody algebra in favorable cases.

The automorphic product is a scalar or section of a line bundle. It may arise as a trace, index, or partition function of a categorical or chiral object. Reconstruction in the opposite direction is not unique: different categories can have the same index. The ordered multiplication, braiding, and extension data are lost under the trace.

33.9 Omega backgrounds and BPS algebras

An Omega background makes rotations equivariantly exact and can reduce a higher-dimensional field theory to a noncommutative algebra of protected operators. In M-theory and gauge-theory examples, the resulting algebras include Yangians and their deformations. The deformation parameters are equivariant weights; line-defect spectral parameters are positions in a residual holomorphic curve.

The algebraic structure is obtained from the factorization OPE of protected operators. Koszul duality appears when local bulk observables are coupled to a defect and one computes derived endomorphisms of the trivial defect. The named BPS algebra follows from a separate comparison with geometric correspondences or representation-theoretic generators.

33.10 Shifted symplectic geometry

Derived moduli stacks of fields or objects often carry shifted symplectic forms. A boundary condition is represented by a Lagrangian morphism. The derived intersection of two Lagrangians has a shifted Poisson structure and may quantize to the algebra of boundary-changing operators.

Let $L_0, L_1 \rightarrow \mathcal{M}$ be Lagrangians in an n -shifted symplectic stack. Their derived intersection is $(n-1)$ -shifted symplectic. Its $(n-1)$ -shifted symplectic form determines a P_n -structure. Whenever an E_n -quantisation of that P_n -structure has been constructed and varies factorisably over a chiral curve, the resulting observables form an E_n algebra in factorisation D -modules.

This gives a geometric source of ordered chiral algebras when the ambient symplectic shift is $n=1$, so that the Lagrangian intersection is 0-shifted symplectic and its quantisation is associative. The

quantization and the factorization family are part of the theorem; the shifted symplectic form alone is classical data.

33.11 A source-to-chiral comparison

Suppose the higher-dimensional factorisation pushforward constructs

$$A_{\text{geom}} \in \text{Alg}_{E_1}(\text{FactD}(X)),$$

and a Hall, BPS, or localization construction independently produces

$$A_{\text{src}} \in \text{Alg}_{E_1}(\text{FactD}(X)).$$

The comparison datum is an E_1 -algebra morphism

$$\phi : A_{\text{src}} \longrightarrow A_{\text{geom}}$$

compatible with the chiral coefficient. Applying the ordered bar gives the commutative square

$$\begin{array}{ccc} A_{\text{src}} & \xrightarrow{\phi} & A_{\text{geom}} \\ \text{Bar}_X^{\text{ord}} \downarrow & & \downarrow \text{Bar}_X^{\text{ord}} \\ \text{Bar}_X^{\text{ord}}(A_{\text{src}}) & \xrightarrow{\text{Bar}(\phi)} & \text{Bar}_X^{\text{ord}}(A_{\text{geom}}). \end{array} \quad (188)$$

The upper objects are associative and the lower objects are coassociative; the functorial bar map is what relates the two types. The construction of A_{src} contains the Hall or BPS correspondences, while the construction of A_{geom} contains factorisation pushforward. Their equality is the theorem that ϕ is an equivalence.

If ϕ is an equivalence on a chosen complete sector, then (188) identifies the corresponding Hall or BPS resolution with the ordered chiral bar. A quantum-group identification still requires the coproduct, pairing, R -matrix, and PBW data of Criterion 33.4.

33.12 The higher-dimensional conclusion

Higher-dimensional geometry enters ordered chiral Koszul duality through factorization pushforward, boundary endomorphisms, and morphisms of deformation complexes. Cyclic traces produce surfaces; Hall correspondences produce associative multiplication; Omega backgrounds produce deformation parameters; shifted symplectic forms produce Poisson brackets. The chiral curve carries the spectral dependence of the resulting operators.

The language remains unchanged: an E_1 algebra in factorization D -modules is the target through which these sources can be compared without erasing their origin.

34 The single language

34.1 The theorem of the construction

Theorem 34.1 (Noncommutative chiral Koszul geometry). *Let X be a smooth complex algebraic curve and let*

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)).$$

Then the following statements hold in the indicated ambients.

- (i) *The operadic Grothendieck construction for $E_1 \otimes \text{Ch}_X$ presents A by ordered sequences of entire finite chiral configurations. Chiral collision occurs inside a block; the associative operadic maps compose adjacent blocks.*
- (ii) *The pointwise tensor and chiral convolution on the Ran category satisfy the duoidal interchange law. The E_1 multiplication of A is a morphism of chiral factorisation coalgebras. Consequently the ordered bar is a coassociative coalgebra object in chiral factorisation coalgebras.*
- (iii) *The internal ordered bar is the derived interval tensor product*

$$\text{Bar}_X^{\text{ord}}(A) \simeq \mathbf{1} \otimes_A^L \mathbf{1}.$$

Its differential contracts adjacent transverse factors, while the complete chiral singularity remains in the coefficient D -module.

- (iv) *On connected positive-weight objects and their complete Mittag–Leffler limits, internal associative bar and cobar reconstruct one another. The boundary Koszul dual $A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1})$ is the endomorphism algebra of the augmentation boundary, and $A^{!!}$ is the derived augmentation completion.*
- (v) *The universal twisting morphism identifies augmentation-complete A -modules with conilpotent comodules over $C = \text{Bar}_X^{\text{ord}}(A)$. Its two-sided form sends the diagonal bimodule to C_Δ , carries relative tensor products to derived cotensor products, and identifies Hochschild homology and cohomology with the corresponding coHochschild theories.*
- (vi) *The cardinality pro-nilpotence used by Francis–Gaitsgory belongs to $D(\text{Ran } X)$ with the chiral convolution product, not to the pointwise coefficient tensor of $\text{FactD}(X)$. In that chiral monoidal category, the enhanced theorem gives the reduced associative bar–cobar equivalence and, in the corresponding Lie case, the chiral Lie–commutative-coalgebra equivalence.*
- (vii) *A cofibrant resolution of $E_1 \otimes \text{Ch}_X$ is described by two-coloured levelled forests. Transverse edges encode associative homotopy, chiral edges encode collision homotopy, and the two edge-contraction differentials anticommute.*
- (viii) *The pole filtration of a collision divisor records every normal symbol of the singular product. On a formal disc, a translation-equivariant chiral coefficient becomes a vertex or nonlocal-vertex operation. In the standard chiral operadic model, its Maurer–Cartan equation is the integral Jacobi identity and hence the Borchers identity.*

- (ix) *The bare E_1 structure contains no braid. An E_2 enhancement or a flat configuration-space connection supplies exchange. A rigid meromorphic line category with a faithful strong monoidal fibre functor reconstructs a coquasitriangular bialgebra, and a Hopf algebra when the antipode exists. A faithful quasi-monoidal fibre functor reconstructs a coquasi-bialgebra, or a quasi-Hopf algebra under the corresponding rigidity hypotheses. A Yangian or quantum loop algebra is obtained only after a separate recognition theorem.*
- (x) *A square-zero BRST derivation extends to the ordered bar and forms a bicomplex with its differential. The comparison with the bar of BRST cohomology is the corresponding convergent spectral sequence; degree-zero Drinfeld–Sokolov reduction is recovered when BRST cohomology is concentrated.*
- (xi) *A transverse Calabi–Yau trace and a chiral modular sewing structure define two edge colours in a completed graph transform. Their structure element satisfies the two-edge master equation (155). Ribbon genus and chiral genus are independent gradings.*
- (xii) *A perturbatively renormalized holomorphic–topological BV theory whose observables satisfy the hypotheses of Theorem 10.5 produces such an A . Its renormalized graph weights give a morphism from the mixed forest operad when they extend to compactified configurations and satisfy the quantum master equation.*

The theorem does not identify unlike constructions. It says that each is obtained from one object by a functor whose source, target, and hypotheses are visible.

34.2 The Platonic object

The invariant object is

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X), \otimes^{\text{pt}}). \quad (189)$$

It is equivalently an algebra over the tensor-product operad

$$E_1 \otimes \text{Ch}_X,$$

or, on the regular-holonomic product-generated analytic locus, a factorization algebra on $X^{\text{an}} \times \mathbb{R}$ that is holomorphic in X and locally constant in $\mathbb{R}$.

This is the minimal language because it remembers exactly the two primitive experiments:

collision on the complex curve, ordered composition on the real line.

Removing the first loses the operator product. Removing the second loses noncommutative fusion. Adding a second real direction produces exchange rather than altering either primitive operation.

34.3 The functorial enrichments

enrichment or realization	mathematical and physical output
formal coordinate and translation	modes and vertex or nonlocal-vertex operations
pole filtration	normal symbols and the Poisson-vertex associated graded
internal interval bar	boundary amplitudes and a conilpotent coassociative coalgebra
augmentation boundary	the endomorphism algebra $A^!$ and derived completion
E_2 transverse enhancement	exchange paths and the Yang–Baxter equation; braided defect categories after categorical extension
strong or quasi-monoidal fibre functor	Tannaka bialgebra or coquasi-bialgebra, followed by quantum-group recognition
transverse Calabi–Yau trace	Hochschild chains, annuli, and ribbon genus
chiral modular sewing	conformal blocks, plumbing, and chiral genus
conformal vector	projective connection and determinant-line anomaly
BRST derivation	derived reduction and W -algebra cohomology
BV quantization	renormalized graph weights and the quantum master equation
Calabi–Yau or Hall source	factorization pushforward and line algebras after a specified functor
trace or index	characters, Verlinde numbers, theta series, and automorphic functions

A scalar appears only in the last row or as a coefficient of an invertible anomaly theory. It cannot replace the structured operations above it.

34.4 The clarity obtained

The full OPE is not one residue; it is the diagonal D -module with its pole filtration. Arnold’s relation is not the Borchers identity; it is a relation among configuration-space forms that may be paired with a coefficient identity. A Fulton–MacPherson screen is not contractible merely because its incidence category has a terminal corolla. The internal E_1 bar is not the Francis–Gaitsgory chiral bar; the two use different monoidal products. An ordered line is not braided. A quantum group is not a kernel dimension; it is a reconstruction of a tensor-compatible exchange category. The modular master equation is not forced by a vertex algebra; it is forced by a constructed sewing action. Curved Koszul data is not projective conformal curvature. The derived centre is not every physical bulk; it is the universal target of central bulk actions.

Each correction strengthens the original intuition. The subject becomes unified not by declaring the outputs equal, but by deriving them from the same local object through exact functors.

34.5 Mathematics and physics

Mathematically, the primitive datum (189) is the algebraic form of one holomorphic and one topological direction. The chiral coefficient is controlled by diagonals and the Ran space. The ordered product is controlled by intervals. Their common cofibrant resolution is controlled by two-coloured forests. Koszul duality is the passage from an augmented algebra to the coalgebra of interval states and to the endomorphisms of its augmentation boundary.

Physically, the same datum is the protected local algebra of a holomorphic–topological system. Chiral collisions are short-distance singularities. The E_1 product is time- or normal-ordered composition. The bar is interval evolution between boundary states. The E_2 centre is the universal algebra of central bulk operations. Braiding is exchange of lines in a transverse plane. Ribbon contraction is an open loop. Deligne–Mumford contraction is worldsheet plumbing. The BV master equation asserts compatibility of renormalized amplitudes with these boundary degenerations.

The two descriptions coincide when a factorization-observable comparison has been constructed. Without that comparison they are parallel realizations of the same operadic geometry, not identical by metaphor.

34.6 No scalar replacement

Corollary 34.2 (No universal scalar replacement). *No invariant obtained from any one of the standard scalarisation procedures used here—ordinary or supertrace, Euler characteristic, conformal-block rank, Grothendieck class, determinant, or numerical index—is conservative on the data consisting of the E_1 multiplication, complete chiral coefficient, interval bar coalgebra, exchange local system, and the two sewing theories. Whenever two scalar expressions agree, the equality is the image of a specified comparison or trace.*

Proof. The commutator forgets the symmetric part of multiplication. A trace forgets noncentral operations. A Grothendieck ring forgets associators and twists. A conformal-block rank forgets monodromy. A central charge forgets the remaining operator products and module category. Each displayed scalarisation has a nontrivial kernel on at least one of these structures, so none is conservative on the full class of complete ordered chiral objects considered in this paper. $\square$

The universal object is therefore not a number and not a single residue. It is the noncommutative chiral algebra (189), together with the functorial bars, centres, exchange structures, and sewing operations that exist for the structures actually supplied.

Coda: the geometry remembers

A field is not determined by its value at a point. Its singular behavior when another field approaches is part of the field. On a curve, that behavior is a D -module near a diagonal.

A product is not determined by the collection of its factors. The order in which intervals meet is part of the product. On a line, that order is an E_1 operation.

An exchange is not determined by an order. It is a path in a two-dimensional configuration space. Its algebraic memory is monodromy, and its universal reconstruction is braided Tannaka theory.

A trace is not determined by a product. It is the permission to close an interval. Its geometric memory is a circle; its algebraic memory is Hochschild homology; its coalgebraic memory is the self-cotensor of the diagonal bar state.

A genus is not determined by a trace alone. Ribbon genus is formed by closing open topological boundaries. Chiral genus is formed by plumbing complex curves. Their graph equations may commute, but their geometric parameters are different.

An anomaly is not a failed differential with an unnamed scalar. It is a class in the deformation theory of an invertible field theory. It becomes a number only after a line, a connection, a surface, and a normalization have been chosen.

A quantum group is not the amount of information lost by averaging. It is the algebra reconstructed from a tensor-compatible theory of exchange. The R -matrix is transport around a collision. The associator is comparison of two transports among three points. The Hopf algebra is the coend that remembers all matrix coefficients at once.

Koszul duality is not a reflection formula. It is the passage from an operation to the coalgebra of its resolved boundary states, and from that coalgebra to the endomorphisms of the augmentation boundary. When the Morita hypotheses hold, the centre is preserved because the category of boundaries is preserved.

The common syntax is factorisation. Disjoint regions multiply. Colliding regions produce singular coefficients. Cutting a region produces a tensor product over an algebra. Gluing a region produces a cotensor product over a coalgebra. Moving a defect produces monodromy. Closing a boundary produces a trace. Degenerating a surface produces a master equation.

The equations are local because their geometry is local. Associativity is the boundary of an interval compactification. Jacobi is the boundary relation among partial diagonals. Yang–Baxter is flatness on three-point configurations. The Batalin–Vilkovisky equation is Stokes’ theorem for renormalized graph weights. The modular equation is the oriented cancellation of codimension-two clutching strata.

The theory therefore has one invariant sentence and many exact realizations:

$$\boxed{A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)).} \quad (190)$$

The diagonal supplies chirality. The interval supplies noncommutativity. The plane supplies exchange. The circle supplies trace. The stable curve supplies sewing. The bar supplies resolution. The centre supplies universal bulk action. The determinant line supplies anomaly. The BV complex supplies quantum consistency.

Nothing is gained by collapsing these structures into one scalar. Everything is gained by placing them in one geometry.

A Orientation lines and the collision differential

The square-zero identities used throughout the paper are identities of oriented correspondences. This appendix fixes one convention and proves the two cancellations on which the collision complex and the corolla-normalized forest resolution rest. A coherent change of convention conjugates every differential by diagonal signs and does not change the resulting complex.

A.1 Cohomological suspension and unordered tensor products

Let (K, d_K) be a cochain complex. We write $sK = K[1]$ in the cohomological convention

$$(sK)^n = K^{n+1}, \quad |sx| = |x| - 1, \quad d_{sK}(sx) = -s(d_K x).$$

For a finite set S , the unordered tensor product is

$$\bigotimes_{s \in S}^{\text{unord}} K_s := \left(\bigoplus_{\alpha: \{1, \dots, |S|\} \xrightarrow{\sim} S} K_{\alpha(1)} \otimes \cdots \otimes K_{\alpha(|S|)} \right)_{\Sigma_{|S|}}, \quad (191)$$

where a permutation acts by the Koszul symmetry. An ordering of S identifies (191) with an ordinary tensor product, but no ordering is part of the object.

Let $\mu : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$ be a degree-zero graded Lie bracket. Its suspended form

$$\bar{\mu} : s\mathfrak{g} \otimes s\mathfrak{g} \longrightarrow s\mathfrak{g}$$

has cohomological degree +1 and is defined by

$$\bar{\mu}(sx, sy) = (-1)^{|x|} s[x, y]. \quad (192)$$

With this convention, graded antisymmetry of $[-, -]$ becomes graded symmetry of $\bar{\mu}$ with respect to the shifted degrees:

$$\bar{\mu}(u, v) = (-1)^{|u||v|} \bar{\mu}(v, u).$$

The Jacobi identity becomes

$$\bar{\mu}(\bar{\mu}(u, v), w) + (-1)^{|u|(|v|+|w|)} \bar{\mu}(\bar{\mu}(v, w), u) + (-1)^{|w|(|u|+|v|)} \bar{\mu}(\bar{\mu}(w, u), v) = 0. \quad (193)$$

The same formulas hold for a chiral Lie bracket after the tensor products are replaced by chiral products and the outputs are pushed forward to the common small diagonal.

A.2 Determinant lines of collision chains

Let I be finite and let $\Pi(I)$ be its partition lattice, ordered by refinement. A cover $\pi \triangleleft \rho$ merges exactly two blocks. Write $e_{\pi, \rho}$ for the one-dimensional conormal direction of that merger. For a saturated chain

$$\gamma = (\pi_0 \triangleleft \pi_1 \triangleleft \cdots \triangleleft \pi_r)$$

define

$$\text{or}(\gamma) = \bigwedge_{i=1}^r (e_{\pi_0, \pi_1} \oplus \cdots \oplus e_{\pi_{r-1}, \pi_r}).$$

Deleting the j -th edge gives the simplicial boundary sign $(-1)^{j-1}$. Interchanging two independent collision directions reverses the determinant orientation.

For a partition π , let

$$C_\pi(\mathfrak{g}) = \Delta_{\pi,!} \left(\bigotimes_{B \in \pi}^{\text{unord}} s\mathfrak{g} \right).$$

If $\pi \leq \rho$ merges blocks B and C , the map $d_{\pi,\rho} : C_\pi(\mathfrak{g}) \rightarrow C_\rho(\mathfrak{g})$ is the identity on the other blocks and $\bar{\mu}$ on the B, C factors, tensored with the oriented conormal line. The total collision differential is

$$d_{\text{coll}} = d_{\mathfrak{g}} + \sum_{\pi \leq \rho} d_{\pi,\rho}. \quad (194)$$

Lemma A.1 (Internal-collision cancellation). *If $d_{\mathfrak{g}}$ is a derivation of the chiral bracket, then*

$$d_{\mathfrak{g}} d_{\text{coll}}^{(1)} + d_{\text{coll}}^{(1)} d_{\mathfrak{g}} = 0,$$

where $d_{\text{coll}}^{(1)}$ is the sum over covers.

Proof. On the two merged factors this is the suspended form of

$$d[x, y] = [dx, y] + (-1)^{|x|} [x, dy].$$

The identity $d(sx) = -s(dx)$ gives the sign required when the degree-+1 map $\bar{\mu}$ is interchanged with the unary differential. Untouched factors contribute the usual tensor-product signs. Because the operation is defined on the unordered tensor product, the calculation is independent of the temporary ordering used to write it. $\square$

A.3 Rank-two intervals in the partition lattice

Every coefficient of $(d_{\text{coll}}^{(1)})^2$ is supported on a rank-two interval $[\pi, \sigma]$ of $\Pi(I)$. Such an interval has exactly one of two forms: two disjoint pairs of blocks are merged, or three blocks are merged into one.

Proposition A.2 (Disjoint-pair cancellation). *Let B, C, D, E be four distinct blocks of π . The two composites that merge B with C and D with E in the two possible orders sum to zero.*

Proof. The two coefficient operations act on disjoint factors. The transposition of the two odd collision directions can be recorded either as the interchange sign of the two degree-one maps or as the reversal

$$e_{BC} \wedge e_{DE} = -e_{DE} \wedge e_{BC}$$

of the determinant line. These are two presentations of the same transposition, not two independent signs. In the total oriented convention the two paths induce opposite maps to the common codimension-two component. $\square$

Proposition A.3 (Triple-collision coefficient). *Let B, C, D be three blocks of π , and let σ be obtained by replacing them with $B \cup C \cup D$. The component $C_\pi(\mathfrak{g}) \rightarrow C_\sigma(\mathfrak{g})$ of $(d_{\text{coll}}^{(1)})^2$ is one half of the suspended chiral Jacobiator.*

Proof. The three intermediate partitions are obtained by first merging B, C , first merging C, D , or first merging D, B . For a temporary ordering u, v, w of the suspended factors, the oriented sum is

$$\bar{\mu}(\bar{\mu}(u, v), w) + (-1)^{|u|(|v|+|w|)}\bar{\mu}(\bar{\mu}(v, w), u) + (-1)^{|w|(|u|+|v|)}\bar{\mu}(\bar{\mu}(w, u), v),$$

which is (193). The factor $1/2$ is the conventional conversion between the square of an odd coderivation and one half of its graded commutator with itself. $\square$

Theorem A.4 (Square of the collision differential). *For a graded chiral antisymmetric operation μ compatible with $d_{\mathfrak{g}}$,*

$$d_{\text{coll}}^2 = \frac{1}{2}d_{[\mu, \mu]}.$$

Consequently $d_{\text{coll}}^2 = 0$ if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal square is zero and the mixed terms vanish by Theorem A.1. Disjoint mergers cancel by Theorem A.2; triple mergers give the Jacobiator by Theorem A.3. Conversely, the component from the discrete three-block partition to the one-block partition is precisely the Jacobiator. $\square$

A.4 Signs in the corolla-normalized forest resolution

For a nonempty block B , let $K(B) = N_*(N \text{Tree}(B))$ be the normalized nerve complex from Section 14. Its vertical differential ∂_B is the alternating simplicial boundary and satisfies $\partial_B^2 = 0$. The augmentation and terminal-corolla section

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C} \xrightarrow{\iota_B} K(B)$$

are chain maps. For disjoint blocks B, C , set

$$m_{B,C} = \iota_{B \cup C}(\epsilon_B \otimes \epsilon_C) : K(B) \otimes K(C) \longrightarrow K(B \cup C). \quad (195)$$

The maps $m_{B,C}$ are symmetric and strictly associative, and commute with the vertical differentials.

The total complex

$$W_I^{\text{FM}}(\mathfrak{g}) = \bigoplus_{\pi \in \Pi(I)} C_{\pi}(\mathfrak{g}) \otimes \bigotimes_{B \in \pi}^{\text{unord}} K(B)$$

has vertical differential $\partial = \sum_B \partial_B$ and horizontal differential obtained by tensoring $d_{\pi, \rho}$ with $m_{B,C}$. The mixed commutator vanishes because (195) is a chain map. Strict associativity of the $m_{B,C}$ reduces the horizontal square to the partition-lattice calculation above. Hence the total square is zero exactly when the chiral Jacobi identity holds.

This construction is intentionally different from an assertion that a single internal-edge contraction merges disconnected roots. Vertical contractions resolve a rooted tree inside a fixed block. Horizontal collision maps merge blocks through the partition differential and insert the terminal corolla of the new block.

A.5 Residues and coorientations

Let Y be smooth and $D_1, \dots, D_r$ a normal-crossing divisor. A local logarithmic form can be written

$$\alpha = \sum_{J \subseteq \{1, \dots, r\}} \left(\bigwedge_{j \in J} \frac{dt_j}{t_j} \right) \wedge \alpha_J.$$

The iterated residue along an ordered tuple is the corresponding coefficient. Therefore

$$\text{Res}_{D_i} \text{Res}_{D_j} = -\text{Res}_{D_j} \text{Res}_{D_i} \quad (196)$$

when the residue maps include their conormal orientation lines. This concerns the logarithmic conormal factors. Higher OPE poles are first paired with normal jets, as in Section 15, and then subjected to the same residue orientation law.

A.6 Stable graphs

For a stable graph Γ , set $\text{or}(\Gamma) = \det \mathbb{R}^{E(\Gamma)}$. A separating clutching inserts an edge joining two vertices; a nonseparating clutching inserts a loop. Ordering the newly inserted edges gives the graph differential. The determinant-line calculation proves that two successive node-forming operations anticommute. The automorphism group of Γ acts on $\text{or}(\Gamma)$, and the modular convolution complex takes coinvariants with this twist.

In a BV presentation, joining different vertices gives the bracket and joining two flags of one vertex gives the loop operator Δ . With the standard modular-operad determinant convention, the codimension-two relations give

$$d^2 = 0, \quad d\Delta + \Delta d = 0, \quad \Delta^2 = 0,$$

together with the second-order identity expressing the bracket as the deviation of Δ from being a derivation. These equations are orientation identities of two node-forming parameters; no scalar curvature enters their proof.

B Low-arity calculations

Low arity is where every abstract object can be seen without compression. This appendix computes the collision complex in arities two, three, and four; identifies the Arnold and infinitesimal-braid cancellations; and records the first examples of pole symbols, BRST descent, and modular sewing.

B.1 Two labels

Let $I = \{1, 2\}$. The partition lattice has two elements,

$$1|2 < 12.$$

For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is

$$\Delta_{1|2,!}(\mathfrak{sg} \boxtimes \mathfrak{sg}) \xrightarrow{\bar{\mu}} \Delta_{12,!}(\mathfrak{sg}). \quad (197)$$

Since $\Delta_{1|2}$ is the identity of X^2 and Δ_{12} is the diagonal, the map is exactly the chiral bracket

$$j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

written after suspension. There is no quadratic condition in arity two. Antisymmetry is the compatibility with the transposition of labels.

On a formal disc, a translation-invariant binary chiral operation is represented by

$$\{a_\lambda b\} + :ab:, \quad \{a_\lambda b\} = \sum_{n \geq 0} \frac{\lambda^n}{n!} a_{(n)} b.$$

The summand $a_{(n)} b$ is the coefficient of the normal principal part of order $n + 1$. Thus the arity-two chain already contains the full singular OPE and the regular product.

Example B.1 (Heisenberg). For the rank-one Heisenberg chiral algebra with field J and

$$J(z)J(w) \sim \frac{k}{(z-w)^2},$$

the binary operation has no simple-pole bracket. Its singular part is the second normal symbol $k N_{\Delta/X^2}^{\otimes 2}$. The logarithmic residue of $d \log(z-w)$ does not produce k ; the coefficient is supplied by the order-two principal-part component of the chiral operation.

Example B.2 (Affine currents). For a finite-dimensional Lie algebra $\mathfrak{g}$ with invariant form $(,)$,

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}.$$

The first normal symbol is the Lie bracket and the second is the invariant form multiplied by the level. They have different tensorial weights and satisfy different identities: Jacobi for the first, invariance for the second.

B.2 Three labels and the Jacobiator

For $I = \{1, 2, 3\}$, the partition lattice is

$$1|2|3 < \{12|3, 13|2, 23|1\} < 123.$$

Suppressing the diagonal pushforwards, the collision complex has the form

$$(s\mathfrak{g})_{\Sigma_0}^{\otimes 3} \xrightarrow{d_1} ((s\mathfrak{g})^{\otimes 2})^{\oplus 3} \xrightarrow{d_2} s\mathfrak{g}. \quad (198)$$

Here the notation Σ_0 only reminds us that the labels are fixed; no symmetric quotient is taken. For homogeneous $u, v, w \in s\mathfrak{g}$,

$$d_1(u \otimes v \otimes w) = \begin{pmatrix} \bar{\mu}(u, v) \otimes w \\ (-1)^{|v||w|} \bar{\mu}(u, w) \otimes v \\ (-1)^{|u|(|v|+|w|)} \bar{\mu}(v, w) \otimes u \end{pmatrix},$$

where the three rows correspond to $12|3$, $13|2$, and $23|1$. With boundary orientations chosen cyclically,

$$d_2(x_{12|3}, x_{13|2}, x_{23|1}) = \bar{\mu}(x_{12}, x_3) - \bar{\mu}(x_{13}, x_2) + \bar{\mu}(x_{23}, x_1),$$

with the Koszul permutations implicit. Multiplication gives the suspended Jacobiator. This is the smallest calculation in which the square of the collision differential contains information.

B.3 Arnold's relation

On $\text{Conf}_3(\mathbb{C})$ put

$$\omega_{ij} = d \log(z_i - z_j).$$

Set $x = z_1 - z_2$ and $y = z_2 - z_3$, so that $z_1 - z_3 = x + y$. Then

$$\omega_{12} = \frac{dx}{x}, \quad \omega_{23} = \frac{dy}{y}, \quad \omega_{13} = \frac{dx + dy}{x + y}.$$

A direct calculation gives

$$\begin{aligned} \omega_{12} \wedge \omega_{23} - \omega_{12} \wedge \omega_{13} - \omega_{13} \wedge \omega_{23} &= \left(\frac{1}{xy} - \frac{1}{x(x+y)} - \frac{1}{y(x+y)} \right) dx \wedge dy \\ &= 0. \end{aligned}$$

Equivalently,

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0. \quad (199)$$

This relation lies in the coefficient algebra of configuration-space forms. It is not the Jacobi identity. A flat connection combines the two by tensoring ω_{ij} with endomorphisms that satisfy the infinitesimal braid relations.

B.4 The KZ curvature in arity three

Let V_1, V_2, V_3 be $\mathfrak{g}$ -modules and let Ω_{ij} denote the invariant Casimir acting in the i -th and j -th factors. Put

$$A = \hbar(\Omega_{12}\omega_{12} + \Omega_{13}\omega_{13} + \Omega_{23}\omega_{23}), \quad \nabla = d - A.$$

Since $d\omega_{ij} = 0$, the curvature is $F_\nabla = A \wedge A$. The terms with identical pairs vanish. After collecting coefficients,

$$\begin{aligned} \frac{1}{\hbar^2} F_\nabla &= [\Omega_{12}, \Omega_{23}] \omega_{12} \wedge \omega_{23} + [\Omega_{12}, \Omega_{13}] \omega_{12} \wedge \omega_{13} \\ &\quad + [\Omega_{13}, \Omega_{23}] \omega_{13} \wedge \omega_{23}. \end{aligned} \quad (200)$$

Invariance of the Casimir gives

$$[\Omega_{12}, \Omega_{13} + \Omega_{23}] = 0, \quad [\Omega_{13}, \Omega_{12} + \Omega_{23}] = 0.$$

Hence the three commutators in (200) are the same up to signs, and the curvature is that common commutator multiplied by (199). Therefore $F_\nabla = 0$.

This calculation displays the division of labor:

$$\text{Arnold relation} \quad \times \quad \text{invariance and Jacobi of } \mathfrak{g} \quad = \quad \text{flatness of KZ.}$$

The geometric and algebraic factors are individually meaningful and neither replaces the other.

B.5 The full vertex identity in arity three

For a vertex algebra, the coefficient operation is not exhausted by Ω_{ij} . Write

$$Y(a, z)Y(b, w) \sim \sum_{n \geq 0} \frac{Y(a_{(n)}b, w)}{(z - w)^{n+1}}.$$

The equality between the expansions in the regions $|z_1 - z_2| < |z_2 - z_3|$ and $|z_2 - z_3| < |z_1 - z_3|$ is the formal-distribution Jacobi identity. Multiplying by powers of the differences and comparing coefficients gives the Borcherds identity

$$\begin{aligned} \sum_{i \geq 0} \binom{m}{i} (a_{(n+i)}b)_{(m+k-i)}c &= \sum_{i \geq 0} (-1)^i \binom{n}{i} (a_{(m+n-i)}b_{(k+i)}c \\ &\quad - (-1)^n b_{(n+k-i)}a_{(m+i)}c), \end{aligned} \quad (201)$$

with the usual parity modification in the super case. The binomial coefficients come from the expansion of translated formal variables. The chiral operad encodes (201) as a single Maurer–Cartan equation. Configuration-space forms can realize the domains and their boundary combinatorics, but they do not manufacture these mode coefficients.

B.6 Four labels

The partition lattice of four labels has ranks 0, 1, 2, 3. The rank-one elements are the six partitions with one pair and two singletons. Rank two has two species:

$$12|34 \quad \text{and its two analogues,} \quad 123|4 \quad \text{and its three analogues.}$$

Rank three is 1234. The two species of rank-two partitions separate the two mechanisms in the square of the differential:

- The interval from $1|2|3|4$ to $12|34$ contains two paths. Their cancellation is the disjoint-pair sign.
- The interval from $1|2|3|4$ to $123|4$ contains three paths. Their sum is the Jacobiator on the first three factors, tensored with the identity on the fourth.

The next differential, from rank two to rank three, tests compatibility of Jacobi with a fourth insertion. No new quadratic identity appears: the higher relation follows because the collision complex is the Chevalley complex of the chiral Lie algebra. In the Fulton–MacPherson resolution, however, the rank-three geometry has nontrivial screens. Their cohomology contributes additional form classes without changing the algebraic Chevalley differential.

B.7 The screen of three points

Consider three points on a complex tangent line modulo translation and dilation. Fix the first two at 0 and 1; the third gives a coordinate

$$u \in \mathbb{P}^1 \setminus \{0, 1, \infty\}.$$

A compactification is $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, with boundary at 0, 1, ∞ . Thus the screen is not contractible. Its open cohomology is

$$H^0 \cong \mathbb{C}, \quad H^1 \cong \mathbb{C} d \log u \oplus \mathbb{C} d \log(1 - u).$$

The relation among the three boundary residues is the global residue theorem. The combinatorial forest category over the triple collision does have a terminal corolla and therefore contractible nerve. This explicit example proves why the incidence resolution and the geometric screen complex must be distinguished.

B.8 Virasoro pole symbols

The Virasoro OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \quad (202)$$

Let D be the pair diagonal. The three singular terms of (202) lie in

$$N_{D/X^2}^{\otimes 4}, \quad N_{D/X^2}^{\otimes 2} \otimes \langle T \rangle, \quad N_{D/X^2} \otimes \langle \partial T \rangle.$$

The regular term lies in filtration degree zero. A change of coordinate acts triangularly on this filtered object; the Schwarzian term records the extension by the fourth-order central symbol. No scalar extracted from one term determines the full filtered operation.

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity for this polynomial is equivalent to the Virasoro Lie conformal algebra identity. The normally ordered product is then part of the corresponding vertex algebra operation. This gives a direct formal-disc test of the chiral selection theorem.

B.9 A BRST bicomplex

Let (A, d_A) be a dg chiral algebra and let $Q : A \rightarrow A$ be a degree-one square-zero derivation anticommuting with d_A :

$$Q^2 = 0, \quad d_A Q + Q d_A = 0, \quad Q\mu = \mu(Q \otimes 1 + 1 \otimes Q).$$

On the collision complex, Q acts on every coefficient factor. Denote this extension by Q_{coll} . The derivation identity gives

$$Q_{\text{coll}} d_{\text{coll}} + d_{\text{coll}} Q_{\text{coll}} = 0.$$

Hence

$$D = d_A + d_{\text{coll}} + Q_{\text{coll}}$$

satisfies $D^2 = 0$. Filter by ghost number. If the spectral sequence is bounded below and its first page is concentrated in ghost degree zero, then

$$H_Q^0(C^{\text{coll}}(A)) \cong C^{\text{coll}}(H_Q^0(A)).$$

This is the low-arity mechanism behind bar–BRST interchange. In Drinfeld–Sokolov reduction the complex is infinite and filtered, so convergence must be proved from the good grading and PBW filtration; the algebraic identity above remains exact at every finite stage.

B.10 The first modular sewings

The stable pairs with smallest complexity are $(0, 3)$ and $(1, 1)$. The space $\overline{\mathcal{M}}_{0,3}$ is a point, so a cyclic three-point operation is a tensor

$$\theta_{0,3} \in (V^{\otimes 3})^{\Sigma_3}$$

with the appropriate degree and determinant twist. Contracting two legs with the inverse pairing produces a genus-one one-point tensor

$$\Delta\theta_{0,3} \in V.$$

The complex compactification $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ has three boundary divisors, and the global residue theorem relates their logarithmic residues. A real associahedral interval inside $\overline{\mathcal{M}}_{0,4}(\mathbb{R})$ has two endpoints; Stokes' theorem on that interval gives equality between the two adjacent parenthesizations. Analytic continuation, or the symmetric operadic action, relates the three complex channels. With coefficient tensors inserted, these statements give the associativity or Jacobi relation appropriate to the chosen operad.

A basic codimension-two modular relation occurs when a loop contraction and a separating contraction meet. Its algebraic image is the BV second-order identity. The relation is unconditional for an algebra over the modular operad: all contractions are already defined. It does not assert that a given chiral algebra possesses a nondegenerate cyclic pairing or convergent sewing amplitudes.

B.11 The free boson determinant

Let q satisfy $|q| < 1$. On the one-particle space of a chiral boson, sewing a cylinder acts diagonally with eigenvalues q^n , $n \geq 1$. Its Fredholm determinant is

$$\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n).$$

Second quantization gives the oscillator partition function

$$\text{Tr}_{\text{Fock}} q^{L_0 - 1/24} = q^{-1/24} \prod_{n \geq 1} (1 - q^n)^{-1} = \eta(\tau)^{-1}, \quad q = e^{2\pi i \tau}.$$

This equality has three exact readings: a trace in the representation, a determinant of the sewing operator, and a Gaussian functional integral. The equality does not identify the central charge with an arbitrary Koszul curvature; it identifies three computations of the same free-field amplitude.

B.12 A diagnostic table

Low-arity datum	Mathematical identity	Physical reading
pair collision	chiral operation supported on Δ	local OPE/contact term
triple collision	chiral Jacobi or Borchers identity	independence of nested short-distance expansion
Arnold form	relation in $H^2(\text{Conf}_3)$	cancellation of configuration-space propagators
Casimir coefficient	infinitesimal braid relation	flat KZ transport
three-point screen	$\mathcal{M}_{0,4}$ geometry	relative configuration in a collision
loop contraction	BV operator Δ	one-loop sewing
anomaly cocycle	failure of the quantum master equation	obstruction to quantization
Fredholm determinant	trace-class sewing identity	Gaussian partition function

Every row compares objects only after the indicated coefficient map or realization has been constructed. This is the low-arity form of the paper's governing rule: geometry supplies the carrier, algebra supplies the local operation, and physics supplies a representation by observables and amplitudes.

C Low-arity calculations for the ordered theory

C.1 Arity zero and one

Let $A \rightarrow \mathbf{1}$ be an augmented ordered chiral algebra and write $\bar{A}$ for the augmentation ideal. The reduced bar begins

$$B_0 = \mathbf{1}, \quad B_1 = \bar{A}[1].$$

The internal differential on B_1 is the suspension of the differential of A . There is no multiplication face in arity one.

The coaugmentation $\mathbf{1} \rightarrow B$ represents the empty ordered configuration. The projection $B \rightarrow \bar{A}[1]$ is the primitive quotient. The reduced coproduct vanishes on B_1 .

C.2 Arity two

In arity two,

$$B_2 = \bar{A}[1] \otimes \bar{A}[1].$$

For homogeneous $a, b \in \bar{A}$, the multiplication face is

$$b_2[a|b] = (-1)^{|a|}[ab]$$

with the suspension convention of Section A. The internal differential is

$$b_1[a|b] = [da|b] + (-1)^{|a|-1}[a|db].$$

The identity $b_1b_2 + b_2b_1 = 0$ is the Leibniz rule for multiplication.

Geometrically, the configuration of two ordered intervals has one collision face. The chiral coefficient

of ab may have a pole along $z_1 = z_2$. The bar face composes the two letters; it does not take a residue of that pole.

C.3 Arity three and associativity

There are two multiplication faces:

$$[a|b|c] \mapsto [ab|c], \quad [a|b|c] \mapsto [a|bc].$$

Applying the multiplication differential twice gives

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

The two boundary points of the compactified one-dimensional moduli of three ordered intervals have opposite orientations. Associativity makes the coefficient operations equal. Hence $b_2^2 = 0$.

For an A_∞ algebra, the compactification is the interval K_4 . Its interior operation is m_3 , and the boundary identity is

$$dm_3 + m_2(m_2 \otimes 1) - m_2(1 \otimes m_2) + \text{terms containing } m_1 = 0.$$

The strict ordered chiral algebra is the case $m_{r \geq 3} = 0$.

C.4 The first mixed square

Take four product disks

$$U_i \times I_i \subset X \times \mathbb{R}, \quad I_1 < I_2 < I_3 < I_4.$$

Suppose the first two chiral points collide while the last two transverse intervals become adjacent. There are two codimension-two paths:

$$\text{chiral collision then interval fusion,} \quad \text{interval fusion then chiral collision.}$$

The coefficient maps agree by the interchange law (75). The conormal orientation lines occur in opposite orders, so the two boundary contributions cancel in the total differential.

This is the smallest calculation proving that the chiral resolution and ordered bar can be totalized without conflating their differentials.

C.5 No braid in arity two

The ordered chamber

$$\{t_1 < t_2\} \subset \text{Conf}_2(\mathbb{R})$$

is contractible. A path remains in the same chamber and returns no monodromy. There is no path to the chamber $t_2 < t_1$ without crossing $t_1 = t_2$.

In two transverse dimensions,

$$\text{Conf}_2(\mathbb{R}^2) \simeq S^1.$$

A half-rotation exchanges the two points; its square is the generator of the pure braid group $P_2 \cong \mathbb{Z}$. Thus the R -matrix first appears after the E_2 thickening.

C.6 The free algebra

Let $A = T_X(M)$. In bar degree two the map

$$M^{\otimes p} \mid M^{\otimes q} \longmapsto M^{\otimes(p+q)}$$

concatenates words. The contracting homotopy splits off the first letter of every word of total tensor length at least two. Consequently only the one-letter class survives. This gives the contraction of Theorem 29.1 explicitly.

C.7 The square-zero algebra

For $A = \mathbf{1} \oplus M$ with $M^2 = 0$, all multiplication faces vanish. Hence

$$H_\bullet \text{Bar}_X^{\text{ord}}(A) \cong \bigoplus_{r \geq 0} H_\bullet(M[1]^{\otimes r}).$$

The deconcatenation coproduct cuts a word at every position. Continuous duality gives the completed free algebra on $M^\vee[-1]$.

C.8 A quiver calculation

For the quiver $1 \xrightarrow{u} 2 \xrightarrow{v} 3$, the relative bar over $S = \mathbf{1}e_1 \oplus \mathbf{1}e_2 \oplus \mathbf{1}e_3$ has

$$B_1 = M_u \oplus M_v, \quad B_2 = M_v \otimes M_u.$$

The multiplication face sends $[v|u]$ to the path vu . If no relation is imposed, the path belongs to the algebra and the two-term resolution is exact. If the relation $vu = 0$ is imposed, $[v|u]$ becomes the quadratic Koszul cogenerator. The calculation is internal to $\text{FactD}(X)$, so M_u and M_v may carry singular chiral coefficients.

C.9 The enveloping algebra

Let $A = U_X(\mathfrak{g})$. Antisymmetrization sends

$$x_1 \wedge \cdots \wedge x_r \longmapsto \sum_{\sigma \in \Sigma_r} \text{sgn}(\sigma) [x_{\sigma(1)} \mid \cdots \mid x_{\sigma(r)}].$$

The bar multiplication differential becomes the Chevalley differential

$$\sum_{i < j} (-1)^{\epsilon_{ij}} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_r.$$

In arity three its square is the Jacobiator. PBW proves that antisymmetrization is a quasi-isomorphism.

C.10 BRST in arity two

Let Q be an odd derivation of A with $Q^2 = 0$. On $[a|b]$,

$$Q_{\text{bar}}[a|b] = [Qa|b] + (-1)^{|a|-1}[a|Qb].$$

The mixed commutator with the bar face is

$$(Q_{\text{bar}}b + bQ_{\text{bar}})[a|b] = \pm(Q(ab) - (Qa)b - (-1)^{|a|}a(Qb)) = 0.$$

Thus $b + Q_{\text{bar}}$ is square-zero. The comparison of its cohomology with the bar of $H_Q(A)$ is the spectral-sequence question, not this identity.

C.11 The annulus and the torus

The transverse annulus amplitude is

$$\text{HH}_\bullet(A) = A \otimes_{A \otimes A^{\text{op}}}^L A.$$

It closes an ordered interval and raises the ribbon genus. The genus-one chiral block is obtained by sewing two marked points of a chiral sphere and raises the curve genus. Their lowest chain models both contain a trace, but they act on different moduli spaces. In the bimodular complex they occupy bidegrees $(0, 1)$ and $(1, 0)$.

C.12 The first Tannaka coefficient

Let a meromorphic line category have exchange

$$R(z) = 1 + \hbar r(z) + O(\hbar^2).$$

The quantum Yang–Baxter equation at order $\hbar^2$ gives

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0.$$

For $r(z) = \Omega/z$, the coefficient of each commutator is paired with the Arnold three-term relation. This is the first nontrivial compatibility between chiral collision geometry and transverse exchange. It requires the E_2 line category; it is absent from the bare E_1 bar.

D The stratified Fulton–MacPherson spectral sequence

The Fulton–MacPherson space carries two different complexes that are easily conflated. The incidence complex remembers how boundary strata meet. The geometric complex also remembers the cohomology of the screens. A spectral sequence relates them and states exactly when the incidence complex is sufficient.

D.1 A filtered space with normal-crossing boundary

Let Y be a smooth complex variety and

$$D = \bigcup_{a \in A} D_a$$

a simple normal-crossing divisor. For $J \subseteq A$, put

$$D_J = \bigcap_{a \in J} D_a, \quad D_J^\circ = D_J \setminus \bigcup_{b \notin J} D_b.$$

Let $j : Y \setminus D \hookrightarrow Y$. The increasing filtration by depth of support on the boundary gives a Cousin-type spectral sequence for $Rj_*\mathbb{C}$ or, algebraically, for the logarithmic de Rham complex. In one standard indexing,

$$E_1^{-p,q} = \bigoplus_{|J|=p} H^{q-2p}(D_J)(-p) \implies H^{q-p}(Y \setminus D). \quad (203)$$

The first differential is the alternating sum of Gysin or residue maps. Its signs are the determinant signs of Section A. Deligne's mixed Hodge theory identifies (203) with the weight spectral sequence when Y is proper.

For coefficients in a complex $\mathcal{L}$ constructible along the boundary stratification, the same filtration gives

$$E_1^{-p,q}(\mathcal{L}) = \bigoplus_{|J|=p} \mathbb{H}^{q-2p}(D_J, \mathcal{L}|_{D_J})(-p) \implies \mathbb{H}^{q-p}(Y \setminus D, \mathcal{L}). \quad (204)$$

Nothing in the construction requires $\mathcal{L}$ to be constant. In chiral applications it is assembled from external tensor powers of the coefficient D -module and from principal-part lines along the normal directions.

D.2 Boundary strata of Fulton–MacPherson space

Let X be a smooth curve and $Y_I = \text{FM}_I(X)$. Boundary divisors are indexed by subsets $S \subseteq I$, $|S| \geq 2$, subject to the convention determined by the chosen compactification. A collection $\mathcal{N}$ of subsets labels a nonempty stratum exactly when it is nested: any two members are disjoint or one contains the other. The nested set is equivalently a rooted collision forest F .

The open stratum associated with F has the form of an iterated bundle whose base records the root configurations on X and whose fibers are screen configuration spaces. On a curve, the screen attached to a vertex with r incoming branches is the compactification of r points on an affine line modulo translations and dilations. Its interior is

$$\text{Conf}_r(\mathbb{A}^1)/(\mathbb{G}_a \rtimes \mathbb{G}_m).$$

For $r = 2$ the screen is a point. For $r = 3$ its interior is $\mathcal{M}_{0,4}$ and its compactification is $\overline{\mathcal{M}}_{0,4}$. For larger r it has nontrivial cohomology in several degrees.

Consequently, a stratum is not determined by the forest alone. The forest gives its incidence type; the screens give its internal topology.

D.3 The forest resolution as the row of constants

Let $p : Y_I \rightarrow X^I$ be the blowdown. Consider the filtration of a geometric coefficient complex $\mathcal{K}_I$ by boundary depth. The E_1 page is a sum over forests,

$$E_1^{p,q} = \bigoplus_{\substack{F \in \text{For}(I) \\ |E(F)|=p}} R^q(p_F)_*(\mathcal{K}_I|_{D_F^\circ}) \otimes \text{or}(F), \quad (205)$$

where p_F is the projection of the stratum to the corresponding partial diagonal. The differential contracts one edge and applies the induced restriction or residue map.

The row $q = 0$ contains the locally constant functions on every screen. If the coefficient system has no higher screen cohomology and no nontrivial local monodromy, then after barycentric or nerve replacement of the face incidence category and blockwise terminal-corolla normalization, this row maps to the combinatorial forest resolution $W_I^{\text{FM}}(\mathfrak{g})$ of Section 14. The augmentation to the partition collision complex is a quasi-isomorphism because every rooted-tree category has a terminal corolla.

Criterion D.1 (Incidence sufficiency). *The incidence forest complex computes the proper pushforward of $\mathcal{K}_I$ if the following two conditions hold:*

- (i) $R^q(p_F)_*(\mathcal{K}_I|_{D_F^\circ}) = 0$ for every $q > 0$ and every forest F ;
- (ii) *the degree-zero pushforwards identify with the coefficient system used in the forest complex, compatibly with the vertical incidence differential and the corolla-normalized horizontal collision merge.*

The criterion is a theorem of spectral sequences: under the two conditions, (205) is concentrated in one row. It is deliberately strong. Constant coefficients on complex screens generally do not satisfy it.

D.4 The first nonconstant row

For a triple-collision vertex, the screen interior is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore the $q = 1$ row of (205) already appears in arity three. Its logarithmic generators may be taken to be

$$d \log u, \quad d \log(1 - u).$$

Their residues at $0, 1, \infty$ satisfy one linear relation. This row carries the first period data of ordered configurations. Integrating its forms along paths gives logarithms; iterating at higher arity gives multiple polylogarithms and, for KZ transport, multiple zeta values.

The chiral Chevalley differential does not require these periods. It uses the diagonal coefficient operation and the incidence of partitions. Configuration-space integral realizations do require them. The spectral sequence locates the exact extra data.

D.5 Orlik–Solomon cohomology

For $X = \mathbb{A}^1$, the cohomology algebra of $\text{Conf}_n(X)$ is the Orlik–Solomon algebra of the braid arrangement. It is generated in degree one by ω_{ij} with relations

$$\omega_{ij} = \omega_{ji}, \quad \omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

A no-broken-circuit basis gives the Poincaré polynomial

$$P_{\text{Conf}_n(\mathbb{C})}(t) = \prod_{j=1}^{n-1} (1 + jt). \quad (206)$$

The coefficient $(n-1)!$ in top degree is geometric cohomology, not the dimension of a bar chain group for an arbitrary chiral algebra. To form a geometric bar model one tensors or convolves (206) with

the algebraic coefficient complex and includes all differentials. Dimensions of the tensor factors alone do not determine its homology.

D.6 Nested residues

On a normal-crossing chart with local boundary parameters t_e indexed by the edges of a collision forest, a logarithmic form has an expansion

$$\alpha = \sum_{J \subseteq E(F)} \left(\bigwedge_{e \in J} \frac{dt_e}{t_e} \right) \wedge \alpha_J.$$

The depth filtration counts $|J|$. The first differential of the residue spectral sequence extracts $\alpha_{\{e\}}$ and restricts it to $t_e = 0$. A second residue extracts $\alpha_{\{e,f\}}$. Because

$$\frac{dt_e}{t_e} \wedge \frac{dt_f}{t_f} = -\frac{dt_f}{t_f} \wedge \frac{dt_e}{t_e},$$

the codimension-two contribution cancels between the two orders. When the coefficient system itself changes under contraction, the remaining coefficient is the associator or Jacobiator of that system.

For higher poles, ordinary Poincare residue still acts only on the logarithmic conormal factor. The order- m_e coefficient is first paired with the dual m_e -jet of the normal line, producing a logarithmic coefficient; ordinary residue then removes dt_e/t_e . Invariantly this is the pairing

$$N_e^{\otimes m_e} \otimes N_e^{\vee \otimes m_e} \longrightarrow \mathcal{O}$$

followed by Poincare residue. Different pole orders remain in different associated-graded normal lines.

D.7 The Rees spectral sequence

Let $F_{\mathbf{m}}$ be the multi-pole filtration along the collision divisors. The multi-Rees object

$$\mathcal{R}(\mathcal{K}_I) = \bigoplus_{\mathbf{m} \geq 0} F_{\mathbf{m}} \mathcal{K}_I \mathbf{t}^{\mathbf{m}}$$

interpolates between the filtered chiral coefficient system and its normal-cone associated graded. Filtering the geometric complex simultaneously by boundary depth and pole order gives a bicomplex. One spectral sequence first resolves collision incidence and then pole order; the other first passes to normal symbols and then resolves the forest geometry. Under boundedness in each finite arity and pole window, the two converge to the same total cohomology.

This interchange explains the relation between chiral and Poisson vertex structures. The pole associated graded is classical; the extension data between pole orders is quantum. Fulton–MacPherson depth is independent of pole degree; it records how many nested collisions occur. The two filtrations must not be collapsed into one integer.

D.8 A comparison theorem for finite windows

Fix a finite set I , a maximum pole order M , and a bounded cohomological range. Let $\mathcal{K}_I^{\leq M}$ be the truncated principal-part coefficient system. Suppose there is a morphism

$$\Phi_I : W_I^{\text{FM}}(\mathfrak{g}; \leq M) \longrightarrow Rp_*\mathcal{K}_I^{\leq M} \quad (207)$$

that is compatible with the vertical incidence differential and the corolla-normalized collision merge, and induces an isomorphism on every stratumwise cohomology group in the chosen range.

Theorem D.2 (Finite-window comparison). *Under the preceding hypotheses, (207) is a quasi-isomorphism in the chosen cohomological range.*

Proof. Filter source and target by boundary depth. The map induces a morphism of the corresponding spectral sequences. By hypothesis it is an isomorphism on E_1 in the range. Since the filtration is finite for fixed I , the comparison theorem for spectral sequences gives an isomorphism on the associated graded of cohomology, hence on cohomology itself. $\square$

The theorem is deliberately local in arity, pole order, and cohomological degree. Passage to an unbounded completed object requires control of inverse limits. Surjective transition maps imply the Mittag-Leffler condition and kill $\varprojlim^1$; absent such control, completion may create cohomology.

D.9 Renormalization and compactified configurations

Let M be a real manifold and consider a perturbative quantum field theory. A Feynman amplitude associated with a graph Γ is initially a differential form or distribution on the configuration space of its vertices away from the diagonals. Pulling it to a compactification separates the asymptotic regimes of every collision forest. Renormalization is an extension across the boundary satisfying locality and compatibility with disjoint unions.

The forest formula of perturbative renormalization has the same incidence category as the Fulton–MacPherson boundary. A divergent subgraph corresponds to a cluster of vertices; nested and disjoint divergent subgraphs correspond to nested and disjoint boundary divisors. Counterterms live on the associated diagonals. Stokes’ theorem on the compactification converts the failure of a Ward identity into boundary integrals. The quantum master equation holds when the resulting local cocycle is exact or vanishes.

This physical realization uses more than the incidence complex:

- (i) a propagator or parametrix;
- (ii) a gauge-fixing operator;
- (iii) an extension of singular distributions;
- (iv) a renormalization prescription;
- (v) a proof that the counterterms preserve the required symmetries.

When these data are fixed, the geometric boundary complex computes the anomaly. The combinatorial identity $\partial^2 = 0$ alone does not establish the absence of an anomaly.

D.10 Degeneration of curves

For a family of curves approaching a node, the relevant compactification contains two kinds of boundary simultaneously: collisions of marked points in the fibers and degeneration of the curve. Logarithmic Fulton–MacPherson spaces resolve the first relative to the second. A local model of the node is

$$xy = q,$$

where q is the smoothing parameter. Collision parameters in the two branches and the nodal parameter q define independent conormal directions. The total residue complex therefore has a multigrading.

Separating degeneration corresponds to two components joined at the node. Nonseparating degeneration identifies two points on one component. The associated maps on conformal blocks are the two sewing operations of a modular operad. The screen cohomology of point collisions and the vanishing-cycle or nearby-cycle data of the node are distinct. A correct global comparison must include both.

D.11 Consequences

The spectral sequence proves four statements that are used repeatedly in the body of the paper.

- (1) The partition collision complex is canonical and computes chiral Chevalley chains.
- (2) The forest incidence complex is a canonical resolution of the partition complex.
- (3) The geometric Fulton–MacPherson complex contains additional screen cohomology.
- (4) An equality between the geometric and algebraic complexes is a comparison theorem, not a consequence of contractibility of the forest nerve.

From the physical side, the same four statements say that locality determines the pattern of counterterms, while the propagator and its periods determine their numerical amplitudes. The first is combinatorial and functorial. The second is analytic and theory-dependent. Chiral Koszul duality is the algebraic local-to-global theorem; configuration integrals are geometric realizations of its coefficient maps.

E Formal neighborhoods, normal jets, and degeneration coordinates

The local geometry of a chiral operation is the geometry of a divisor and its formal neighborhood. The formulas in this appendix make explicit how pole coefficients, delta derivatives, coordinate changes, and sewing parameters are different expressions of normal geometry.

E.1 The formal completion of a diagonal

Let X be a smooth curve and let $\Delta \subset X \times X$ be the diagonal. Its ideal sheaf $\mathcal{I}_\Delta$ satisfies

$$\mathcal{I}_\Delta / \mathcal{I}_\Delta^2 \cong \Omega_X^1, \quad N_{\Delta/X^2} \cong T_X.$$

The m -th infinitesimal neighborhood is

$$\Delta^{(m)} = \text{Spec}(\mathcal{O}_{X^2}/\mathcal{I}_\Delta^{m+1}).$$

The associated graded algebra of the formal completion is

$$\text{gr}_{\mathcal{I}} \widehat{\mathcal{O}_{X^2}_\Delta} \cong \text{Sym}_{\mathcal{O}_X}(\mathcal{I}_\Delta/\mathcal{I}_\Delta^2) \cong \text{Sym}_{\mathcal{O}_X}(\Omega_X^1). \quad (208)$$

Dualizing gives the algebra of normal differential operators

$$\text{Sym}_{\mathcal{O}_X}(T_X).$$

A coordinate difference $t = z_1 - z_2$ trivializes the conormal line and identifies the formal completion with $\mathcal{O}_X[[t]]$. A different coordinate changes t by a formal power series with invertible linear term. The formal neighborhood, not the chosen series, is invariant.

E.2 Principal parts

Let $D \subset Y$ be a smooth divisor with local equation $t = 0$. For $m \geq 0$, the sheaf $\mathcal{O}_Y(mD)$ consists locally of functions with poles of order at most m . There is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \xrightarrow{\sigma_{m+1}} N_{D/Y}^{\otimes(m+1)} \longrightarrow 0. \quad (209)$$

To verify the target, write a section as

$$f = t^{-(m+1)}a_{-(m+1)} + t^{-m}a_{-m} + \cdots.$$

Modulo $\mathcal{O}_Y(mD)$ only the leading coefficient remains. If $t' = ut + O(t^2)$, then

$$(t')^{-(m+1)} = u^{-(m+1)}t^{-(m+1)} + O(t^{-m}),$$

so $a_{-(m+1)}$ transforms as the $(m+1)$ -st power of the normal line. This proves (209) without choosing a global equation.

The pole filtration

$$F_m \mathcal{O}_Y(*D) = \mathcal{O}_Y(mD)$$

has associated graded

$$\text{gr}_m^F \mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes m}, \quad m \geq 1. \quad (210)$$

A complete OPE is a filtered morphism with coefficients in $\mathcal{O}_Y(*D)$. Its mode of order $m-1$ is the image under σ_m .

E.3 Differential operators and delta derivatives

Let $i : D \hookrightarrow Y$ be a smooth divisor. Distributions supported on D are generated by normal derivatives of the delta distribution. Algebraically, the Kashiwara equivalence identifies right D_Y -modules

supported on D with right D_D -modules. Locally, a supported distribution has the form

$$T = \sum_{m=0}^M T_m \partial_t^m \delta(t). \quad (211)$$

The coefficient pairing of principal parts with normal jets is the formal residue identity

$$\text{Res}_{t=0}(t^n t^{-m-1} dt) = \delta_{mn}. \quad (212)$$

The dual jet functional may be represented distributionally by a normalized derivative of $\delta(t)$. In complex analysis, (212) is Cauchy's coefficient formula; in D -module theory it is the pairing between local cohomology along the diagonal and normal differential operators.

For a vertex algebra, the singular commutator is

$$[Y(a, z), Y(b, w)] = \sum_{m \geq 0} Y(a_{(m)} b, w) \frac{1}{m!} \partial_w^m \delta(z - w). \quad (213)$$

Equation (213) is the distributional form of the chiral operation. It contains every singular mode. The normally ordered product is recovered from the regular extension, not from the supported commutator alone.

E.4 The Rees deformation

The Rees algebra of the ideal of a divisor is

$$\mathcal{R}(\mathcal{I}_D) = \bigoplus_{m \geq 0} \mathcal{I}_D^m u^{-m}.$$

Its relative spectrum over $\mathbb{A}_u^1$ is the deformation to the normal cone. In a local chart one may write

$$t = ux. \quad (214)$$

For $u \neq 0$, the fiber is isomorphic to the original neighborhood. For $u = 0$, the coordinate x is the normal coordinate. The corresponding construction for the pole filtration uses inverse powers and interpolates between meromorphic kernels and normal symbols.

For several normal-crossing divisors D_e with local equations t_e , the multi-Rees space has equations

$$t_e = u_e x_e.$$

Its deepest special fiber is the direct sum of the normal lines. On a Fulton–MacPherson stratum indexed by a forest F , the indices e are the internal edges of F . This proves the normal-bundle splitting

$$N_{D_F/\text{FM}_I(X)} \cong \bigoplus_{e \in E(F)} N_e. \quad (215)$$

Operadic substitution on the associated graded is substitution of the normal coordinates x_e along contracted subtrees.

E.5 Coordinate changes and triangularity

Let $w = f(z)$ with $f'(z) \neq 0$. Near $z_1 = z_2 = z$, put $t = z_1 - z_2$. Taylor expansion gives

$$f(z+t) - f(z) = f'(z)t + \frac{1}{2}f''(z)t^2 + \frac{1}{6}f'''(z)t^3 + O(t^4), \quad (216)$$

$$\frac{1}{f(z+t) - f(z)} = \frac{1}{f'(z)t} - \frac{f''(z)}{2f'(z)^2} + O(t), \quad (217)$$

$$\frac{1}{(f(z+t) - f(z))^2} = \frac{1}{f'(z)^2 t^2} - \frac{f''(z)}{f'(z)^3 t} + O(1). \quad (218)$$

A pole of order m therefore transforms into a linear combination of poles of orders at most m , with leading factor $f'(z)^{-m}$. The associated graded is tensorial, while the filtered object is triangular.

This is exactly the transformation law of vertex modes. The top pole of an OPE between primary fields transforms by the tensor power prescribed by its conformal weights. Lower poles acquire corrections from derivatives of f . The complete chiral operation is coordinate-free because these corrections are the transition functions of the filtered principal-parts sheaf.

E.6 The Schwarzian derivative

The Schwarzian derivative is

$$\{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)} \right)^2.$$

It obeys the cocycle identity

$$\{f \circ g, z\} = \{f, g(z)\}g'(z)^2 + \{g, z\}. \quad (219)$$

A stress tensor of central charge c transforms as

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12} \{f, z\}. \quad (220)$$

The first term is the tensorial transformation of a quadratic differential. The second is an affine cocycle. Thus stress tensors form a torsor for projective connections rather than the vector space of quadratic differentials.

Equation (220) follows from the fourth-order central term of the Virasoro OPE. The central term is a fourth normal symbol, while the Schwarzian is the extension class that mixes it into filtration degree zero under coordinate change. This is a precise example of why the associated graded and the filtered chiral operation contain different information.

E.7 The affine connection and the shifted level

For affine currents, the invariant form gives the second normal symbol. The KZ connection contains instead the Casimir divided by $k + h^\vee$:

$$\nabla = d - \frac{1}{k + h^\vee} \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

The level k and the shifted denominator $k + h^\vee$ arise in different calculations. The first is the coefficient of the central extension in the current OPE. The second is the coefficient of the Sugawara stress tensor and the KZ transport. Their relation is a theorem of the current-algebra representation, not a normalization identity that allows one to replace either by the other.

At $k = -h^\vee$, the Sugawara construction is singular. The current OPE and its level remain defined. The KZ connection in this normalization is singular. The Feigin–Frenkel centre appears. These facts are compatible because they concern different pieces of structure.

E.8 Prime form and normalized bidifferential

Let X be a compact Riemann surface of genus g , choose a symplectic basis of homology, and let Ω be the period matrix. The prime form $E(p, q)$ is a $(-1/2, -1/2)$ -form with a simple zero on the diagonal. Locally,

$$E(p, q) = \frac{z(p) - z(q)}{\sqrt{dz(p)}\sqrt{dz(q)}}(1 + O((z(p) - z(q))^2)).$$

The normalized fundamental bidifferential is

$$B(p, q) = d_p d_q \log E(p, q) + \sum_{i,j=1}^g c_{ij} \omega_i(p) \omega_j(q), \quad (221)$$

where the coefficients c_{ij} are chosen, according to convention, so that the A -periods vanish. It has a double pole with biresidue one on the diagonal. Changing the symplectic basis acts by a modular transformation and changes the decomposition into prime-form and holomorphic pieces, while the normalized bidifferential transforms covariantly.

The prime form is not a global replacement for the coordinate difference as an ordinary function. It is a section of a line bundle with quasi-periodicity. A propagator constructed from it requires a choice of normalization, spin structure, or Green function appropriate to the theory. Its local singular symbol is universal; its harmonic completion is global.

E.9 Plumbing coordinates

A nodal curve has local equation $xy = 0$. A smoothing is

$$xy = q, \quad |q| < 1. \quad (222)$$

Removing discs around $x = 0$ and $y = 0$ and identifying the annuli by $y = q/x$ produces the smooth fiber. The parameter q is a normal coordinate to the Deligne–Mumford boundary divisor.

For a conformal field theory, sewing inserts a complete set of states:

$$\mathcal{A}_q = \sum_{\alpha} q^{L_0 - c/24} |\alpha\rangle \langle \alpha|, \quad (223)$$

with the dual basis defined by the two-point pairing. In a rational theory the sum is organized by finitely many simple modules and convergent graded traces. In a general vertex algebra the expression may be formal. The algebraic clutching map and the analytic convergence of (223) are separate assertions.

The factor $q^{-c/24}$ is the vacuum energy on the cylinder. Under a modular change of plumbing

coordinate it contributes to the projective anomaly. The same central charge appears in the Virasoro OPE because both are representations of the same central extension of vector fields. The equality is specific and exact; it does not identify central charge with unrelated levels or arithmetic weights.

E.10 Separating and nonseparating nodes

A separating node joins curves of genera g_1 and g_2 . Sewing gives a bilinear composition

$$\mathbb{V}_{g_1, n_1+1} \otimes \mathbb{V}_{g_2, n_2+1} \longrightarrow \mathbb{V}_{g_1+g_2, n_1+n_2}.$$

A nonseparating node identifies two marked points of one curve and contracts them with the inverse pairing:

$$\mathbb{V}_{g-1, n+2} \longrightarrow \mathbb{V}_{g, n}.$$

These are the two modular-operad compositions. Their codimension-two compatibility is the geometric source of the bracket and loop terms in the quantum master equation.

The normal parameter q belongs to the base of the family of curves. Collision parameters t_e belong to relative configuration spaces in the fibers. In a degeneration with colliding insertions, both occur. A superconnection may mix their differentials, but an equation that equates a fiberwise endomorphism with dq/q or with a Hodge class is ill-typed unless the total complex and its base degrees have been defined.

E.11 Tangential base points

The spaces $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ and $\mathcal{M}_{0, n}$ are not simply connected. Iterated integrals between boundary points require tangential base points. A tangent vector at 0 specifies the asymptotic direction of approach and regularizes logarithmic divergences. The KZ associator is the regularized parallel transport from the tangent vector at 0 to the tangent vector at 1.

The choice is geometric: it is a nonzero vector in the normal line to a boundary divisor. Changing it conjugates the regularized monodromy by an exponential of the residue. This is the local origin of gauge dependence of associators. The braid category is invariant up to braided equivalence; a particular power-series associator is a representative.

E.12 Reality and positivity

Algebraic chiral operations are defined over a field and do not include a Hilbert-space structure. A unitary conformal field theory adds an antilinear involution and a positive Hermitian form compatible with the modes. The adjoint relation is

$$L_n^\dagger = L_{-n}$$

for the Virasoro modes in radial quantization, with analogous formulas for primary fields. Reflection positivity is an analytic inequality, not a consequence of the chiral Jacobi identity.

Similarly, a Euclidean Green function depends on a real structure and metric, while the singular chiral kernel depends only on complex geometry. The chiral algebra determines protected holomorphic quantities. Reconstruction of a unitary full conformal field theory requires left-right pairing, modular invariance, and positivity. These are additional structures.

E.13 Local geometry summarized

The formal neighborhood of a diagonal contains the jets. The pole filtration contains the singular modes. The Rees deformation contains the classical limit. The Fulton–MacPherson normal bundle contains nested collision scales. The plumbing parameter contains nodal sewing. A projective connection contains the stress-tensor anomaly. Each object is a line or filtered sheaf attached to a geometric normal direction. This is the local geometry in which the coordinate formulas of chiral algebra and the short-distance formulas of quantum field theory become intrinsic.

F Field-theoretic derivations

This appendix derives the physical structures used in the paper from actions and Ward identities. Every identification is stated after the boundary observables, pairings, and comparison maps have been specified.

F.1 Classical BV theory

A classical gauge theory in the Batalin–Vilkovisky formalism consists of a graded space of fields $\mathcal{E}$, an odd symplectic pairing of cohomological degree -1 , and an action $S \in \mathcal{O}(\mathcal{E})$ of degree zero. The odd symplectic form induces a Poisson bracket $\{-, -\}$ of degree one. The classical master equation is

$$QS + \frac{1}{2}\{S, S\} = 0, \quad (224)$$

where Q is the linearized differential. Equivalently, the Hamiltonian vector field

$$d_S = Q + \{S, -\}$$

satisfies $d_S^2 = 0$.

The Taylor coefficients of S define cyclic multilinear operations. Equation (224) is their L_∞ relation. If the fields are local sections of bundles on a manifold, the operations are supported on the small diagonal. Thus the local L_∞ algebra of fields is the field-theoretic analogue of a diagonal-supported chiral operation.

Gauge fixing chooses a Lagrangian complement and a parametrix for the kinetic operator. This choice is needed to define Feynman amplitudes. The classical local operations and their master equation do not depend on the choice up to canonical homotopy.

F.2 Quantum master equation and anomaly

A regularized BV Laplacian Δ is a second-order operator of degree one. The quantum master equation is

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \quad (225)$$

or equivalently

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0. \quad (226)$$

In an interacting field theory the action is replaced by a scale-dependent effective interaction and Δ by a scale-dependent regularized operator. Renormalization constructs a compatible family of solutions

when the obstruction vanishes [14, 15].

Suppose a solution has been constructed modulo $\hbar^N$. The coefficient of $\hbar^N$ in the failure of (226) is closed for the classical differential d_S . Its cohomology class is independent of changes of counterterm. This is the anomaly class. It vanishes exactly when the solution extends to the next order. This obstruction theory is the physical counterpart of the filtered Maurer–Cartan obstruction theory of Section 25.

The equality between the stable-graph master equation and the BV quantum master equation is exact for a cyclic local field theory: vertices are Taylor coefficients of S , internal edges are contractions by the propagator, and loops are contractions of two flags of one connected graph. The equality is an equality of graph expansions. It does not assert that an arbitrary chiral algebra is the boundary of a given field theory.

F.3 Factorization observables

For an open set U , let $\text{Obs}^{\text{cl}}(U)$ be the Chevalley–Eilenberg cochains of compactly supported fields in U . Extension by zero and disjoint union define structure maps

$$\text{Obs}^{\text{cl}}(U_1) \otimes \cdots \otimes \text{Obs}^{\text{cl}}(U_n) \longrightarrow \text{Obs}^{\text{cl}}(V)$$

for disjoint $U_i \subset V$. The cosheaf and descent properties make Obs^{cl} a factorization algebra. A perturbative quantization deforms it to Obs^q over $\mathbb{C}[[\hbar]]$.

If translations in some directions are homotopically trivial, the factorization algebra is locally constant in those directions and determines an E_n algebra. If dependence on one complex direction is holomorphic, restriction to a holomorphic curve yields a chiral factorization algebra. Boundary conditions replace the bulk field complex by a relative complex and produce a boundary factorization algebra.

A comparison with an algebraic chiral algebra is a quasi-isomorphism

$$\Phi : U^{\text{ch}}(\mathfrak{g}) \longrightarrow \text{Obs}_{\partial}^q$$

compatible with factorization. Once Φ is given, the algebraic chiral bracket equals the renormalized boundary OPE in cohomology.

F.4 Chern–Simons theory and affine currents

Let M be an oriented three-manifold, G a compact Lie group with invariant form $(\cdot, \cdot)$, and $k \in \mathbb{Z}$. The Chern–Simons action is

$$S_{\text{CS}}(A) = \frac{k}{4\pi} \int_M \left((A, dA) + \frac{2}{3} (A, A \wedge A) \right). \quad (227)$$

On a manifold with boundary, variation gives

$$\delta S_{\text{CS}} = \frac{k}{2\pi} \int_M (\delta A, F_A) + \frac{k}{4\pi} \int_{\partial M} (A, \delta A).$$

A boundary condition chooses a Lagrangian subspace for the boundary symplectic form. In complex coordinates on the boundary, fixing one component produces a chiral boundary theory. The residual

gauge symmetry has current algebra

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w}. \quad (228)$$

The double pole is the boundary central extension determined by the coefficient of (227). The simple pole is the Lie bracket. The one-loop shift by $h^\vee$ enters the effective coupling and the Sugawara construction; it is not a change in the classical coefficient of the central extension.

Wilson lines are labelled by representations of G . Their expectation values depend on framing. Canonical quantization on a surface gives the space of conformal blocks of the affine algebra. Witten's path-integral derivation of knot invariants identifies the braiding and ribbon trace with quantum-group data [74, 63]. The Drinfeld–Kohno theorem supplies the algebraic comparison of braid categories.

F.5 One-loop exactness in a holomorphic gauge

A partially holomorphic gauge can make Chern–Simons theory a deformation of a topological–holomorphic BF theory. In the gauge constructed by Gwilliam and Williams, the perturbative quantization is one-loop exact and the bulk–boundary relation with chiral WZW theory is explicit [45]. One-loop exactness means that higher-loop counterterms vanish in that gauge after the relevant local cohomology calculation. It does not mean that every boundary correlator is Gaussian; tree-level nonlinearity and representation-theoretic sums remain.

The exactness result supplies a rigorous model in which the boundary current OPE, the anomaly shift, and the factorization algebra of observables can be compared at chain level. It is one of the clearest physical realizations of chiral Koszul duality: the local Lie algebra of gauge transformations passes to Chevalley cochains, while boundary factorization organizes their states and operators.

F.6 Four-dimensional Chern–Simons theory

Let Σ be an oriented real surface and C a complex curve with meromorphic one-form ω . Four-dimensional Chern–Simons theory has action

$$S_{4d}(A) = \frac{1}{2\pi\hbar} \int_{\Sigma \times C} \omega \wedge \left((A, dA) + \frac{2}{3}(A, A \wedge A) \right). \quad (229)$$

The theory is topological along Σ and holomorphic along C . Wilson lines supported on curves in Σ and located at points of C possess a spectral parameter. Crossing two lines gives an R -matrix. The quantum Yang–Baxter equation follows from isotopy invariance of three line crossings. Rational, trigonometric, and elliptic kernels arise from the geometry of (C, ω) [19, 20].

The line-operator OPE gives a tensor category with spectral parameters. A Yangian or quantum loop algebra is recognized when the category is generated by evaluation representations and the R -matrix and relations agree with the corresponding presentation. The field theory provides the monodromy and operator products; the algebraic recognition identifies the named Hopf algebra.

F.7 The Polyakov anomaly

For a two-dimensional conformal field theory on a Riemann surface with metric g , the partition function is not generally invariant under a Weyl rescaling $g \mapsto e^{2\sigma}g$. The infinitesimal anomaly is

$$\delta_\sigma \log Z[g] = \frac{c}{24\pi} \int_X \sigma R_g d\text{vol}_g + \text{boundary terms.} \quad (230)$$

Integrating gives the Polyakov action, up to local counterterms [65]. The coefficient c is the Virasoro central charge. On the moduli space of complex structures, the same anomaly is the curvature of a determinant or Hodge line.

For a free scalar, Gaussian integration gives

$$Z[g] \propto (\det' \Delta_g)^{-1/2}$$

with zero-mode factors. The variation of the zeta-regularized determinant reproduces (230) with $c = 1$. For free fermions, ghosts, and systems of several fields, determinants and Pfaffians give the corresponding central charges. The equality between OPE central charge and determinant-line curvature is a local index theorem for that field theory.

F.8 BRST quantization of the bosonic string

In conformal gauge, the bosonic string contains a matter CFT of central charge c_m and a (b, c) ghost system of central charge -26 . The BRST current has the form

$$j_{\text{BRST}} = c \left(T_m + \frac{1}{2} T_{gh} \right) + \text{derivative term.}$$

The BRST charge $Q = \oint j_{\text{BRST}}$ satisfies

$$Q^2 \propto (c_m - 26) \sum_{n \in \mathbb{Z}} (n^3 - n) c_{-n} c_n. \quad (231)$$

Therefore $Q^2 = 0$ precisely when the total Virasoro central charge vanishes, apart from the intercept condition in the state-space formulation. This is anomaly cancellation. It is an exact equality of central extensions in the coupled matter–ghost chiral algebra.

The number 26 is not the value of a universal Koszul conductor. It is the negative central charge of the reparametrization ghost system of weights $(2, -1)$. Similar BRST systems in other reductions have different ghost contents and different anomaly polynomials.

F.9 Conformal blocks as states of quantization

Canonical quantization of Chern–Simons theory on a surface Σ produces a Hilbert space depending on the complex structure used in geometric quantization. Projective flatness identifies these spaces as the complex structure varies. The resulting projective bundle is identified with affine conformal blocks. Sewing surfaces corresponds to gluing three-manifolds and summing over intermediate representations.

For a modular tensor category with simple objects i , the dimension of the genus- g state space is

$$\dim \mathcal{H}_g = \sum_i S_{0i}^{2-2g}. \quad (232)$$

This is the Verlinde formula. The full modular functor contains the mapping-class-group representation and gluing maps, not only the dimensions [73, 47].

F.10 Cardy asymptotics

Let

$$Z(\tau, \bar{\tau}) = \text{Tr}_{\mathcal{H}} q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24}$$

be a modular-invariant torus partition function with a discrete spectrum, nonnegative degeneracies, and a unique vacuum that dominates the low-temperature limit. Applying the modular transformation $\tau \mapsto -1/\tau$ and a saddle-point estimate gives the asymptotic density

$$\log \rho(h, \bar{h}) \sim 2\pi \sqrt{\frac{c_{\text{eff}} h}{6}} + 2\pi \sqrt{\frac{\bar{c}_{\text{eff}} \bar{h}}{6}}. \quad (233)$$

The assumptions are essential. Logarithmic theories, continuous spectra, noncompact targets, and additional light states can change the asymptotics.

For asymptotically AdS_3 gravity with Brown–Henneaux boundary conditions,

$$c = \frac{3\ell}{2G}.$$

When the gravitational spectrum satisfies the hypotheses leading to (233), the Cardy entropy agrees with the Bekenstein–Hawking entropy of the BTZ black hole. The equality identifies a regime of a gravitational theory with a regime of a boundary CFT; it is not a formal consequence of the Virasoro algebra alone [12].

F.11 Entanglement entropy

For the vacuum of a unitary two-dimensional CFT on the line, the replica method computes the entropy of an interval of length L with ultraviolet cutoff ϵ :

$$S_A = \frac{c + \bar{c}}{6} \log \frac{L}{\epsilon} + s_0. \quad (234)$$

The proof uses twist fields of dimensions

$$h_n = \frac{c}{24} \left(n - \frac{1}{n} \right), \quad \bar{h}_n = \frac{\bar{c}}{24} \left(n - \frac{1}{n} \right),$$

and analytic continuation in the replica number. Equation (234) depends on the state, geometry, and full left–right CFT. A holomorphic chiral algebra supplies the central charge and twist-field constraints but not by itself the density matrix.

Relative entropy, modular Hamiltonians, and topological entanglement contain further information. They cannot be classified by the depth of a chiral obstruction tower without a theorem connecting that tower to replica correlators.

F.12 Boundary algebras in supersymmetric gauge theory

Topological or holomorphic twists of supersymmetric gauge theories often produce protected boundary operator algebras. Costello and Gaiotto construct vertex algebras associated with boundary conditions in three-dimensional $\mathcal{N} = 4$ gauge theories and relate their modules to bulk line defects [16]. Costello, Dimofte, and Gaiotto construct boundary chiral algebras for holomorphic twists of three-dimensional $\mathcal{N} = 2$ theories.

These constructions supply examples of the comparison map Φ . Mirror symmetry or duality of the bulk can induce equivalences of boundary chiral algebras or their module categories. The precise statement depends on the chosen twist and boundary condition. A boundary chiral algebra is a protected sector; it is not the complete untwisted quantum field theory.

F.13 Celestial and twistor chiral algebras

In suitable four-dimensional gauge theories, scattering amplitudes can be organized by a chiral algebra on the celestial sphere or on twistor space. Costello and Paquette derive form-factor integrands from correlators of a chiral algebra obtained from a six-dimensional holomorphic theory [17]. The OPE associativity constrains loop corrections; anomalies of the chiral algebra correspond to obstructions in the bulk theory.

The exact object is a protected holomorphic observable algebra. Mellin transformation converts momentum variables to conformal weights when passing to the celestial basis. Soft theorems provide current insertions. Equality with a complete gravitational S -matrix requires control of infrared divergences, reality conditions, and the reconstruction transform; no such equality follows from chiral algebra alone.

F.14 Calabi–Yau categories and closed strings

A smooth and proper Calabi–Yau dg category has a nondegenerate cyclic trace. Its Hochschild chains carry the Connes operator, and its Hochschild cochains carry a Gerstenhaber structure. Costello constructs a topological conformal field theory from Calabi–Yau categorical data [18]. The cyclic trace permits the contraction of graph edges, exactly as in the modular envelope.

To obtain a chiral algebra on a curve from a higher-dimensional Calabi–Yau geometry one needs a field theory or factorization algebra, a compactification or defect, and a restriction map to the curve. The resulting morphism of deformation complexes may carry cyclic classes to chiral operations. The construction is geometry-dependent. Numerical invariants such as the Euler characteristic or a Borchers-product weight do not determine the chiral operation without this morphism.

F.15 The physical synthesis

The equations derived above form one coherent structure:

$$\begin{aligned} \text{local action} &\longrightarrow \text{BV Maurer–Cartan element} \longrightarrow \text{factorization observables,} \\ &\longrightarrow \text{boundary chiral operation.} \end{aligned}$$

A propagator resolves internal edges. Fulton–MacPherson geometry resolves collisions. The quantum master equation governs boundary strata of Feynman graphs. KZ transport governs exchange of

current-algebra insertions. Conformal blocks govern sewing of states. Determinant lines govern projective anomalies. Each passage is exact when its comparison map has been constructed.

The resulting unity is physical in Feynman's sense and mathematical in Beilinson's sense. One begins with local interactions, follows their support under collision and gluing, and never equates objects merely because their formulas resemble one another.

G Exact models seen from every side

A general language is convincing only when its principal objects agree in examples by independent calculations. This appendix treats five models. In each case the chiral operation, collision coalgebra, ordered transport, modular amplitude, and field-theoretic realization are kept distinct and then compared by explicit formulas.

G.1 The rank-one Heisenberg algebra

Let H_k be generated by a field J with OPE

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \quad (235)$$

Equivalently,

$$[J_\lambda J] = k\lambda.$$

The chiral Jacobi identity is immediate because the lambda bracket is central. The full vertex algebra is the symmetric algebra on the negative modes J_{-n} , $n \geq 1$, applied to the vacuum.

Normal geometry. The singular operation in (235) is the second normal symbol of the diagonal. Under $z \mapsto f(z)$, a weight-one current transforms as a one-form, and the coefficient k pairs the two normal lines. Let $\mathfrak{h}_{\text{ch}}$ denote the abelian current chiral Lie algebra and let

$$0 \longrightarrow \omega_X \mathbf{K} \longrightarrow \widehat{\mathfrak{h}}_{\text{ch}} \longrightarrow \mathfrak{h}_{\text{ch}} \longrightarrow 0$$

be its central extension. The bracket of two current sections lands in the central line through the second normal symbol. Hence the Chevalley differential of $\widehat{\mathfrak{h}}_{\text{ch}}$ is not zero when the central cogenerator is retained; the quotient $\mathfrak{h}_{\text{ch}}$ is abelian.

Fock modules. For a momentum α , the Fock module $\mathcal{F}_\alpha$ is generated by a vector $|\alpha\rangle$ such that

$$J_n |\alpha\rangle = 0 \quad (n > 0), \quad J_0 |\alpha\rangle = \alpha |\alpha\rangle.$$

The oscillator modes satisfy

$$[J_m, J_n] = km \delta_{m+n,0}.$$

After choosing a conformal vector

$$T = \frac{1}{2k} : JJ :$$

for $k \neq 0$, the central charge is one. The character is

$$\text{ch } \mathcal{F}_\alpha(\tau) = \frac{q^{\alpha^2/(2k)-1/24}}{\eta(\tau)}. \quad (236)$$

The level k controls the momentum quadratic form, while the oscillator determinant has central charge one. This is an explicit demonstration that level and Virasoro anomaly are different invariants.

Sewing. Let T_q be the one-particle sewing operator with eigenvalues q^n . Then

$$\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n).$$

The bosonic Fock functor turns a one-particle determinant into its inverse, giving (236). On a compact Riemann surface of genus g , the nonzero-mode Gaussian integral gives a zeta-regularized determinant of the scalar Laplacian, while zero modes give a volume factor. A compactified boson with momentum lattice Λ multiplies the oscillator determinant by a Siegel theta series.

Field theory. Abelian Chern–Simons theory with action

$$\frac{k}{4\pi} \int_M A \wedge dA$$

and a chiral boundary condition has boundary current algebra (235). A Wilson line of charge q acquires the braiding phase determined by the bilinear form $q_1 q_2 / k$ after the standard normalization of charges and level. The boundary algebra, line category, and Gaussian determinant therefore arise from different operations in one free field theory.

Koszul reading. The quotient $\mathfrak{h}_{\text{ch}}$ has a cofree commutative Chevalley coalgebra with zero bracket differential. For the central extension $\widehat{\mathfrak{h}}_{\text{ch}}$, the Chevalley coderivation contracts two current cogenerators to the central cogenerator. In the level- k vacuum representation the central element acts as $k\mathbf{1}$; after reducing by the vacuum line, this contraction is represented as curvature in the reduced bar model. The completed cobar reconstructs the corresponding chiral enveloping algebra. Linear duality exchanges symmetric algebras and symmetric coalgebras under finite-weight completion.

G.2 Affine Kac–Moody currents

Let $\mathfrak{g}$ be simple and let $(,)$ be an invariant form. The universal affine vertex algebra at level k has currents with OPE

$$J^a(z)J^b(w) \sim \frac{k(a, b)}{(z - w)^2} + \frac{J^{[a, b]}(w)}{z - w}. \quad (237)$$

The lambda bracket is

$$[J_\lambda^a J^b] = J^{[a, b]} + k\lambda(a, b).$$

Its Jacobi identity has two coefficients. The coefficient independent of λ is Jacobi in $\mathfrak{g}$. The coefficient linear in the formal variables is invariance of $(,)$. The central term is not needed to repair Jacobi of the simple-pole bracket; it is a separate cocycle compatible with it.

Collision complex. For the centrally extended chiral Lie algebra, the partition collision differential has two components: the simple-pole bracket merges currents to $J^{[a,b]}$, and the central cocycle merges them to the central line. The simple-pole component squares to zero by Jacobi; compatibility with the central component is exactly invariance of $(,)$. A geometric coefficient model that includes all principal parts carries both the first and second normal symbols.

Sugawara tensor. For $k \neq -h^\vee$, the stress tensor is

$$T(z) = \frac{1}{2(k + h^\vee)} \sum_a : J^a J_a : (z), \quad (238)$$

with central charge

$$c = \frac{k \dim \mathfrak{g}}{k + h^\vee}. \quad (239)$$

The shift by $h^\vee$ comes from normal ordering. Equation (239) is rational in k ; any linear expression in $k + h^\vee$ is not the affine conformal anomaly.

KZ transport. For modules $V_1, \dots, V_n$, conformal blocks satisfy the KZ equation

$$\left(\partial_{z_i} - \frac{1}{k + h^\vee} \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j} \right) F = 0.$$

Flatness is the calculation of Section B. The monodromy is a braid-group representation. Under the Drinfeld–Kohno comparison, it agrees with the braiding of the corresponding quantum-group category at a parameter fixed by the normalization of the invariant form and the exponentiation of $(k + h^\vee)^{-1}$.

Chern–Simons theory. Three-dimensional Chern–Simons theory at integral level produces the same conformal blocks after quantization on a surface. Wilson-line braiding gives the same ribbon category. The equality is a bulk–boundary theorem involving the current algebra, not a consequence of the numerical level alone. Framing dependence corresponds to the ribbon twist and the gravitational Chern–Simons counterterm.

Critical level. At $k = -h^\vee$, (238) is undefined. The affine algebra itself remains defined and develops the Feigin–Frenkel centre. The KZ equation in the displayed normalization is singular. The centre is identified with functions on opers for the Langlands dual group. These statements explain criticality without assigning the value zero to the current-algebra level.

Koszul and representation-theoretic dualities. Chiral bar–cobar reconstruction applies in the pro-nilpotent Ran category. Quadratic PBW recognition can be tested after choosing a presentation. Kazhdan–Lusztig equivalence, Drinfeld–Kohno monodromy, Feigin–Frenkel duality, and quadratic Koszul duality are different theorems. They can be composed only when their hypotheses and categories match.

G.3 The Virasoro chiral algebra

The Virasoro algebra is the central extension of vector fields on the circle. Its OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \quad (240)$$

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity is a polynomial identity in two formal variables. It expresses the cocycle identity of the Gelfand–Fuchs central extension and the Lie bracket of vector fields.

Normal symbols. The three singular terms of (240) occupy normal degrees four, two, and one. The normally ordered product occupies degree zero. A logarithmic residue detects only the degree-one component after a logarithmic form has been supplied. The chiral operation remembers all four pieces and their extensions.

Projective connections. The transformation law

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12} \{f, z\}$$

shows that the stress tensor is an affine projective connection. The Schwarzian cocycle is the coordinate form of the Virasoro central extension. On moduli, it gives a projective connection on conformal blocks. The Hodge coefficient is proportional to c , while insertions and boundary factorization contribute further classes.

Verma modules and null vectors. A highest-weight module $M(c, h)$ is generated by v with

$$L_0 v = h v, \quad L_n v = 0 \quad (n > 0).$$

The Kac determinant detects singular vectors. Passing from the universal Virasoro algebra to a minimal-model quotient changes the homological algebra because relations generated by null vectors are imposed. Koszulness of the universal object cannot be transferred to every simple quotient without a separate calculation.

The bosonic string. The (b, c) reparametrization ghosts have central charge -26 . Coupling them to a matter theory gives a nilpotent BRST charge when the total central charge and intercept anomaly vanish. The number 26 is therefore the cancellation value of a specific ghost system. The value 13 is the midpoint of the reflection $c \mapsto 26 - c$; it is not by itself a theorem of Virasoro self-duality.

Polyakov and determinant geometry. The Weyl anomaly has coefficient c . For free fields it is computed from determinants. The curvature of the determinant line agrees with the central extension represented by (240). This is the rigorous bridge between local OPE and global moduli curvature. It does not imply that the Virasoro algebra determines a full nonchiral CFT or a gravitational path integral.

Three-dimensional gravity. Brown–Henneaux boundary conditions produce two Virasoro algebras with

$$c = \bar{c} = \frac{3\ell}{2G}.$$

Under modular invariance, a sparse light spectrum, and vacuum dominance, Cardy asymptotics reproduce the BTZ entropy. The chiral algebra supplies the asymptotic symmetry and central extension. The spectrum and modular invariant are additional physical data.

G.4 Free fermions and ghost systems

A free complex fermion has fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w}.$$

The associated Clifford algebra is quadratic. Wick’s theorem expresses every correlator as a determinant. On a Riemann surface, a choice of spin structure enters the Szego kernel. The partition function is a determinant or Pfaffian of the Dirac operator.

A (b, c) system with conformal weights $(\lambda, 1 - \lambda)$ has stress tensor

$$T = (1 - \lambda):(\partial b)c: - \lambda:b\partial c:$$

and central charge

$$c_{bc} = 1 - 3(2\lambda - 1)^2. \quad (241)$$

A bosonic (β, γ) system with the same weights has the opposite determinant statistics and central charge

$$c_{\beta\gamma} = -c_{bc}.$$

The cancellation in a tensor product is a determinant identity and a Virasoro central-charge identity. The associated graded polynomial and exterior algebras are ordinary Koszul duals. The Weyl and Clifford algebras are PBW deformations of these two algebras and are related by the corresponding curved or inhomogeneous Koszul formalism when their constant quadratic terms are retained. These facts do not imply equality of their module categories as conformal theories.

Ghost number provides a second grading. BRST reduction uses the ghost differential and the vertex differential simultaneously. Spectral-sequence concentration can make the reduction exact. When concentration fails, higher BRST cohomology remains and cannot be discarded.

G.5 Lattice vertex algebras

Let Λ be an even lattice with bilinear form $(,)$ and let ε be a normalized two-cocycle on Λ . The lattice vertex algebra is

$$V_\Lambda = M(1) \otimes_{\mathbb{C}} [\Lambda].$$

For $\alpha, \beta \in \Lambda$, the vertex operators have leading product

$$Y(e^\alpha, z)Y(e^\beta, w) = (z - w)^{(\alpha, \beta)} \varepsilon(\alpha, \beta) Y(e^{\alpha+\beta}, w) + \cdots. \quad (242)$$

Evenness gives locality. The cocycle controls the ordered phase. Replacing ε by a cohomologous cocycle gives an isomorphic algebra; replacing it by its inverse reverses the commutator phase.

The genus-one character is

$$\text{ch } V_\Lambda(\tau) = \frac{\Theta_\Lambda(\tau)}{\eta(\tau)^{\text{rk } \Lambda}}. \quad (243)$$

For an even unimodular positive-definite lattice, the chiral central charge is $\text{rk } \Lambda$ and the theory is holomorphic. At genus g , the zero-mode sum is the degree- g Siegel theta series

$$\Theta_\Lambda^{(g)}(\Omega) = \sum_{x_1, \dots, x_g \in \Lambda} \exp \left(\pi i \sum_{i,j} (x_i, x_j) \Omega_{ij} \right).$$

The oscillator and lattice pieces are independent factors.

For the two even unimodular rank-sixteen lattices $E_8 \oplus E_8$ and D_{16}^+ , the Siegel theta series agree in degrees $g = 1, 2, 3$. In degree four their difference is a nonzero scalar multiple of the Schottky form on $\mathfrak{H}_4$; its restriction to the genus-four Jacobian locus vanishes [59, 66]. Thus genus two does not separate the two lattice theories through their theta series, and even the degree-four difference disappears after restriction to smooth genus-four curves.

The Leech lattice VOA and the Moonshine VOA both have central charge 24. Their weight-one spaces differ: the Leech theory has a rank-twenty-four Heisenberg sector, while Moonshine has no weight-one states. Therefore central charge does not determine the chiral algebra, its OPE, or its automorphism group.

G.6 Principal W -algebras

Let $\mathfrak{g}$ be simple and f principal nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from the affine algebra, charged ghosts, and neutral fields when required. Its degree-zero cohomology is the principal W -algebra $\mathcal{W}^k(\mathfrak{g}, f)$ under the standard hypotheses.

For the universal principal W -algebra at generic level, the standard strong generators have conformal weights $d_i + 1$, where d_i are the exponents of $\mathfrak{g}$; special levels may impose decoupling relations or collapse generators. Their OPEs contain nonlinear normally ordered polynomials. These terms are transferred operations of the BRST contraction; they are not captured by the simple-pole Lie bracket alone.

Feigin–Frenkel duality relates principal W -algebras for Langlands dual Lie algebras at levels satisfying

$$r^\vee(k + h^\vee)(\ell + {}^L h^\vee) = 1, \quad (244)$$

with the lacing convention included. The relation is multiplicative in shifted levels. It is not the affine reflection $k \mapsto -k - 2h^\vee$.

If a filtered BRST contraction is compatible with the chiral bar construction and its spectral sequence converges, then bar and BRST cohomology commute. This transports collision operations and Koszul comparisons through the reduction. The conclusion is a theorem about the specified contraction. It is not automatic for every nilpotent orbit or every completion.

G.7 Five comparisons in one table

Model	highest local pole	ordered datum	modular datum	physical realization
Heisenberg	double pole k	abelian phase	oscillator determinant and zero modes	abelian Chern–Simons boundary
affine	double pole plus Lie pole	KZ braid monodromy	affine conformal blocks	nonabelian Chern–Simons/WZW
Virasoro	fourth-order central pole	monodromy of Virasoro blocks	projective connection and determinant line	conformal anomaly; AdS_3 asymptotic symmetry
free ghosts	simple contraction	fermionic or bosonic exchange	determinant/Pfaffian	gauge fixing and BRST
lattice	integral power and cocycle	cocycle phase	Siegel theta series	compactified free boson
W -algebra	nonlinear higher-spin OPE	module-category braiding when constructed	W -conformal blocks	Hamiltonian reduction

The columns are not functions of the first column alone. The local pole controls the singular operation. Ordered transport requires modules and a connection. Modular data requires sewing. A physical realization requires an action and boundary condition. The exact models show that the same chiral operation can feed all four constructions without making them identical.

G.8 What the examples establish

The examples test the common language by independent constructions whose comparison maps are known.

For Heisenberg theory, the boundary quantization map identifies the current OPE with the abelian Chern–Simons boundary bracket, while Gaussian sewing identifies the oscillator trace with the determinant formula.

For affine theory, the Ward identities produce the KZ connection; Drinfeld–Kohno identifies its braid category with the corresponding quantum-group category, and the Chern–Simons/WZW comparison identifies the same braid data with Wilson-line exchange.

For Virasoro theory, the fourth normal symbol gives the central cocycle, the Ward identity gives the Schwarzian transformation, and the family index theorem identifies its moduli-space image with determinant-line curvature and the Weyl anomaly.

For ghost systems, the Clifford or Weyl quadratic relations determine the parity signs, the corresponding determinant ratio gives the central charge, and the BRST square measures the same total anomaly after the BRST comparison is made.

For lattice theories, the group-algebra cocycle determines ordered phases, Poisson summation gives the modular transformation of the theta series, and the compactified-boson path integral yields the same theta factor after zero-mode and oscillator separation.

Each agreement is mediated by an explicit construction. This is the standard of evidence appropriate to chiral Koszul duality: one operation, calculated in several faithful realizations, with every passage preserving the category of the object.

H Exact distinctions, historical lineage, and the common language

A unifying language is useful only when it preserves distinctions. The purpose of this appendix is to state the logical relations among the objects of the paper, to place them in their historical setting, and to formulate the mathematical and physical readings without changing the type of any construction.

H.1 The primitive object

Fix a smooth complex curve X . The primitive datum of this paper is

$$A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}). \quad (245)$$

Thus A consists of a factorisable D -module together with a unital associative multiplication

$$\star : A \otimes^{\mathrm{pt}} A \longrightarrow A$$

and an augmentation $A \rightarrow \mathbf{1}$. The factorisable D -module carries the holomorphic collision data; the E_1 multiplication carries transverse ordered composition. These structures satisfy the interchange law because $\star$ is a morphism in $\mathrm{FactD}(X)$.

A diagonal-supported chiral Lie operation

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

is a separate and important specialization of the chiral coefficient. It gives the Francis–Gaitsgory Chevalley branch, but it is not the source of the transverse E_1 multiplication and does not replace (245).

Construction	Output	Additional datum or hypothesis
ordered interval bar	conilpotent coassociative coalgebra in $\text{FactD}(X)$	augmentation; completion when infinite sums occur
boundary endomorphisms	$A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1})$	compactness and completion for double duality
quadratic comparison	presentation coalgebra $A^i \rightarrow \text{Bar}_X^{\text{ord}}(A)$	chosen quadratic or filtered presentation
chiral Chevalley	commutative factorisation coalgebra on $\text{Ran } X$	a chiral Lie coefficient and Ran pronilpotence
formal-disc realization	vertex, quantum-vertex, or nonlocal-vertex operations	coordinate, translation, and when present exchange data
Verdier dual	dual constructible or holonomic object	dualisability, continuity, and monoidal base change
E_2 enhancement	exchange paths and Yang–Baxter coherence	a transverse plane or equivalent flat exchange structure
Tannaka reconstruction	bialgebra/Hopf or coquasi/quasi-Hopf object	rigid tensor category and strong or quasi-monoidal fibre functor
two sewing structures	bicoloured stable-graph operations	transverse cyclic trace and chiral modular sewing
conformal blocks	sheaf or complex on moduli	modules, finiteness, projective connection, and sewing
internal Hochschild cochains	E_2 derived centre	homotopically correct diagonal bimodule resolution
physical observables	analytic holomorphic–topological factorisation algebra	fields, action, gauge fixing, renormalisation, and an algebraisation comparison

An equality between two rows requires a comparison morphism. The existence of a suggestive analogy does not supply that morphism.

H.2 Bar, Koszul dual, Verdier dual, and centre

Four neighboring words refer to four different constructions.

Bar. For an augmented algebraic object A , the bar construction is a coalgebraic resolution. Its universal property is expressed by twisting morphisms. It exists under augmentation and conilpotence conditions whether or not A is quadratic or Koszul.

Quadratic Koszul dual. A quadratic presentation $A = T(V)/(R)$ defines a quadratic coalgebra A^i and, when finite duals exist, an algebra $A^! = (A^i)^\vee$. The canonical map $A^i \rightarrow B(A)$ is a quasi-isomorphism exactly on the Koszul locus. The quadratic dual depends on the presentation until that comparison has been proved invariant.

Verdier dual. Verdier duality is a contravariant functor on sheaves or D -modules. Applied to a factorization coalgebra it may produce a factorization algebra when duality commutes with the

relevant tensor products, diagonals, and limits. This is a geometric duality, not the universal bar–cobar counit.

Derived centre. The derived centre is Hochschild cochains of an associative object. For a chiral Lie algebra $\mathfrak{g}$, the typed definition passes through its chiral enveloping factorization algebra:

$$Z_{\text{ch}}^{\text{der}}(\mathfrak{g}) = \text{HH}_{\text{Fact}}^{\bullet}(U^{\text{ch}}(\mathfrak{g}), U^{\text{ch}}(\mathfrak{g})).$$

It carries an E_2 structure. It is neither the bar coalgebra nor a scalar trace of the bar coalgebra.

The four constructions can be related on favorable finite and perfect loci, but none is a synonym for another.

H.3 Four squares and four curvatures

The notation d^2 occurs in several theories. The symbols are equal only when the underlying operators are equal.

- (i) An internal differential on a dg object satisfies $d^2 = 0$.
- (ii) In a curved A_{∞} algebra, m_1^2 is an inner operation determined by m_0 together with higher terms.
- (iii) A connection satisfies $\nabla^2 = F_{\nabla}$, a two-form on the base with values in endomorphisms.
- (iv) A BV quantum differential has square equal to the anomaly operator unless the quantum master equation holds.

A central scalar curvature of a projective connection does not make an internal chiral differential curved. A central element has zero commutator, so the formula $d^2 = [m_0, -]$ cannot represent nonzero scalar projective curvature. An anomaly class can be cancelled by a counterterm when it is exact. A projective curvature can be cancelled by tensoring with a line of opposite Atiyah class when that line exists. These are different cancellation mechanisms.

H.4 Residue, principal part, and propagator

Three local objects are also distinct.

Residue. Poincare residue extracts the coefficient of a logarithmic normal form. It is a map of line-bundle weight one.

Principal part. The coefficient of a pole of order m is a section of $N_{D/Y}^{\otimes m}$. The direct sum over all m is the associated graded of the pole filtration. It contains the singular modes of a chiral operation.

Propagator. A propagator is a Green kernel or parametrix selected by a kinetic operator and a gauge-fixing condition. It is analytic data. Its singular part may represent a principal part, but its regular and harmonic parts depend on global geometry.

The logarithmic form $d\log(z - w)$ is a canonical generator in the cohomology of a two-point configuration on an affine screen. It is not the universal propagator of every chiral theory and does not extract arbitrary OPE poles.

H.5 Ordering, descent, and braiding

Labelled configurations remember the pure braid group. Passing to unlabelled configurations is descent along a finite cover. Over a field of characteristic zero, homotopy invariants and homotopy coinvariants for the finite symmetric group are related by averaging, but the local system on the labelled configuration space can have nontrivial braid monodromy before descent.

A vector-space splitting of an averaging map is not a Drinfeld associator. An associator is a group-like series satisfying the pentagon and hexagon equations. It compares different parenthesized monodromies. The Grothendieck–Teichmüller group acts on associators, not on arbitrary linear splittings.

Verdier duality sends a local system with monodromy ρ to the dual local system with monodromy ρ^{-T} . An equality $R \mapsto R^{-1}$ follows when the chosen bilinear form identifies dual and original representations and when conventions convert inverse transpose to inverse. An equality of named quantum groups requires a Tannakian or presentation-theoretic recognition theorem.

H.6 Modular sewing and the master equation

The compactification $\overline{\mathcal{M}}_{g,n}$ has boundary divisors indexed by stable graphs with one edge. Codimension-two strata correspond to graphs with two edges. Their oriented incidence relation gives the differential in the Feynman transform. If a cyclic object assigns compatible operations to every clutching morphism, the total operation satisfies the quantum master equation

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0.$$

The equation follows from the compatibility maps. It does not construct them.

For conformal blocks, sewing is a theorem under the finiteness and representation-theoretic assumptions of the modular-functor construction. The connection is projectively flat. Its first Chern class contains a Hodge contribution proportional to the central charge, insertion contributions from conformal weights, and boundary contributions from factorization channels. Therefore no single scalar classifies the modular theory.

H.7 The mathematical history

The language assembled here was formed by a sequence of exact advances.

Historical note H.1 (Distributions). Dirac’s delta function made an operation supported on a submanifold calculable. Schwartz’s distribution theory made support and differentiation intrinsic. A chiral bracket is the holomorphic, D -module-valued descendant of this idea: the product is defined away from the diagonal and its singular commutator is supported on the diagonal.

Historical note H.2 (Operator products). Wilson separated short-distance singularity from long-distance state dependence. The operator product expansion is a local asymptotic expansion whose coefficients are local fields. Vertex algebra theory replaces convergence questions by a formal identity. Chiral algebra theory replaces a chosen coordinate by a geometric correspondence.

Historical note H.3 (Borcherds and Beilinson–Drinfeld). Borcherds recognized that locality, commutator formulas, and associativity are coefficient forms of one formal-distribution identity. Beilinson and Drinfeld placed that identity on a curve: poles are allowed away from diagonals, while operations land on diagonals. The formal variable becomes the normal direction to a geometric collision.

Historical note H.4 (Fulton–MacPherson and Kontsevich). Fulton and MacPherson resolved simultaneous collisions by adding screens that record relative configurations. Kontsevich used compactified configuration spaces to organize deformation quantization. In perturbative field theory the same spaces separate ultraviolet regions and make the forest structure of counterterms geometric.

Historical note H.5 (Ran space and chiral Koszul duality). The Ran space regards a finite subset of a curve as one variable. Factorization is multiplication over disjoint unions of finite subsets. Francis and Gaitsgory proved that, in the pro-nilpotent chiral monoidal category, chiral Lie algebras and conilpotent commutative factorization coalgebras are related by Chevalley chains and primitives. This is the categorical core of chiral Koszul duality.

Historical note H.6 (Braids and quantum groups). Knizhnik and Zamolodchikov produced a flat connection from current algebra. Kohno and Drinfeld identified its braid monodromy with the braided tensor structure of quantum groups. Witten’s Chern–Simons theory gave the same braid representations as Wilson-line amplitudes. Costello–Witten–Yamazaki obtained rational, trigonometric, and elliptic integrable systems from four-dimensional gauge theory. These are exact comparisons of monodromy categories, not consequences of symmetrization alone.

Historical note H.7 (Polyakov, Segal, and modular geometry). Polyakov’s path integral made the conformal anomaly a curvature on the space of metrics. Segal formulated conformal field theory by sewing Riemann surfaces. Tsuchiya–Ueno–Yamada constructed conformal blocks in families. The determinant and Hodge lines record the projective anomaly. Getzler–Kapranov’s modular operads encode the topology of sewing independently of the analytic convergence of amplitudes.

Historical note H.8 (Feynman and BV). Feynman’s graphs are the perturbative expansion of an action. The Batalin–Vilkovisky formalism turns gauge invariance into the classical and quantum master equations. Costello’s renormalization theory constructs effective interactions on compactified configuration spaces and identifies anomalies as local cohomology classes. The stable-graph master equation of a cyclic chiral object has the same algebraic shape; an equality of theories requires a map identifying their fields, pairings, and vertices.

H.8 The physical dictionary

Let Obs be a factorization algebra of quantum observables on a manifold whose protected boundary geometry is $X^{\text{an}} \times \mathbb{R}$. Suppose there is a quasi-isomorphism of $E_1 \otimes \text{Ch}_X$ factorization objects

$$\Phi : A^{\text{an}} \xrightarrow{\sim} \text{Obs}_{\partial}. \quad (246)$$

For an enveloping example one may take $A = U^{\text{ch}}(\mathfrak{g})$ with its supplied transverse product. The following identifications are then exact consequences of Φ .

Chiral geometry	Quantum-field-theoretic realization
D -module of fields	cochain complex of boundary local observables
chiral operation	renormalized boundary OPE
partial diagonal	coincidence locus of insertions
principal-part filtration	scaling hierarchy of contact singularities
partition collision complex	local short-distance factorization complex
Fulton–MacPherson screen	relative ultraviolet configuration
chiral Chevalley coalgebra	coalgebra of factorized insertions
ordered local system	exchange transport of defects or lines
cyclic pairing	two-point function used to sew legs
modular graph operation	perturbative amplitude with fixed topology
BV anomaly class	obstruction to preserving the quantum master equation
Hochschild centre map	bulk observable acting centrally on the boundary

Without (246), the two columns are mathematically analogous constructions. With it, they are two models of the same operations.

H.9 Exact statements that do not follow

The following implications are false without additional hypotheses.

No-go theorem H.9 (No quantum group from dimension loss). *The dimension of the kernel of $V^{\otimes n} \rightarrow \text{Sym}^n V$ does not determine an r -matrix, associator, or quantum group. It contains no multiplication, coproduct, monodromy, or coherence equation.*

No-go theorem H.10 (No modular object from topology alone). *The equality $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ does not assign coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been given.*

No-go theorem H.11 (No bulk reconstruction from a boundary centre alone). *The derived centre is terminal among algebraic central actions on the boundary. It does not detect a decoupled bulk sector. Therefore a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is proved to be an equivalence.*

No-go theorem H.12 (No classification by central charge). *Central charge determines the Virasoro anomaly coefficient. It does not determine the full chiral operation, representation category, braid monodromy, or higher-genus conformal blocks. Holomorphic theories at the same central charge provide immediate examples.*

No-go theorem H.13 (No identification of anomalies of different types). *A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borchers-product weight are invariants in different theories. They may be related by a theorem in a particular family. They are not one universal scalar.*

H.10 The common language as a commutative diagram

The subject can be expressed by the following diagram. Solid arrows are universal for the stated input; arrows labelled by extra structure exist only when that structure has been supplied.

$$\begin{array}{ccccc}
& A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)) & & & \\
& \swarrow \text{Bar}_X^{\text{ord}} \quad \searrow \text{HH}_{\text{FactD}}^\bullet & & & \\
\text{Coalg}_{\text{coAss}}^{\text{conil}}(\text{FactD}(X)) & \xleftarrow{\text{Cobar}_X^{\text{ord}}} A' \xrightarrow{\text{RHom}_A(1, -)} Z_{\text{ord, ch}}(A) & \xrightarrow{E_2/E_3 \text{ extension}} \mathcal{L}_A & & (247) \\
& \downarrow \text{RHom}_A(1, 1) & \nearrow \text{bulk action} & & \\
& \text{Obs}_{\partial}(X \times \mathbb{R}) & & & \\
& & & \downarrow \text{Tannaka} & \\
& & & \mathcal{H}_A &
\end{array}$$

A chiral Lie coefficient $\mathfrak{g}$ supplies the independent branch

$$\mathfrak{g} \xrightarrow{\text{coChev}^{\text{ch}}} \text{CoFact}_{\text{conil}}(\text{Ran } X),$$

and a comparison with the ordered interval bar must be a monoidal theorem, not an identification by notation. A transverse cyclic trace and a chiral modular sewing system add the bicoloured Feynman transform and its master element Θ_A^{bi} . A field-theoretic realization adds propagators, renormalisation, and states. The common source is the ordered chiral algebra together with the extra structures displayed on the relevant arrows.

H.11 The final principle

The mathematical phenomenon is the coexistence of holomorphic collision and transverse associative composition. The physical phenomenon is the short-distance product of observables together with their ordered fusion along a topological direction. The geometric carrier is the operadic product of chiral collision spaces and oriented intervals. The algebraic laws are the chiral identities of the coefficient, associativity of the E_1 multiplication, and their interchange. Chiral Koszul duality passes to factorised coalgebraic states in the Ran direction; ordered bar–cobar duality passes to interval states in the transverse direction. Ordering remembers composition, exchange data remembers transport, cyclicity permits loops, moduli record sewing, and the derived centre records universal central action.

These clauses form one statement because every noun has a category and every verb is a functor or a correspondence. The unity of the subject is not the equality of all its invariants. It is the fact that each invariant is obtained from the same ordered chiral object, with explicitly named additional structures, by an exact geometric construction.

I Historical lineage of the language

I.1 Distributions and local fields

Dirac’s delta distribution made support a mathematical datum rather than a limiting metaphor [27]. Quantum field theory inherited this lesson: products of fields are defined away from coincidence and acquire contact terms on diagonals. The chiral D -module operation is the algebraic form of that distributional fact. It records the singular product and its support before a coordinate expansion is chosen.

Wilson’s operator product expansion separated short-distance coefficients from local operators

[75]. Borchers and the vertex-algebra tradition encoded the consistency of those coefficients in formal distributions. Beilinson and Drinfeld replaced the formal coordinate by a curve and the formal delta function by a morphism of D -modules supported on the diagonal. The present paper uses the Beilinson–Drinfeld object as the chiral coefficient of a genuinely associative ordered product.

I.2 Time ordering and the associative direction

Feynman’s time-ordered product is not commutative multiplication [39]. It depends on an ordering of insertion times, and its perturbative expansion is indexed by graphs. Stasheff’s associahedra later supplied the homotopy geometry of all parenthesizations [68, 69]. Factorization homology identifies the same geometry with the bar construction on an interval.

The ordered chiral algebra combines these two historical lines. Dirac and Beilinson govern support on the complex diagonal. Feynman and Stasheff govern composition along the ordered real direction.

I.3 Configuration spaces and renormalization

Arnold computed the cohomology of configuration spaces and isolated the three-term relation that underlies the flatness of logarithmic connections [4]. Fulton and MacPherson compactified configurations by replacing collisions with screens [40]. Axelrod–Singer and Kontsevich used related compactifications to make perturbative gauge-theory and deformation-quantization integrals finite and compatible with boundary strata [9, 53].

The compactification has two roles that must remain distinct. Its face category resolves collision incidence. Its screen spaces carry nontrivial cohomology and propagator integrals. The first is combinatorial; the second contains the analytic weight of Feynman graphs.

I.4 Chiral algebras and Koszul duality

Koszul, Priddy, Quillen, and the operadic tradition developed the passage between algebraic operations and coalgebraic resolutions. Beilinson–Drinfeld placed local Lie operations on the Ran space. Francis–Gaitsgory proved the corresponding chiral Koszul duality in a pro-nilpotent monoidal category.

The associative extension considered here uses the same Ran geometry with one additional ordered operadic direction. Its bar is the interval bar internal to factorization D -modules. Its dual is the derived endomorphism algebra of the augmentation boundary. This is the form of Koszul duality that appears in defect physics.

I.5 Anomalies and geometry

Polyakov identified the Weyl anomaly with a local action for the conformal factor [65]. Quillen and the family index theorem identified the same anomaly with curvature of a determinant line [62]. Belavin and Knizhnik related determinant geometry to string measures on moduli [11]. These are different realizations of a central extension of conformal transformations.

The language of the present paper preserves the chain of maps: local stress-tensor cocycle, Ward identity, Atiyah algebra of conformal blocks, determinant-line curvature. It does not replace the chain by a universal scalar.

I.6 Gauge theory, knots, and quantum groups

Witten’s Chern–Simons theory connected three-dimensional gauge fields, two-dimensional current algebras, conformal blocks, braid groups, and knot invariants [74]. Drinfeld and Jimbo supplied the quantum groups [29, 48]; Kohno identified their braid representations with KZ monodromy [51]. The decisive structure is the category of lines and its exchange, not the dimension of an invariant subspace.

Costello, Witten, and Yamazaki added a holomorphic spectral curve to topological gauge theory [19, 20]. Wilson-line OPEs then produced rational, trigonometric, and elliptic quantum groups. Gauge-theoretic constructions of Gaiotto, Moore, and Neitzke and of Nekrasov placed line defects, framed BPS states, Omega backgrounds, and BPS/CFT Ward identities in the same protected sector [41, 57]; Coulomb-branch convolution gives a complementary algebraic realization [10]. The spectral curve carries chiral dependence; transverse topology carries fusion and exchange.

I.7 Factorization algebras and BV theory

Costello and Gwilliam formulated perturbative quantum field theory as a factorization algebra of observables [14, 15]. The BV classical and quantum master equations became local cohomological identities. Boundary and defect observables became factorization modules. This made the passage from an action functional to a chiral or topological operator algebra a functor with a defined source and target.

The present construction is compatible with that formalism. An ordered chiral algebra is the algebraic local observable structure on a holomorphic–topological product. A BV theory is one realization of it, together with fields, pairing, integration, and renormalization.

I.8 Open–closed theories and centres

Hochschild cochains have long governed deformations of associative algebras. Deligne’s conjecture and the Swiss-cheese operad identified their E_2 structure and their universal action on an E_1 algebra. In open–closed field theory, the centre is the universal closed algebra acting on the open sector.

This universal property is exact but limited. It reconstructs the protected bulk from a boundary only when the bulk-to-centre map is an equivalence. Witten’s open-string field theory and Costello’s Calabi–Yau categories supply settings in which Hochschild chains and cyclic operations are physical closed states [76, 18]. Twisted holography supplies settings in which a boundary algebra determines a protected bulk sector.

I.9 The resulting synthesis

The historical constructions meet in one typed sentence. A noncommutative chiral algebra is an E_1 algebra in factorization D -modules; its chiral coefficient is a diagonal-supported distribution, its multiplication is ordered fusion, its bar is interval factorization homology, its centre is the universal exchange algebra, its quantum group is Tannaka reconstruction of a braided line category, its ribbon loops come from a cyclic trace, its worldsheet sewing comes from conformal blocks, and its anomalies are invertible theories identified by transgression.

Every clause has an independent historical origin. Their unity is the geometry of support, order, exchange, and sewing.

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